Answers to Questions from the RMS

Contents

1 Question Q565

Informal verdict. **A — Complete solution**, with **high confidence**.

Lean coverage. Lean proves the unconditional first relative correction for Bell numbers, including the missing saddle-window and tail estimates and exact normalization.

1.1 Audited informal answer

1.1.1 Faithful restatement

Let $T(n)$ be the n-th Bell number, written in Dobinski form as

$$T(n) = e^{-1} \sum_{k=0}^{\infty} \frac{k^n}{k!}.$$

Let $w : [0, \infty$\to[1,\infty)) be the inverse of $y \mapsto y \log y$; thus

$$w(x) \log w(x) = x.$$

The earlier response R506 established

$$T(n) \sim f(n), \qquad f(x) = \frac{w(x)^x e^{,w(x)-x-1}}{\sqrt{\log x}}.$$

Q565 asks for the next term in this asymptotic expansion. There are no subparts or footnotes. There is no apparent mathematical typo; the only implicit domain point is that the displayed real-valued formula for $f(x)$ requires $x > 1$, which is harmless in an asymptotic question as $n \to \infty$.

Throughout, log denotes the natural logarithm.

1.1.2 Main result

Set

$$L = \log n, \qquad \ell = \log \log n.$$

Then, as $n \to \infty$ through the positive integers,

$$\boxed{T(n) = f(n) \left[1 + \frac{\ell - 1}{2L} + \frac{3\ell^2 - 10\ell + 3}{8L^2} + O\left(\frac{\ell^3}{L^3} \right) \right].}$$

In particular, the requested next term is

$$\boxed{T(n) = f(n) + \frac{\log \log n - 1}{2 \log n}, f(n) + O\left(f(n) \frac{(\log \log n)^2}{(\log n)^2} \right).}$$

Consequently,

$$\boxed{T(n) - f(n) \sim \frac{\log\log n}{2\log n}, f(n).}$$

The expression with $\log\log n - 1$ is the more accurate first correction; the last display gives its simpler asymptotic equivalent.

A stronger saddle-point form is also valid. If

$$r = \log w(n),$$

so that $re^r = n$ and $w(n) = e^r = n/r$, then

$$\boxed{T(n) = \frac{w(n)^n e^{,w(n)-n-1}}{\sqrt{1+r}} \left[1 - \frac{r^2(2r^2+7r+10)}{24n(r+1)^3} + O\left(\left(\frac{r}{n}\right)^{3/2}\right)\right].}$$

All O-constants here are absolute, and the estimates hold uniformly as $n \to \infty$ through the integers.

1.2 Proof

1.2.1 1. Exponential generating function

Write $B_n = T(n)$. From Dobinski's formula,

$$\begin{aligned}\sum_{n=0}^{\infty} B_n \frac{z^n}{n!} &= e^{-1} \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{n=0}^{\infty} \frac{(kz)^n}{n!} \\ &= e^{-1} \sum_{k=0}^{\infty} \frac{e^{kz}}{k!} == e^{-1} e^{e^z} \exp(e^z - 1).\end{aligned}$$

The interchange is justified by absolute and locally uniform convergence. Hence

$$\frac{B_n}{n!} = [z^n] \exp(e^z - 1).$$

Let $r > 0$ be the unique solution of

$$re^r = n.$$

This is the saddle-point equation. Put

$$b = n(r+1).$$

Cauchy's coefficient formula on the circle $|z| = r$ gives

$$\frac{B_n}{n!} = \frac{e^{e^r - 1}}{2\pi r^n} J_n,$$

where

$$J_n = \int_{-\pi}^{\pi} e^{\Phi(\theta)}, d\theta, \qquad \Phi(\theta) = e^{re^{i\theta}} - e^r - in\theta.$$

The choice $re^r = n$ ensures $\Phi'(0) = 0$.

1.2.2 2. Negligibility of the noncentral arc

For $-\pi \le \theta \le \pi$,

$$\begin{aligned} \Re\Phi(\theta) &= e^{r\cos\theta}\cos(r\sin\theta) - e^r \\ &\le e^{r\cos\theta} - e^r \\ &= -e^r\left(1 - e^{-r(1-\cos\theta)}\right). \end{aligned}$$

There is an absolute $c > 0$ such that

$$1 - e^{-x} \ge c\min(x, 1) \qquad (x \ge 0),$$

and

$$1 - \cos\theta \ge \frac{2\theta^2}{\pi^2} \qquad (|\theta| \le \pi).$$

Since $e^r = n/r$, it follows that

$$\Re\Phi(\theta) \le -c\min(n\theta^2, n/r). \tag{1}$$

Define

$$q = \frac{r}{n} = e^{-r} = \frac{1}{w(n)}$$

and choose

$$T = q^{-1/20}, \qquad \theta_0 = \frac{T}{\sqrt{b}}.$$

Then

$$n\theta_0^2 = \frac{T^2}{r+1} = \frac{q^{-1/10}}{r+1} \longrightarrow \infty.$$

Consequently, by 1,

$$\int_{\theta_0 \le |\theta| \le \pi} \left| e^{\Phi(\theta)} \right|, d\theta \le 2\pi \exp\left(-c\frac{q^{-1/10}}{r+1} \right). \tag{2}$$

This is smaller than $b^{-1/2}q^M$ for every fixed $M > 0$. Thus only the arc $|\theta| \le \theta_0$ contributes to any algebraic order in q.

1.2.3 3. Expansion near the saddle

Direct differentiation gives

$$\Phi''(0) = -n(r+1) = -b,$$

$$\Phi'''(0) = -i, n(r^2 + 3r + 1),$$

and

$$\Phi^{(4)}(0) = n(r^3 + 6r^2 + 7r + 1).$$

Moreover, on the central arc,

$$|\Phi^{(5)}(\theta)| \le Cnr^4$$

for an absolute constant C. Taylor's formula therefore yields

$$\Phi(\theta) = -\frac{b}{2}\theta^2 - \frac{ic}{6}\theta^3 + \frac{d}{24}\theta^4 + O(nr^4|\theta|^5), \tag{3}$$

where

$$c = n(r^2 + 3r + 1), \qquad d = n(r^3 + 6r^2 + 7r + 1).$$

Set $t = \sqrt{b}, \theta$. Then 3 becomes

$$\Phi\left(\frac{t}{\sqrt{b}}\right) = -\frac{t^2}{2} - i\lambda_3 t^3 + \lambda_4 t^4 + O(q^{3/2}|t|^5), \tag{4}$$

where

$$\lambda_3 = \frac{c}{6b^{3/2}} = O(q^{1/2}), \qquad \lambda_4 = \frac{d}{24b^2} = O(q).$$

For $|t| \le T = q^{-1/20}$, all the perturbation terms in 4 tend uniformly to zero. Expanding the exponential gives, for some fixed polynomial P,

$$\exp\Big(-i\lambda_3 t^3 + \lambda_4 t^4 + O(q^{3/2}|t|^5)\Big) = 1 - i\lambda_3 t^3 + \lambda_4 t^4 - \frac{1}{2}\lambda_3^2 t^6 + O\big(q^{3/2}P(|t|)\big). \tag{5}$$

The term in t^3 integrates to zero over the symmetric interval. The standard Gaussian moments, obtained by integration by parts, are

$$\frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} t^4 e^{-t^2/2}, dt = 3, \qquad \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} t^6 e^{-t^2/2}, dt = 15.$$

Using 2–5, extending the Gaussian integral from ([-T,T]) to $\mathbb{R}$, and noting that the resulting tails are exponentially small, we obtain

$$J_n = \sqrt{\frac{2\pi}{b}}\left[1 + 3\lambda_4 - \frac{15}{2}\lambda_3^2 + O(q^{3/2})\right]. \tag{6}$$

Now

$$3\lambda_4 = \frac{r^3 + 6r^2 + 7r + 1}{8n(r+1)^2},$$

whereas

$$\frac{15}{2}\lambda_3^2 = \frac{5(r^2 + 3r + 1)^2}{24n(r+1)^3}.$$

Putting these over a common denominator gives

$$3\lambda_4 - \frac{15}{2}\lambda_3^2 = -\frac{2r^4 + 9r^3 + 16r^2 + 6r + 2}{24n(r+1)^3}. \tag{7}$$

Consequently,

$$\frac{B_n}{n!} = \frac{e^{e^r - 1}}{r^n\sqrt{2\pi n(r+1)}}\left[1 - \frac{2r^4 + 9r^3 + 16r^2 + 6r + 2}{24n(r+1)^3} + O(q^{3/2})\right]. \tag{8}$$

1.2.4 4. Incorporating Stirling's expansion

We use Stirling's formula with its first correction:

$$n! = \sqrt{2\pi n}\left(\frac{n}{e}\right)^n \left[1 + \frac{1}{12n} + O(n^{-2})\right]. \tag{9}$$

For completeness, 9 follows from the Euler–Maclaurin expansion

$$\log n! = \left(n + \frac{1}{2}\right)\log n - n + \frac{1}{2}\log(2\pi) + \frac{1}{12n} + O(n^{-3});$$

the constant is identified by Wallis's product, and exponentiation yields 9.

Multiplying 8 by 9, and using $n/r = e^r = w(n)$, gives

$$B_n = \frac{w(n)^n e^{w(n)-n-1}}{\sqrt{r+1}}\left[1 + \frac{1}{12n} - \frac{2r^4 + 9r^3 + 16r^2 + 6r + 2}{24n(r+1)^3} + O(q^{3/2})\right].$$

The two displayed correction terms combine as follows:

$$\begin{aligned}\frac{1}{12n} - \frac{2r^4 + 9r^3 + 16r^2 + 6r + 2}{24n(r+1)^3}& \\ = -\frac{r^2(2r^2 + 7r + 10)}{24n(r+1)^3}&.\end{aligned}$$

Thus

$$B_n = \frac{w(n)^n e^{w(n)-n-1}}{\sqrt{r+1}}\left[1 - \frac{r^2(2r^2 + 7r + 10)}{24n(r+1)^3} + O\Big(q^{3/2}\Big)\right],$$

which proves the stronger saddle-point formula.

Recall that

$$f(n) = \frac{w(n)^n e^{w(n)-n-1}}{\sqrt{L}}, \qquad L = \log n.$$

Therefore

$$\frac{B_n}{f(n)} = \sqrt{\frac{L}{r+1}}\left[1 + O\left(\frac{r}{n}\right)\right]. \tag{10}$$

The $O(r/n)$ term is smaller than every fixed negative power of $\log n$, so the logarithmic correction comes entirely from the factor $\sqrt{L/(r+1)}$.

Since $re^r = n$, taking logarithms gives

$$r + \log r = L. \tag{11}$$

Let $\ell = \log L$. We next expand the solution of 11.
Because $r < L$ and $\log r \le \log L = \ell$,

$$L - \ell \le r < L.$$

Write

$$r = L - \ell + \delta.$$

Then $0 \le \delta \le \ell$, and 11 gives

$$\delta + \log\left(1 - \frac{\ell - \delta}{L}\right) = 0.$$

Since $0 \le (\ell - \delta)/L \le \ell/L \to 0$,

$$\delta = \frac{\ell - \delta}{L} + O\left(\frac{\ell^2}{L^2}\right).$$

Hence

$$\delta = \frac{\ell}{L} + O\left(\frac{\ell^2}{L^2}\right),$$

and therefore

$$r = L - \ell + \frac{\ell}{L} + O\left(\frac{\ell^2}{L^2}\right). \tag{12}$$

It follows that

$$\frac{r+1}{L} = 1 + \frac{1-\ell}{L} + \frac{\ell}{L^2} + O\left(\frac{\ell^2}{L^3}\right). \tag{13}$$

Apply

$$(1+u)^{-1/2} = 1 - \frac{u}{2} + \frac{3u^2}{8} + O(u^3)$$

to the right-hand side of 13. Since $u = O(\ell/L)$, we find

$$\sqrt{\frac{L}{r+1}} = 1 + \frac{\ell - 1}{2L} + \frac{3\ell^2 - 10\ell + 3}{8L^2} + O\left(\frac{\ell^3}{L^3}\right). \tag{14}$$

Finally, $r/n = O(L/n)$, so the saddle correction in 10 is

$$O(L/n) = o\left(\frac{\ell^3}{L^3}\right).$$

Combining 10 and 14 proves

$$T(n) = f(n)\left[1 + \frac{\ell - 1}{2L} + \frac{3\ell^2 - 10\ell + 3}{8L^2} + O\left(\frac{\ell^3}{L^3}\right)\right].$$

This completes the proof.

1.2.5 Boundary cases and numerical checks

The asymptotic statement concerns integers $n \to \infty$. The function $f(n)$ is not finite at $n = 1$, because its denominator is $\sqrt{\log 1} = 0$; thus no assertion is made there. The Bell numbers themselves satisfy $T(0) = T(1) = 1$, with the usual convention $0^0 = 1$ in Dobinski's formula.

For sufficiently large n, $\ell - 1 > 0$, and the error is smaller than the first correction. Hence the expansion also proves

$$T(n) > f(n)$$

for all sufficiently large n. No such finite-n inequality is implicit in the asymptotic statement.

The following values are an independent numerical check, not part of the proof. Here

$$R_1 = 1 + \frac{\ell - 1}{2L}, \qquad R_2 = R_1 + \frac{3\ell^2 - 10\ell + 3}{8L^2}.$$

n	$T(n)/f(n)$	R_1	R_2
20	0.956182538	1.016221193	0.955487159
100	1.021429523	1.057237801	1.026146557
1000	1.050833863	1.067507077	1.054091807

The convergence is logarithmically slow; the L^{-2} term is therefore still significant at moderate n.

A reproducible Python check is:

```
from math import comb
import mpmath as mp
```

```
mp.mp.dps = 80


def bell_numbers(n_max: int) -> list[int]:
    """Return B_0, ..., B_n_max using B_{m+1} = sum_k binom(m,k) B_k."""
    values = [1]
    for m in range(n_max):
        values.append(sum(comb(m, k) * values[k] for k in range(m + 1)))
    return values


def check(n_values: list[int]) -> None:
    bells = bell_numbers(max(n_values))

    print("n        T(n)/f(n)              R1                      R2")
    for n in n_values:
        L = mp.log(n)
        ell = mp.log(L)
        r = mp.re(mp.lambertw(n))
        w = n / r

        log_f = n * mp.log(w) + w - n - 1 - mp.log(L) / 2
        exact_ratio = mp.mpf(bells[n]) * mp.exp(-log_f)

        R1 = 1 + (ell - 1) / (2 * L)
        R2 = R1 + (3 * ell**2 - 10 * ell + 3) / (8 * L**2)

        print(
            f"{n:<8d} "
            f"{mp.nstr(exact_ratio, 14):<24} "
            f"{mp.nstr(R1, 14):<24} "
            f"{mp.nstr(R2, 14)}"
        )


check([20, 100, 1000])
```

Thus the precise first correction requested by Q565 is

$$\boxed{\frac{\log\log n - 1}{2\log n}, f(n),}$$

with relative remainder $O((\log\log n)^2/(\log n)^2)$.

Formal verification. Lean source; online type-check.

2 Question Q587

Informal verdict. A — Complete solution, with **high confidence** for every substantive case $n \geq 1$ and $1 \leq p < n$.

Lean coverage. The audited retry archive closes canonical Q587 for every nondegenerate order-connected real interval and f : I -> R. It gives literal relative coefficient/divided-difference DIT with unique values, strictly increasing tuple/Finset compatibility, part a and its limiting polynomial, part b as a relative epsilon-delta and nhdsWithin little-o expansion, part c with ContDiffOn/iterated derivWithin, and part d from DITp everywhere, all at finite endpoints. The part-b public theorem takes the full lower-limit family; exists_ddLimOn_of_le derives that family from DITn, so the canonical implication is kernel-connected without an extra mathematical hypothesis. Singleton intervals are classified as vacuous/nonunique above order zero, and the continuity versus punctured order-zero conventions are separated with a relative counterexample.

2.1 Audited informal answer

2.1.1 1. Faithful restatement and conventions

Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$. For distinct nodes

$$x_0 < x_1 < \cdots < x_n$$

in I, let $P_{x_0,\ldots,x_n}$ be the Lagrange interpolation polynomial of f at those nodes, written in the ordinary monomial basis as

$$P_{x_0,\ldots,x_n}(X) = \sum_{q=0}^{n} c_q(x_0,\ldots,x_n)X^q.$$

The function f is said to admit an interpolated Taylor expansion of order n, abbreviated DIT(n), at $x \in I$, when every $c_q(x_0,\ldots,x_n)$ has a finite limit as all nodes tend to x. The notation $f^{[q]}(x)$ denotes (q!) times that limit. The four questions are:

1. Show that it is enough to assume convergence of the highest coefficient c_n.
2. Show that DIT(n) implies an ordinary asymptotic expansion of order n.
3. If $p < n$ and $f \in C^p$, prove that f has DIT(n) at x if and only if $f^{(p)}$ has DIT($n - p$) there.
4. If f has DIT(p) at every point, determine whether DIT(n) for f is equivalent to DIT($n - p$) for the resulting function $f^{[p]}$.

There are three minor conventions to make explicit.

- "Increasing order" must mean **strictly** increasing, since ordinary Lagrange interpolation requires distinct nodes.
- Limits at an endpoint of I are relative, hence one-sided. Likewise, $C^p(I)$ means C^p on the interior, with the corresponding one-sided continuous extensions at finite endpoints.
- Because the polynomial is expanded in powers of X, rather than $X - x$, the lower quantities $f^{[q]}(x)$, $q < n$, depend on the ambient order n. When necessary, they will be denoted $f^{[q]}_{(n)}(x)$. The top quantity $f^{[n]}(x)$ is intrinsic.

There is also a genuine order-zero convention issue, addressed at the end.

2.1.2 2. Divided differences

For pairwise distinct $t_0, \ldots, t_r \in I$, write

$$[t_0, \ldots, t_r]f := \sum_{i=0}^{r} \frac{f(t_i)}{\prod\limits_{\substack{0 \le j \le r \\ j \ne i}} (t_i - t_j)}.$$

This is the divided difference of order r. It is symmetric in its arguments.

From the Lagrange formula,

$$P_{t_0,\dots,t_r}(X) = \sum_{i=0}^{r} f(t_i) \prod_{j\neq i} \frac{X - t_j}{t_i - t_j},$$

the leading coefficient is

$$c_r(t_0, \dots, t_r) = [t_0, \dots, t_r]f.$$

We shall repeatedly use the replacement identity

$$[A, u]f - [A, v]f = (u - v)[A, u, v]f, \tag{2.1}$$

where A is any list of distinct common nodes and all displayed nodes are pairwise distinct. It follows immediately from the usual recursion for divided differences, or from the explicit formula above.

Lemma 2.1 — Descent of divided differences Let $r \geq 1$, and suppose that the order-r divided differences of f are bounded near x: there exist $M, \rho_0 > 0$ such that

$$|[t_0, \dots, t_r]f| \leq M$$

whenever the nodes are distinct and belong to

$$U_{\rho_0} := I \cap (x - \rho_0, x + \rho_0).$$

Then the order-$(r-1)$ divided differences have a finite limit as their r nodes tend to x.

Proof Fix $0 < \rho < \rho_0$, and let

$$A = (a_0, \dots, a_{r-1}), \qquad B = (b_0, \dots, b_{r-1})$$

be two r-tuples of distinct nodes in U_ρ.

Choose distinct auxiliary nodes

$$C = (c_0, \dots, c_{r-1}) \subset U_\rho$$

disjoint from all the nodes occurring in A and B. This is possible because a nondegenerate real interval contains infinitely many points in every relative neighborhood of x.

Replace the nodes of A one at a time by the corresponding nodes of C. At each step, (2.1) gives

$$\left|[A^{(j+1)}]f - [A^{(j)}]f\right| \leq |c_j - a_j|M \leq 2\rho M.$$

There are r replacements, hence

$$|[A]f - [C]f| \leq 2r\rho M.$$

Similarly,

$$|[B]f - [C]f| \leq 2r\rho M.$$

Therefore

$$|[A]f - [B]f| \leq 4r\rho M. \tag{2.2}$$

Thus the oscillation of the order-$(r-1)$ divided difference over all r-tuples in U_ρ tends to zero with ρ. The Cauchy criterion gives a unique finite limit as all nodes tend to x. $\square$

Consequently, if the order-n divided difference has a finite limit at x, then successively all lower divided differences do. We write

$$\lambda_k(f;x) := \lim_{t_0,\ldots,t_k \to x} [t_0, \ldots, t_k]f, \qquad 0 \leq k \leq n. \tag{2.3}$$

For $n \geq 1$, the limit $\lambda_0(f;x)$ equals $f(x)$. Indeed,

$$f(t) - f(x) = (t - x)[x,t]f,$$

and the first divided differences remain bounded near x.

2.2 3. Part a

2.2.1 Theorem 3.1

For $n \geq 1$, the following are equivalent:

1. f has DIT(n) at x;
2. the highest coefficient $c_n(x_0, \ldots, x_n)$ has a finite limit as all nodes tend to x;
3. the order-n divided difference

$$[x_0, \ldots, x_n]f$$

has a finite limit as all nodes tend to x.

Moreover, if the limits $\lambda_k(f;x)$ are defined by (2.3), the interpolation polynomials converge coefficientwise to

$$Q_{n,x}(X) := \sum_{k=0}^{n} \lambda_k(f;x)(X-x)^k. \tag{3.1}$$

Proof The equivalence of 2 and 3 follows from

$$c_n(x_0, \ldots, x_n) = [x_0, \ldots, x_n]f.$$

Assume that this highest coefficient has a finite limit. It is then bounded near the diagonal. Lemma 2.1, applied successively, shows that all limits

$$\lambda_k(f;x), \qquad k = 0, \ldots, n,$$

exist.

For any ordering of the distinct nodes, Newton's interpolation formula is

$$P_{x_0,\ldots,x_n}(X) = \sum_{k=0}^{n} [x_0, \ldots, x_k]f \prod_{j=0}^{k-1}(X - x_j). \tag{3.2}$$

As all nodes tend to x,

$$[x_0, \ldots, x_k]f \longrightarrow \lambda_k(f;x)$$

and

$$\prod_{j=0}^{k-1}(X - x_j) \longrightarrow (X-x)^k$$

coefficientwise. Hence (3.2) converges coefficientwise to $Q_{n,x}$. In particular, all ordinary monomial coefficients c_q have finite limits.

The converse is immediate because $\mathrm{DIT}(n)$ includes convergence of c_n. $\square$

Expanding (3.1) in powers of X, we obtain the explicit answer

$$\lim c_q(x_0, \ldots, x_n) = \sum_{k=q}^{n} \binom{k}{q} (-x)^{k-q} \lambda_k(f;x). \tag{3.3}$$

Thus

$$f_{(n)}^{[q]}(x) = q! \sum_{k=q}^{n} \binom{k}{q} (-x)^{k-q} \lambda_k(f;x). \tag{3.4}$$

In particular,

$$f^{[n]}(x) = n!\lambda_n(f;x). \tag{3.5}$$

Formula (3.4) also explains why the lower $f^{[q]}_{(n)}$ generally depend on n.

2.3 4. Part b

2.3.1 Theorem 4.1

If f has DIT(n) at x, with $n \geq 1$, then

$$f(x+h) = \sum_{k=0}^{n} \lambda_k(f;x)h^k + o(|h|^n) \qquad (h \to 0,\ x+h \in I). \tag{4.1}$$

Equivalently,

$$f(t) = Q_{n,x}(t) + o(|t-x|^n), \tag{4.2}$$

where $Q_{n,x}$ is the coefficientwise limit of the interpolation polynomials.

Proof Fix $h \neq 0$, with $x+h \in I$. Choose n pairwise distinct auxiliary nodes

$$u_0^{(m)}, \ldots, u_{n-1}^{(m)} \in I \setminus x+h$$

such that

$$u_j^{(m)} \longrightarrow x \qquad (m \to \infty)$$

for every j.

Apply Newton's formula to the nodes

$$u_0^{(m)}, \ldots, u_{n-1}^{(m)}, x+h$$

and evaluate at $X = x+h$:

$$\begin{aligned} f(x+h) = &\sum_{k=0}^{n-1} [u_0^{(m)}, \ldots, u_k^{(m)}]f \prod_{j=0}^{k-1} (x+h-u_j^{(m)}) \\ &+ [u_0^{(m)}, \ldots, u_{n-1}^{(m)}, x+h]f \prod_{j=0}^{n-1} (x+h-u_j^{(m)}). \end{aligned} \tag{4.3}$$

As $m \to \infty$, the terms of order $k < n$ tend to

$$\lambda_k(f;x)h^k,$$

and the final product tends to $h^n \neq 0$. Hence the last divided difference in (4.3) has a limit, namely

$$R(h) := \frac{f(x+h) - \sum_{k=0}^{n-1} \lambda_k(f;x)h^k}{h^n}. \tag{4.4}$$

We claim that

$$R(h) \longrightarrow \lambda_n(f;x).$$

Given $\varepsilon > 0$, choose $\delta > 0$ such that

$$\big|[t_0, \dots, t_n]f - \lambda_n(f;x)\big| < \varepsilon \tag{4.5}$$

whenever all the distinct t_i lie in $I \cap (x-\delta, x+\delta)$.

If $0 < |h| < \delta$, then, for sufficiently large m, all the auxiliary nodes in (4.3), as well as $x+h$, lie in this neighborhood. Thus the last divided difference in (4.3) differs from $\lambda_n(f;x)$ by less than ε. Passing to its limit gives

$$|R(h) - \lambda_n(f;x)| \leq \varepsilon.$$

This proves (4.1). $\square$

In terms of the coefficients from the statement,

$$Q_{n,x}(X) = \sum_{q=0}^{n} \frac{f_{(n)}^{[q]}(x)}{q!} X^q,$$

so another exact formulation of the requested expansion is

$$f(t) = \sum_{q=0}^{n} \frac{f_{(n)}^{[q]}(x)}{q!} t^q + o(|t-x|^n). \tag{4.6}$$

The centered coefficients $\lambda_k(f;x)$ are the more intrinsic ones.

2.4 5. Differentiating a DIT

The central result for parts c and d is the following one-step equivalence.

2.4.1 Lemma 5.1 — Differentiation theorem

Let $r \geq 1$, and let f be C^1 on a relative neighborhood of x. Then

$$f \text{ has DIT}(r) \text{ at } x \quad \Longleftrightarrow \quad f' \text{ has DIT}(r-1) \text{ at } x. \tag{5.1}$$

When these conditions hold,

$$\lambda_{r-1}(f';x) = r, \lambda_r(f;x). \tag{5.2}$$

In fact, continuity of (f') is more than is needed; differentiability on a neighborhood suffices.

Proof: from f to (f') Assume

$$\lambda_r(f;x) = L.$$

Let

$$y_0 < \cdots < y_{r-1}$$

be distinct nodes near x, and define

$$G(t) := [y_0 + t, \ldots, y_{r-1} + t]f.$$

Because all differences $(y_i + t) - (y_j + t)$ are independent of t, the explicit divided-difference formula gives

$$G'(0) = [y_0, \ldots, y_{r-1}]f'. \tag{5.3}$$

For sufficiently small nonzero h, telescope by shifting the nodes one at a time:

$$G(h) - G(0) = \sum_{j=0}^{r-1} \Big([y_0 + h, \ldots, y_j + h, y_{j+1}, \ldots, y_{r-1}]f$$

$$- - - - - - - - - - - - - - - [y_0 + h, \ldots, y_{j-1} + h, y_j, \ldots, y_{r-}$$

By the replacement identity (2.1), each summand equals

$$h[Z_j(h)]f,$$

where $Z_j(h)$ is the $(r+1)$-tuple consisting of the common $r-1$ nodes together with y_j and $y_j + h$. Hence

$$\frac{G(h) - G(0)}{h} = \sum_{j=0}^{r-1} [Z_j(h)]f. \tag{5.4}$$

Let $\varepsilon > 0$. If all y_i are sufficiently close to x, then for all sufficiently small h, every divided difference on the right of (5.4) differs from L by at most ε/r. Therefore

$$\left| \frac{G(h) - G(0)}{h} - rL \right| < \varepsilon.$$

Letting $h \to 0$ and using (5.3),

$$\big| [y_0, \ldots, y_{r-1}] f' - rL \big| \le \varepsilon.$$

Thus

$$[y_0, \ldots, y_{r-1}] f' \longrightarrow rL,$$

which, by part a, is precisely DIT$(r-1)$ for (f').

Proof: from (f') to f Conversely, assume

$$\lambda_{r-1}(f'; x) = M.$$

Take arbitrary nodes

$$z_0 < z_1 < \cdots < z_r$$

near x, and let P be the degree-$\le r$ interpolation polynomial of f at these nodes. The function

$$H := f - P$$

vanishes at each z_i. By Rolle's theorem, for every $i = 0, \ldots, r-1$, there is a point

$$\xi_i \in (z_i, z_{i+1})$$

such that

$$H'(\xi_i) = 0.$$

Thus

$$P'(\xi_i) = f'(\xi_i).$$

The points satisfy

$$\xi_0 < \cdots < \xi_{r-1},$$

and (P'), of degree at most $r-1$, is therefore the interpolation polynomial of (f') at these r points.

The leading coefficient of (P') is

$$r[z_0, \ldots, z_r] f,$$

while the leading coefficient of the interpolation polynomial of (f') is

$$[\xi_0, \ldots, \xi_{r-1}]f'.$$

Consequently,

$$r[z_0, \ldots, z_r]f = [\xi_0, \ldots, \xi_{r-1}]f'. \tag{5.5}$$

As all $z_i \to x$, all $\xi_i \to x$. Therefore the right-hand side of (5.5) tends to M, and

$$[z_0, \ldots, z_r]f \longrightarrow \frac{M}{r}.$$

Part a now gives DIT(r) for f, with

$$\lambda_r(f; x) = \frac{M}{r}.$$

This completes the proof. □

2.5 6. Part c

Let $0 \leq p < n$ and $f \in C^p(I)$.

For $p = 0$, the assertion is tautological. Suppose $p \geq 1$. Apply Lemma 5.1 successively to

$$f,\ f',\ f'', \ldots, f^{(p-1)}$$

with orders

$$n,\ n-1,\ n-2, \ldots, n-p+1.$$

We obtain

$$\begin{aligned} f \text{ has DIT}(n) \text{ at } x &\Longleftrightarrow f' \text{ has DIT}(n-1) \text{ at } x \\ &\Longleftrightarrow f'' \text{ has DIT}(n-2) \text{ at } x \\ &\ \ \vdots \\ &\Longleftrightarrow f^{(p)} \text{ has DIT}(n-p) \text{ at } x. \end{aligned}$$

This proves the required equivalence.

The limiting highest coefficients satisfy the precise identity

$$\lambda_{n-p}(f^{(p)}; x) == n(n-1)\cdots(n-p+1)\lambda_n(f; x)\frac{n!}{(n-p)!}\lambda_n(f; x). \tag{6.1}$$

Multiplying by $(n - p!)$, we get the particularly simple relation

$$\boxed{\left(f^{(p)}\right)^{[,n-p,]}(x) = f^{[n]}(x).} \tag{6.2}$$

Thus part c holds, and (6.1) gives the exact transformation of the top divided-difference limit.

2.6 7. Part d

The substantive answer is affirmative for every $1 \le p < n$.

The point is that having DIT(p) at every point automatically forces f to be C^p, and $f^{[p]}$ is then exactly the ordinary derivative $f^{(p)}$.

2.6.1 Lemma 7.1 — DIT(1) everywhere implies C^1

Suppose g has DIT(1) at every point of I. Then $g \in C^1(I)$, and

$$g'(x) = \lambda_1(g; x) = g^{[1]}(x).$$

Proof At a fixed x,

$$\frac{g(t) - g(x)}{t - x} = [x, t]g \longrightarrow \lambda_1(g; x),$$

so g is differentiable and

$$g'(x) = \lambda_1(g; x).$$

It remains to prove continuity of (g'). Given $\varepsilon > 0$, the strict secant limit at x provides $\delta > 0$ such that

$$|[u, v]g - g'(x)| < \varepsilon$$

whenever $u \ne v$ and both (u,v) lie within δ of x.

For y sufficiently close to x, let $v \to y$. All secants ([y,v]g) are then within ε of (g'x). Passing to the limit $v \to y$ gives

$$|g'(y) - g'(x)| \le \varepsilon.$$

Thus (g') is continuous. □

2.6.2 Proposition 7.2

Let $p \ge 1$. If f has DIT(p) at every point of I, then

$$f \in C^p(I)$$

and

$$\boxed{f^{[p]} = f^{(p)}.} \tag{7.1}$$

Proof Since DIT(p) implies, by the descent argument in part a, existence of the first divided-difference limit, f has DIT(1) at every point. Lemma 7.1 therefore gives

$$f \in C^1(I).$$

By Lemma 5.1, because f has DIT(p) everywhere,

$$f'$$

has DIT($p-1$) everywhere. If $p-1 \geq 1$, Lemma 7.1 applied to (f') gives $f' \in C^1$, hence $f \in C^2$.

Continuing inductively,

$$f^{(j)}$$

has DIT($p-j$) everywhere and is C^1 for $j = 0, \ldots, p-1$. Hence

$$f \in C^p(I).$$

The factor relation (5.2), iterated p times, gives

$$f^{(p)}(x) = p!\lambda_p(f;x).$$

Since c_p is the order-p divided difference,

$$f^{[p]}(x) = p!\lambda_p(f;x).$$

Therefore

$$f^{[p]}(x) = f^{(p)}(x)$$

at every point. □

We may now apply part c:

$$\begin{aligned} f \text{ has DIT}(n) \text{ at } x &\iff f^{(p)} \text{ has DIT}(n-p) \text{ at } x \\ &\iff f^{[p]} \text{ has DIT}(n-p) \text{ at } x. \end{aligned}$$

Hence:

$$\boxed{\text{For every } 1 \leq p < n, \text{ the answer to part (d) is yes.}} \tag{7.2}$$

The global assumption "at every point of I" can be weakened: for the equivalence at a fixed x, it is enough that f have DIT(p) throughout a relative neighborhood of x.

2.6.3 8. The order-zero ambiguity

If $p = 0$, the answer depends on what “DIT(0)” means.

For $n = 0$, the interpolation polynomial is simply

$$P_{x_0}(X) = f(x_0).$$

Thus the literal condition in the PDF is merely the existence of

$$\lim_{t \to x} f(t).$$

Continuity convention If DIT(0) is understood to require

$$\lim_{t \to x} f(t) = f(x),$$

then

$$f^{[0]} = f,$$

and part d for $p = 0$ is tautologically true.

Punctured-limit convention If only the finite punctured limit is required, part d is false for $p = 0$.

Take

$$I = (-1, 1), \qquad f(t) = \begin{cases} 1, & t = 0, \\ 0, & t \neq 0. \end{cases}$$

At every point, the punctured limit exists, and

$$f^{[0]} \equiv 0.$$

Thus $f^{[0]}$ has DIT(1) at 0. But f does not, since for $t > 0$,

$$[0, t]f = \frac{f(t) - f(0)}{t} = -\frac{1}{t},$$

which diverges.

Therefore the fully literal conclusion is:

$$\boxed{\text{Part (d) is affirmative for every } p \geq 1.[2mm]\text{For } p = 0, \text{ it is affirmative under the continuity convent} \\ \text{but false under the punctured-limit convention.}}$$

The same convention affects part b when $n = 0$: a punctured finite limit gives an expansion with a constant equal to that limit, but not necessarily with constant $f(x)$.

2.7 9. Checks and sharpness

9.1 The case $n = 1$ DIT(1) at x means

$$[u, v]f = \frac{f(v) - f(u)}{v - u}$$

has a limit as $u, v \to x$. This is strict differentiability at x. The results give

$$f(x + h) = f(x) + \lambda_1(f; x)h + o(|h|),$$

and, if this condition holds everywhere, $f \in C^1$.

9.2 Polynomial check For a polynomial f, all the limits exist. The limiting interpolation polynomial is its Taylor polynomial of degree n at x:

$$Q_{n,x}(X) = \sum_{k=0}^{n} \frac{f^{(k)}(x)}{k!}(X - x)^k,$$

with the sum truncated automatically if $\deg f < n$. Thus

$$\lambda_k(f; x) = \frac{f^{(k)}(x)}{k!},$$

and (6.1) becomes

$$\frac{f^{(n)}(x)}{(n-p)!} = \frac{n!}{(n-p)!}\frac{f^{(n)}(x)}{n!},$$

as required.

9.3 The converse to part b is false An ordinary asymptotic expansion need not imply a DIT.

Let

$$g(0) = 0, \qquad g(t) = t^2 \sin(1/t) \quad (t \neq 0).$$

Then

$$g(t) = o(t),$$

so g has a first-order expansion at 0 with derivative 0. But

$$g'(t) = 2t \sin(1/t) - \cos(1/t)$$

oscillates between values close to 1 and (-1) as $t \to 0$.

At points $t_k \to 0$ where $g'(t_k) \to 1$, choose $h_k \to 0$ so that

$$[t_k, t_k + h_k]g \to 1.$$

At another sequence $s_k \to 0$, choose secants tending to (-1). Hence the two-point divided differences do not have a unique limit as both nodes tend to 0. Thus g does not have DIT(1) there.

So DIT is strictly stronger than the existence of an ordinary Peano–Taylor expansion.

9.4 Role of the hypotheses

- Distinctness of the nodes is indispensable for ordinary Lagrange interpolation. Repeated nodes would require Hermite interpolation.
- The interval assumption is used essentially in the Rolle-theorem argument: each consecutive pair of interpolation nodes determines an intervening point.
- The C^p assumption in part c is sufficient but not optimal. The proof only needs p-fold differentiability on a neighborhood of x.
- In part d, the "every point" hypothesis is what upgrades the strict divided-difference regularity to C^p. Locally around x is sufficient.
- Endpoint cases require no separate argument beyond taking relative limits and one-sided derivatives.

2.8 Final classification

For $n \geq 1$, let

$$\Lambda_n(f; x) = \lim_{x_0, \ldots, x_n \to x} [x_0, \ldots, x_n]f.$$

Then:

$$\boxed{f \text{ has DIT}(n) \text{ at } x \iff \Lambda_n(f; x) \in \mathbb{R}.}$$

When this holds, all lower limits $\lambda_k(f; x)$ exist, the interpolation polynomials converge to

$$Q_{n,x}(X) = \sum_{k=0}^{n} \lambda_k(f; x)(X - x)^k,$$

and

$$\boxed{f(x+h)=\sum_{k=0}^{n}\lambda_k(f;x)h^k+o(|h|^n).}$$

For $0\le p<n$ and $f\in C^p$,

$$\boxed{f \text{ has DIT}(n) \text{ at } x \iff f^{(p)} \text{ has DIT}(n-p) \text{ at } x,}$$

with

$$\boxed{\lambda_{n-p}(f^{(p)};x)=\frac{n!}{(n-p)!}\lambda_n(f;x).}$$

Finally, if $p\ge 1$ and f has DIT(p) at every point, then

$$\boxed{f\in C^p, \qquad f^{[p]}=f^{(p)},}$$

and therefore

$$\boxed{f \text{ has DIT}(n) \text{ at } x \iff f^{[p]} \text{ has DIT}(n-p) \text{ at } x.}$$

Formal verification. Lean source; online type-check.

3 Question Q604

Informal verdict. A — Complete solution, with **very high confidence**.

Lean coverage. The Lean development proves the printed normalized Bezout-pair theorem, its uniqueness, the universal binomial identity, the classification of all pairs, the worked example, and the constant-polynomial boundary case over an arbitrary field.

3.1 Audited informal answer

3.1.1 1. Faithful restatement

Let $\mathbb{K}$ be a field. Let $A,B\in\mathbb{K}[X]$ be coprime, and let $U,V\in\mathbb{K}[X]$ satisfy

$$\deg U<\deg B, \qquad AU+BV=1.$$

For positive integers (m,n), determine explicitly the pair $(U_{m,n},V_{m,n})\in\mathbb{K}[X]^2$ such that

$$\deg U_{m,n} < \deg B^n \qquad \text{and} \qquad A^m U_{m,n} + B^n V_{m,n} = 1.$$

This is exactly Q604 on the rendered PDF page. The notation d° denotes polynomial degree. There is no apparent typo. The statement leaves implicit the standard convention that $\mathbb{K}$ is a field; this is the setting used below. We use $\deg 0 = -\infty$.

3.1.2 2. Explicit answer

For integers $r \geq 1$ and $s \geq 1$, define

$$S_{r,s}(T) := \sum_{j=0}^{s-1} \binom{r+j-1}{j} T^j \in \mathbb{K}[T],$$

where the integer binomial coefficients are mapped into $\mathbb{K}$ through the canonical homomorphism $\mathbb{Z} \to \mathbb{K}$.

Set

$$\widehat{U}_{m,n} := U^m S_{m,n}(BV) = U^m \sum_{j=0}^{n-1} \binom{m+j-1}{j} (BV)^j$$

and

$$\widehat{V}_{m,n} := V^n S_{n,m}(AU) = V^n \sum_{i=0}^{m-1} \binom{n+i-1}{i} (AU)^i.$$

Perform the Euclidean division

$$\widehat{U}_{m,n} = B^n Q_{m,n} + R_{m,n}, \qquad \deg R_{m,n} < \deg B^n.$$

Then the required pair is

$$\boxed{U_{m,n} = R_{m,n}, \qquad V_{m,n} = \widehat{V}_{m,n} + A^m Q_{m,n}.}$$

Equivalently, writing $\mathrm{rem}_C(P)$ for the Euclidean remainder of P modulo a nonzero polynomial C,

$$\boxed{U_{m,n} = \mathrm{rem}_{B^n}\left(U^m \sum_{j=0}^{n-1} \binom{m+j-1}{j} (BV)^j \right),}$$

and then

$$\boxed{V_{m,n} = \frac{1 - A^m U_{m,n}}{B^n}.}$$

The quotient in the last formula is a polynomial, as proved below.

3.1.3 3. The universal binomial Bézout identity

The formula rests on the following identity.

Lemma Let R be any commutative ring, and let $x, y \in R$ satisfy $x + y = 1$. For all $r, s \geq 1$,

$$\boxed{x^r S_{r,s}(y) + y^s S_{s,r}(x) = 1.}$$

Thus explicitly,

$$x^r \sum_{j=0}^{s-1} \binom{r+j-1}{j} y^j + y^s \sum_{i=0}^{r-1} \binom{s+i-1}{i} x^i = 1.$$

Proof For convenience put $S_{r,0} = 0$. Pascal's identity gives, for $r \geq 2$,

$$S_{r,s}(T) - S_{r-1,s}(T) = T S_{r,s-1}(T).$$

Indeed, for $j \geq 1$,

$$\binom{r+j-1}{j} - \binom{r+j-2}{j} = \binom{r+j-2}{j-1}.$$

Since $x = 1 - y$,

$$\begin{aligned}
S_{r-1,s}(y) - x S_{r,s}(y) &= S_{r-1,s}(y) - (1-y) S_{r,s}(y) \\
&= S_{r-1,s}(y) - S_{r,s}(y) + y S_{r,s}(y) \\
&= -y S_{r,s-1}(y) + y S_{r,s}(y) \\
&= \binom{r+s-2}{s-1} y^s.
\end{aligned}$$

Consequently,

$$x^{r-1} S_{r-1,s}(y) - x^r S_{r,s}(y) = \binom{r+s-2}{s-1} x^{r-1} y^s.$$

On the other hand,

$$S_{s,r}(x) - S_{s,r-1}(x) = \binom{s+r-2}{r-1} x^{r-1}.$$

Because

$$\binom{r+s-2}{s-1} = \binom{r+s-2}{r-1},$$

we obtain

$$x^r S_{r,s}(y) + y^s S_{s,r}(x) = x^{r-1} S_{r-1,s}(y) + y^s S_{s,r-1}(x).$$

Iterating this reduction in r gives

$$x^r S_{r,s}(y) + y^s S_{s,r}(x) = x S_{1,s}(y) + y^s.$$

Finally,

$$S_{1,s}(y) = 1 + y + \cdots + y^{s-1},$$

so

$$x S_{1,s}(y) + y^s == (1-y)(1 + y + \cdots + y^{s-1}) + y^s 1.$$

This proves the lemma. Since it is an identity with integer coefficients, it remains valid in every characteristic. □

3.1.4 4. Proof of the formula

Apply the lemma with

$$x = AU, \qquad y = BV.$$

The given Bézout identity says precisely that $x + y = 1$. Therefore

$$(AU)^m S_{m,n}(BV) + (BV)^n S_{n,m}(AU) = 1.$$

Factoring out A^m and B^n gives

$$A^m \left(U^m S_{m,n}(BV)\right) + B^n \left(V^n S_{n,m}(AU)\right) = 1,$$

that is,

$$A^m\widehat{U}_{m,n} + B^n\widehat{V}_{m,n} = 1. \tag{1}$$

Thus $(\widehat{U}_{m,n}, \widehat{V}_{m,n})$ is already an explicit Bézout pair for A^m and B^n. It need not, however, satisfy the required degree condition.

Now divide

$$\widehat{U}_{m,n} = B^n Q_{m,n} + R_{m,n}, \qquad \deg R_{m,n} < \deg B^n.$$

Substitution into 1 yields

$$\begin{aligned} 1 &= A^m(B^n Q_{m,n} + R_{m,n}) + B^n\widehat{V}_{m,n} \\ &= A^m R_{m,n} + B^n\big(\widehat{V}_{m,n} + A^m Q_{m,n}\big). \end{aligned}$$

Hence

$$U_{m,n} = R_{m,n}, \qquad V_{m,n} = \widehat{V}_{m,n} + A^m Q_{m,n}$$

has all the required properties.

In particular,

$$B^n \mid 1 - A^m U_{m,n},$$

so

$$V_{m,n} = \frac{1 - A^m U_{m,n}}{B^n}$$

is indeed a polynomial.

3.1.5 5. Uniqueness and classification of all pairs

Suppose that two pairs ((P,Q)) and ((P',Q')) satisfy

$$A^m P + B^n Q = 1, \qquad A^m P' + B^n Q' = 1.$$

Subtracting gives

$$A^m(P - P') = -B^n(Q - Q'). \tag{2}$$

Because 1 is itself a Bézout identity for A^m and B^n, equation 2 implies that $B^n \mid P - P'$. Indeed,

$$\begin{aligned} P - P' &= \big(A^m\widehat{U}_{m,n} + B^n\widehat{V}_{m,n}\big)(P - P') \\ &= \widehat{U}_{m,n}, A^m(P - P') + B^n\widehat{V}_{m,n}(P - P'), \end{aligned}$$

and both terms on the right are divisible by B^n.

If B is nonconstant and

$$\deg P, \deg P' < \deg B^n,$$

then either $P - P' = 0$, or

$$\deg(P - P') \geq \deg B^n,$$

which is impossible. Thus $P = P'$, and then 2 gives $Q = Q'$.

If B is a nonzero constant, the degree condition is $\deg P < 0$, hence $P = 0$, so uniqueness is again immediate.

Therefore the normalized pair $(U_{m,n}, V_{m,n})$ is unique.

Moreover, all Bézout pairs without the degree restriction are exactly

$$\boxed{\big(U_{m,n} + B^n T, ; V_{m,n} - A^m T\big), \qquad T \in \mathbb{K}[X].}$$

Thus the condition $\deg U_{m,n} < \deg B^n$ selects precisely one pair from this affine family.

3.1.6 6. Symmetric remainder formula

Assume for this subsection that A and B are both nonconstant. Then the normalized pair automatically satisfies

$$\deg V_{m,n} < m \deg A.$$

Indeed, if $V_{m,n} \neq 0$, the highest-degree terms in

$$A^m U_{m,n} + B^n V_{m,n} = 1$$

must cancel, so

$$m \deg A + \deg U_{m,n} = n \deg B + \deg V_{m,n}.$$

Since $\deg U_{m,n} < n \deg B$, it follows that

$$\deg V_{m,n} < m \deg A.$$

Now $V_{m,n} \equiv \widehat{V}_{m,n} \pmod{A^m}$, because

$$V_{m,n} = \widehat{V}_{m,n} + A^m Q_{m,n}.$$

Consequently,

$$\boxed{V_{m,n} = \mathrm{rem}_{A^m}\left(V^n \sum_{i=0}^{m-1} \binom{n+i-1}{i} (AU)^i\right).}$$

Thus, in the generic nonconstant case, the answer has the completely symmetric form

$$\boxed{\begin{aligned} U_{m,n} &= \mathrm{rem}_{B^n}\left(U^m S_{m,n}(BV)\right), \\ V_{m,n} &= \mathrm{rem}_{A^m}\left(V^n S_{n,m}(AU)\right). \end{aligned}}$$

3.1.7 7. A directly degree-normalized B-adic formula

The remainder modulo B^n can itself be expanded explicitly using only successive divisions by B.

Assume $\deg B = b \geq 1$. Define

$$\rho_B(P) := \mathrm{rem}_B(P), \qquad \delta_B(P) := \frac{P - \rho_B(P)}{B}.$$

Thus

$$P = \rho_B(P) + B\delta_B(P).$$

Iterating gives the finite B-adic expansion

$$P = \sum_{r \geq 0} B^r \rho_B\big(\delta_B^r(P)\big).$$

Since

$$\widehat{U}_{m,n} = \sum_{k=0}^{n-1} \binom{m+k-1}{k} B^k U^m V^k,$$

its terms of B-adic order strictly below n give

$$\boxed{U_{m,n} = \sum_{r=0}^{n-1} B^r C_r,}$$

where

$$\boxed{C_r = \sum_{k=0}^{r} \binom{m+k-1}{k}, \rho_B! \left(\delta_B^{,r-k}(U^m V^k)\right).}$$

Every C_r has degree $< b$, so the displayed formula directly gives

$$\deg U_{m,n} < nb = \deg B^n.$$

This is equivalent to the single-remainder formula but may be preferable for an implementation based on B-adic digits.

3.1.8 8. Small and boundary cases

Case $m = n = 1$ Here $S_{1,1} = 1$, so

$$\widehat{U}_{1,1} = U, \qquad \widehat{V}_{1,1} = V.$$

Because $\deg U < \deg B$, no reduction is needed:

$$(U_{1,1}, V_{1,1}) = (U, V).$$

Case $n = 1$ Since $S_{m,1} = 1$,

$$\boxed{U_{m,1} = \operatorname{rem}_B(U^m).}$$

The corresponding second coefficient is

$$V_{m,1} = \frac{1 - A^m U_{m,1}}{B}.$$

Before normalization, the symmetric pair is

$$\widehat{U}_{m,1} = U^m, \qquad \widehat{V}_{m,1} = V \sum_{i=0}^{m-1} (AU)^i.$$

Case $m = 1$ Because $S_{1,n}(T) = 1 + T + \cdots + T^{n-1}$,

$$U_{1,n} = \operatorname{rem}_{B^n} \left(U \sum_{j=0}^{n-1} (BV)^j \right),$$

while

$$\widehat{V}_{1,n} = V^n.$$

The raw identity is simply

$$A, U \sum_{j=0}^{n-1} (BV)^j + (BV)^n = 1.$$

Constant B If $B \in \mathbb{K}^\times$, then $\deg U < 0$ forces $U = 0$, and $AU + BV = 1$ gives

$$V = B^{-1}.$$

The unique answer is therefore

$$\boxed{U_{m,n} = 0, \qquad V_{m,n} = B^{-n}.}$$

This also covers the possible degenerate case $A = 0$.

Constant A**, nonconstant** B If $A \in \mathbb{K}^\times$ and B is nonconstant, the canonical initial pair is

$$U = A^{-1}, \qquad V = 0.$$

Hence

$$\boxed{U_{m,n} = A^{-m}, \qquad V_{m,n} = 0.}$$

Positive characteristic There are no exceptional characteristics. The coefficients

$$\binom{m+j-1}{j}$$

are simply reduced modulo $\operatorname{char} \mathbb{K}$. The proof uses no factorial denominators and is an identity over $\mathbb{Z}$.

3.1.9 9. Why the Euclidean reduction is genuinely necessary

The raw pair $(\widehat{U}_{m,n}, \widehat{V}_{m,n})$ need not satisfy the degree condition.

For example, over $\mathbb{Q}$, take

$$A = X^2, \qquad B = X + 1, \qquad U = 1, \qquad V = 1 - X.$$

Then

$$AU + BV = X^2 + (X + 1)(1 - X) = 1.$$

For $m = 1, n = 2$,

$$\widehat{U}_{1,2} == 1 + BV2 - X^2.$$

But

$$\deg(2 - X^2) = 2 = \deg(X + 1)^2,$$

so the strict degree condition fails. Dividing,

$$2 - X^2 = -(X + 1)^2 + (2X + 3).$$

Thus

$$U_{1,2} = 2X + 3.$$

Also,

$$\widehat{V}_{1,2} = V^2 = (1 - X)^2, \qquad Q_{1,2} = -1,$$

so

$$V_{1,2} == (1 - X)^2 - X^2 1 - 2X.$$

Indeed,

$$X^2(2X + 3) + (X + 1)^2(1 - 2X) = 1.$$

3.1.10 10. Hypothesis audit

The identity $AU + BV = 1$ already implies that A and B are coprime, so the explicit coprimality hypothesis is logically redundant once (U,V) are given.

The initial condition $\deg U < \deg B$ is not needed for the construction: the same formula works starting from any Bézout pair $AU + BV = 1$. Its role is to make the given initial pair canonical.

The target condition

$$\deg U_{m,n} < \deg B^n$$

is essential for uniqueness. Without it, all solutions are obtained by adding

$$(B^n T, -A^m T).$$

Finally, the field hypothesis is used for unrestricted Euclidean division in $\mathbb{K}[X]$. The universal binomial identity itself is valid over every commutative ring; the normalized construction also extends, for example, to coefficient rings over which division by the chosen B is available, such as when B is monic.

Formal verification. Lean source; online type-check.

4 Question Q668

Informal verdict. C — Substantial progress, with **high confidence in the central classification**.

Lean coverage (partial). For arbitrary nonempty n, Lean classifies all nonnegative periodic points, proves existence of a limiting norm and existential phase-by-phase shadowing by a periodic binary orbit in S n, and proves that every nonnegative orbit tends to zero exactly when n is a power of two. For n=5 it proves the exact unique nonzero binary 15-cycle, an existential zero-or-positive asymptotic dichotomy, and a nonconstant zero-basin eigenray. It does not recover, from independent data attached to an arbitrary input, whether the scale is zero or positive or the exact positive scale and binary attractor/phase; solving n=5 alone would still not cover the canonical all-n question. The response's additional Section 6 examples are optional, not canonical completion requirements.

4.1 Audited informal answer

4.2 Partial result

I give below:

1. a complete classification of all possible periodic attractors for every n;
2. a proof that every real orbit is asymptotic, phase by phase, to one of them;
3. an exact finite algorithm for rational data;
4. a complete description of the possible asymptotic cycles for $n = 5$;
5. an explicit nonconstant 5-tuple converging to 0;
6. an exact algebraic 5-tuple which approaches the nonzero 15-cycle but never enters any periodic orbit.

What remains unresolved by this analysis is a direct criterion, for an arbitrary exact real X, deciding which attractor basin contains X and computing the limiting scale without iterating indefinitely.

4.2.1 1. Faithful restatement

Let $n \geq 3$. For

$$X = (x_1, \ldots, x_n) \in \mathbb{R}^n_+,$$

define

$$P(X) = \big(|x_1 - x_2|, |x_2 - x_3|, \ldots, |x_{n-1} - x_n|, |x_n - x_1|\big).$$

Study the orbit

$$X,\ P(X),\ P^2(X), \ldots$$

for a prescribed starting vector X.

The editorial note says that $n = 4$ is understood: every orbit eventually reaches 0, except for one algebraic family, up to elementary transformations, represented after one iteration by

$$(1, a, a^2, a^3),$$

where $a-1$ is the real root of

$$u^3 + 2u^2 - 2 = 0.$$

Equivalently, a satisfies

$$a^3 - a^2 - a - 1 = 0.$$

The note explicitly welcomes any partial result encompassing $n = 5$.
There are two minor notational points.

- Here $\mathbb{R}_+$ must mean $[0, \infty)$, since $P(X)$ can have zero coordinates.
- The phrase "up to a simple transformation" in the editorial note is informal and does not specify the transformations, but this does not affect the formal problem.

Throughout, write

$$X^{(q)} = P^q(X), \qquad M_q = |X^{(q)}|_\infty.$$

4.3 2. Main classification theorem

Write

$$n = hm, \qquad h = 2^{\nu_2(n)}, \qquad m \text{ odd}.$$

Thus h is the largest power of 2 dividing n.
For a binary vector $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_n) \in 0, 1^n$, put

$$E_\varepsilon(z) = \sum_{j=1}^{n} \varepsilon_j z^{j-1} \in \mathbb{F}_2[z].$$

Define

$$\mathcal{S}_n = \left\{ \varepsilon \in 0, 1^n : 1 + z^h \mid E_\varepsilon(z) \text{ in } \mathbb{F}_2[z] \right\}.$$

Equivalently,

$$\boxed{\varepsilon \in \mathcal{S}_n \iff \sum_{k=0}^{m-1} \varepsilon_{r+kh} \equiv 0 \pmod 2 \quad (1 \le r \le h).} \tag{2.1}$$

Thus $\mathcal{S}_n$ is given by h independent parity conditions and

$$|\mathcal{S}_n| = 2^{n-h}.$$

4.3.1 Theorem 2.1

For every $X \in \mathbb{R}^n_+$:

1. The sequence M_q is nonincreasing, and therefore

$$c(X) := \lim_{q\to\infty} M_q$$

exists.

1. There exists a periodic binary sequence

$(\varepsilon^{(q)})_{q\geq 0}$, with every $\varepsilon^{(q)} \in \mathcal{S}_n$ and

$$\varepsilon^{(q+1)} = P(\varepsilon^{(q)}),$$

such that

$$\boxed{\lim_{q\to\infty} \left|P^q(X) - c(X)\varepsilon^{(q)}\right|_\infty = 0.} \tag{2.2}$$

1. The periodic points of P in $\mathbb{R}^n_+$ are exactly

$$\boxed{c\varepsilon, \qquad c \geq 0, \quad \varepsilon \in \mathcal{S}_n.} \tag{2.3}$$

1. The map P permutes the finite set $\mathcal{S}_n$. Hence all possible periodic cycles can be computed by taking the cycle decomposition of this permutation.

1. Every real orbit converges to 0 if and only if n is a power of 2.

An equivalent limit-point and periodic-orbit classification appears in the classical work of Misiurewicz and Schinzel; the proof below is self-contained and also makes the phase-by-phase convergence explicit. ([hrj.episciences.org][1])

4.4 3. Proof of Theorem 2.1

4.4.1 3.1. Monotonicity

If $Y = (y_1, \ldots, y_n) \in [0, M]^n$, then

$$|y_i - y_{i+1}| \leq M.$$

Consequently,

$$|P(Y)| * \infty \leq |Y| * \infty.$$

Thus $M_{q+1} \leq M_q$, proving the existence of $c(X)$.
We shall need the following strict form.

4.4.2 3.2. Strict decrease unless all entries are extremal

For $u_0, \ldots, u_k \in [0, M]$, define recursively

$$\Delta_0(u_0) = u_0$$

and

$$\Delta_k(u_0, \ldots, u_k) = |\Delta_{k-1}(u_0, \ldots, u_{k-1}) ------------------------------$$

Then, with cyclic indices,

$$\left(P^k(Y)\right)_i = \Delta_k(y_i, y_{i+1}, \ldots, y_{i+k}). \tag{3.1}$$

Lemma 3.1 Suppose $u_0, \ldots, u_k \in [0, M]$. If

$$\Delta_k(u_0, \ldots, u_k) = M,$$

then

$$u_j \in 0, M \qquad (0 \leq j \leq k).$$

Proof First observe the following endpoint fact.

If

$$u_0, \dots, u_{k-1} \in 0, M, \qquad u_k = t \in [0, M],$$

then

$$\Delta_k(u_0, \dots, u_k) \in t, M - t. \tag{3.2}$$

Indeed, after taking one row of absolute differences, all entries except possibly the last are again in ({0,M}), while the last is either t or $M - t$. Induction proves (3.2). The analogous assertion holds when the exceptional entry is the first one.

We now prove the lemma by induction on k. The case $k = 0$ is immediate. Set

$$A = \Delta_{k-1}(u_0, \dots, u_{k-1}), \qquad B = \Delta_{k-1}(u_1, \dots, u_k).$$

Both A and B belong to ([0,M]). Since

$$|A - B| = M,$$

we must have

$$A, B = 0, M.$$

Suppose, for example, that $A = M$. By the induction hypothesis,

$$u_0, \dots, u_{k-1} \in 0, M.$$

Since $B = 0$, the endpoint observation applied to $u_1, \dots, u_k$ implies that $u_k \in 0, M$. The other case is symmetric. □

Corollary 3.2 If $Y \in [0, M]^n$ has at least one coordinate strictly between 0 and M, then

$$\boxed{|P^{n-1}(Y)|_\infty < M.} \tag{3.3}$$

Proof Every coordinate of $P^{n-1}(Y)$ is an $(n-1)$-st absolute difference involving all n original coordinates once. Lemma 3.1 shows that none of these coordinates can equal M. □

4.4.3 3.3. Every limit point is binary up to scale

Let $\Omega(X)$ be the set of limit points of the orbit.

Take $Y \in \Omega(X)$. There is a sequence $q_j \to \infty$ such that

$$X^{(q_j)} \longrightarrow Y.$$

Since $M_{q_j} \to c(X)$,

$$|Y|_\infty = c(X).$$

Suppose $c(X) > 0$ and that some coordinate of Y lies strictly between 0 and $c(X)$. Corollary 3.2 gives

$$|P^{n-1}(Y)|_\infty < c(X).$$

On the other hand, continuity of P yields

$$P^{n-1}(X^{(q_j)}) \longrightarrow P^{n-1}(Y),$$

while

$$\left|P^{n-1}(X^{(q_j)})\right|\Big|_\infty = M_{q_j+n-1} \longrightarrow c(X),$$

a contradiction.

Therefore every limit point has the form

$$Y = c(X)\varepsilon, \qquad \varepsilon \in 0,1^n. \tag{3.4}$$

When $c(X) = 0$, the orbit itself converges to 0.

4.4.4 3.4. Algebra over $\mathbb{F}_2$

On binary vectors, absolute difference agrees with addition modulo 2:

$$|a-b| = a+b \pmod 2 \qquad (a,b \in 0,1).$$

Let $L : \mathbb{F}_2^n \to \mathbb{F}_2^n$ denote the resulting linear map

$$(L\varepsilon)_i = \varepsilon_i + \varepsilon_{i+1}.$$

Identify $\mathbb{F}_2^n$ with

$$R_n = \mathbb{F}_2[z]/(z^n - 1)$$

by $\varepsilon \mapsto E_\varepsilon(z)$. Then

$$E_{L\varepsilon}(z) = z^{-1}(1+z)E_\varepsilon(z) \quad \text{in } R_n. \tag{3.5}$$

Since multiplication by z is invertible in R_n, the image of L^h is the ideal generated by

$$(1+z)^h = 1 + z^h,$$

where we used that h is a power of 2.

Moreover $1 + z^h \mid z^n - 1$ in $\mathbb{F}_2[z]$. It follows that

$$\operatorname{im} L^h = \mathcal{S}_n. \tag{3.6}$$

Reducing E_ε modulo $1 + z^h$ gives

$$E_\varepsilon(z) \equiv \sum_{r=1}^{h} \left(\sum_{k=0}^{m-1} \varepsilon_{r+kh} \right) z^{r-1},$$

which proves the parity formulation (2.1).

4.4.5 3.5. Why limit points belong to $\mathcal{S}_n$

The omega-limit set satisfies

$$P^k(\Omega(X)) = \Omega(X) \qquad (k \geq 1). \tag{3.7}$$

Indeed, the inclusion from left to right follows by continuity. Conversely, if

$$Y = \lim_j X^{(q_j)},$$

take a convergent subsequence of $X^{(q_j - k)}$; its limit $Z \in \Omega(X)$ satisfies $P^k(Z) = Y$.

Now let $Y = c\varepsilon \in \Omega(X)$, with $c > 0$. By (3.7), there exists $c\delta \in \Omega(X)$ such that

$$P^h(c\delta) = c\varepsilon.$$

Hence

$$\varepsilon = L^h \delta \in \operatorname{im} L^h = \mathcal{S}_n.$$

Thus

$$\Omega(X) \subseteq c(X)\mathcal{S}_n. \tag{3.8}$$

4.4.6 3.6. P is a permutation of $\mathcal{S}_n$

Since

$$\mathcal{S}_n = \operatorname{im} L^h,$$

we have $L(\mathcal{S}_n) \subseteq \mathcal{S}_n$.

It remains to prove injectivity. The kernel of L consists of the two constant binary vectors:

$$\ker L = 0, \mathbf{1}, \qquad \mathbf{1} = (1, \ldots, 1).$$

We claim that $\mathbf{1} \notin \mathcal{S}_n$. Its polynomial is

$$J_n(z) = 1 + z + \cdots + z^{n-1} = \frac{z^n + 1}{z + 1} \quad \text{in characteristic 2.}$$

Because $n = hm$ with m odd,

$$z^n + 1 = (z^m + 1)^h.$$

The root $z = 1$ has multiplicity exactly one in $z^m + 1$, since

$$\frac{d}{dz}(z^m + 1)\Big|_{z=1} = m \equiv 1 \pmod 2.$$

Thus $z = 1$ has multiplicity h in $z^n + 1$, and multiplicity $h - 1$ in J_n. Therefore

$$(1 + z)^h = 1 + z^h \nmid J_n(z).$$

So $\mathbf{1} \notin \mathcal{S}_n$.

If $L\varepsilon = L\delta$ with $\varepsilon, \delta \in \mathcal{S}_n$, then

$$\varepsilon + \delta \in \ker L.$$

It cannot equal $\mathbf{1}$, because $\mathcal{S}_n$ is a vector space and $\mathbf{1} \notin \mathcal{S}_n$. Hence $\varepsilon = \delta$.

Thus L, and therefore P, permutes $\mathcal{S}_n$.

It also follows that the binary periodic points are precisely the elements of $\mathcal{S}_n$. Indeed:

- every element of $\mathcal{S}_n$ is periodic because L permutes this finite set;
- if $L^r\varepsilon = \varepsilon$, choose a multiple $s \geq h$ of r; then

$$\varepsilon = L^s\varepsilon = L^h(L^{s-h}\varepsilon),$$

so $\varepsilon \in \operatorname{im} L^h = \mathcal{S}_n$.

4.4.7 3.7. The orbit shadows one periodic cycle

We know that all limit points lie in the finite set

$$F = c(X)\mathcal{S}_n.$$

Consequently,

$$\operatorname{dist}(X^{(q)}, F) \longrightarrow 0. \tag{3.9}$$

Otherwise a subsequence staying a positive distance from F would have a limit point outside F.

Assume $c(X) > 0$. Distinct points of F are separated in the supremum norm by at least $c(X)$.

The map P is 2-Lipschitz:

$$|P(U) - P(V)| * \infty \leq 2|U - V| * \infty. \tag{3.10}$$

For all sufficiently large q, there is therefore a unique $f_q \in F$ with

$$|X^{(q)} - f_q|_\infty < \frac{c(X)}{10}.$$

Furthermore,

$$\begin{aligned}|P(f_q) - f_{q+1}| * \infty &\leq |P(f_q) - P(X^{(q)})| * \infty + |X^{(q+1)} - f_{q+1}|_\infty \\ &< \frac{3c(X)}{10}.\end{aligned}$$

Since two distinct elements of F are at distance at least $c(X)$, this forces

$$f_{q+1} = P(f_q).$$

Hence, from some point onward, the nearest elements f_q follow one periodic cycle. Together with (3.9), this proves (2.2).

4.4.8 3.8. All real periodic points

Let Y be periodic. The sequence $|P^q(Y)|_\infty$ is both periodic and nonincreasing, hence constant.

If some coordinate of Y were strictly between 0 and $M = |Y|_\infty$, Corollary 3.2 would give

$$|P^{n-1}(Y)|_\infty < M,$$

a contradiction. Thus

$$Y = M\varepsilon$$

for a binary vector ε. The binary classification proves $\varepsilon \in \mathcal{S}_n$.

Conversely, $M\varepsilon$ is periodic whenever $\varepsilon \in \mathcal{S}_n$, by positive homogeneity:

$$P(M\varepsilon) = MP(\varepsilon).$$

This proves (2.3).

Finally, if n is a power of 2, then $h = n$, so the only degree-$< n$ polynomial divisible by $1 + z^n$ is zero. Hence

$$\mathcal{S}_n = 0$$

and every orbit converges to 0.

If n is not a power of 2, then $h < n$, so

$$|\mathcal{S}_n| = 2^{n-h} > 1,$$

and nonzero periodic orbits exist. □

4.5 4. Complete asymptotic description for $n = 5$

For $n = 5$,

$$h = 1, \qquad m = 5.$$

Condition (2.1) becomes

$$\varepsilon_1 + \cdots + \varepsilon_5 \equiv 0 \pmod 2.$$

Thus

$$\boxed{\mathcal{S}_5 = \varepsilon \in 0, 1^5 : \varepsilon \text{ has even Hamming weight.}} \tag{4.1}$$

There are 16 such vectors.

4.5.1 Proposition 4.1

The zero vector is fixed, and the other 15 vectors of $\mathcal{S}_5$ form one cycle of exact length 15.

Proof Set

$$H(z) = 1 + z + z^2 + z^3 + z^4.$$

This polynomial is irreducible over $\mathbb{F}_2$:

- it has no root in $\mathbb{F}_2$;
- if it factored without a linear factor, it would be a product of two irreducible quadratics;
- the only monic irreducible quadratic over $\mathbb{F}_2$ is

$z^2 + z + 1$, whose square is $z^4 + z^2 + 1 \neq H(z)$.

Consequently,

$$K = \mathbb{F}_2[z]/(H)$$

is the field with 16 elements.

The space $\mathcal{S}_5 = (1 + z)R_5$ is naturally isomorphic, as a vector space, to K. Under this identification, the binary Ducci map acts as multiplication by

$$\mu = 1 + \alpha,$$

where α is a nontrivial fifth root of unity in K.

Since

$$1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 = 0,$$

we get

$$\mu^3 = (1 + \alpha)^3 = 1 + \alpha + \alpha^2 + \alpha^3 = \alpha^4.$$

Thus μ^3 has order 5. Since $K^\times$ has order 15, the order of μ is either 5 or 15.

But

$$\mu^5 = (1 + \alpha)^5 = (1 + \alpha^4)(1 + \alpha) = \alpha + \alpha^4.$$

If $\mu^5 = 1$, then

$$1 + \alpha + \alpha^4 = 0.$$

Adding $1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 = 0$ gives

$$\alpha^2 + \alpha^3 = \alpha^2(1 + \alpha) = 0,$$

which is impossible. Therefore μ has order 15.

Multiplication by μ consequently acts transitively on the 15 nonzero elements of K. $\square$

One explicit orientation of the cycle is

$$\begin{aligned}
e_0 &= (1,1,0,0,0),\\
e_1 &= (0,1,0,0,1),\\
e_2 &= (1,1,0,1,1),\\
e_3 &= (0,1,1,0,0),\\
e_4 &= (1,0,1,0,0),\\
e_5 &= (1,1,1,0,1),\\
e_6 &= (0,0,1,1,0),\\
e_7 &= (0,1,0,1,0),\\
e_8 &= (1,1,1,1,0),\\
e_9 &= (0,0,0,1,1),\\
e_{10} &= (0,0,1,0,1),\\
e_{11} &= (0,1,1,1,1),\\
e_{12} &= (1,0,0,0,1),\\
e_{13} &= (1,0,0,1,0),\\
e_{14} &= (1,0,1,1,1),
\end{aligned}$$

with

$$P(e_q) = e_{q+1}, \qquad e_{15} = e_0.$$

4.5.2 Corollary 4.2

For every $X \in \mathbb{R}^5_+$, exactly one of the following holds.

Zero-attractor case

$$\boxed{P^q(X) \longrightarrow 0.}$$

Nonzero-attractor case There are $c > 0$ and a phase $r \in 0, \dots, 14$ such that

$$\boxed{\left| P^q(X) - c, e_{q+r \pmod{15}} \right|_\infty \longrightarrow 0.} \tag{4.2}$$

Thus the possible asymptotic shapes for $n = 5$ are completely classified.

4.6 5. A nonconstant 5-tuple converging to zero

The zero-attractor case is not confined to constant vectors.

Let $\rho \in (0, 1)$ be a root of

$$\rho^4 + 3\rho^3 + 2\rho^2 - 2\rho - 3 = 0. \tag{5.1}$$

Such a root exists because the polynomial is negative at 0 and positive at 1. Numerically,

$$\rho \approx 0.9275619755.$$

Define

$$V = \big(1, 1 - \rho, 1 - \rho^2, (1 + \rho)(1 - \rho^2), (1 + \rho)^2(1 - \rho^2)\big).$$

The identity

$$(1 + \rho)^3(1 - \rho^2) - 1 = -\rho\big(\rho^4 + 3\rho^3 + 2\rho^2 - 2\rho - 3\big)$$

shows that

$$(1 + \rho)^3(1 - \rho^2) = 1.$$

The coordinates satisfy

$$0 < v_2 < v_3 < v_4 < v_5 < v_1.$$

Moreover,

$$\begin{aligned} v_1 - v_2 &= \rho v_1, \\ v_3 - v_2 &= \rho v_2, \\ v_4 - v_3 &= \rho v_3, \\ v_5 - v_4 &= \rho v_4, \\ v_1 - v_5 &= \rho v_5. \end{aligned}$$

Therefore

$$\boxed{P(V) = \rho V.}$$

Consequently,

$$P^q(V) = \rho^q V \longrightarrow 0,$$

but $P^q(V) \neq 0$ for every finite q.

4.7 6. An exact non-eventually-periodic $n = 5$ orbit with nonzero attractor

This is the more delicate phenomenon. Earlier work explicitly asked whether the $n = 5$ situation permits an orbit which approaches a nontrivial periodic cycle without ever becoming periodic. ([fq.math.ca][2])

The following gives such an orbit exactly.

4.7.1 Theorem 6.1

There exists $X \in \mathbb{R}_+^5$ such that

$$|P^q(X) - e_{q \bmod 15}|_\infty \longrightarrow 0,$$

but no $P^q(X)$ is periodic.

Construction Let

$$p(t) = t^3 - 19t^2 + 139t - 1.$$

We have

$$p\left(\frac{1}{139}\right) = -\frac{2640}{2685619} < 0,$$

and

$$p\left(\frac{1}{138}\right) = \frac{16423}{2628072} > 0.$$

Also $p'(t) > 0$ on this interval. Let

$$\lambda \in \left(\frac{1}{139}, \frac{1}{138}\right)$$

be its unique root. Numerically,

$$\lambda \approx 0.007201330580038611.$$

Define

$$w_0 = \begin{pmatrix} 19960 \\ 83\lambda^2 - 424\lambda + 18633 \\ 22\lambda^2 - 1916\lambda + 6502 \\ 112\lambda^2 - 2496\lambda + 9512 \\ 7\lambda^2 - 156\lambda + 4337 \end{pmatrix}. \tag{6.1}$$

For a sign word $\sigma = (\sigma_1, \dots, \sigma_5) \in +, -^5$, let B_σ be the matrix defined by

$$(B_\sigma v)_i = s_i(v_i - v_{i+1}), \qquad s_i = \begin{cases} 1, & \sigma_i = +, \\ -1, & \sigma_i = -, \end{cases}$$

with cyclic indices.
Use the following 15 sign words:

$$\begin{array}{c|c c c|c c c|c} q & \sigma_q & & q & \sigma_q & & q & \sigma_q \\ \hline 0 & ++-+- & & 5 & +-+-- & & 10 & +-+-+ \\ 1 & -+--+ & & 6 & --++- & & 11 & -+-++ \\ 2 & ++--+ & & 7 & -+-++ & & 12 & +++-- \\ 3 & --+++ & & 8 & ++-+- & & 13 & +--+- \\ 4 & +-+-- & & 9 & +---+ & & 14 & +--+- \end{array} \tag{6.2}$$

Define recursively

$$w_{q+1} = B_{\sigma_q} w_q \qquad (0 \le q < 15).$$

The product is

$$A = B_{\sigma_{14}} \cdots B_{\sigma_0} = \begin{pmatrix} 2 & -1 & -15 & 3 & 11 \\ -9 & 10 & -13 & 5 & 7 \\ 11 & -11 & 10 & -7 & -3 \\ 1 & -1 & 9 & -4 & -5 \\ 1 & -1 & -3 & 1 & 2 \end{pmatrix}. \tag{6.3}$$

If $w_0(t)$ denotes the vector in (6.1) with λ replaced by t, direct multiplication gives

$$Aw_0(t) - tw_0(t) = p(t) \left(083221127 \right). \tag{6.4}$$

Hence

$$w_{15} = Aw_0 = \lambda w_0. \tag{6.5}$$

The finite exact verification following the proof establishes two further facts:

$$w_{q,i} > 0 \qquad (0 \le q < 15,\ 1 \le i \le 5), \tag{6.6}$$

and whenever $e_{q,i} = e_{q,i+1}$, the corresponding sign in σ_q is the sign of

$$w_{q,i} - w_{q,i+1}. \tag{6.7}$$

Let

$$K = \max_{\substack{0 \le q < 15 \\ 1 \le i \le 5}} |w_{q,i} - w_{q,i+1}|, \qquad W = \max_{\substack{0 \le q < 15 \\ 1 \le i \le 5}} w_{q,i}.$$

Choose $\varepsilon > 0$ so small that

$$\varepsilon K < \frac{1}{2}, \qquad \varepsilon W < \frac{1}{2}.$$

Set

$$X = e_0 + \varepsilon w_0. \tag{6.8}$$

Proof of the claimed dynamics For every q, consider

$$e_q + \delta w_q$$

with $0 < \delta \le \varepsilon$.

If $e_{q,i} \ne e_{q,i+1}$, their difference is ± 1, and $\delta K < 1/2$ ensures that the perturbation cannot change its sign.

If $e_{q,i} = e_{q,i+1}$, condition (6.7) determines the sign of the perturbed difference.

Therefore

$$P(e_q + \delta w_q) = e_{q+1} + \delta w_{q+1}. \tag{6.9}$$

Using $w_{15} = \lambda w_0$ and induction gives the exact formula

$$\boxed{P^{15k+q}(X) = e_q + \varepsilon \lambda^k w_q, \qquad k \ge 0, \quad 0 \le q < 15.} \tag{6.10}$$

Since $0 < \lambda < 1$,

$$P^{15k+q}(X) \longrightarrow e_q.$$

Thus the orbit approaches the nonzero 15-cycle.

On the other hand, every coordinate in (6.10) is strictly positive by (6.6). Every nonzero periodic point for $n = 5$ is a scalar multiple of an even-weight binary vector, and hence has at least one zero coordinate. Thus no iterate in (6.10) is periodic.

This proves Theorem 6.1. □

4.7.2 Exact finite verifier

This is not a floating-point experiment. All inequalities are certified using rational interval arithmetic on

$$\lambda \in \left[\frac{1}{139}, \frac{1}{138}\right]$$

and reductions modulo $p(\lambda) = 0$.

```
from fractions import Fraction
import sympy as sp

t = sp.symbols("t")
p = t**3 - 19*t**2 + 139*t - 1
lo, hi = Fraction(1, 139), Fraction(1, 138)

def peval(x):
    return x**3 - 19*x**2 + 139*x - 1

# Existence and uniqueness of the small positive root.
assert peval(lo) < 0 < peval(hi)
assert 3*lo*lo - 38*hi + 139 > 0

cycle = [
    (1,1,0,0,0),
    (0,1,0,0,1),
    (1,1,0,1,1),
    (0,1,1,0,0),
    (1,0,1,0,0),
    (1,1,1,0,1),
    (0,0,1,1,0),
```

```
    (0,1,0,1,0),
    (1,1,1,1,0),
    (0,0,0,1,1),
    (0,0,1,0,1),
    (0,1,1,1,1),
    (1,0,0,0,1),
    (1,0,0,1,0),
    (1,0,1,1,1),
]

words = [
    "++-+-", "-+--+", "++--+", "--+++",
    "+-+--", "+-+--", "--++-", "-+-++",
    "++-+-", "+---+", "+-+-+", "-+-++",
    "+++--", "+--+-", "+--+-",
]

def branch_matrix(word):
    B = sp.zeros(5)
    for i, symbol in enumerate(word):
        s = 1 if symbol == "+" else -1
        B[i, i] = s
        B[i, (i + 1) % 5] = -s
    return B

def reduce_mod_p(expr):
    return sp.rem(
        sp.Poly(sp.expand(expr), t),
        sp.Poly(p, t),
    ).as_expr()

def reduce_vector(v):
    return sp.Matrix([
        sp.expand(reduce_mod_p(x)) for x in v
    ])

w0 = sp.Matrix([
    19960,
    83*t**2 - 424*t + 18633,
    22*t**2 - 1916*t + 6502,
```

```
    112*t**2 - 2496*t + 9512,
    7*t**2 - 156*t + 4337,
])

ws = [w0]
A = sp.eye(5)

for q, word in enumerate(words):
    # Check the binary cycle and fixed signs on unequal edges.
    assert tuple(
        abs(cycle[q][i] - cycle[q][(i + 1) % 5])
        for i in range(5)
    ) == cycle[(q + 1) % 15]

    for i, symbol in enumerate(word):
        if cycle[q][i] != cycle[q][(i + 1) % 5]:
            s = 1 if symbol == "+" else -1
            assert s * (
                cycle[q][i] - cycle[q][(i + 1) % 5]
            ) == 1

    B = branch_matrix(word)
    A = B * A
    ws.append(reduce_vector(B * ws[-1]))

A_expected = sp.Matrix([
    [ 2, -1, -15,  3, 11],
    [-9, 10, -13,  5,  7],
    [11,-11,  10, -7, -3],
    [ 1, -1,   9, -4, -5],
    [ 1, -1,  -3,  1,  2],
])

assert A == A_expected
assert all(reduce_mod_p(x) == 0 for x in A*w0 - t*w0)
assert all(reduce_mod_p(x) == 0 for x in ws[15] - t*w0)

def coefficients(expr):
    poly = sp.Poly(expr, t, domain=sp.QQ)
    out = []
```

```
    for degree in range(3):
        c = poly.coeff_monomial(t**degree)
        out.append(Fraction(int(c.p), int(c.q)))
    return out

def interval(expr):
    powers = [
        (Fraction(1), Fraction(1)),
        (lo, hi),
        (lo*lo, hi*hi),
    ]

    lower = Fraction(0)
    upper = Fraction(0)

    for coefficient, (a, b) in zip(coefficients(expr), powers):
        if coefficient >= 0:
            lower += coefficient * a
            upper += coefficient * b
        else:
            lower += coefficient * b
            upper += coefficient * a

    return lower, upper

lower_bounds = []

for q in range(15):
    for i in range(5):
        # Positivity of every perturbation coordinate.
        lower_bounds.append(interval(ws[q][i])[0])

        # Correct sign whenever the binary edge is zero.
        j = (i + 1) % 5
        if cycle[q][i] == cycle[q][j]:
            s = 1 if words[q][i] == "+" else -1
            lower_bounds.append(
                interval(s * (ws[q][i] - ws[q][j]))[0]
            )
```

```
minimum = min(lower_bounds)

print("Return matrix:")
print(A)
print("Certified minimum:", minimum, float(minimum))

assert minimum == Fraction(28286630, 1333149)
assert minimum > 21
```

The certified lower bound is

$$\frac{28286630}{1333149} > 21,$$

so all required inequalities are strict with a wide margin.

4.8 7. Exact algorithm for rational starting vectors

Suppose

$$X = (x_1, \ldots, x_n) \in \mathbb{Q}_+^n.$$

Choose a common denominator D and put

$$A = DX \in \mathbb{Z}_+^n.$$

Positive homogeneity gives

$$P^q(X) = \frac{1}{D} P^q(A).$$

Let

$$R = \max_i A_i - \min_i A_i.$$

Every coordinate of $P(A)$ lies in $0, \ldots, R$, and the same is true for every later iterate. Thus after the first step there are at most

$$(R + 1)^n$$

possible states.

The following exact algorithm therefore terminates.

```
INPUT: rational vector X of length n

1. Let D be the least common multiple of all denominators.
2. Set A := D X, an integer vector.
3. Set Y := P(A).
4. Initialize an empty dictionary seen.
5. For q = 1,2,3,...:
       if Y is already in seen:
           μ := seen[Y]
           ℓ := q - μ
           return transient length μ,
                  cycle length ℓ,
                  exact cycle divided by D
       seen[Y] := q
       Y := P(Y)
```

Each iteration takes $O(n)$ integer operations on integers having $O(\log R)$ bits.

The crude worst-case bounds are therefore

$$O\big(n(R+1)^n\big)$$

arithmetic operations and

$$O\big(n(R+1)^n \log R\big)$$

bit storage. Floyd's cycle-finding method reduces the memory to $O(n \log R)$ bits, at the cost of rerunning part of the orbit to output the cycle.

The same procedure applies whenever there exist $a \neq 0$ and $b \in \mathbb{R}$ such that

$$a(X + b\mathbf{1}) \in \mathbb{Q}^n,$$

because

$$P(X + b\mathbf{1}) = P(X), \qquad P(aX) = |a|P(X).$$

4.8.1 Consequence for rational 5-tuples

For odd n, no positive constant vector lies in the image of P. Indeed, if

$$P(Y) = c\mathbf{1}, \qquad c > 0,$$

then there are signs $s_i \in \pm 1$ such that

$$y_i - y_{i+1} = s_i c.$$

Summing cyclically gives

$$0 = c \sum_{i=1}^{n} s_i,$$

which is impossible when n is odd.

It follows that for odd n, only constant starting vectors can ever reach zero exactly.

Therefore, for $n = 5$:

$$\boxed{X \in \mathbb{Q}_+^5,\ X \text{ nonconstant} \quad \Longrightarrow \quad \text{the orbit eventually enters the exact period-15 cycle.}}$$

4.9 8. Small and boundary cases

Constant vectors If $X = c\mathbf{1}$, then $P(X) = 0$.

$n = 3$ Here $h = 1$, so $\mathcal{S}_3$ consists of the even-weight binary vectors:

$$000, \quad 110, \quad 011, \quad 101.$$

The three nonzero vectors form the cycle

$$110 \longmapsto 011 \longmapsto 101 \longmapsto 110.$$

Every real 3-orbit is asymptotic either to 0 or to a scalar multiple of this 3-cycle.

$n = 4$ Here $h = 4$, so

$$\mathcal{S}_4 = 0000.$$

Thus every real orbit converges to zero. This does not imply finite extinction: the algebraic exceptional family mentioned in the editorial note converges to zero without reaching it.

$n = 5$ There is one nonzero 15-cycle up to scale and phase. Both nonperiodic possibilities occur:

- the eigenray of Section 5 approaches zero;
- the construction of Section 6 approaches the nonzero 15-cycle.

Necessity of the power-of-two hypothesis If n is not a power of 2, then

$$|\mathcal{S}_n| = 2^{n-h} > 1,$$

so nonzero periodic orbits exist. Hence "every orbit converges to zero" is equivalent to n being a power of 2.

Nonnegativity The restriction to $\mathbb{R}^n_+$ is not essential: P sends every vector in $\mathbb{R}^n$ into $\mathbb{R}^n_+$ after one iteration.

4.10 9. Exact remaining obstruction

The asymptotic shape is now determined for every n, and completely explicit for $n = 5$. What is not supplied by Theorem 2.1 is a finite intrinsic criterion for an arbitrary real input X deciding whether

$$c(X) = \lim_{q\to\infty} |P^q(X)|_\infty$$

is zero or positive, nor a closed formula for $c(X)$ and the eventual phase.

For $n = 5$, the exact obstruction can be formulated as follows.

Near the 15-cycle, the 15-step return map is piecewise linear. Each branch is a product of 15 signed difference matrices. A complete basin classification would require describing all admissible infinite paths through the resulting finite polyhedral branch graph and determining which associated matrix products contract the perturbation while preserving a positive limiting scale.

The construction in Section 6 gives one exact contracting return branch. A natural next lemma would be:

Proposed next lemma. Construct the finite directed graph of admissible polyhedral cones for the 15-step return map at e_0, and classify all strongly connected components whose matrix cocycle has a nontrivial stable direction.

That would turn the remaining $n = 5$ basin problem into a finite exact computation followed by a matrix-semigroup argument.

Thus the strongest justified status is:

$$\boxed{\begin{array}{l}\text{Complete classification of all attractors for every } n;\\ \text{complete asymptotic classification for } n=5;\\ \text{exact rational-input algorithm;}\\ \text{explicit nonperiodic } n=5 \text{ examples for both } c=0 \text{ and } c>0;\\ \text{general real basin classification still remaining.}\end{array}}$$

[1]: https://hrj.episciences.org/105/pdf ”https://hrj.episciences.org/105/pdf”
[2]: https://www.fq.math.ca/Papers1/45-2/quartzerr02_2007.pdf ”https://www.fq.math.ca/Papers1/45-2/quartzerr02_2007.pdf”

Formal verification. Lean source; online type-check.

5 Question Q701

Informal verdict. **A — Complete solution**, with **high confidence**.

Lean coverage. Lean proves the full Q701 classification: for every function f in the anchored real Holder ball on [0,1], f is extreme if and only if the path infimum of the pairwise Holder slack vanishes from 0 to every x, equivalently if and only if every x is joined to 0 by finite chains with arbitrarily small total deficit. Both implications are unconditional, using the exact scalar perturbation criterion; convexity and explicit extreme and non-extreme examples are also proved. The theorem is stated for arbitrary real alpha and therefore specializes directly to every printed 0 < alpha < 1 case.

5.1 Audited informal answer

5.1.1 1. Faithful restatement

Fix an exponent $\alpha \in (0,1)$. Let

$$\mathcal{C}_\alpha = \{f : [0,1] \to \mathbb{R} : f(0) = 0, \quad |f(x) - f(y)| \le |x-y|^\alpha \text{ for all } x, y \in [0,1]\}.$$

Equipped with the uniform topology, $\mathcal{C}_\alpha$ is a compact convex subset of $C([0,1])$. The problem asks for simple properties and, ideally, a characterization of its extreme points. There are no subparts or additional qualifications in the PDF.

The phrase "α-Hölder of ratio 1" must mean Hölder constant **at most** 1. If it meant that the optimal constant were exactly 1, the set would not be convex: x^α and $-x^\alpha$ would belong to it, whereas their midpoint 0 would not.

The answer below gives a complete intrinsic characterization.

5.1.2 2. The slack pseudometric

Write

$$d_\alpha(x, y) = |x - y|^\alpha.$$

Since $0 < \alpha < 1$, the inequality

$$(r + s)^\alpha \le r^\alpha + s^\alpha$$

shows that d_α is a metric.

For $f \in \mathcal{C}_\alpha$, define its **slack** on a pair ((x,y)) by

$$\sigma_f(x, y) == d_\alpha(x, y) - |f(x) - f(y)||x - y|^\alpha - |f(x) - f(y)|.$$

Thus $\sigma_f(x, y) \ge 0$, and $\sigma_f(x, y) = 0$ precisely when the Hölder inequality is saturated at ((x,y)).

For a finite chain

$$P = (x_0, x_1, \ldots, x_n)$$

define its total slack by

$$\ell_f(P) = \sum_{i=1}^{n} \sigma_f(x_{i-1}, x_i).$$

Finally, define

$$\rho_f(x, y) = \inf \left\{ \ell_f(x_0, \ldots, x_n) : n \ge 1,\ x_0 = x,\ x_n = y \right\},$$

and put $\rho_f(x, x) = 0$.

The function ρ_f is a pseudometric. Indeed, symmetry follows by reversing chains, and the triangle inequality follows by concatenating chains. Moreover, using the one-edge chain,

$$0 \le \rho_f(x, y) \le \sigma_f(x, y) \le d_\alpha(x, y). \tag{2.1}$$

It is also the largest pseudometric bounded above by σ_f.

5.1.3 3. Complete characterization

Theorem For $f \in \mathcal{C}_\alpha$, the following conditions are equivalent.

1. f is an extreme point of $\mathcal{C}_\alpha$.

1. For every $x \in [0,1]$,

$$\rho_f(0,x) = 0. \tag{3.1}$$

1. For every $x, y \in [0,1]$,

$$\rho_f(x,y) = 0. \tag{3.2}$$

1. For every $x \in [0,1]$ and every $\varepsilon > 0$, there exist an integer $n \geq 1$ and points

$$0 = x_0, x_1, \ldots, x_n = x$$

such that

$$\boxed{\sum_{i=1}^{n} (|x_i - x_{i-1}|^\alpha -|f(x_i) - f(x_{i-1})|) < \varepsilon.} \tag{3.3}$$

Equivalently, f is extreme precisely when every point can be joined to 0 by finite chains along which the accumulated deficit in the Hölder inequalities is arbitrarily small.

A general pointed-metric-space version of this criterion appears in a theorem of Jeff D. Farmer; the proof below is self-contained. ([SciSpace][1])

5.1.4 4. Proof of the characterization

We first identify all possible midpoint perturbations of f.

Lemma 4.1: midpoint perturbations Let $u : [0,1] \to \mathbb{R}$ satisfy $u(0) = 0$. Then

$$f + u \in \mathcal{C}_\alpha \quad \text{and} \quad f - u \in \mathcal{C}_\alpha$$

if and only if

$$|u(x) - u(y)| \le \sigma_f(x, y) \qquad \text{for all } x, y \in [0,1]. \tag{4.1}$$

Proof For real numbers (A,B),

$$\max |A + B|,\ |A - B| = |A| + |B|. \tag{4.2}$$

Indeed, if A and B have the same sign, the first absolute value equals (|A|+|B|); if they have opposite signs, the second one does.

Apply this with

$$A = f(x) - f(y), \qquad B = u(x) - u(y).$$

The two conditions

$$|A + B| \le d_\alpha(x, y), \qquad |A - B| \le d_\alpha(x, y)$$

are therefore equivalent to

$$|A| + |B| \le d_\alpha(x, y),$$

which is exactly

$$|u(x) - u(y)| \le d_\alpha(x, y) - |f(x) - f(y)| = \sigma_f(x, y).$$

This proves the lemma. □

Lemma 4.2: path formulation Condition (4.1) is equivalent to

$$|u(x) - u(y)| \le \rho_f(x, y) \qquad \text{for all } x, y. \tag{4.3}$$

Proof If (4.1) holds, then for every chain $x = x_0, \ldots, x_n = y$,

$$\begin{aligned} |u(x) - u(y)| &\le \sum_{i=1}^{n} |u(x_i) - u(x_{i-1})| \\ &\le \sum_{i=1}^{n} \sigma_f(x_{i-1}, x_i). \end{aligned}$$

Taking the infimum over all chains gives (4.3).
Conversely, (2.1) gives $\rho_f(x, y) \le \sigma_f(x, y)$, so (4.3) implies (4.1). $\square$
Thus all symmetric decompositions

$$f = \frac{(f + u) + (f - u)}{2}$$

are described exactly by the 1-Lipschitz functions for the pseudometric ρ_f, anchored at 0.

Proof that 2 implies extremality Suppose that $\rho_f(0, x) = 0$ for every x.

Assume

$$f = \frac{g + h}{2}, \qquad g, h \in \mathcal{C}_\alpha.$$

Set

$$u = \frac{g - h}{2}.$$

Then $g = f + u$, $h = f - u$, and $u(0) = 0$. By Lemma 4.2,

$$|u(x)| = |u(x) - u(0)| \le \rho_f(0, x) = 0.$$

Hence $u = 0$, so $g = h = f$. Therefore f is extreme.

Proof that failure of 2 implies non-extremality Suppose that

$$\rho_f(0, x_*) > 0$$

for some $x_* \in [0, 1]$. Define

$$u(x) = \rho_f(0, x).$$

The reverse triangle inequality for the pseudometric ρ_f gives

$$|u(x) - u(y)| = |\rho_f(0, x) - \rho_f(0, y)| \leq \rho_f(x, y).$$

By (2.1),

$$|u(x) - u(y)| \leq \rho_f(x, y) \leq \sigma_f(x, y).$$

Consequently, Lemma 4.1 gives

$$f + u,\ f - u \in \mathcal{C}_\alpha.$$

Moreover, $u(0) = 0$, while $u(x_*) > 0$, so $u \neq 0$. Thus

$$f = \frac{(f + u) + (f - u)}{2}$$

is a nontrivial midpoint decomposition, and f is not extreme.
This proves the equivalence of 1 and 2.
Conditions 2 and 3 are equivalent because

$$\rho_f(x, y) \leq \rho_f(x, 0) + \rho_f(0, y).$$

Finally, 2 and 4 are equivalent by the definition of the infimum $\rho_f(0, x)$. The theorem is proved. □

5.1.5 5. Quotient-space form of the answer

Define an equivalence relation by

$$x \sim_f y \quad \Longleftrightarrow \quad \rho_f(x, y) = 0.$$

The pseudometric ρ_f induces a genuine metric on the quotient

$$X_f = [0, 1]/! \sim_f .$$

Every nontrivial decomposition of f is obtained by choosing a nonzero anchored 1-Lipschitz function v on X_f, pulling it back to $u = v \circ q_f$, and writing

$$f = \frac{(f + u) + (f - u)}{2}.$$

Hence another exact formulation is

$$\boxed{f \text{ is extreme} \quad \Longleftrightarrow \quad X_f \text{ consists of a single point.}}$$

This not only determines whether f is extreme; it describes all its possible symmetric midpoint decompositions.

5.1.6 6. Saturated pairs and a simple sufficient condition

Call ({x,y}) a **tight pair** if

$$|f(x) - f(y)| = |x - y|^\alpha.$$

Let G_f be the graph with vertex set ([0,1]) and an edge between every tight pair.

If every point can be connected to 0 by a finite path in G_f, then every edge in the path has zero slack, so $\rho_f(0, x) = 0$. Thus f is extreme.

More generally, let R_f be the set of points connected to 0 by finite tight paths. If R_f is dense in ([0,1]), then f is extreme. Indeed, $x \mapsto \rho_f(0, x)$ is α-Hölder by

$$|\rho_f(0, x) - \rho_f(0, y)| \le \rho_f(x, y) \le |x - y|^\alpha,$$

and it vanishes on the dense set R_f.

A stronger conclusion: exposed points If R_f is dense, then f is in fact an **exposed point** of $\mathcal{C}_\alpha$.

To see this, choose a dense sequence $(z_n) \subset R_f$. For each n, choose a finite tight path from 0 to z_n. Enumerate all its oriented edges as (p_m, q_m), orienting each one so that

$$\varepsilon_m\big(f(q_m) - f(p_m)\big) = |q_m - p_m|^\alpha, \qquad \varepsilon_m \in -1, 1.$$

Define the continuous linear functional

$$\Lambda(g) = \sum_{m=1}^{\infty} 2^{-m} \varepsilon_m \big(g(q_m) - g(p_m)\big).$$

For every $g \in \mathcal{C}_\alpha$,

$$\Lambda(g) \le \sum_{m=1}^{\infty} 2^{-m} |q_m - p_m|^\alpha = \Lambda(f).$$

Equality forces equality on every oriented tight edge, hence $g(z_n) = f(z_n)$ for all n. Continuity and density then give $g = f$. Thus f is the unique maximizer of Λ.

5.1.7 7. The finite-set classification

Let

$$F = t_0, t_1, \dots, t_N, \qquad t_0 = 0,$$

be finite, and let $\mathcal{C}_\alpha(F)$ be the analogous finite-dimensional polytope.

For $f \in \mathcal{C}_\alpha(F)$, form the graph whose edges are precisely the saturated pairs

$$|f(t_i) - f(t_j)| = |t_i - t_j|^\alpha.$$

Then

$$\boxed{f \in \operatorname{Ext} \mathcal{C}_\alpha(F) \quad \Longleftrightarrow \quad \text{the saturated-pair graph is connected.}}$$

Indeed, in a finite graph the zero weighted-path distance from 0 to every vertex is equivalent to the existence of a path made entirely of zero-weight edges. If no such path exists, the minimum over the finitely many simple paths is strictly positive.

Equivalently, a finite extreme point is characterized by the existence of a spanning tree all of whose edge constraints are saturated.

This yields an exact $O(N^2)$ recognition algorithm: construct all saturated edges and run a graph search from t_0.

5.1.8 8. Explicit extreme points

8.1 Distance cones For $p \in [0, 1]$, define

$$k_p(x) = p^\alpha - |x - p|^\alpha.$$

Then $k_p(0) = 0$. Since d_α is a metric,

$$|k_p(x) - k_p(y)| = \left| d_\alpha(p, x) - d_\alpha(p, y) \right| \leq d_\alpha(x, y),$$

so $k_p \in \mathcal{C}_\alpha$.

Moreover,

$$k_p(p) - k_p(x) = |x-p|^\alpha.$$

Thus every x is tightly connected to p, and p is tightly connected to 0. Therefore k_p is extreme, indeed exposed. The same holds for $-k_p$.

Special cases include

$$k_0(x) = -x^\alpha, \qquad -k_0(x) = x^\alpha,$$

and

$$k_1(x) = 1-(1-x)^\alpha.$$

8.2 A one-turn family For $c \in [0,1]$, define

$$T_c(x) = \begin{cases} x^\alpha, & 0 \le x \le c, \\ c^\alpha - (x-c)^\alpha, & c \le x \le 1. \end{cases} \tag{8.1}$$

We verify that $T_c \in \mathcal{C}_\alpha$.

If both points lie on the same side of c, this follows from

$$|r^\alpha - s^\alpha| \le |r-s|^\alpha.$$

Now let $x \le c \le y$. Then

$$\begin{aligned} T_c(x) - T_c(y) &= x^\alpha - c^\alpha + (y-c)^\alpha \\ &\le (y-c)^\alpha \le (y-x)^\alpha, \end{aligned}$$

while

$$\begin{aligned} T_c(y) - T_c(x) &= c^\alpha - x^\alpha - (y-c)^\alpha \\ &\le c^\alpha - x^\alpha \le (c-x)^\alpha \le (y-x)^\alpha. \end{aligned}$$

Hence $|T_c(x) - T_c(y)| \le |x-y|^\alpha$.

Every $x \le c$ is tightly connected to 0, since $T_c(x) = x^\alpha$. Every $x \ge c$ is tightly connected to c, since

$$T_c(c) - T_c(x) = (x-c)^\alpha.$$

Thus T_c is extreme and exposed.

5.1.9 9. McShane-type construction of further extreme points

Let $A \subset [0,1]$ be compact, with $0 \in A$, and let $g : A \to \mathbb{R}$ be an extreme anchored 1-Lipschitz function for d_α.

Define the two canonical extensions

$$M^+g(x) = \inf_{y \in A}\bigl(g(y) + |x-y|^\alpha\bigr),$$

and

$$M^-g(x) = \sup_{y \in A}\bigl(g(y) - |x-y|^\alpha\bigr).$$

Both are elements of $\mathcal{C}_\alpha$ and agree with g on A.

Because A is compact, the infimum or supremum is attained. Thus for every x, there is some $y_x \in A$ such that either

$$M^+g(x) - g(y_x) = |x - y_x|^\alpha$$

or

$$g(y_x) - M^-g(x) = |x - y_x|^\alpha.$$

Consequently x is joined to A by a zero-slack edge. Within A, extremality of g supplies arbitrarily small-slack chains to 0. Concatenating them proves, by the main theorem, that both M^+g and M^-g are extreme.

In particular, beginning with a finite extreme configuration—equivalently, a connected saturated graph—and applying either extension gives a large explicit class of extreme functions on the whole interval.

5.1.10 10. Necessary conditions, and why they are not sufficient

Let

$$[f]_\alpha = \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x-y|^\alpha}.$$

Every extreme point lies on the Hölder sphere If $[f]_\alpha = L < 1$, set

$$u(x) = (1-L)x^\alpha.$$

Then $[u]_\alpha = 1 - L$, and therefore

$$[f \pm u] * \alpha \leq [f] * \alpha + [u]_\alpha = 1.$$

Thus $f \pm u \in \mathcal{C}_\alpha$, $u \neq 0$, and f is not extreme. Hence

$$f \in \operatorname{Ext} \mathcal{C}_\alpha \quad \Longrightarrow \quad [f]_\alpha = 1. \tag{10.1}$$

Hölder norm 1 is not sufficient Consider

$$f(x) = x.$$

Because $0 \leq |x - y| \leq 1$,

$$|x - y| \leq |x - y|^\alpha,$$

so $f \in \mathcal{C}_\alpha$, and $[f]_\alpha = 1$, with equality at ((0,1)).
Nevertheless f is not extreme. Define

$$u(x) = (1 - \alpha)x(1 - x).$$

For $x < y$, put $r = y - x$. Then

$$|u(y) - u(x)| = (1 - \alpha)r, |1 - x - y|.$$

Since $0 \leq x \leq 1 - r$,

$$|1 - x - y| \leq 1 - r,$$

and hence

$$|u(y) - u(x)| \leq (1 - \alpha)r(1 - r).$$

The convexity inequality

$$r^{-(1-\alpha)} \geq 1 + (1 - \alpha)(1 - r)$$

gives, after multiplication by r,

$$r^\alpha - r \geq (1 - \alpha)r(1 - r).$$

Therefore

$$|u(y) - u(x)| \leq r^\alpha - r = |x - y|^\alpha - |f(x) - f(y)|.$$

Lemma 4.1 now yields

$$f + u, \quad f - u \in \mathcal{C}_\alpha.$$

Since $u \neq 0$, f is not extreme.

Thus the condition $[f]_\alpha = 1$ is necessary but far from sufficient.

5.1.11 11. Strong extremality

A point $f \in \mathcal{C}_\alpha$ is called strongly extreme in the uniform norm if

$$\left\| \frac{g_n + h_n}{2} - f \right\|_\infty \longrightarrow 0 \quad \Longrightarrow \quad |g_n - h_n| * \infty \longrightarrow 0$$

for all sequences $g_n, h_n \in \mathcal{C}_\alpha$.

Because $\mathcal{C}_\alpha$ is compact in the uniform norm,

$$\boxed{f \text{ is extreme} \quad \Longleftrightarrow \quad f \text{ is strongly extreme.}}$$

Only the nontrivial implication needs proof. If it failed, there would be sequences with midpoint converging to f but

$$|g_n - h_n|_\infty \geq \varepsilon > 0.$$

By compactness, a subsequence would satisfy

$$g_n \to g, \qquad h_n \to h$$

uniformly. Then

$$f = \frac{g+h}{2}, \qquad |g - h|_\infty \geq \varepsilon,$$

contradicting extremality.

5.1.12 12. Boundary cases and necessity of the hypotheses

The case $\alpha = 1$ The same slack-pseudometric theorem remains valid. It reduces to the classical concrete characterization

$$\boxed{f \in \operatorname{Ext} \mathcal{C}_1 \quad \Longleftrightarrow \quad |f'(t)| = 1 \text{ for almost every } t \in [0, 1].}$$

Here every 1-Lipschitz function is absolutely continuous and $|f'| \leq 1$ almost everywhere.

Indeed, for $x < y$,

$$\rho_f(x,y) = \int_x^y \bigl(1 - |f'(t)|\bigr), dt. \tag{12.1}$$

For the upper bound, take increasingly fine monotone partitions. The sum of the absolute increments converges upward to the total variation

$$\mathrm{Var}_{[x,y]}(f) = \int_x^y |f'(t)|, dt.$$

For the reverse inequality, every edge $[z_{i-1}, z_i]$ of an arbitrary chain satisfies

$$|z_i - z_{i-1}| - |f(z_i) - f(z_{i-1})| \geq \int_{[z_{i-1},z_i]} (1 - |f'|).$$

Every point between x and y is crossed by at least one edge of a chain from x to y, so summation gives the reverse inequality in (12.1).

Notice the sharp change at the boundary: $f(x) = x$ is extreme for $\alpha = 1$, but, as shown above, is not extreme for any $0 < \alpha < 1$.

The case $\alpha > 1$ Every such Hölder function on an interval is constant. Indeed, subdividing ([x,y]) into n equal pieces gives

$$|f(y) - f(x)| \leq n\left(\frac{y-x}{n}\right)^{\alpha} = (y-x)^{\alpha} n^{1-\alpha} \longrightarrow 0.$$

With $f(0) = 0$, the set consists only of the zero function, which is trivially extreme.

The case $\alpha = 0$ The stated compactness fails. Even among continuous functions, the family with oscillation at most 1 is not equicontinuous; for example,

$$f_n(x) = \min(1, nx)$$

has no uniformly convergent subsequence.

The anchoring condition The condition $f(0) = 0$ is essential. Without it, arbitrary constants may be added:

$$f = \frac{(f+c) + (f-c)}{2}.$$

Thus the unanchored set has no extreme points and is not bounded in the uniform norm.

The real-valued condition The scalar identity

$$\max(|A+B|,|A-B|)=|A|+|B|$$

is specifically real. For complex- or vector-valued Hölder maps, the midpoint perturbation condition and therefore the characterization require modification.

5.1.13 13. Compactness, separation, countability, and choice audit

The ambient space is the real Banach space $C([0,1])$ with the uniform norm, hence a Hausdorff metric space.

For every $f \in \mathcal{C}_\alpha$,

$$|f(x)| \leq x^\alpha \leq 1,$$

and the family has the common modulus of continuity $r \mapsto r^\alpha$. It is closed under uniform convergence. The metric form of Arzelà–Ascoli therefore gives compactness. In this setting compactness and sequential compactness are equivalent.

The main extremality theorem itself is algebraic and metric; it uses neither compactness nor separability and works for arbitrary pointed metric spaces. Compactness is used only for the strong-extremality consequence and for attainment in the extension construction.

No general net argument, transfinite induction, or nonconstructive choice principle is required. The exposed-point argument uses only the separability of ([0,1]), through a countable dense sequence.

5.1.14 Final classification

The extreme points of the compact convex set in Q701 are exactly the functions satisfying

$$\boxed{\forall x \in [0,1]\ \forall \varepsilon > 0 \exists, 0 = x_0, x_1, \ldots, x_n = x \quad \sum_{i=1}^{n} (|x_i - x_{i-1}|^\alpha ------------------}$$

Equivalently, the largest pseudometric lying below the pairwise Hölder slack

$$|x-y|^\alpha - |f(x)-f(y)|$$

must vanish identically. This criterion is necessary and sufficient and also gives a canonical nonzero perturbation $u(x) = \rho_f(0, x)$ whenever f is not extreme.

[1]: https://scispace.com/pdf/extreme-points-of-the-unit-ball-of-the-space-of-lipschitz-3l772juamw.pdf "Extreme points of the unit ball of the space of Lipschitz functions"

Formal verification. Lean source; online type-check.

6 Question Q706

Informal verdict. C — Substantial progress, with **high confidence in the stated exact special cases**.

Lean coverage (partial). The published/imported generation-1 source remains the verified 3,644-line partial corpus. The latest archive reports compiling 5,366 Lean lines across Q706, Q706Ext, and Q706Second, adding general compactness and attainment, singleton boxes, global bounds and reductions, interlacing, new signed second/penultimate cells, all endpoints for n<=3, homogeneity, translation, and etaConst_penultimate. General intermediate endpoints remain unevaluated; q706_aggregate uses sInf/sSup and conditional known cells, and the audited n<=5 plus eta_(6,2)=4 browser results are not in Lean.

6.1 Audited informal answer

6.2 Partial result

6.2.1 1. Faithful restatement

For $a \leq b$, let $S_n([a, b])$ be the set of real symmetric $n \times n$ matrices all of whose entries belong to ([a,b]). For a real symmetric matrix A, write

$$\lambda_1(A) \geq \lambda_2(A) \geq \cdots \geq \lambda_n(A)$$

for its eigenvalues in decreasing order. The question asks for the lower and upper endpoints of

$$\{\lambda_k(A) : A \in S_n([a, b])\}$$

as functions of (n,a,b). The statement implicitly assumes $1 \leq k \leq n$; there is no other ambiguity or apparent typo.

Define

$$L_k(n;a,b) := \min_{A\in S_n([a,b])} \lambda_k(A), \qquad U_k(n;a,b) := \max_{A\in S_n([a,b])} \lambda_k(A).$$

The set $S_n([a,b])$ is compact and connected, and $A \mapsto \lambda_k(A)$ is continuous. Hence the set asked for in Q706 is exactly the compact interval

$$\boxed{[L_k(n;a,b), U_k(n;a,b)]}.$$

I obtain complete closed formulas for $k = 1, n$, complete formulas for $k = 2$ when $0 \le a \le b$, and—by sign reversal—for $k = n-1$ when $a \le b \le 0$. For every k, I also give an exact invariant minimax characterization and explicit universal bounds.

6.3 2. Main results

Throughout, put

$$c := \frac{a+b}{2}, \qquad r := \frac{b-a}{2}, \qquad \rho := \max|a|, |b|.$$

6.3.1 2.1 Exact answer for the largest and smallest ordered eigenvalues

For every $n \ge 2$,

$$\boxed{L_1(n;a,b) = \begin{cases} na, & a \ge 0, \\ a, & a \le 0 \le b, \\ a-b, & b \le 0, \end{cases}} \tag{2.1}$$

and

$$\boxed{U_n(n;a,b) = \begin{cases} b-a, & a \ge 0, \\ b, & a \le 0 \le b, \\ nb, & b \le 0. \end{cases}} \tag{2.2}$$

The other two outer endpoints are

$$\boxed{U_1(n;a,b)=\begin{cases} nb, & a+b\ge 0,\\ \dfrac{n(b-a)}{2}, & a+b\le 0,\ n \text{ even},\\ \dfrac{nb+\sqrt{(n^2-1)a^2+b^2}}{2}, & a+b\le 0,\ n \text{ odd},\end{cases}} \tag{2.3}$$

and

$$\boxed{L_n(n;a,b)=\begin{cases} na, & a+b\le 0,\\ \dfrac{n(a-b)}{2}, & a+b\ge 0,\ n \text{ even},\\ \dfrac{na-\sqrt{a^2+(n^2-1)b^2}}{2}, & a+b\ge 0,\ n \text{ odd}.\end{cases}} \tag{2.4}$$

The formulas agree on the boundary $a+b=0$.
Thus the ranges of λ_1 and λ_n are completely determined:

$$\lambda_1(A) : A\in S_n([a,b]) = [L_1(n;a,b), U_1(n;a,b)],$$

$$\lambda_n(A) : A\in S_n([a,b]) = [L_n(n;a,b), U_n(n;a,b)].$$

6.3.2 2.2 Exact answer for λ_2 when the interval is nonnegative

Assume

$$0\le a\le b.$$

Then

$$\boxed{L_2(n;a,b)=a-b} \tag{2.5}$$

and

$$\boxed{U_2(n;a,b)=\begin{cases} \dfrac{n(b-a)}{2}, & n \text{ even},\\ \dfrac{nb-\sqrt{(n^2-1)a^2+b^2}}{2}, & n \text{ odd}.\end{cases}} \tag{2.6}$$

Consequently,

$$\boxed{\lambda_2(A) : A \in S_n([a,b]) = \left[a-b, \begin{cases} \dfrac{n(b-a)}{2}, & n \text{ even}, \\ \dfrac{nb - \sqrt{(n^2-1)a^2+b^2}}{2}, & n \text{ odd} \end{cases} \right].}$$

In particular, for $a = 0$,

$$\boxed{U_2(n;0,b) = b \left\lfloor \frac{n}{2} \right\rfloor.} \tag{2.7}$$

6.3.3 2.3 Exact answer for λ_{n-1} when the interval is nonpositive

By applying (2.5)–(2.6) to (-A), if

$$a \le b \le 0,$$

then

$$\boxed{U_{n-1}(n;a,b) = b-a} \tag{2.8}$$

and

$$\boxed{L_{n-1}(n;a,b) = \begin{cases} \dfrac{n(a-b)}{2}, & n \text{ even}, \\ \dfrac{na + \sqrt{a^2+(n^2-1)b^2}}{2}, & n \text{ odd}. \end{cases}} \tag{2.9}$$

6.3.4 2.4 Exact invariant characterization for every k

Let $\mathbf{1} = (1,\dots,1)^T$. For a symmetric matrix $X = (x_{ij})$, write

$$|X|_{1,e} := \sum_{i,j=1}^{n} |x_{ij}|$$

for its entrywise 1-norm.

For a subspace $V \subseteq \mathbb{R}^n$, define

$$\mathcal{D}(V) := \{X \in S_n(\mathbb{R}) : X \succeq 0,\ \mathrm{tr}\, X = 1,\ \mathrm{Ran}\, X \subseteq V\}.$$

Then for every $1 \le k \le n$,

$$\boxed{U_k(n;a,b) = \max_{V \le \mathbb{R}^n} \max_{\dim V = k} \min_{X \in \mathcal{D}(V)} \left(c, \mathbf{1}^T X \mathbf{1} + r|X|_{1,e}\right).} \tag{2.10}$$

Putting $q = n - k + 1$,

$$\boxed{L_k(n;a,b) = \min_{W \le \mathbb{R}^n} \min_{\dim W = q} \max_{X \in \mathcal{D}(W)} \left(c, \mathbf{1}^T X \mathbf{1} - r|X|_{1,e}\right).} \tag{2.11}$$

These are exact formulas, not relaxations.
For the centered interval ([-r,r]), they simplify to

$$\boxed{U_k(n;-r,r) = r, \eta_{n,k}, \qquad \eta_{n,k} := \max_{\dim V = k} \min_{X \in \mathcal{D}(V)} |X|_{1,e}.} \tag{2.12}$$

Furthermore,

$$L_k(n;-r,r) = -U_{n-k+1}(n;-r,r). \tag{2.13}$$

The constants satisfy

$$\eta_{n,1} = n, \qquad \eta_{n,n} = 1.$$

Evaluating $\eta_{n,k}$ explicitly for all intermediate k is the central remaining obstruction.

6.3.5 2.5 General explicit bounds for all k

For every $1 \le k \le n$,

$$\boxed{U_k(n;a,b) \le b + \rho\sqrt{\frac{(n-k)(n-1)}{k}}} \tag{2.14}$$

and

$$\boxed{L_k(n;a,b) \ge a - \rho\sqrt{\frac{(k-1)(n-1)}{n-k+1}}.} \tag{2.15}$$

A second family of bounds follows from the geometry of orthogonal projections. Let

$$c_+ := \max(c, 0), \qquad c_- := \min(c, 0).$$

An elementary estimate gives

$$\boxed{U_k(n; a, b) \le \frac{n}{k}\big(c_+ + r\sqrt{k}\big)} \tag{2.16}$$

and, with $q = n - k + 1$,

$$\boxed{L_k(n; a, b) \ge \frac{n}{q}\big(c_- - r\sqrt{q}\big).} \tag{2.17}$$

A sharper known projection estimate replaces $\sqrt{s}$ by

$$\beta_s := \frac{s + \sqrt{s+2}}{1 + \sqrt{s+2}},$$

giving

$$\boxed{U_k(n; a, b) \le \frac{n}{k}\big(c_+ + r\beta_k\big)} \tag{2.18}$$

and

$$\boxed{L_k(n; a, b) \ge \frac{n}{n-k+1}\big(c_- - r\beta_{,n-k+1}\big).} \tag{2.19}$$

The external input here is the theorem that every rank-s real orthogonal projection P of order n satisfies

$$\sum_{i,j} |p_{ij}| \le \beta_s n.$$

Its hypotheses match exactly the spectral projections used below. ([arXiv][1])

6.4 3. Preliminary reductions

6.4.1 3.1 Sign reversal

If $A \in S_n([a, b])$, then

$$-A \in S_n([-b, -a])$$

and

$$\lambda_k(-A) = -\lambda_{n-k+1}(A).$$

Therefore

$$\boxed{L_k(n;a,b) = -U_{n-k+1}(n;-b,-a).} \tag{3.1}$$

This will repeatedly halve the work.

6.4.2 3.2 The diagonal may be fixed at an endpoint

Suppose (A') is obtained from A by increasing some diagonal entries. Then

$$A' - A \succeq 0.$$

By the Courant–Fischer principle,

$$\lambda_k(A') \geq \lambda_k(A)$$

for every k. Consequently:

- in computing $U_k(n;a,b)$, all diagonal entries may be taken equal to b;
- in computing $L_k(n;a,b)$, all diagonal entries may be taken equal to a.

This observation is used throughout.

6.5 4. Proof of the exact outer formulas

6.5.1 4.1 Maximizing λ_1

For a fixed unit vector $x = (x_i)$,

$$x^T A x = \sum_{i,j} a_{ij} x_i x_j.$$

Each coefficient a_{ij} can be chosen independently, subject only to symmetry. It is optimal to choose b when $x_i x_j \geq 0$ and a when $x_i x_j < 0$. Thus

$$\max_{A \in S_n([a,b])} x^T A x = c \left(\sum_i x_i \right)^2 + r \left(\sum_i |x_i| \right)^2 . \tag{4.1}$$

Since the two maximizations commute,

$$U_1(n;a,b) = \max_{|x|_2=1} \left[c(\mathbf{1}^T x)^2 + r|x|_1^2 \right] . \tag{4.2}$$

Case $a+b \geq 0$ Here $c \geq 0$. Since

$$|\mathbf{1}^T x| \leq |x|_1 \leq \sqrt{n},$$

we obtain

$$c(\mathbf{1}^T x)^2 + r|x|_1^2 \leq (c+r)|x|_1^2 = b|x|_1^2 \leq nb.$$

Equality is attained for

$$x = \frac{1}{\sqrt{n}}\mathbf{1}, \qquad A = bJ_n.$$

Hence

$$U_1 = nb.$$

Case $a+b<0$ In this case $a<0$ and $a^2>b^2$.

For a sign vector $\varepsilon \in \pm 1^n$, define the endpoint matrix

$$B_\varepsilon = (b_{ij}), \qquad b_{ij} = \begin{cases} b, & \varepsilon_i = \varepsilon_j, \\ a, & \varepsilon_i \neq \varepsilon_j. \end{cases}$$

For each x, choosing $\varepsilon_i = \operatorname{sgn} x_i$ gives the right-hand side of (4.1), so

$$U_1 = \max_\varepsilon \lambda_1(B_\varepsilon). \tag{4.3}$$

Suppose ε has p plus signs and $q = n-p$ minus signs. Up to permutation,

$$B_\varepsilon = \begin{pmatrix} bJ_p & aJ_{p,q} \\ aJ_{q,p} & bJ_q \end{pmatrix}. \tag{4.4}$$

It has $n-2$ zero eigenvalues, and its remaining two eigenvalues are those of

$$\begin{pmatrix} bp & a\sqrt{pq} \\ a\sqrt{pq} & bq \end{pmatrix}.$$

Thus

$$\lambda_1(B_\varepsilon) = \frac{bn + \sqrt{b^2(p-q)^2 + 4a^2pq}}{2}. \tag{4.5}$$

Since

$$b^2(p-q)^2 + 4a^2pq = a^2n^2 + (b^2-a^2)(p-q)^2$$

and $b^2 - a^2 < 0$, the expression is maximized when (|p-q|) is as small as possible.

If n is even, $p = q = n/2$, giving

$$U_1 = \frac{n(b-a)}{2}.$$

If n is odd, $|p-q| = 1$ and $4pq = n^2 - 1$, giving

$$U_1 = \frac{nb + \sqrt{(n^2-1)a^2 + b^2}}{2}.$$

This proves (2.3).

6.5.2 4.2 Minimizing λ_1

The map $A \mapsto \lambda_1(A)$ is convex and invariant under simultaneous permutation of rows and columns.

Given A, average PAP^T over all permutation matrices P. The result has the form

$$\overline{A} = (d-e)I + eJ,$$

where $d, e \in [a, b]$. Convexity gives

$$\lambda_1(\overline{A}) \leq \lambda_1(A).$$

Hence the minimum is attained among such two-parameter matrices.

Their eigenvalues are

$$d + (n-1)e$$

on $\mathbb{R}\mathbf{1}$, and

$$d - e$$

with multiplicity $n-1$. Increasing d increases every eigenvalue, so $d = a$. We must therefore minimize

$$f(e) = \max a + (n-1)e,\ a - e, \qquad e \in [a, b]. \tag{4.6}$$

If $e \geq 0$, the first term is larger. If $e \leq 0$, the second is larger.

- If $a \geq 0$, the minimum occurs at $e = a$, and equals na.
- If $a \leq 0 \leq b$, the minimum occurs at $e = 0$, and equals a.
- If $b \leq 0$, the minimum occurs at $e = b$, and equals $a - b$.

This proves (2.1).

The extremizing matrices may be chosen respectively as

$$aJ, \qquad aI, \qquad (a-b)I + bJ.$$

6.5.3 4.3 The formulas for L_n and U_n

Apply the sign-reversal identity (3.1) to the formulas just proved:

$$L_n(n; a, b) = -U_1(n; -b, -a),$$

$$U_n(n; a, b) = -L_1(n; -b, -a).$$

This gives exactly (2.2) and (2.4).

For example, when $a + b > 0$, a minimizing matrix for λ_n is obtained from a balanced partition $p + q = n$:

$$\begin{pmatrix} aJ_p & bJ_{p,q} \\ bJ_{q,p} & aJ_q \end{pmatrix}.$$

6.6 5. Exact λ_2-range for nonnegative intervals

Assume $0 \leq a \leq b$.

6.6.1 5.1 The lower endpoint

In computing L_2, set every diagonal entry equal to a.

For any two indices $i \neq j$, the corresponding principal 2×2 submatrix is

$$\begin{pmatrix} a & t \\ t & a \end{pmatrix}, \qquad t \in [a, b].$$

Its smaller eigenvalue is $a - t \geq a - b$.

By Courant–Fischer, restricting the two-dimensional maximizing subspace to the coordinate plane generated by e_i, e_j gives the interlacing inequality

$$\lambda_2(A) \geq a - t \geq a - b.$$

Conversely,

$$A = (a-b)I + bJ$$

belongs to $S_n([a,b])$ and has eigenvalues

$$a + (n-1)b, \qquad a-b, \ldots, a-b.$$

Hence

$$L_2 = a - b.$$

6.6.2 5.2 The upper endpoint when $a > 0$

By diagonal monotonicity, take $a_{ii} = b$.

Because every entry of A is strictly positive, the top eigenvalue is simple and admits an eigenvector with all coordinates strictly positive. Indeed, replacing a vector by its coordinatewise absolute value does not decrease its Rayleigh quotient, and strict positivity of the entries gives strict improvement for a vector having both signs.

Let x be an eigenvector for $\lambda = \lambda_2(A)$, orthogonal to the positive top eigenvector. Hence x has both positive and negative coordinates.

Set

$$P = i : x_i > 0, \qquad N = i : x_i < 0,$$

$$p = |P|, \qquad q = |N|,$$

and

$$U = \sum_{i \in P} x_i > 0, \qquad V = -\sum_{j \in N} x_j > 0.$$

For $i \in P$,

$$\begin{aligned} \lambda x_i &= \sum_{j \in P} a_{ij} x_j + \sum_{j \in N} a_{ij} x_j \\ &\leq bU - aV. \end{aligned}$$

Summing over $i \in P$ gives

$$\lambda U \leq p(bU - aV). \tag{5.1}$$

Similarly, for $j \in N$,

$$\lambda(-x_j) \leq bV - aU,$$

and hence

$$\lambda V \leq q(bV - aU). \tag{5.2}$$

If $\lambda \leq 0$, the desired positive upper bound is immediate. Assume $\lambda > 0$. Equations (5.1)–(5.2) imply

$$(bp - \lambda)U \geq apV,$$

$$(bq - \lambda)V \geq aqU.$$

Multiplying and cancelling $UV > 0$,

$$(bp - \lambda)(bq - \lambda) \geq a^2 pq. \tag{5.3}$$

Moreover $bp - \lambda > 0$ and $bq - \lambda > 0$. Therefore λ is no larger than the smaller root of

$$(bp - t)(bq - t) - a^2 pq = 0.$$

Thus

$$\lambda \leq t_-(p,q) := \frac{b(p+q) - \sqrt{b^2(p-q)^2 + 4a^2pq}}{2}. \tag{5.4}$$

Write $m = p + q \leq n$. Since

$$b^2(p-q)^2 + 4a^2pq = a^2m^2 + (b^2 - a^2)(p-q)^2,$$

for fixed m the right-hand side of (5.4) is maximized by making (|p-q|) as small as possible.

Define

$$\phi_m := \begin{cases} \dfrac{m(b-a)}{2}, & m \text{ even}, \\ \dfrac{mb - \sqrt{(m^2-1)a^2 + b^2}}{2}, & m \text{ odd}. \end{cases}$$

Then $t_-(p,q) \le \phi_m$.

It remains to note that ϕ_m is nondecreasing in m. For even m,

$$\phi_{m+1} - \phi_m = \frac{b + ma - \sqrt{b^2 + a^2 m(m+2)}}{2} \ge 0,$$

because

$$(b+ma)^2 - \big(b^2 + a^2 m(m+2)\big) = 2am(b-a) \ge 0.$$

For odd m,

$$\phi_{m+1} - \phi_m = \frac{b - (m+1)a + \sqrt{a^2(m^2-1) + b^2}}{2} \ge 0.$$

If $(m+1)a \le b$, this is immediate. Otherwise it follows by squaring, since

$$a^2(m^2-1) + b^2 - \big((m+1)a - b\big)^2 = 2a(m+1)(b-a) \ge 0.$$

Consequently,

$$\lambda_2(A) \le \phi_n.$$

6.6.3 5.3 The case $a = 0$

For $0 < \varepsilon < 1$, put

$$A_\varepsilon = (1-\varepsilon)A + \varepsilon b J.$$

Then A_ε has diagonal b and all its entries lie in $[\varepsilon b, b]$. The preceding result gives

$$\lambda_2(A_\varepsilon) \le \phi_n(\varepsilon b, b).$$

Letting $\varepsilon \downarrow 0$ and using continuity of the eigenvalues yields the formula for $a = 0$.

6.6.4 5.4 Attainment

Take a partition with

$$p = \left\lfloor \frac{n}{2} \right\rfloor, \qquad q = \left\lceil \frac{n}{2} \right\rceil$$

and define

$$A_{p,q} := \begin{pmatrix} bJ_p & aJ_{p,q} \\ aJ_{q,p} & bJ_q \end{pmatrix}. \tag{5.5}$$

It has $n-2$ zero eigenvalues, while its two remaining eigenvalues are

$$\frac{bn \pm \sqrt{b^2(p-q)^2 + 4a^2pq}}{2}.$$

Both are nonnegative, and the smaller one is therefore $\lambda_2(A_{p,q})$. It equals exactly the right-hand side of (2.6).

This proves the complete nonnegative-interval result.

6.7 6. Exact minimax characterization for arbitrary k

We now prove (2.10)–(2.11).

6.7.1 6.1 A bilinear endpoint computation

Let $X \succeq 0$ with $\operatorname{tr} X = 1$. In particular $x_{ii} \geq 0$.

Since

$$\operatorname{tr}(AX) = \sum_i a_{ii}x_{ii} + 2\sum_{i<j} a_{ij}x_{ij},$$

entrywise maximization gives

$$\max_{A \in S_n([a,b])} \operatorname{tr}(AX) = c, \mathbf{1}^T X \mathbf{1} + r|X|_{1,e}. \tag{6.1}$$

Similarly,

$$\min_{A \in S_n([a,b])} \operatorname{tr}(AX) = c, \mathbf{1}^T X \mathbf{1} - r|X|_{1,e}. \tag{6.2}$$

6.7.2 6.2 Upper endpoint

Courant–Fischer gives

$$\lambda_k(A) = \max_{\dim V = k} \min_{x \in V \|x\|=1} x^T Ax.$$

For fixed V,

$$\min_{x \in V \|x\|=1} x^T Ax = \min_{X \in \mathcal{D}(V)} \operatorname{tr}(AX),$$

because a linear functional on the compact convex set $\mathcal{D}(V)$ attains its minimum at a rank-one density matrix xx^T.

Therefore

$$U_k = \max_{\dim V = k} \max_{A \in S_n([a,b])} \min_{X \in \mathcal{D}(V)} \operatorname{tr}(AX).$$

For fixed V, the sets of admissible A and X are compact and convex, and the function $(A, X) \mapsto \operatorname{tr}(AX)$ is bilinear. The finite-dimensional minimax theorem thus gives

$$\max_A \min_X \operatorname{tr}(AX) = \min_X \max_A \operatorname{tr}(AX).$$

Using (6.1) proves (2.10).

6.7.3 6.3 Lower endpoint

Let $q = n - k + 1$. The second form of Courant–Fischer is

$$\lambda_k(A) = \min_{\dim W = q} \max_{x \in W \|x\|=1} x^T Ax.$$

As above,

$$\max_{x \in W \|x\|=1} x^T Ax = \max_{X \in \mathcal{D}(W)} \operatorname{tr}(AX).$$

Thus

$$L_k = \min_{\dim W = q} \min_A \max_{X \in \mathcal{D}(W)} \operatorname{tr}(AX).$$

Applying minimax in the opposite order and then (6.2) gives (2.11).

6.8 7. Ky Fan envelope and projection bounds

Let $\mathcal{P}_s(n)$ denote the rank-s orthogonal projections in $\mathbb{R}^n$.

The Ky Fan maximum principle gives

$$\sum_{i=1}^{s} \lambda_i(A) = \max_{P \in \mathcal{P}_s(n)} \operatorname{tr}(PA).$$

Consequently,

$$\boxed{\max_{A \in S_n([a,b])} \sum_{i=1}^{s} \lambda_i(A) = \max_{P \in \mathcal{P}_s(n)} \bigl(c\,\mathbf{1}^T P \mathbf{1} + r|P|_{1,e}\bigr)\,.} \tag{7.1}$$

Similarly,

$$\boxed{\min_{A \in S_n([a,b])} \sum_{i=n-s+1}^{n} \lambda_i(A) = \min_{P \in \mathcal{P}_s(n)} \bigl(c\,\mathbf{1}^T P \mathbf{1} - r|P|_{1,e}\bigr)\,.} \tag{7.2}$$

Since

$$s\lambda_s(A) \le \sum_{i=1}^{s} \lambda_i(A),$$

equation (7.1) yields an upper bound for U_s. Likewise, with $q = n-k+1$,

$$\sum_{i=k}^{n} \lambda_i(A) \le q\lambda_k(A),$$

so (7.2) yields a lower bound for L_k.

For an orthogonal projection P,

$$0 \le \mathbf{1}^T P \mathbf{1} = |P\mathbf{1}|^2 \le n, \tag{7.3}$$

and, by Cauchy–Schwarz,

$$\begin{aligned} |P|_{1,e} &\le n \left(\sum_{i,j} p_{ij}^2 \right)^{1/2} \\ &= n\bigl(\operatorname{tr} P^2\bigr)^{1/2} \\ &= n\sqrt{s}. \end{aligned} \tag{7.4}$$

Equations (2.16)–(2.17) follow. Replacing (7.4) by the sharper projection theorem stated after (2.19) proves (2.18)–(2.19).

For the centered interval ([-r,r]), (7.1) becomes

$$\max_A \sum_{i=1}^{s} \lambda_i(A) = r \max_{P\in\mathcal{P}_s(n)} |P|_{1,e}. \tag{7.5}$$

Thus even the optimal sum of the first s eigenvalues is exactly a finite-dimensional projection-constant problem.

6.9 8. Proof of the elementary variance bounds

We use the following order-statistic lemma.

Lemma Let

$$\mu_1 \geq \cdots \geq \mu_n, \qquad \overline{\mu} = \frac{1}{n}\sum_i \mu_i, \qquad V = \sum_i (\mu_i - \overline{\mu})^2.$$

Then

$$\mu_k \leq \overline{\mu} + \sqrt{\frac{n-k}{kn}V}, \tag{8.1}$$

and

$$\mu_k \geq \overline{\mu} - \sqrt{\frac{k-1}{n(n-k+1)}V}. \tag{8.2}$$

Proof Put $\delta = \mu_k - \overline{\mu}$. If $\delta \leq 0$, (8.1) is immediate. Otherwise the first k deviations are at least δ, and the sum of the remaining deviations is at most $-k\delta$. Hence, by Cauchy–Schwarz,

$$V \geq k\delta^2 + \frac{k^2\delta^2}{n-k} = \frac{kn}{n-k}\delta^2.$$

This proves (8.1). Formula (8.2) follows by applying (8.1) to $-\mu_n, \ldots, -\mu_1$.
□

For a maximizer of U_k, take all diagonal entries equal to b. Then

$$\operatorname{tr} A = nb$$

and

$$\begin{aligned}\sum_{i=1}^{n}(\lambda_i(A)-b)^2 &= \operatorname{tr}(A-bI)^2\\ &= 2\sum_{i<j}a_{ij}^2\\ &\le n(n-1)\rho^2.\end{aligned}$$

Applying (8.1) gives (2.14).

For a minimizer of L_k, all diagonal entries equal a, and the same argument with (8.2) gives (2.15).

6.10 9. Boundary cases and necessity of the hypotheses

6.10.1 9.1 The degenerate interval $a = b = t$

There is only one matrix, tJ. Its spectrum is

$$nt, 0, \ldots, 0.$$

If $t > 0$, nt is the largest eigenvalue; if $t < 0$, it is the smallest. All formulas above reduce correctly to this spectrum.

6.10.2 9.2 Dimension $n = 2$

There are no genuinely intermediate eigenvalues. Formulas (2.1)–(2.4) give the complete answer for both eigenvalues.

For instance, if $0 \le a \le b$,

$$\lambda_2(A) : A \in S_2([a,b]) = [a-b, b-a].$$

6.10.3 9.3 The sign assumption in the λ_2 theorem is essential

For $n = 5$, let $G = C_5$, and put

$$S = J - I - 2A(G), \qquad M = I + S = J - 2A(G).$$

Then $M \in S_5([-1,1])$, and its spectrum is

$$1+\sqrt{5}, \quad 1+\sqrt{5}, \quad 1, \quad 1-\sqrt{5}, \quad 1-\sqrt{5}.$$

Thus

$$\lambda_2(M) = 1 + \sqrt{5}.$$

The nonnegative-interval proof cannot be extended by simply substituting $a < 0$: negative cross-entries can reinforce, rather than suppress, the two sign parts of an eigenvector.

6.10.4 9.4 Endpoint matrices do not always suffice

For the centered interval ([-1,1]),

$$U_n(n; -1, 1) = 1,$$

attained by I. Its off-diagonal entries are interior points of the interval. Thus a brute-force enumeration of matrices with all entries in ({-1,1}) cannot solve Q706 in general.

This is one reason the density-matrix minimax formula (2.10) is needed.

6.11 10. Exact remaining obstruction

The results above do **not** give closed formulas for every intermediate pair ((n,k)). Already in the centered case, the exact problem becomes

$$\eta_{n,k} = \max_{\dim V = k} \min_{\substack{X \succeq 0,\ \operatorname{tr} X = 1 \\ \operatorname{Ran} X \subseteq V}} |X|_{1,e}. \tag{10.1}$$

The Ky Fan relaxation replaces the inner minimum by the particular choice

$$X = \frac{1}{k} P_V$$

and gives

$$\eta_{n,k} \leq \frac{1}{k} \max_{P \in \mathcal{P}_k(n)} |P|_{1,e}, \tag{10.2}$$

but equality is not automatic: it would require the optimized endpoint matrix to have its first k eigenvalues equal on the relevant subspace.

The same projection geometry, together with Seidel matrices and equiangular tight frames, is central in current work on extremizing intermediate graph eigenvalues. Recent results still formulate this as a difficult family of extremal problems and obtain sharp bounds and equality classifications only in special ranks. ([arXiv][1])

A concrete next lemma would be:

Target lemma. Characterize the subspaces V for which the inner minimizer in (10.1) is a normalized orthogonal projection P_V/k, and characterize when an entrywise endpoint matrix A satisfies

$$P_V A P_V = \eta_{n,k} P_V.$$

Such a result would convert the exact Ky Fan envelope into an exact formula for the individual k-th eigenvalue and reduce extremizers to structured two-eigenvalue sign or block matrices.

At present, (10.1) is the precise unresolved core left by this solution.

[1]: https://arxiv.org/html/2608.03196v1 "Extremal graphs for the -th eigenvalue"

Formal verification. Lean source; online type-check.

7 Question Q728

Informal verdict. A — Complete solution, with high confidence, for the meaningful range $n \geq 3$.

Lean coverage. Lean formalizes every legal distinct positive-integer split, termination, terminal payoff, and the minimax predicates by well-founded recursion. It proves unconditionally for every meaningful n >= 3 that A wins for n mod 3 = 2, B wins for n mod 3 = 1, and both players have non-losing strategies exactly in the draw case n mod 3 = 0. The lower bound is the necessary domain repair because the printed last-mover payoff is undefined for n = 1, 2.

7.1 Audited informal answer

7.1.1 1. Faithful restatement and a domain issue

A position is a finite collection of sticks of positive integer lengths. A legal move replaces one stick of length m by two sticks of distinct positive integer lengths $a < b$ satisfying $a + b = m$. Hence a stick is splittable exactly when $m \geq 3$.

Player A makes the first move, then B, and they alternate until every remaining stick has length 1 or 2. If the terminal position contains x sticks of length 1 and y sticks of length 2, then:

- if $x > y$, the last mover wins;
- if $x < y$, the last mover loses;
- if $x = y$, the game is drawn.

The question asks whether, under optimal play, the outcome depends only on $n \bmod 3$ in the stated way. This is the complete statement of Q728 in the canonical PDF.

There is one genuine omission: the PDF does not explicitly assume $n \geq 3$. For $n = 1, 2$, A has no legal first move and there is no "last player", so the payoff rule is undefined. I therefore solve the intended problem for $n \geq 3$.

Theorem. For every integer $n \geq 3$:

$$\begin{cases} A \text{ wins}, & n \equiv 2 \pmod 3, \\ B \text{ wins}, & n \equiv 1 \pmod 3, \\ \text{the game is drawn}, & n \equiv 0 \pmod 3. \end{cases}$$

Thus the proposed assertion is true after inserting the necessary hypothesis $n \geq 3$.

The proof is constructive and gives explicit optimal strategies.

7.1.2 2. Elementary terminal identities

Let a terminal position contain x pieces of length 1 and y pieces of length 2. Conservation of total length gives

$$x + 2y = n.$$

Consequently,

$$x + y = n - y.$$

After A's opening move there are exactly two pieces, and B is to move. Since every subsequent move increases the number of pieces by one, the number of moves remaining after the opening is

$$R = (x + y) - 2 = n - y - 2.$$

Therefore:

- B is the overall last mover exactly when R is odd;
- A is the overall last mover exactly when R is even, including $R = 0$, since A then made the opening and final move.

The main task is thus to control the possible terminal value of y.

7.2 3. The control interval of a position

For a positive integer m, define

$$\beta(m) = \begin{cases} 0, & m = 1, \\ 1, & m = 2, \\ \lfloor m/3 \rfloor, & m \geq 3, \end{cases}$$

and

$$\eta(m) = \begin{cases} 1, & m \geq 5 \text{ and } m \equiv 2 \pmod 3, \\ 0, & \text{otherwise.} \end{cases}$$

We call a piece counted by η a **hot piece**.
For a position $P = (m_1, \ldots, m_k)$, put

$$b(P) = \sum_{i=1}^{k} \beta(m_i), \qquad h(P) = \sum_{i=1}^{k} \eta(m_i),$$

and define its **control interval**

$$J(P) = \left\{ b(P) + \left\lfloor \frac{h(P)}{2} \right\rfloor, b(P) + \left\lceil \frac{h(P)}{2} \right\rceil \right\}.$$

Thus $J(P)$ is either a singleton or a pair of consecutive integers.
If P is terminal, every piece is 1 or 2. Hence $h(P) = 0$, while $b(P)$ is precisely the number y of pieces of length 2. Therefore

$$\boxed{P \text{ terminal} \quad \Longrightarrow \quad J(P) = y.}$$

This is why $J(P)$ is useful.

7.2.1 4. Effect of one move

Suppose a piece m is split into $a + b = m$, with $1 \le a < b$. Write

$$\Delta b = \beta(a) + \beta(b) - \beta(m), \qquad \Delta h = \eta(a) + \eta(b) - \eta(m).$$

The following table exhausts all possibilities. In the table, residue classes of the children are unordered.

m mod 3	Child residues	Additional condition	$(\Delta b, \Delta h)$
0	$0 + 0$	—	((0,0))
0	$1 + 2$	the 2 mod 3 child is 2	((0,0))
0	$1 + 2$	the 2 mod 3 child is at least 5	((-1,1))
1	$0 + 1$	—	((0,0))
1	$2 + 2$	exactly one child is 2	((0,1))
1	$2 + 2$	both children are at least 5	((-1,2))
2	$1 + 1$	—	((0,-1))
2	$0 + 2$	the 2 mod 3 child is 2	((1,-1))
2	$0 + 2$	the 2 mod 3 child is at least 5	((0,0))

Here a splittable piece congruent to 2 mod 3 is necessarily at least 5, so it is itself hot.

For completeness, the table follows immediately from

$$\begin{aligned} (\beta(3r), \eta(3r)) &= (r, 0), \\ (\beta(3r+1), \eta(3r+1)) &= (r, 0), \\ (\beta(2), \eta(2)) &= (1, 0), \\ (\beta(3r+2), \eta(3r+2)) &= (r, 1), \qquad r \ge 1, \end{aligned}$$

together with the possible residue decompositions

$$\begin{aligned} 0 &= 0 + 0 = 1 + 2, \\ 1 &= 0 + 1 = 2 + 2, \qquad \pmod 3. \\ 2 &= 0 + 2 = 1 + 1 \end{aligned}$$

Define the half-integral centre

$$c(P) = b(P) + \frac{h(P)}{2}.$$

The table shows that under every move $P \to P'$,

$$c(P') - c(P) \in \left\{ -\frac{1}{2}, 0, \frac{1}{2} \right\}.$$

Since

$$J(P) = \lfloor c(P) \rfloor, \lceil c(P) \rceil,$$

we obtain the following overlap property.

Lemma 1 — Overlap under every move For every legal move $P \to P'$,

$$\boxed{J(P) \cap J(P') \neq \varnothing.}$$

Indeed, two half-integers differing by at most $(1/2)$ have integer hulls sharing at least one integer.

7.3 5. The steering lemma

Fix two consecutive integers

$$T_t = t, t+1.$$

Lemma 2 — One-move steering Suppose P is nonterminal and

$$J(P) \cap T_t \neq \varnothing.$$

Then there is a legal move $P \to P'$ such that

$$\boxed{J(P') \subseteq T_t.}$$

Proof Write $b = b(P)$ and $h = h(P)$.

Case 1: h is odd Write $h = 2r + 1$. Then

$$J(P) = b + r, b + r + 1.$$

Because $h > 0$, there is a hot piece $m \equiv 2 \pmod 3$, $m \geq 5$. Both of the following splits are legal:

$$m = 1 + (m-1), \qquad m = 2 + (m-2).$$

From the move table:

- the first split has $(\Delta b, \Delta h) = (0, -1)$, and therefore

$$J(P') = b + r;$$

- the second has $(\Delta b, \Delta h) = (1, -1)$, and therefore

$$J(P') = b + r + 1.$$

At least one of the two endpoints of $J(P)$ belongs to T_t. Choose the corresponding split. The new interval is then a singleton contained in T_t.

Case 2: $h > 0$ is even Write $h = 2r$. Then

$$J(P) = s, \qquad s = b + r.$$

Since $J(P) \cap T_t \neq \varnothing$, either $s = t$ or $s = t + 1$.
Again choose a hot piece m.

- Splitting $m = 1 + (m - 1)$ gives

$$J(P') = s - 1, s.$$

- Splitting $m = 2 + (m - 2)$ gives

$$J(P') = s, s + 1.$$

Thus:

- if $s = t$, use $(2{+}m - 2)$;
- if $s = t + 1$, use $(1{+}m - 1)$.

In either case $J(P') = T_t$.

Case 3: $h = 0$ Then $J(P) = b$, and $b \in T_t$.

Because P is nonterminal, it contains a splittable piece $m \geq 3$. Since $h = 0$, no splittable piece can be congruent to 2 mod 3. Hence $m \equiv 0$ or 1 mod 3.

- If $m = 3$, split it as $1 + 2$.
- If $m \equiv 0 \pmod 3$ and $m \geq 6$, split it as

$$m = 2 + (m - 2).$$

- If $m \equiv 1 \pmod 3$, split it as

$$m = 1 + (m-1).$$

Each of these moves has $(\Delta b, \Delta h) = (0,0)$, so

$$J(P') = J(P) = b \subseteq T_t.$$

This proves the lemma. □

7.4 6. The interval-control theorem

The previous two lemmas combine into a robust strategy.

Theorem 3 — A designated player can control the terminal value Let P_0 be a position satisfying

$$J(P_0) \subseteq T_t = t, t+1.$$

Designate either of the two players as the **controller**. It does not matter whether the controller or the opponent is next to move.

Then the controller has a strategy forcing the terminal number y of length-2 pieces to satisfy

$$\boxed{y \in t, t+1.}$$

Proof The controller maintains the following invariant:

- immediately after every controller move,

$$J(P) \subseteq T_t;$$

- immediately after every opponent move,

$$J(P) \cap T_t \neq \varnothing.$$

Initially $J(P_0) \subseteq T_t$.

Suppose the controller has just moved and $J(P) \subseteq T_t$. If the opponent makes a move $P \to Q$, Lemma 1 gives

$$J(P) \cap J(Q) \neq \varnothing.$$

Since $J(P) \subseteq T_t$, it follows that

$$J(Q) \cap T_t \neq \varnothing.$$

If Q is terminal, then $J(Q) = y$, so $y \in T_t$. If Q is nonterminal, Lemma 2 gives the controller a move $Q \to Q'$ with

$$J(Q') \subseteq T_t.$$

The same argument applies when the opponent is initially first to move.

Finally, the game necessarily terminates: every move increases the number of pieces by one, while the number of pieces is at most n. $\square$

This theorem is stronger than what is needed: either player can be designated as controller, independently of whose turn it is.

7.5 7. Control intervals for two initial pieces

Suppose the current position consists of two pieces whose total length is N, and write

$$N = 3q + r, \qquad r \in 0, 1, 2.$$

For residue bookkeeping, distinguish:

- type 0: length $3i$;
- type 1: length $3i + 1$;
- type 2_0: the exceptional length 2;
- type 2_1: length $3i + 2 \geq 5$.

Direct substitution in the definitions of β and η gives:

N mod 3	Types of the two pieces	$J(P)$
0	$0 + 0$	({q})
0	$1 + 2_0$	({q})
0	$1 + 2_1$	({q-1,q})
1	$0 + 1$	({q})
1	$2_0 + 2_0$	({q+1})
1	$2_0 + 2_1$	({q,q+1})
1	$2_1 + 2_1$	({q})
2	$1 + 1$	({q})
2	$0 + 2_0$	({q+1})
2	$0 + 2_1$	({q,q+1})

For example, in the last row the pieces are $3i$ and $3j+2$, with $j \geq 1$. Then $q = i + j$,

$$b(P) = i + j = q, \qquad h(P) = 1,$$

so $J(P) = q, q+1$. The other rows follow identically.

We therefore have the two key inclusions:

$$\boxed{N \equiv 1, 2 \pmod 3 \quad \Longrightarrow \quad J(P) \subseteq q, q+1,}$$

and

$$\boxed{N \equiv 0 \pmod 3 \quad \Longrightarrow \quad J(P) \subseteq q-1, q.}$$

These statements hold for every two-piece position of total length N, whether or not the two lengths are distinct.

7.6 8. Determination of the winner

After A's opening move, there are two pieces and B is next.

7.6.1 Case 1: $n = 3q + 1$

Every opening produces a two-piece position P satisfying

$$J(P) \subseteq q, q+1.$$

Designate B as controller. By Theorem 3, B can force

$$y \in q, q+1.$$

We check both possibilities.

If $y = q$ Then

$$x = n - 2y = 3q + 1 - 2q = q + 1 > q = y.$$

Also,

$$R = n - y - 2 = 3q + 1 - q - 2 = 2q - 1$$

is odd, so B is the last mover. Since $x > y$, the last mover wins. Thus B wins.

If $y = q+1$ Then

$$x = 3q + 1 - 2(q+1) = q - 1 < q + 1 = y.$$

Moreover,

$$R = 3q + 1 - (q+1) - 2 = 2q - 2$$

is even, so A is the last mover. Since $x < y$, the last mover loses. Thus B again wins.

Therefore

$$\boxed{n \equiv 1 \pmod 3 \quad \Longrightarrow \quad B \text{ wins.}}$$

7.6.2 Case 2: $n = 3q + 2$

Here $n \geq 5$, since $n = 2$ lies outside the legal range.

Every opening gives a position satisfying

$$J(P) \subseteq q, q+1.$$

This time designate A as controller. Although B is next to move, Theorem 3 applies regardless of who moves next. Hence A can force

$$y \in q, q+1.$$

If $y = q$ Then

$$x = 3q + 2 - 2q = q + 2 > q = y,$$

and

$$R = 3q + 2 - q - 2 = 2q$$

is even. Therefore A is the last mover and wins.

If $y = q + 1$ Then

$$x = 3q + 2 - 2(q + 1) = q < q + 1 = y,$$

while

$$R = 3q + 2 - (q + 1) - 2 = 2q - 1$$

is odd. Thus B is the last mover, but because $x < y$, the last mover loses. Hence A wins.

Therefore

$$\boxed{n \equiv 2 \pmod 3 \quad \Longrightarrow \quad A \text{ wins.}}$$

In fact, after any legal opening, A can use the interval-control strategy.

7.6.3 Case 3: $n = 3q$

We prove separately that B cannot lose and that A cannot lose.

B has a non-losing strategy Whatever opening A chooses, the resulting two-piece position satisfies

$$J(P) \subseteq q - 1, q.$$

Designate B as controller. Then B can force

$$y \in q - 1, q.$$

- If $y = q$, then $x = q$, so the game is drawn.
- If $y = q - 1$, then

$$x = 3q - 2(q - 1) = q + 2 > q - 1 = y,$$

and

$$R = 3q - (q - 1) - 2 = 2q - 1$$

is odd. Hence B is the last mover and wins.

Thus B can force either a win or a draw.

A has a non-losing strategy Player A chooses the following opening:

$$\begin{cases} 1+2, & n=3, \\ 2+4, & n=6, \\ 3+(n-3), & n\geq 9. \end{cases}$$

All these splits are legal and have distinct positive parts. Their control interval is exactly

$$J(P) = q.$$

Indeed:

- for $n=3$, $b(1)+b(2)=1=q$;
- for $n=6$, $b(2)+b(4)=1+1=2=q$;
- for $n=3q\geq 9$,

$$b(3)+b(3q-3)=1+(q-1)=q,$$

and there are no hot pieces.

Now designate A as controller with target

$$q, q+1.$$

Although B moves next, Theorem 3 allows A to force

$$y \in q, q+1.$$

- If $y=q$, the game is drawn.
- If $y=q+1$, then, whenever this value is feasible,

$$x = 3q-2(q+1) = q-2 < q+1 = y,$$

and

$$R = 3q-(q+1)-2 = 2q-3$$

is odd. Thus B is the last mover and loses, so A wins.

Hence A can force either a win or a draw.

Since both players possess non-losing strategies, neither can force a win against optimal opposition. Therefore

$$\boxed{n \equiv 0 \pmod 3 \quad \Longrightarrow \quad \text{draw.}}$$

If both players use the displayed control strategies, B forces $y \in q-1, q$ while A forces $y \in q, q+1$; consequently $y = q$, so the resulting play is necessarily a draw.

7.7 9. Explicit optimal strategy and complexity

The proof gives a direct algorithm.

Suppose a player has been designated as controller for a target

$$T_t = t, t+1.$$

At each controller turn, compute $b(P)$, $h(P)$, and $J(P)$.

When $h(P)$ is odd Choose any hot piece m.

- Splitting $m = 1 + (m-1)$ makes the new interval the lower endpoint of $J(P)$.
- Splitting $m = 2 + (m-2)$ makes it the upper endpoint.

Choose an endpoint lying in T_t.

When $h(P) > 0$ is even Then $J(P) = s$, where $s = t$ or $s = t+1$. Choose a hot piece m.

- If $s = t$, use $m = 2 + (m-2)$.
- If $s = t+1$, use $m = 1 + (m-1)$.

The resulting interval is exactly T_t.

When $h(P) = 0$ Choose any splittable piece m.

- If $m = 3$, play $1 + 2$.
- If $m \equiv 0 \pmod 3$ and $m \geq 6$, play $(2{+}m-2)$.
- If $m \equiv 1 \pmod 3$, play $(1{+}m-1)$.

This preserves $J(P)$.

The appropriate controllers and targets are:

n mod 3	controller after the opening	target
1	B	$q, q+1$
2	A	$q, q+1$
0	B	$q-1, q$

For $n \equiv 0 \pmod 3$, A additionally uses the special opening above and controls ({q,q+1}).

If the multiset of pieces is maintained together with $b(P)$, $h(P)$, and buckets for hot and non-hot active pieces, every update and every strategic choice takes $O(1)$ word operations. There are at most $n-1$ moves, since every move increases the number of pieces and there can be at most n pieces. Thus the full strategy is implementable in

$$O(n)$$

word operations and $O(n)$ storage. A naive implementation that scans all current pieces at every turn has $O(n^2)$ worst-case running time.

7.8 10. Boundary checks

$n = 3$ The only move is

$$3 \longrightarrow 1 + 2.$$

Thus $x = y = 1$, and the game is drawn.

$n = 4$ The only opening is

$$4 \longrightarrow 1 + 3.$$

Then B must split $3 \to 1 + 2$. The terminal position has $x = 2, y = 1$, and B is the last mover, so B wins.

$n = 5$ Player A may play

$$5 \longrightarrow 2 + 3.$$

Player B must split $3 \to 1 + 2$. The terminal pieces are (1,2,2), so $x = 1 < 2 = y$. Since B is the last mover, B loses and A wins.

$n = 6$ Player A uses

$$6 \longrightarrow 2 + 4.$$

Then the continuation is forced:

$$4 \longrightarrow 1 + 3, \qquad 3 \longrightarrow 1 + 2.$$

The terminal position has $x = y = 2$, hence a draw.
These are exactly the first four cases predicted by the theorem.

7.9 11. Necessity of the hypotheses

- The restriction $n \geq 3$ is necessary merely to make the stated payoff rule well-defined: for $n = 1, 2$, no one ever plays, so there is no last player.
- The requirement that the two new lengths be **distinct** is essential. If equal splits were permitted, then for $n = 3$ one would have

$$3 \to 1 + 2 \to 1 + 1 + 1,$$

so B, rather than producing a draw, would be the last mover with $x > y$ and would win.

- Integrality is used both for termination at pieces of lengths (1,2) and for the residue-3 control invariant.

7.10 12. Exact computational verification

The proof above is independent of computation. The following exact dynamic program was used only as a discovery and consistency check. It verifies the claimed pattern for $3 \leq n \leq 40$.

```
from functools import cache


def legal_moves(position: tuple[int, ...]) -> set[tuple[int, ...]]:
    """All legal successor positions."""
    successors: set[tuple[int, ...]] = set()
```

```
    for i, m in enumerate(position):
        # a < m-a, so equal splits are automatically excluded.
        for a in range(1, (m + 1) // 2):
            b = m - a
            if a < b:
                nxt = list(position)
                nxt.pop(i)
                nxt.extend((a, b))
                successors.add(tuple(sorted(nxt)))

    return successors


@cache
def value(position: tuple[int, ...]) -> int:
    """
    Minimax value for the player whose turn it is:
      +1 = that player can force a win,
       0 = that player can force a draw,
      -1 = that player must lose under optimal play.
    """
    successors = legal_moves(position)

    if not successors:
        x = position.count(1)
        y = position.count(2)

        # At a genuine terminal position, the player to move is
        # the opponent of the last mover. That player wins iff x < y.
        return (y > x) - (y < x)

    # Roles are reversed after every move.
    return max(-value(nxt) for nxt in successors)


expected = {
    0: 0,    # draw
    1: -1,   # A loses, hence B wins
    2: 1,    # A wins
}
```

```
for n in range(3, 41):
    assert value((n,)) == expected[n % 3]

print("Verified for every n from 3 through 40.")
```

For a fixed n, the cached verifier visits at most the $p(n)$ integer partitions of n, where p is the partition function. There are $O(n)$ potential splits per state, so the conceptual transition count is (O(n,pn)). This is only a finite verification method; the invariant proof establishes the result for all $n \geq 3$.

7.10.1 Final conclusion

The answer to Q728 is **yes**, with the necessary clarification that the game is defined for $n \geq 3$:

$$\boxed{\begin{aligned} n \equiv 2 \pmod 3 &\implies A \text{ wins}, \\ n \equiv 1 \pmod 3 &\implies B \text{ wins}, \\ n \equiv 0 \pmod 3 &\implies \text{draw}. \end{aligned}}$$

Formal verification. Lean source; online type-check.

8 Question Q730

Informal verdict. **A — Complete solution**, with high confidence.

Lean coverage. Lean proves the complete necessary-and-sufficient Q730 classification, including prescribed diagonal order, rank-deficient prefix inequalities, and the full-rank determinant equality.

8.1 Audited informal answer

8.1.1 1. Faithful restatement

Let $M \in M_n(\mathbb{C})$ have rank r, and let $t_1, \ldots, t_r > 0$. Determine when there exist unitary matrices $U, V \in U(n)$ such that

$$UMV = \begin{pmatrix} T & B \\ 0 & 0 \end{pmatrix},$$

where $T \in M_r(\mathbb{C})$ is upper triangular with diagonal

$$\operatorname{diag}(T) = (t_1, \dots, t_r),$$

and $B \in M_{r,n-r}(\mathbb{C})$. The lower-left and lower-right blocks are zero matrices of sizes $(n-r) \times r$ and $(n-r) \times (n-r)$, respectively.

The PDF adds that an earlier answer treated the full-rank case $|\det M| = 1$ with $t_1 = \cdots = t_n = 1$.

The phrase "Soit des réels $t_1, \dots, t_r$" is a minor grammatical typo; mathematically it clearly means "Let $t_1, \dots, t_r$ be positive real numbers." The use of UMV, rather than the more usual UMV^*, causes no issue because $V \mapsto V^*$ is a bijection of the unitary group.

8.1.2 2. Final classification

Let

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$$

be the nonzero singular values of M. Let

$$\tau_1 \geq \tau_2 \geq \cdots \geq \tau_r > 0$$

be the decreasing rearrangement of $t_1, \dots, t_r$.

Theorem The required unitary matrices (U,V) exist precisely under the following conditions.

Rank-deficient case $r < n$ They exist if and only if

$$\boxed{\prod_{i=1}^{k} \tau_i \leq \prod_{i=1}^{k} \sigma_i \qquad (1 \leq k \leq r).} \tag{1}$$

Thus (τ_i) is weakly log-majorized by (σ_i).

Full-rank case $r = n$ They exist if and only if

$$\boxed{\prod_{i=1}^{k} \tau_i \leq \prod_{i=1}^{k} \sigma_i \qquad (1 \leq k < n),} \tag{2}$$

and

$$\boxed{\prod_{i=1}^{n} t_i == \prod_{i=1}^{n} \sigma_i |\det M|.} \tag{3}$$

Thus (τ_i) is log-majorized by (σ_i).

The order $t_1, \ldots, t_r$ requested on the diagonal imposes no additional condition: once the eigenvalues have been realized, the unitary Schur decomposition can put them on the diagonal in any prescribed order.

The central eigenvalue–singular-value criterion is traditionally called the Weyl–Horn theorem. A complete proof sufficient for the present problem is included below rather than used as a black box. ([Journals de l'Université du Wyoming][1])

8.2 3. Preparatory results

8.2.1 3.1. Two-sided unitary equivalence

Lemma 1 Two matrices $A, B \in M_n(\mathbb{C})$ have the same singular values, counted with multiplicity, if and only if there exist unitary matrices (P,Q) such that

$$B = PAQ.$$

Proof The forward implication follows from the singular value decompositions

$$A = P_A \Sigma Q_A^*, \qquad B = P_B \Sigma Q_B^*,$$

with the same nonnegative diagonal matrix Σ. Then

$$B = (P_B P_A^*), A, (Q_A Q_B^*).$$

Both factors are unitary.

Conversely, if $B = PAQ$ with (P,Q) unitary, then

$$B^* B = Q^* A^* A Q,$$

so A^*A and B^*B have the same eigenvalues. Hence A and B have the same singular values. □

8.2.2 3.2. Ordered unitary triangularization

Lemma 2 Let $A \in M_n(\mathbb{C})$, and let $\lambda_1, \ldots, \lambda_n$ be its eigenvalues, listed with algebraic multiplicity in any prescribed order. There exists a unitary W such that W^*AW is upper triangular with diagonal

$$(\lambda_1, \ldots, \lambda_n).$$

Proof Choose a unit eigenvector x_1 for λ_1 and extend it to an orthonormal basis. In that basis,

$$A = \begin{pmatrix} \lambda_1 & * \\ 0 & A_1 \end{pmatrix}.$$

The characteristic polynomial factors as

$$\chi_A(z) = (z - \lambda_1)\chi_{A_1}(z),$$

so the eigenvalues of A_1 are $\lambda_2, \ldots, \lambda_n$, with multiplicity. Apply the same argument recursively to A_1. □

8.3 4. Eigenvalues versus singular values

We now prove the exact form of the Weyl–Horn theorem needed here.

8.3.1 Theorem 3 — Weyl–Horn criterion

Let

$$s_1 \geq \cdots \geq s_n \geq 0$$

and let $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$, ordered so that

$$|\lambda_1| \geq \cdots \geq |\lambda_n|.$$

There exists a matrix $A \in M_n(\mathbb{C})$ whose singular values are $s_1, \ldots, s_n$ and whose eigenvalues are $\lambda_1, \ldots, \lambda_n$ if and only if

$$\prod_{i=1}^{k} |\lambda_i| \leq \prod_{i=1}^{k} s_i \qquad (1 \leq k < n), \tag{4}$$

and

$$\prod_{i=1}^{n} |\lambda_i| = \prod_{i=1}^{n} s_i. \tag{5}$$

8.3.2 Proof: necessity

For $1 \le k \le n$, consider the induced operator

$$\bigwedge^k A$$

on $\bigwedge^k \mathbb{C}^n$.

By upper triangularizing A, one sees that the eigenvalues of $\bigwedge^k A$ are all products

$$\lambda_{i_1} \cdots \lambda_{i_k}, \qquad 1 \le i_1 < \cdots < i_k \le n.$$

Because the moduli of the λ_i are decreasing,

$$\rho\left(\bigwedge^k A\right) = \prod_{i=1}^{k} |\lambda_i|,$$

where ρ denotes the spectral radius.

On the other hand, applying the exterior-power functor to a singular value decomposition of A shows that the singular values of $\bigwedge^k A$ are the products

$$s_{i_1} \cdots s_{i_k}.$$

Consequently,

$$\left|\bigwedge^k A\right| = \prod_{i=1}^{k} s_i.$$

Since the spectral radius does not exceed the operator norm,

$$\prod_{i=1}^{k} |\lambda_i| \le \prod_{i=1}^{k} s_i.$$

This proves 4. Finally,

$$\prod_{i=1}^{n} |\lambda_i| == |\det A| = \sqrt{\det(A^*A)} \prod_{i=1}^{n} s_i,$$

which is 5.

8.3.3 Proof: sufficiency

We argue by induction on n.

For $n = 1$, condition 5 says $s_1 = |\lambda_1|$, so the matrix (λ_1) works.

Assume the result known in dimension $n - 1$.

Case 1: $\lambda_1 = 0$ Then all the λ_i vanish. Equation 5 implies

$$s_n = 0.$$

Consider the weighted shift

$$A = \begin{pmatrix} 0 & s_1 & 0 & \cdots & 0 \\ 0 & 0 & s_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 0 & s_{n-1} \\ 0 & \cdots & \cdots & 0 & 0 \end{pmatrix}.$$

It is nilpotent, so all its eigenvalues are zero. Moreover,

$$A^*A = \mathrm{diag}(0, s_1^2, \ldots, s_{n-1}^2),$$

so its singular values are

$$s_1, \ldots, s_{n-1}, 0 = s_n.$$

Case 2: $\lambda_1 \neq 0$ Put

$$a = |\lambda_1| > 0.$$

Condition 4 for $k = 1$ gives $a \leq s_1$. We also have $a \geq s_n$: otherwise

$$|\lambda_i| \leq a < s_n \leq s_i$$

for every i, contradicting the product equality 5.

Hence there is an index $j \in 2, \ldots, n$ such that

$$s_{j-1} \geq a \geq s_j. \tag{6}$$

Define

$$d = \frac{s_1 s_j}{a}. \tag{7}$$

Consider

$$H = \begin{pmatrix} \lambda_1 & \mu \\ 0 & d \end{pmatrix},$$

where

$$\mu^2 = s_1^2 + s_j^2 - a^2 - d^2.$$

Using 7,

$$\begin{aligned} \mu^2 &= s_1^2 + s_j^2 - a^2 - \frac{s_1^2 s_j^2}{a^2} \\ &= \frac{(s_1^2 - a^2)(a^2 - s_j^2)}{a^2} \geq 0 \end{aligned}$$

by 6. We take $\mu \geq 0$.
Now

$$\operatorname{tr}(H^* H) == a^2 + d^2 + \mu^2 s_1^2 + s_j^2$$

and

$$\det(H^* H) = |\det H|^2 = a^2 d^2 = s_1^2 s_j^2.$$

Thus the two eigenvalues of $H^* H$ are s_1^2, s_j^2, and the singular values of H are s_1, s_j.

Consider the following multiset of $n-1$ nonnegative numbers:

$$\mathcal{S}' = \{d, s_2, \ldots, s_{j-1}, s_{j+1}, \ldots, s_n\}.$$

Let

$$y_1 \geq \cdots \geq y_{n-1}$$

be its decreasing rearrangement. We claim that the singular-value data $y_1, \ldots, y_{n-1}$ and eigenvalue data $\lambda_2, \ldots, \lambda_n$ satisfy the induction hypotheses.

First,

$$
\begin{aligned}
\prod_{i=2}^{n} |\lambda_i| &= \frac{1}{a} \prod_{i=1}^{n} s_i \\
&= d \prod_{\substack{2 \le i \le n \\ i \ne j}} s_i \\
&= \prod_{\ell=1}^{n-1} y_\ell .
\end{aligned} \tag{8}
$$

It remains to prove the partial-product inequalities.

Let $1 \le m \le j-2$. Since $s_2, \ldots, s_{m+1}$ are members of $\mathcal{S}'$, and each of them is at least a, we have

$$
\begin{aligned}
\prod_{i=2}^{m+1} |\lambda_i| &\le a^m \\
&\le \prod_{i=2}^{m+1} s_i \\
&\le \prod_{\ell=1}^{m} y_\ell .
\end{aligned} \tag{9}
$$

Now let $j-1 \le m \le n-2$. Applying 4 with $k = m+1$ gives

$$
a \prod_{i=2}^{m+1} |\lambda_i| \le \prod_{i=1}^{m+1} s_i .
$$

Therefore

$$
\begin{aligned}
\prod_{i=2}^{m+1} |\lambda_i| &\le \frac{1}{a} \prod_{i=1}^{m+1} s_i \\
&= d \prod_{\substack{2 \le i \le m+1 \\ i \ne j}} s_i \\
&\le \prod_{\ell=1}^{m} y_\ell .
\end{aligned} \tag{10}
$$

The final inequality holds because the preceding expression is the product of m members of $\mathcal{S}'$, while $y_1 \cdots y_m$ is the product of its m largest members.

By the induction hypothesis, there exists $C \in M_{n-1}(\mathbb{C})$ with singular values $y_1, \ldots, y_{n-1}$ and eigenvalues $\lambda_2, \ldots, \lambda_n$.

Let

$$D = \operatorname{diag}\big(d, s_2, \ldots, s_{j-1}, s_{j+1}, \ldots, s_n\big).$$

Since C and D have the same singular values, Lemma 1 gives unitary $P, Q \in U(n-1)$ such that

$$C = PDQ.$$

Define

$$A_0 = \begin{pmatrix} \lambda_1 & \mu e_1^* \\ 0 & D \end{pmatrix},$$

where e_1 is the first standard basis vector of $\mathbb{C}^{n-1}$. The matrix A_0 is the direct sum, up to the displayed ordering of coordinates, of the 2×2 matrix H and the diagonal entries s_i for $i \neq 1, j$. Hence its singular values are $s_1, \ldots, s_n$.

Finally set

$$A = (1 \oplus P)A_0(1 \oplus Q).$$

Its singular values remain $s_1, \ldots, s_n$, and

$$A = \begin{pmatrix} \lambda_1 & \mu e_1^* Q \\ 0 & C \end{pmatrix}.$$

Thus its eigenvalues are λ_1 together with the eigenvalues $\lambda_2, \ldots, \lambda_n$ of C. The induction is complete. □

8.3.4 5.1. Necessity

Suppose that the requested unitary matrices exist, and set

$$A = UMV = \begin{pmatrix} T & B \\ 0 & 0 \end{pmatrix}.$$

Because multiplication on the left and right by unitaries preserves singular values, the singular values of A are

$$\sigma_1, \ldots, \sigma_r, \underbrace{0, \ldots, 0}_{n-r}.$$

Since A is block upper triangular and T is upper triangular with diagonal $t_1, \dots, t_r$, the eigenvalues of A are

$$t_1, \dots, t_r, \underbrace{0, \dots, 0}_{n-r}.$$

Order the positive eigenvalues decreasingly as $\tau_1 \geq \cdots \geq \tau_r$. The necessary part of Theorem 3 gives

$$\prod_{i=1}^{k} \tau_i \leq \prod_{i=1}^{k} \sigma_i, \qquad 1 \leq k \leq r.$$

If $r = n$, the determinant identity additionally gives

$$\prod_{i=1}^{n} t_i == |\det A| = |\det M| \prod_{i=1}^{n} \sigma_i.$$

This proves necessity.

8.3.5 5.2. Sufficiency

Assume first that $r > 0$, and define the full lists

$$s = (\sigma_1, \dots, \sigma_r, 0, \dots, 0)$$

and

$$\lambda = (\tau_1, \dots, \tau_r, 0, \dots, 0).$$

If $r < n$, the assumed inequalities give the Weyl–Horn conditions for $k \leq r$. For every $k \geq r + 1$,

$$\prod_{i=1}^{k} |\lambda_i| == 0 \prod_{i=1}^{k} s_i.$$

In particular, the full products agree.

If $r = n$, the assumed partial-product inequalities and the determinant equality are exactly the Weyl–Horn conditions.

Theorem 3 therefore provides a matrix $A_0 \in M_n(\mathbb{C})$ with the same singular values as M and with eigenvalues

$$t_1, \dots, t_r, 0, \dots, 0$$

as a multiset.

By Lemma 1, there exist unitary matrices U_0, V_0 such that

$$A_0 = U_0 M V_0. \tag{11}$$

Use Lemma 2 to choose a unitary W such that

$$R = W^* A_0 W$$

is upper triangular with diagonal in the particular order

$$t_1, \ldots, t_r, 0, \ldots, 0.$$

Partition R as

$$R = \begin{pmatrix} T & B \\ 0 & C \end{pmatrix},$$

where $T \in M_r(\mathbb{C})$. Then T is upper triangular with diagonal $t_1, \ldots, t_r$, hence T is invertible.

The rank of R equals the rank of M, namely r. But right multiplication by the invertible matrix

$$\begin{pmatrix} I_r & -T^{-1}B \\ 0 & I_{n-r} \end{pmatrix}$$

gives

$$R \begin{pmatrix} I_r & -T^{-1}B \\ 0 & I_{n-r} \end{pmatrix} = \begin{pmatrix} T & 0 \\ 0 & C \end{pmatrix}.$$

Therefore

$$\operatorname{rank} R == \operatorname{rank} T + \operatorname{rank} C r + \operatorname{rank} C.$$

Since $\operatorname{rank} R = r$, necessarily $C = 0$.

Combining this with 11,

$$R = (W^* U_0), M, (V_0 W).$$

Both $W^* U_0$ and $V_0 W$ are unitary, and R has exactly the required form.

If $r = 0$, then $M = 0$, the list $t_1, \ldots, t_r$ is empty, and the assertion is trivially true.

This completes the proof. $\square$

8.4 6. Boundary cases and checks

8.4.1 Rank one

Suppose $r = 1 < n$, and let σ_1 be the unique nonzero singular value. The condition reduces to

$$t_1 \leq \sigma_1.$$

This is visibly sufficient: the matrix

$$\begin{pmatrix} t_1 & \sqrt{\sigma_1^2 - t_1^2} & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \end{pmatrix}$$

has singular values $\sigma_1, 0, \ldots, 0$ and the required form.

For $n = r = 1$, there is no B-block, and the condition becomes the equality

$$t_1 = \sigma_1 = |M|.$$

This shows explicitly why the product equality disappears as soon as a zero singular value is available.

8.4.2 Higher partial products are genuinely necessary

It is not enough to require only $t_i \leq \sigma_1$. For example, suppose the nonzero singular values are

$$(\sigma_1, \sigma_2) = (3, 1)$$

and the desired diagonal is ((2,2)). Each desired diagonal entry is at most 3, but

$$2 \cdot 2 = 4 > 3 \cdot 1.$$

Hence no such unitary reduction exists.

8.4.3 In full rank, the determinant condition alone is not enough

Take singular values

$$(4, \tfrac{1}{4})$$

and desired diagonal

$$(5, \tfrac{1}{5}).$$

Both products equal 1, but

$$5 > 4,$$

so the first partial-product inequality fails. Thus determinant equality must be supplemented by all the inequalities 2.

8.4.4 Recovery of the earlier special case

Suppose $r = n$, $|\det M| = 1$, and $t_1 = \cdots = t_n = 1$. Then

$$\prod_{i=1}^{n} \sigma_i = 1.$$

For every $k < n$,

$$\prod_{i=1}^{k} \sigma_i \geq 1.$$

Indeed, if this product were < 1, then $\sigma_k < 1$, so every later singular value would also be < 1, forcing the total product to be < 1, a contradiction. Consequently,

$$1 = \prod_{i=1}^{k} t_i \leq \prod_{i=1}^{k} \sigma_i,$$

and equality holds for $k = n$. The theorem therefore recovers exactly the affirmative case mentioned in the PDF.

8.4.5 Final answer

Let $\sigma_1 \geq \cdots \geq \sigma_r > 0$ be the nonzero singular values of M, and let $\tau_1 \geq \cdots \geq \tau_r$ be the decreasing rearrangement of $t_1, \ldots, t_r$.

$$\boxed{\begin{aligned} r<n: &\quad \prod_{i=1}^{k}\tau_i \le \prod_{i=1}^{k}\sigma_i \quad \text{for every } 1\le k\le r;\\ r=n: &\quad \prod_{i=1}^{k}\tau_i \le \prod_{i=1}^{k}\sigma_i \quad \text{for } 1\le k<n,\\ &\quad \prod_{i=1}^{n} t_i = |\det M|. \end{aligned}}$$

These conditions are both necessary and sufficient, and the diagonal entries can be placed in the exact prescribed order $t_1, \ldots, t_r$.

[1]: https://journals.uwyo.edu/index.php/ela/article/download/823/823/823 "A. Horn's result on matrices with prescribed singular values and eigenvalues"

Formal verification. Lean source; online type-check.

9 Question Q748

Informal verdict. C — Substantial progress, with high confidence in the stated partial results.

Lean coverage (partial). Verified Lean covers the 667-line intrinsic-distance/chord core, generic cone lower bounds, Clifford coordinates, and the exact all-pairs singular 2-by-2 formula across seven imported modules (3,143 lines). It does not prove generic cone upper equalities, equality with the PDF finite piecewise-C1 distance, complete positive chord/boundary theory, arbitrary-n generic distances in GL_n^+ or Sigma_n, or the canonical attainment classification.

9.1 Audited informal answer

9.1.1 Faithful restatement

Equip $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$ with its standard Euclidean structure, hence with the Frobenius inner product and norm

$$\langle X, Y\rangle_F = \operatorname{tr}(X^\mathsf{T} Y), \qquad |X| * F = \sqrt{\operatorname{tr}(X^\mathsf{T} X)}.$$

For a piecewise C^1-arc-connected subset $E \subset M_n(\mathbb{R})$, its intrinsic length distance is

$$d_E(A, B) = \inf *\gamma \int_0^1 |\gamma'(t)|_F, dt,$$

where γ ranges over piecewise C^1 curves in E joining A to B.

The problem asks for this distance first on

$$\Omega_n := GL_n^+(\mathbb{R}) = X : \det X > 0,$$

and then on

$$\Sigma_n := X : \det X = 0,$$

the set called “matrices non régulières” in the PDF. The duplicated phrase “x et y” in the first sentence is only a typographical error; there is no mathematical ambiguity. ([Luc Pommeret][1])

9.1.2 Status

Partial result. I give:

1. an exact metric-cone reduction valid for every n;
2. a complete characterization of all pairs for which the answer is simply the Frobenius distance;
3. quantitative strict lower bounds when it is not;
4. explicit constructive upper bounds;
5. a closed formula for all pairs of singular 2×2 matrices;
6. an exact scalar variational determination for $GL_2^+(\mathbb{R})$.

What remains unresolved here is an explicit evaluation, for arbitrary $n \geq 3$, of certain intrinsic angular distances on the unit determinant links. This is the exact obstruction, not merely a technical gap in one estimate.

9.2 1. Exact reduction to angular distances on the unit sphere

Let

$$S_F = X \in M_n(\mathbb{R}) : |X| * F = 1,$$

and define the links

$$Y_n^+ := \Omega_n \cap S_F, \qquad Z_n := \Sigma_n \cap S_F.$$

Write $\delta * +$ and δ_Σ for their intrinsic spherical length distances. Thus curve lengths are still computed with the ambient Frobenius metric, restricted to S_F.

For a nonzero matrix A, put

$$r_A = |A|_F, \qquad \widehat{A} = \frac{A}{|A|_F}.$$

9.2.1 Theorem 1 — Metric-cone formula

For $A, B \in \Omega_n$,

$$\boxed{d_{\Omega_n}(A,B)^2 = r_A^2 + r_B^2 - 2r_A r_B \cos\Big(\min \pi, \delta_+(\widehat{A}, \widehat{B})\Big).} \tag{1}$$

For nonzero $A, B \in \Sigma_n$,

$$\boxed{d_{\Sigma_n}(A,B)^2 = r_A^2 + r_B^2 - 2r_A r_B \cos\Big(\min \pi, \delta_\Sigma(\widehat{A}, \widehat{B})\Big).} \tag{2}$$

If $A = 0$ or $B = 0$, then

$$\boxed{d_{\Sigma_n}(A,B) = |A - B|_F.} \tag{3}$$

Consequently, in both sets,

$$\boxed{|A - B|_F \leq d(A,B) \leq |A|_F + |B|_F.} \tag{4}$$

The upper bound in Σ_n is attained by the broken radial path $A \to 0 \to B$. In GL_n^+, it is in general only an infimum, obtained by passing arbitrarily close to 0.

9.2.2 Proof of the metric-cone formula

We prove a general lemma.

Lemma 2 — Distance in a Euclidean cone Let $Y \subset S^{N-1}$ be piecewise C^1-arc-connected, and let

$$C^\times(Y) = ru : r > 0,\ u \in Y.$$

Denote by δ_Y the intrinsic spherical distance on Y. For $x = ru$ and $y = sv$,

$$d_{C^\times(Y)}(x,y)^2 = r^2 + s^2 - 2rs\cos(\min \pi, \delta_Y(u,v))\,. \tag{5}$$

The same formula holds for the closed cone

$$C(Y) = ru : r \geq 0,\ u \in Y,$$

with the evident convention when an endpoint is 0.

Proof Let $\gamma(t) = \rho(t)u(t)$ be a piecewise C^1 curve in the punctured cone, where $\rho(t) > 0$ and $u(t) \in Y$. Since $|u(t)| = 1$,

$$\langle u, u' \rangle = 0,$$

and hence

$$|\gamma'(t)|^2 = \rho'(t)^2 + \rho(t)^2 |u'(t)|^2. \tag{6}$$

Define the accumulated angular length

$$q(t) = \int_0^t |u'(\tau)|, d\tau$$

and the developed planar curve

$$\widetilde{\gamma}(t) = \rho(t)\big(\cos q(t), \sin q(t)\big) \in \mathbb{R}^2.$$

By 6,

$$|\widetilde{\gamma}'(t)| = |\gamma'(t)|$$

almost everywhere. Thus the two curves have the same length.

Let $L = q(1)$. Necessarily

$$L \geq \delta_Y(u,v).$$

If $L \leq \pi$, the planar chord inequality gives

$$\text{length}(\gamma) \geq \sqrt{r^2 + s^2 - 2rs\cos L}.$$

The right-hand side is increasing in $L \in [0, \pi]$, so

$$\operatorname{length}(\gamma) \geq \sqrt{r^2 + s^2 - 2rs\cos\delta_Y(u,v)}$$

whenever $\delta_Y(u,v) \leq L \leq \pi$.

If $L \geq \pi$, let t_0 be the first time at which $q(t_0) = \pi$, and put $a = \rho(t_0)$. In the developed plane, $\widetilde{\gamma}(t_0) = (-a, 0)$. Therefore

$$\operatorname{length}(\gamma|_{[0,t_0]}) \geq r + a,$$

while, by the reverse triangle inequality,

$$\operatorname{length}(\gamma|_{[t_0,1]}) \geq |s - a|.$$

Hence

$$\operatorname{length}(\gamma) \geq r + a + |s - a| \geq r + s.$$

This proves the lower bound in 5.

For the reverse inequality, first suppose

$$\delta_Y(u,v) < \pi.$$

Choose a unit-speed curve

$$\sigma : [0, L] \longrightarrow Y, \qquad \sigma(0) = u, \quad \sigma(L) = v,$$

with $L < \pi$ and L arbitrarily close to $\delta_Y(u,v)$.

The straight segment in $\mathbb{R}^2$ joining

$$r(1,0) \quad \text{to} \quad s(\cos L, \sin L)$$

can be written in polar form

$$\theta \longmapsto \rho(\theta)(\cos\theta, \sin\theta), \qquad 0 \leq \theta \leq L,$$

where

$$\rho(\theta) = \frac{rs\sin L}{s\sin(L-\theta) + r\sin\theta} > 0.$$

Lift this planar segment to the cone by

$$\Gamma(\theta) = \rho(\theta)\sigma(\theta).$$

Its length equals the length of the planar chord:

$$\sqrt{r^2 + s^2 - 2rs\cos L}.$$

Letting $L \downarrow \delta_Y(u,v)$ proves the upper bound.

If $\delta_Y(u,v) \geq \pi$, fix any finite-length curve in Y from u to v, of length L. For $\varepsilon > 0$, concatenate:

$$ru \longrightarrow \varepsilon u, \qquad \varepsilon u \longrightarrow \varepsilon v, \qquad \varepsilon v \longrightarrow sv.$$

Its length is

$$(r-\varepsilon) + \varepsilon L + (s-\varepsilon) \longrightarrow r+s.$$

This proves 5. If the apex belongs to the cone, the path through 0 gives length exactly $r+s$. $\square$

Applying this lemma to Y_n^+ and Z_n proves Theorem 1.

For completeness, Y_n^+ is piecewise C^1-arc-connected by polar decomposition. Indeed,

$$A = QP, \qquad Q \in SO(n), \quad P > 0,$$

and one connects P to I through positive-definite matrices, then Q to I inside $SO(n)$.

For $n \geq 2$, Z_n is also piecewise C^1-arc-connected. A nonzero singular matrix can, using a singular-value decomposition, be deformed on the unit sphere to a unit rank-one matrix. The unit rank-one locus is the continuous image of

$$S^{n-1} \times S^{n-1}, \qquad (u,v) \longmapsto uv^{\mathsf{T}},$$

and is therefore connected.

9.3 2. Exactly when the distance is the Frobenius chord

The answer is frequently, but not always, $|A-B|_F$. There is a complete algebraic criterion.

9.3.1 Theorem 3 — Chord criterion in GL_n^+

For $A, B \in GL_n^+(\mathbb{R})$, the following are equivalent:

$$\boxed{d_{\Omega_n}(A,B) = |A-B|_F;} \tag{7}$$

$$\boxed{\det((1-t)A + tB) \geq 0 \quad \text{for every } t \in [0,1];} \tag{8}$$

$$\boxed{\text{every negative real eigenvalue of } A^{-1}B \text{ has even algebraic multiplicity.}} \tag{9}$$

Notice that condition 8 allows the segment to touch the singular locus. Thus the infimum can equal the Euclidean chord even though the straight segment itself is not an admissible curve in GL_n^+.

9.3.2 Theorem 4 — Chord criterion in the singular locus

For $A, B \in \Sigma_n$, the following are equivalent:

$$\boxed{d_{\Sigma_n}(A, B) = |A - B|_F;} \tag{10}$$

$$\boxed{\det((1-t)A + tB) = 0 \quad \text{for every } t \in [0,1];} \tag{11}$$

$$\boxed{\det(A + \lambda B) \equiv 0 \quad \text{as a polynomial in } \lambda.} \tag{12}$$

Equivalently, the pencil $A + \lambda B$ is singular over $\mathbb{R}(\lambda)$. This is equivalent to the existence of a nonzero polynomial vector

$$v(\lambda) \in \mathbb{R}[\lambda]^n$$

such that

$$(A + \lambda B)v(\lambda) = 0. \tag{13}$$

A common right or left null vector is sufficient for 11, but is not necessary. The proofs use two elementary geometric lemmas.

9.3.3 2.1 A quantitative obstacle lemma

Lemma 5 Let $x, y \in \mathbb{R}^N$, let $D = |x - y|$, and set

$$c_t = (1-t)x + ty, \qquad 0 < t < 1.$$

Suppose a continuous curve γ from x to y avoids the open ball

$$B(c_t, \rho).$$

Then

$$\boxed{\text{length}(\gamma) \geq \sqrt{t^2D^2 + \rho^2} + \sqrt{(1-t)^2D^2 + \rho^2}.} \tag{14}$$

Proof Let

$$e = \frac{y - x}{D}.$$

The continuous function

$$s \longmapsto \langle \gamma(s) - x, e \rangle$$

takes the values 0 and D. Thus at some point p on the curve,

$$\langle p - x, e \rangle = tD.$$

Write

$$p = c_t + v, \qquad v \perp e.$$

Since the curve avoids $B(c_t, \rho)$,

$$|v| \geq \rho.$$

The two subarcs have lengths at least their chord lengths, so

$$\text{length}(\gamma) \geq |p - x| + |y - p|,$$

which gives 14. $\square$

9.3.4 2.2 Distance to the singular locus

For an invertible matrix C, let $\sigma_{\min}(C)$ denote its least singular value.

Lemma 6

$$\boxed{\text{dist}*F(C, \Sigma_n) = \sigma_{\min}(C).} \tag{15}$$

If $\det C < 0$, then also

$$\boxed{\text{dist}*F(C, \det \geq 0) = \sigma_{\min}(C).} \tag{16}$$

Proof Let S be singular and choose a unit vector $v \in \ker S$. Then

$$|C - S|_F \geq |(C - S)v| = |Cv| \geq \sigma_{\min}(C).$$

Conversely, let $Cv = \sigma_{\min}(C)u$ be a least-singular-value pair. Then

$$S = C - \sigma_{\min}(C)uv^{\mathsf{T}}$$

satisfies $Sv = 0$ and

$$|C - S| * F = \sigma_{\min}(C).$$

This proves 15.

If $\det C < 0$ and P has nonnegative determinant, the segment from C to P meets Σ_n. Hence

$$|C - P|_F \geq \operatorname{dist}_F(C, \Sigma_n).$$

Since $\Sigma_n \subset \det \geq 0$, equality of the two distances follows. □

Combining Lemmas 5 and 6 yields the following useful strict estimates.

9.3.5 Corollary 7 — Quantitative strictness

Let

$$C_t = (1-t)A + tB, \qquad D = |A - B|_F.$$

If $A, B \in GL_n^+$ and $\det C_t < 0$, then

$$\boxed{d_{\Omega_n}(A,B) \geq \sqrt{t^2 D^2 + \sigma_{\min}(C_t)^2} + \sqrt{(1-t)^2 D^2 + \sigma_{\min}(C_t)^2} D.} \tag{17}$$

If $A, B \in \Sigma_n$ and C_t is invertible, then

$$\boxed{d_{\Sigma_n}(A,B) \geq \sqrt{t^2 D^2 + \sigma_{\min}(C_t)^2} + \sqrt{(1-t)^2 D^2 + \sigma_{\min}(C_t)^2} D.} \tag{18}$$

The lower bounds may of course be maximized over all admissible t.

9.3.6 2.3 A local bypass lemma

To prove the sufficiency of 8, one must show that isolated contacts of the segment with Σ_n can be bypassed at arbitrarily small cost.

Lemma 8 — Local positive-determinant bypass For each $C \in \Sigma_n$, there exist constants $h_0, L, R > 0$ such that, whenever $0 < h < h_0$ and

$$X, Y \in GL_n^+ \cap B(C, h),$$

there is a piecewise C^1 path in

$$GL_n^+ \cap B(C, Rh)$$

joining X to Y, of length at most Lh.

Proof Let $k = \operatorname{rank} C$, $m = n - k$. By left and right orthogonal transformations that preserve Frobenius distances and determinant sign, we may suppose

$$C = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix},$$

where D is a positive diagonal $k \times k$ matrix.

The determinant-sign-preserving reduction deserves one detail. Starting with a singular-value decomposition, if the product of the determinants of the two orthogonal factors is (-1), one flips one basis vector in a zero singular direction. This does not alter C, but changes that product to (+1).

Write matrices near C in block form

$$M = \begin{pmatrix} P & F \\ G & H \end{pmatrix}.$$

For M sufficiently close to C, P is invertible and $\det P > 0$. Introduce the Schur complement

$$W = H - GP^{-1}F.$$

The map

$$M \longmapsto (P, F, G, W)$$

is a smooth local diffeomorphism with inverse

$$(P, F, G, W) \longmapsto \begin{pmatrix} P & F \\ G & W + GP^{-1}F \end{pmatrix}.$$

On a sufficiently small fixed neighborhood, this map and its inverse are bi-Lipschitz. Moreover,

$$\det M = \det P, \det W,$$

so in these coordinates

$$\det M > 0 \iff \det W > 0.$$

It remains to connect two small matrices $W_0, W_1 \in GL_m^+$ by a path of size and length $O(h)$. Write their polar decompositions

$$W_i = Q_i H_i, \qquad Q_i \in SO(m), \quad H_i > 0.$$

Fix $a = h$. Connect W_i to aQ_i through

$$Q_i\big((1-s)H_i + saI\big).$$

This stays in GL_m^+ and has length

$$|H_i - aI|_F = O(h).$$

Next connect aQ_0 to aQ_1 in $aSO(m)$. If

$$Q_0^\top Q_1 = e^K, \qquad K^\top = -K,$$

then

$$s \longmapsto aQ_0 e^{sK}$$

has length $a|K|_F = O(h)$. Such a skew-symmetric logarithm always exists for an element of $SO(m)$; one may take the usual block logarithm on its planar rotation blocks, with all angles in $[-\pi, \pi]$.

Interpolate the free coordinates (P,F,G) linearly while following this path in W. The bi-Lipschitz inverse coordinate map gives the asserted path and estimates. The case $k = 0$ is the same argument with the (P,F,G) blocks absent. □

9.3.7 Proof of Theorem 3

Set

$$p(t) = \det((1-t)A + tB).$$

If $p(t) < 0$ for some t, Corollary 7 gives

$$d_{\Omega_n}(A, B) > |A - B|_F.$$

Thus 7 implies 8.

Conversely, suppose $p(t) \geq 0$ on ([0,1]). Since $p(0), p(1) > 0$, p is not the zero polynomial and has only finitely many zeros

$$0 < t_1 < \cdots < t_N < 1.$$

Outside arbitrarily small disjoint intervals around the t_j, the straight segment lies in GL_n^+.

Inside an interval around t_j, its two endpoints lie in GL_n^+ and tend to the singular matrix

$$C_j = (1 - t_j)A + t_j B.$$

Lemma 8 replaces that short portion by a path in GL_n^+ whose length tends to 0 with the interval size. Replacing all such portions gives curves in GL_n^+ whose lengths tend to

$$|A - B| * F.$$

Since every curve has length at least its Euclidean chord,

$$d_{\Omega_n}(A,B) = |A-B|_F.$$

This proves the equivalence of 7 and 8.

To prove the spectral formulation, put

$$C = A^{-1}B.$$

Then

$$p(t) = \det A \cdot \det((1-t)I + tC).$$

Over $\mathbb{C}$,

$$\det((1-t)I + tC) = \prod_{\lambda\in\sigma(C)} (1-t+t\lambda),$$

with eigenvalues repeated according to algebraic multiplicity.

A positive real eigenvalue contributes a positive factor on ([0,1]). A nonreal conjugate pair $\lambda, \bar{\lambda}$ contributes

$$|1-t+t\lambda|^2 > 0.$$

A negative real eigenvalue λ contributes a zero at

$$t = \frac{1}{1-\lambda} \in (0,1),$$

whose multiplicity is precisely the algebraic multiplicity of λ. Hence the determinant is nonnegative throughout ([0,1]) exactly when each such multiplicity is even. □

9.3.8 Proof of Theorem 4

If

$$\det((1-t)A + tB) \equiv 0,$$

the straight segment lies in Σ_n, and therefore

$$d_{\Sigma_n}(A,B) = |A-B|_F.$$

If the determinant polynomial is not identically zero, there is some $t \in (0,1)$ for which C_t is invertible. Corollary 7 then gives strict inequality. This proves 10–11.

The equivalence with 12 follows by homogenizing the determinant polynomial. Finally, a square matrix over the field $\mathbb{R}(\lambda)$ has zero determinant exactly when it has a nonzero kernel vector over that field. Clearing denominators gives 13. □

9.4 3. Constructive upper bounds in arbitrary dimension

The cone formula is exact but leaves an angular metric. The following bounds give concrete finite-dimensional constructions.

9.4.1 3.1 A polar-decomposition route in GL_n^+

Let

$$A = Q_A P_A, \qquad B = Q_B P_B,$$

be the polar decompositions, with

$$Q_A, Q_B \in SO(n), \qquad P_A, P_B > 0.$$

Let the eigenvalues of $Q_A^\top Q_B$ be

$$e^{\pm i\theta_1}, \ldots, e^{\pm i\theta_m}, \qquad 0 \le \theta_j \le \pi,$$

together with possible eigenvalues 1, and put

$$\vartheta(Q_A, Q_B) = \sqrt{2\sum_{j=1}^{m} \theta_j^2}.$$

This is the length of the standard shortest rotation path in $SO(n)$ for the induced Frobenius metric.

For every $\varepsilon > 0$, the concatenation

$$A \longrightarrow \varepsilon Q_A \longrightarrow \varepsilon Q_B \longrightarrow B$$

stays in GL_n^+, and gives

$$\boxed{d_{\Omega_n}(A, B) \le \inf_{\varepsilon > 0} \left(|P_A - \varepsilon I|_F + \varepsilon \vartheta(Q_A, Q_B) + |P_B - \varepsilon I|_F \right).} \tag{19}$$

Taking $\varepsilon \downarrow 0$ recovers the bound

$$d_{\Omega_n}(A, B) \le |A|_F + |B|_F.$$

Exact special case: distance to $SO(n)$ The polar factor is the unique closest element of $SO(n)$, even for the intrinsic distance:

$$\boxed{\operatorname{dist}_{d_{\Omega_n}}(A, SO(n)) = |P_A - I|_F,} \tag{20}$$

and the unique minimizer is Q_A.

Indeed, the segment

$$Q_A\big((1-t)P_A + tI\big)$$

lies in GL_n^+ and has length $|P_A - I|_F$. Conversely, intrinsic distance dominates Euclidean distance, and Q_A is the unique Frobenius-nearest rotation.

9.4.2 3.2 A common-kernel construction in the singular locus

For a unit vector x, put

$$P_x = I - xx^\mathsf{T}.$$

Then AP_x and BP_x both annihilate x. The three paths

$$A \longrightarrow AP_x, \qquad AP_x \longrightarrow BP_x, \qquad BP_x \longrightarrow B$$

all lie in Σ_n. Their total length is

$$|Ax| + |(A-B)P_x| * F + |Bx|.$$

Therefore

$$\boxed{d_{\Sigma_n}(A,B) \le \min_{|x|=1} \left(|Ax| + |(A-B)(I - xx^\mathsf{T})|_F + |Bx|\right).} \tag{21}$$

By applying this to transposes, one also gets

$$\boxed{d_{\Sigma_n}(A,B) \le \min_{|y|=1} \left(|A^\mathsf{T}y| + |(I - yy^\mathsf{T})(A-B)|_F + |B^\mathsf{T}y|\right).} \tag{22}$$

If A and B have a common right or left null vector, these bounds give the Euclidean chord exactly.

9.5 4. Complete solution for singular 2×2 matrices

For

$$A = \begin{pmatrix} a & b\, c & d \end{pmatrix},$$

define

$$\Phi(A) = (z, w) \in \mathbb{C}^2$$

by

$$z = \frac{a + d + i(c-b)}{\sqrt{2}}, \qquad w = \frac{a - d + i(c+b)}{\sqrt{2}}. \tag{23}$$

A direct computation gives

$$|z|^2 + |w|^2 = |A|_F^2, \tag{24}$$

and

$$\det A = \frac{|z|^2 - |w|^2}{2}. \tag{25}$$

Thus Φ is a real-linear isometry, and

$$\Sigma_2 \quad \longleftrightarrow \quad (z, w) : |z| = |w|.$$

The unit link is therefore the Clifford torus

$$T = \left\{ \frac{1}{\sqrt{2}}(e^{i\alpha}, e^{i\beta}) : \alpha, \beta \in \mathbb{R} \right\}.$$

Its induced metric is

$$ds_T^2 = \frac{1}{2}(d\alpha^2 + d\beta^2). \tag{26}$$

For nonzero complex numbers ζ, ξ, define their principal angular separation by

$$\operatorname{ang}(\zeta, \xi) = \min_{k \in \mathbb{Z}} \left| \arg \xi - \arg \zeta + 2\pi k \right| \in [0, \pi].$$

9.5.1 Theorem 9 — Exact distance on Σ_2

Let $A, B \in \Sigma_2 \setminus 0$, with

$$\Phi(A) = (z_A, w_A), \qquad \Phi(B) = (z_B, w_B).$$

Define

$$\Theta(A, B) = \sqrt{\frac{\operatorname{ang}(z_A, z_B)^2 + \operatorname{ang}(w_A, w_B)^2}{2}}. \tag{27}$$

Then $0 \le \Theta(A, B) \le \pi$, and

$$\boxed{d_{\Sigma_2}(A, B) = \sqrt{|A|_F^2 + |B|_F^2 - - - - - - - - - - - - - - - - 2|A|_F |B|_F \cos \Theta(A, B)}.} \tag{28}$$

If one endpoint is 0, the distance is the Frobenius norm of the other.

Proof The universal cover of T is $\mathbb{R}^2$ with metric

$$\frac{1}{2}(d\alpha^2 + d\beta^2).$$

Hence the intrinsic torus distance between the normalized directions of A and B is exactly $\Theta(A, B)$. Since $\Theta \le \pi$, formula 28 follows directly from the metric-cone formula. $\square$

Example Let

$$A = E_{11}, \qquad B = E_{22}.$$

Then

$$\Phi(A) = \frac{1}{\sqrt{2}}(1,1), \qquad \Phi(B) = \frac{1}{\sqrt{2}}(1,-1).$$

Thus

$$\Theta(A,B) = \frac{\pi}{\sqrt{2}},$$

and

$$\boxed{d_{\Sigma_2}(E_{11},E_{22}) = \sqrt{2 - 2\cos(\pi/\sqrt{2})} \approx 1.79204.}$$

This is strictly greater than

$$|E_{11} - E_{22}|_F = \sqrt{2}.$$

At the midpoint, $\frac{1}{2}(E_{11} + E_{22}) = \frac{1}{2}I$ is invertible. The general obstacle bound already gives

$$d_{\Sigma_2}(E_{11},E_{22}) \geq \sqrt{3},$$

while 28 gives the exact value.

9.6 5. Exact variational determination for $GL_2^+(\mathbb{R})$

Under the same isometry Φ,

$$GL_2^+(\mathbb{R}) \quad \longleftrightarrow \quad (z,w) \in \mathbb{C}^2 : |z| > |w|.$$

Its unit link is the open solid Clifford torus

$$H^\circ = (z,w) \in S^3 : |z| > |w|.$$

Use Hopf coordinates

$$z = \cos\eta, e^{i\alpha}, \qquad w = \sin\eta, e^{i\beta}, \qquad 0 \leq \eta < \frac{\pi}{4}. \tag{29}$$

The round metric on S^3 is

$$ds^2 = d\eta^2 + \cos^2\eta, d\alpha^2 + \sin^2\eta, d\beta^2. \tag{30}$$

Let $p, q \in H^\circ$ have coordinates

$$(\eta_0, \alpha_0, \beta_0), \qquad (\eta_1, \alpha_1, \beta_1).$$

Let

$$a = \min_{k\in\mathbb{Z}} |\alpha_1 - \alpha_0 + 2\pi k| \in [0,\pi].$$

If both w-coordinates are nonzero, let

$$b = \min_{\ell\in\mathbb{Z}} |\beta_1 - \beta_0 + 2\pi \ell| \in [0,\pi].$$

If one w-coordinate is zero, its β-coordinate is immaterial, and we set $b = 0$.

For

$$\eta \in H^1([0,1]), \qquad \eta(0) = \eta_0, \quad \eta(1) = \eta_1, \qquad 0 \le \eta(t) \le \frac{\pi}{4},$$

define

$$I_c(\eta) = \int_0^1 \sec^2 \eta(t), dt, \qquad I_s(\eta) = \int_0^1 \csc^2 \eta(t), dt, \tag{31}$$

allowing $I_s(\eta) = +\infty$, and using the convention

$$\frac{b^2}{+\infty} = 0.$$

9.6.1 Theorem 10 — Scalar variational formula for the link

The intrinsic spherical distance in H° satisfies

$$\boxed{\delta_H(p,q)^2 = \inf_\eta \left[\int_0^1 \eta'(t)^2, dt + \frac{a^2}{I_c(\eta)} + \frac{b^2}{I_s(\eta)} \right].} \tag{32}$$

Consequently, for $A, B \in GL_2^+$, with

$$r = |A|_F, \quad s = |B| * F, \qquad p = \frac{\Phi(A)}{r}, \quad q = \frac{\Phi(B)}{s},$$

one has the exact formula

$$\boxed{d_{\Omega_2}(A,B) = \sqrt{r^2 + s^2 - - - - - - - - 2rs\cos(\min \pi, \delta_H(p,q))}.} \tag{33}$$

Thus the 2×2 problem reduces exactly to a one-variable obstacle variational problem.

Proof of 32 For a unit-time curve in H°, the energy is

$$E = \int_0^1 \left(\eta'^2 + \cos^2\eta, \alpha'^2 + \sin^2\eta, \beta'^2\right) dt. \tag{34}$$

The infimum of E is the square of the length distance: every curve can be reparameterized at constant speed, and Cauchy–Schwarz gives $L^2 \le E$.

Fix η. By Cauchy–Schwarz,

$$\left(\int_0^1 \alpha', dt\right)^2 \le \left(\int_0^1 \cos^2\eta, \alpha'^2 dt\right)\left(\int_0^1 \sec^2\eta, dt\right).$$

Therefore the minimum phase energy for a prescribed net change a is

$$\frac{a^2}{I_c(\eta)},$$

attained when

$$\alpha'(t) = \frac{a, \sec^2\eta(t)}{I_c(\eta)}$$

with the appropriate sign.

Similarly,

$$\inf_\beta \int_0^1 \sin^2\eta, \beta'^2 dt = \frac{b^2}{I_s(\eta)}.$$

If $I_s = +\infty$, the phase change can be concentrated arbitrarily near a point where η is arbitrarily small, at vanishing energy cost. Geometrically, at $\eta = 0$ the w-coordinate vanishes and its phase can be reset.

The winding numbers in α and β are minimized by the principal differences (a,b). This proves 32. Allowing the boundary value $\eta = \pi/4$ does not alter the infimum: a boundary path can be pushed to $\eta < \pi/4$, fixing its endpoints and changing its length by an arbitrarily small amount. $\square$

On any interval on which a minimizing profile satisfies

$$0 < \eta < \frac{\pi}{4},$$

its Euler–Lagrange equation is

$$\eta'' + p^2 \frac{\sin\eta}{\cos^3\eta} ---------------------------- - q^2 \frac{\cos\eta}{\sin^3\eta} = 0, \tag{35}$$

where

$$p = \frac{a}{I_c(\eta)}, \qquad q = \frac{b}{I_s(\eta)}.$$

At $\eta = \pi/4$ this is supplemented by the usual obstacle variational inequality; contact with $\eta = 0$ allows a change of the β-phase.

9.7 6. Boundary and sanity checks

9.7.1 Dimension $n = 1$

Here

$$GL_1^+(\mathbb{R}) = (0, \infty),$$

which is convex, so

$$\boxed{d_{\Omega_1}(a, b) = |a - b|.}$$

The singular locus is the singleton ($\{0\}$).

This also confirms that the relevant metric is the induced Euclidean metric, not a left-invariant logarithmic metric.

9.7.2 Positive scalar multiples

If $B = cA$ with $c > 0$, the radial segment is admissible in either cone, and

$$d(A, B) = |c - 1|, |A|_F.$$

9.7.3 Equal polar factors

If

$$A = QP, \qquad B = QR,$$

with $Q \in SO(n)$ and $P, R > 0$, then

$$(1 - t)A + tB = Q((1 - t)P + tR) \in GL_n^+.$$

Hence

$$d_{\Omega_n}(A, B) = |A - B|_F.$$

9.7.4 Equality without an admissible straight segment

For even n,

$$A = I, \qquad B = -I$$

both belong to GL_n^+. The only negative eigenvalue of $A^{-1}B = -I$ is (-1), with even multiplicity n. Therefore

$$\boxed{d_{\Omega_n}(I, -I) = 2\sqrt{n}.}$$

The straight segment passes through 0, so no curve in GL_n^+ attains this infimum. Indeed, any Euclidean curve whose length equals its endpoint chord must itself be the straight segment.

This shows that "geodesic distance" here is genuinely an infimum and need not be realized.

9.7.5 Strict inequality in GL_2^+

Take

$$A = I, \qquad B = \mathrm{diag}(-1, -2).$$

Then

$$\det((1-t)A + tB) = (1-2t)(1-3t),$$

which is negative for $1/3 < t < 1/2$. Hence the distance is strictly greater than

$$|A - B| * F = \sqrt{13}.$$

At $t = 2/5$,

$$C_t = \mathrm{diag}(1/5, -1/5), \qquad \sigma_{\min}(C_t) = \frac{1}{5}.$$

Formula 17 gives

$$d_{\Omega_2}(A, B) \geq \frac{\sqrt{53} + \sqrt{118}}{5} \approx 3.62858\sqrt{13} \approx 3.60555.$$

9.7.6 A singular pencil without a common kernel

Let

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

Then

$$\ker A = \mathbb{R}e_1, \qquad \ker B = \mathbb{R}e_2,$$

so there is no common right kernel. Nevertheless every linear combination of A and B is a 3×3 skew-symmetric matrix, hence singular. Therefore

$$d_{\Sigma_3}(A, B) = |A - B|_F.$$

This shows why the correct condition is singularity of the whole pencil, not existence of a fixed common null vector.

9.8 7. Topological and foundational assumptions

All arguments take place in the finite-dimensional Euclidean space $M_n(\mathbb{R})$.

- **Separation and countability.** The spaces are Hausdorff, metrizable, first countable, second countable and separable automatically. No nonmetrizable or net-based argument occurs.

- **Path-connectedness.** GL_n^+ is piecewise C^1-arc-connected by polar decomposition. The singular locus is star-shaped about 0, hence path-connected; its punctured unit link is connected for $n \geq 2$.
- **Compactness.** No compactness of GL_n^+ is assumed. Compactness is used only in standard finite-dimensional forms, such as compactness of orthogonal groups or unit links. In particular, GL_n^+ is not closed and the infimum need not be attained.
- **Sequential versus general topology.** Every limiting argument can be formulated sequentially because all spaces are metric. The main strictness proofs above avoid limiting compactness altogether by using the explicit obstacle bound.
- **Regularity of curves.** All constructed curves are piecewise C^1. In the 2×2 variational formula, H^1 or absolutely continuous curves give the same infimum by piecewise-smooth approximation.
- **Choice.** No nontrivial choice principle is used; only finite-dimensional singular-value and polar decompositions and finite choices of orthonormal bases occur.
- **Norm.** The Hilbertian Frobenius norm is essential to the cosine-law cone formula. A different norm would give a different intrinsic metric.
- **Determinant component.** Restricting to GL_n^+ is essential. The two determinant-sign components of $GL_n(\mathbb{R})$ cannot be joined by an invertible path.

9.9 8. Exact remaining obstruction

For arbitrary n, Theorem 1 reduces the problem exactly to evaluating

$$\delta_+ \quad \text{on} \quad Y_n^+ = X \in S_F : \det X > 0,$$

and

$$\delta_\Sigma \quad \text{on} \quad Z_n = X \in S_F : \det X = 0.$$

For $n = 2$, Z_2 is a flat Clifford torus, which gives the closed formula 28, while Y_2^+ is a solid Clifford torus and yields the scalar problem 32.

For $n \geq 3$, the singular link is stratified by rank:

$$Z_n = \bigcup_{k=1}^{n-1} X \in S_F : \operatorname{rank} X = k.$$

A shortest path can use lower-rank strata as shortcuts, so solving the ordinary geodesic equations only on the smooth rank-$(n-1)$ stratum is insufficient. Likewise, paths in Y_n^+ may approach several boundary rank strata.

A relevant result of Katz, Katz, Kerner and Liokumovich proves global bi-Lipschitz comparison between the intrinsic and ambient metrics on GL_n^+, and their rank-stratification argument establishes the corresponding normal-embedding property for the determinantal variety. Its conclusion is quantitative comparability $d_{\text{int}} \leq C(n)d_{\text{ext}}$, rather than an exact distance formula. ([arXiv][2])

The most promising next lemma would be a **stratified geodesic normal form**: prove that a minimizing curve on the closed unit determinant link can be decomposed into finitely many smooth geodesic pieces lying in fixed-rank strata, with explicitly stated tangent-cone transmission conditions and a number of rank transitions bounded in terms of n. Such a result would convert the remaining angular problem into a finite-dimensional boundary-value optimization.

[1]: https://lucpommeret.com/assets/Qsansreponse260405.pdf "https://lucpommeret.com/assets/Qsansreponse260405.pdf" [2]: https://arxiv.org/pdf/1602.01227 "https://arxiv.org/pdf/1602.01227"

Formal verification. Lean source; online type-check.

10 Question Q756

Informal verdict. **A — Complete solution**, with high confidence.

Lean coverage. Lean proves the complete polynomial classification and, for every real beta and every real lambda with absolute value greater than one (including lambda less than minus one), constructs a nonzero smooth solution flat at the origin and hence nonpolynomial. It also proves that the full solution space is infinite-dimensional.

10.1 Audited informal answer

10.1.1 1. Faithful restatement

Let $\beta, \lambda \in \mathbb{R}$, with $|\lambda| > 1$, and define

$$F(\beta, \lambda) = \left\{ g \in C^1(\mathbb{R}, \mathbb{R}) : g'(t) = g(\lambda t) - \beta g(t) \quad \text{for every } t \in \mathbb{R} \right\}.$$

The source observes that, for every nonnegative integer p, there is a nonzero polynomial of degree p in $F(\lambda^p, \lambda)$, and asks, for every admissible pair (β, λ), whether $F(\beta, \lambda)$ contains nonpolynomial functions. There are no additional subparts or footnotes attached to Q756.

There is no substantive typo or ambiguity. The only conventional point is whether "natural integer" includes 0; below,

$$\mathbb{N}_0 = 0, 1, 2, \ldots,$$

and the case $p = 0$ is the constant-polynomial case $\beta = 1$.

10.1.2 2. Final theorem

Theorem For every $\beta, \lambda \in \mathbb{R}$ satisfying $|\lambda| > 1$, the space $F(\beta, \lambda)$ contains an infinite-dimensional vector subspace of nonzero C^∞ functions that are flat at the origin:

$$g^{(n)}(0) = 0 \qquad (n \geq 0).$$

Consequently,

$$\boxed{\text{For every } \beta \in \mathbb{R} \text{ and every } |\lambda| > 1, \text{ there exist nonpolynomial functions in } F(\beta, \lambda).}$$

Moreover, the polynomial solutions are completely classified:

- if $\beta \neq \lambda^p$ for every $p \in \mathbb{N}_0$, the zero function is the only polynomial solution;
- if $\beta = \lambda^p$, the polynomial solution space is one-dimensional and is generated by

$$P_p(t) = \sum_{n=0}^{p} \frac{1}{n!} \left(\prod_{k=0}^{n-1} (\lambda^k - \lambda^p) \right) t^n,$$

where the empty product, for $n = 0$, equals 1.

The constructed nonpolynomial solutions require no boundedness or growth assumption at infinity.

10.2 3. Automatic regularity

We first record that the C^1 assumption automatically improves to C^∞.

Lemma 1 Every $g \in F(\beta, \lambda)$ is C^∞, and for every $n \geq 0$,

$$g^{(n+1)}(t) = \lambda^n g^{(n)}(\lambda t) - \beta g^{(n)}(t). \tag{3.1}$$

In particular,

$$g^{(n+1)}(0) = (\lambda^n - \beta) g^{(n)}(0). \tag{3.2}$$

Proof Suppose $g \in C^m$, with $m \geq 1$. Then

$$t \longmapsto g(\lambda t) - \beta g(t)$$

belongs to C^m. Since this function equals (g'), it follows that $g' \in C^m$, hence $g \in C^{m+1}$. Induction gives $g \in C^\infty$.

Differentiating the defining equation n times gives

$$g^{(n+1)}(t) == \frac{d^n}{dt^n} g(\lambda t) - \beta g^{(n)}(t) \lambda^n g^{(n)}(\lambda t) - \beta g^{(n)}(t).$$

Putting $t = 0$ proves (3.2). □

A useful consequence is immediate.

Corollary 2 If $g \in F(\beta, \lambda)$ and $g(0) = 0$, then g is flat at the origin:

$$g^{(n)}(0) = 0 \qquad (n \geq 0).$$

Proof Starting with $g(0) = 0$, recurrence (3.2) gives successively

$$g'(0) = g''(0) = \cdots = 0.$$

□

Thus it will suffice to construct nonzero global solutions whose two one-sided limits at 0 are both zero.

10.3 4. Polynomial solutions

Let

$$P(t) = \sum_{k=0}^{m} a_k t^k, \qquad a_m \neq 0,$$

be a polynomial solution of degree m.

The coefficient of t^m in (P') is zero, whereas the coefficient of t^m in

$$P(\lambda t) - \beta P(t)$$

is

$$(\lambda^m - \beta) a_m.$$

Therefore

$$\beta = \lambda^m. \tag{4.1}$$

More generally, comparison of the coefficient of t^k gives

$$(k+1) a_{k+1} = (\lambda^k - \beta) a_k, \qquad k \geq 0. \tag{4.2}$$

If $a_0 = 0$, recurrence (4.2) gives $a_1 = a_2 = \cdots = 0$, so every nonzero polynomial solution has $a_0 \neq 0$.

Suppose now that $\beta = \lambda^p$. Taking $a_0 = 1$, recurrence (4.2) yields

$$a_n = \frac{1}{n!} \prod_{k=0}^{n-1} (\lambda^k - \lambda^p).$$

For $n \leq p$, none of the factors vanishes, because $|\lambda| > 1$ implies that the numbers $\lambda^0, \lambda^1, \ldots$ are pairwise distinct. At the next step,

$$a_{p+1} == \frac{\lambda^p - \lambda^p}{p+1} a_p 0,$$

and all subsequent coefficients vanish. This gives the polynomial P_p stated in the theorem.

Conversely, the preceding argument shows that any nonzero polynomial solution of degree m necessarily has $\beta = \lambda^m$. Thus the polynomial classification is complete. $\square$

10.4 5. Extension from a fundamental annulus

Set

$$\rho := |\lambda| > 1$$

and, for $n \in \mathbb{Z}$, define the closed annuli

$$A_n = \left\{t \in \mathbb{R} : \rho^n \leq |t| \leq \rho^{n+1}\right\}.$$

The fundamental annulus is

$$A_0 = t : 1 \leq |t| \leq \rho.$$

Let

$$\mathcal{S} = C_c^\infty\big(t : 1 < |t| < \rho\big).$$

Thus every $\varphi \in \mathcal{S}$ vanishes on neighborhoods of the four boundary points

$$-\rho, \quad -1, \quad 1, \quad \rho.$$

We show that any such seed φ can be extended uniquely to a smooth solution on $\mathbb{R} \setminus 0$.

10.4.1 5.1. Extension toward infinity

Set

$$G_\varphi(t) = \varphi(t), \qquad t \in A_0.$$

Assume that G_φ has been defined on A_n, for some $n \geq 0$. For $x \in A_{n+1}$, define

$$G_\varphi(x) = G'_\varphi!\left(\frac{x}{\lambda}\right) + \beta G_\varphi\left(\frac{x}{\lambda}\right). \tag{5.1}$$

Indeed, $x/\lambda \in A_n$.

By construction,

$$G_\varphi(\lambda t) = G'_\varphi(t) + \beta G_\varphi(t)$$

on A_n, which is exactly the desired equation.

Furthermore,

$$\operatorname{supp}\big(G_\varphi|_{A_{n+1}}\big) \subset \lambda, \operatorname{supp}\big(G_\varphi|_{A_n}\big). \tag{5.2}$$

Differentiation does not enlarge support. Since the seed is supported strictly inside A_0, induction shows that every outward piece is supported strictly inside its annulus. Hence adjacent pieces vanish on neighborhoods of their common boundary and glue C^∞-smoothly.

This defines G_φ on $|t| \geq 1$.

10.4.2 5.2. Extension toward the origin

Suppose that G_φ has already been defined for

$$|t| \geq \rho^{-m},$$

where $m \geq 0$. We define it on the next inner shell

$$A_{-(m+1)} = \left\{ t : \rho^{-(m+1)} \leq |t| \leq \rho^{-m} \right\}.$$

Fix a sign $\sigma \in -1, 1$, and put

$$a = \sigma \rho^{-m}.$$

For t in the corresponding component of $A_{-(m+1)}$, define

$$G_\varphi(t) = e^{-\beta(t-a)} G_\varphi(a) + \int_a^t e^{-\beta(t-s)} G_\varphi(\lambda s), ds. \tag{5.3}$$

For such s,

$$\rho^{-m} \leq |\lambda s| \leq \rho^{-m+1},$$

so $G_\varphi(\lambda s)$ is already known.

Differentiating (5.3) gives

$$G'_\varphi(t) = -\beta G_\varphi(t) + G_\varphi(\lambda t),$$

as required.

At $t = a$, formula (5.3) reproduces the previously assigned value $G_\varphi(a)$. The old and new pieces solve the same first-order inhomogeneous differential equation near the boundary, with the same boundary value. Since the forcing

$$t \longmapsto G_\varphi(\lambda t)$$

is already C^∞ there, the pieces glue C^∞-smoothly.

At the initial boundaries $t = \pm 1$, the seed and the forcing both vanish in a neighborhood, so the same argument applies.

Induction defines a function

$$G_\varphi \in C^\infty(\mathbb{R} \setminus 0)$$

satisfying

$$G'_\varphi(t) = G_\varphi(\lambda t) - \beta G_\varphi(t), \qquad t \neq 0. \tag{5.4}$$

All operations in the construction are linear in φ.

10.5 6. Uniform boundedness near the origin

We next prove that every annular extension has finite one-sided limits at 0.

For $n \geq 0$, set

$$M_n = \sup_{t \in A_{-n}} |G_\varphi(t)|.$$

Thus $M_0 = |\varphi|_\infty$.

The width of either connected component of A_{-n}, for $n \geq 1$, is

$$\ell_n == \rho^{-n+1} - \rho^{-n}(\rho - 1)\rho^{-n}.$$

Let $t \in A_{-n}$ and let a denote the outer endpoint of its connected component. From (5.3),

$$|G_\varphi(t)| \leq e^{|\beta|\ell_n}|G_\varphi(a)| + e^{|\beta|\ell_n} \int_a^t |G_\varphi(\lambda s)|, |ds|.$$

Both $G_\varphi(a)$ and $G_\varphi(\lambda s)$ are bounded in absolute value by M_{n-1}. Hence

$$M_n \leq e^{|\beta|\ell_n}(1 + \ell_n)M_{n-1} \leq e^{(|\beta|+1)\ell_n} M_{n-1}. \tag{6.1}$$

But

$$\sum_{n=1}^{\infty} \ell_n == \sum_{n=1}^{\infty} (\rho^{-n+1} - \rho^{-n})1.$$

Iterating (6.1) gives

$$M_n \leq e^{|\beta|+1} M_0 \qquad (n \geq 1). \tag{6.2}$$

Thus G_φ is uniformly bounded on a punctured neighborhood of 0. Since $G_\varphi(\lambda t)$ is also bounded there, equation (5.4) gives a uniform derivative bound:

$$|G'_\varphi(t)| \leq |G_\varphi(\lambda t)| + |\beta|, |G_\varphi(t)| \leq C_{\beta,\lambda,\varphi}, \qquad 0 < |t| \leq 1. \tag{6.3}$$

Consequently G_φ is uniformly Lipschitz on each side of the origin. Therefore the finite limits

$$L_+(\varphi) = \lim_{t \downarrow 0} G_\varphi(t), \qquad L_-(\varphi) = \lim_{t \uparrow 0} G_\varphi(t) \tag{6.4}$$

exist.

The map

$$\Lambda : \mathcal{S} \longrightarrow \mathbb{R}^2, \qquad \Lambda(\varphi) = \big(L_-(\varphi), L_+(\varphi)\big), \tag{6.5}$$

is linear.

10.6 7. Killing both one-sided limits

The seed space $\mathcal{S}$ is infinite-dimensional. More concretely, inside the nonempty interval $(1, \rho)$ one may choose arbitrarily many pairwise disjoint intervals and place a nonzero smooth bump function in each one.

For example, on an interval $(a, b) \subset (1, \rho)$, one can take

$$\psi_{a,b}(t) = \begin{cases} \exp\left(-\dfrac{1}{(t-a)(b-t)}\right), & a < t < b, \\ 0, & \text{otherwise.} \end{cases}$$

Choose three linearly independent seeds

$$\varphi_1, \varphi_2, \varphi_3 \in \mathcal{S}.$$

Their images under Λ are three vectors in the two-dimensional space $\mathbb{R}^2$. Hence there exist real coefficients, not all zero, such that

$$c_1 \Lambda(\varphi_1) + c_2 \Lambda(\varphi_2) + c_3 \Lambda(\varphi_3) = 0.$$

Set

$$\varphi = c_1 \varphi_1 + c_2 \varphi_2 + c_3 \varphi_3.$$

Since the seeds are linearly independent, $\varphi \neq 0$, while

$$L_-(\varphi) = L_+(\varphi) = 0. \tag{7.1}$$

Let G_φ be its annular extension, and define

$$g(t) = \begin{cases} G_\varphi(t), & t \neq 0, \\ 0, & t = 0. \end{cases} \tag{7.2}$$

By (7.1), g is continuous at 0. Furthermore, from the equation on the punctured line,

$$G'_\varphi(t) = G_\varphi(\lambda t) - \beta G_\varphi(t) \longrightarrow 0 \qquad (t \to 0), \tag{7.3}$$

because both one-sided limits of G_φ are zero.

For $t \neq 0$, the mean value theorem applied between 0 and t gives some ξ_t between 0 and t such that

$$\frac{g(t) - g(0)}{t} = G'_\varphi(\xi_t).$$

By (7.3), the right-hand side tends to zero. Thus

$$g'(0) = 0.$$

Moreover (g') is continuous at 0, and the differential equation holds there as well:

$$g'(0) = 0 = g(0) - \beta g(0).$$

Therefore

$$g \in F(\beta, \lambda).$$

By Lemma 1, $g \in C^\infty$. Since $g(0) = 0$, Corollary 2 gives

$$g^{(n)}(0) = 0 \qquad (n \geq 0).$$

But g is nonzero, because it agrees with the nonzero seed φ on A_0. A polynomial flat at a point must be the zero polynomial. Hence g is nonpolynomial.

This proves existence for every admissible (β, λ).

10.6.1 Infinite-dimensionality

The same argument gives more than one solution. Given any $N \geq 1$, choose $N + 2$ linearly independent seeds and restrict Λ to their $(N + 2)$-dimensional span. Its rank is at most 2, so its kernel has dimension at least N. Since restriction to A_0 recovers the seed, the corresponding global extensions remain linearly independent.

Thus the subspace of flat solutions in $F(\beta, \lambda)$ is infinite-dimensional. □

10.7 8. Analytic solutions

The construction also explains the difference between the C^1 and analytic categories.

Proposition A solution $g \in F(\beta, \lambda)$ that is real analytic in a neighborhood of 0 is necessarily polynomial. More precisely:

- if $\beta \neq \lambda^p$ for every $p \in \mathbb{N}_0$, then $g = 0$;
- if $\beta = \lambda^p$, then g is a scalar multiple of P_p.

Proof From (3.2),

$$g^{(n)}(0) = g(0) \prod_{k=0}^{n-1} (\lambda^k - \beta). \tag{8.1}$$

If $g(0) = 0$, all derivatives at 0 vanish. Analyticity gives $g = 0$ on a neighborhood of 0. If g vanishes on $(-\varepsilon, \varepsilon)$, then

$$g(\lambda t) = g'(t) + \beta g(t) = 0$$

for $|t| < \varepsilon$, so g vanishes on

$$(-|\lambda|\varepsilon, |\lambda|\varepsilon).$$

Iteration gives $g = 0$ on all of $\mathbb{R}$.

Now suppose $g(0) \neq 0$. If no factor in (8.1) vanishes, then for all sufficiently large k,

$$|\lambda^k - \beta| \geq \frac{1}{2}|\lambda|^k.$$

Consequently the Taylor coefficient

$$\frac{g^{(n)}(0)}{n!}$$

grows in n-th root faster than a constant multiple of

$$\frac{|\lambda|^{n/2}}{n},$$

which tends to $+\infty$. The Taylor series would therefore have radius of convergence zero, contradicting analyticity.

Hence some factor vanishes:

$$\beta = \lambda^p.$$

Then recurrence (8.1) terminates at order p, and the Taylor series is a scalar multiple of P_p. The same outward propagation argument shows equality on all of $\mathbb{R}$. $\square$

Thus all the nonpolynomial solutions constructed above are necessarily nonanalytic at 0; in fact, they are nonzero and flat there.

10.8 9. Boundary and small-case checks

The case $p = 0$ Here $\beta = \lambda^0 = 1$, and

$$P_0(t) = 1.$$

Indeed,

$$P_0'(t) = 0 = P_0(\lambda t) - P_0(t).$$

The theorem nevertheless supplies infinitely many nonpolynomial flat solutions in $F(1, \lambda)$.

The case $p = 1$ Here $\beta = \lambda$, and

$$P_1(t) = 1 + (1 - \lambda)t.$$

Directly,

$$P_1'(t) = 1 - \lambda$$

and

$$P_1(\lambda t) - \lambda P_1(t) = 1 - \lambda.$$

Negative dilations When $\lambda < -1$, the map $t \mapsto \lambda t$ exchanges the two half-lines. This is why the construction imposes the two independent conditions

$$L_-(\varphi) = L_+(\varphi) = 0.$$

No parity assumption and no sign assumption on β are needed.

Why the strict inequality $|\lambda| > 1$ matters The proof uses the geometrically shrinking annuli and the summability

$$\sum_{n \geq 1} \left(\rho^{-n+1} - \rho^{-n}\right) = 1.$$

The universal conclusion fails at the boundary.

If $\lambda = 1$, then

$$g' = (1 - \beta)g.$$

For $\beta = 1$, all solutions are constant, so there are no nonpolynomial solutions.

If $\lambda = -1$, differentiating the equation gives

$$g'' = (\beta^2 - 1)g, \qquad g'(0) = (1 - \beta)g(0).$$

For $\beta = 1$, all solutions are constant. For $\beta = -1$, all solutions are of the form

$$g(t) = c(1 + 2t),$$

again polynomial. Thus the strict condition $|\lambda| > 1$ cannot simply be replaced by $|\lambda| \geq 1$.

10.8.1 Conclusion

For every real β and every real λ with $|\lambda| > 1$, nonpolynomial solutions not only exist: the space $F(\beta, \lambda)$ contains an infinite-dimensional family of nonzero C^∞ solutions flat at the origin. The polynomial solutions occur exactly at the resonant parameters $\beta = \lambda^p$, but even at those parameters they form only a one-dimensional subspace inside a much larger solution space.

Formal verification. Lean source; online type-check.

11 Question Q759

Informal verdict. **A — Complete solution**, with high confidence.

Lean coverage. Lean proves the complete canonical Q759 statement: for every fixed derivative order, both displayed sequences converge uniformly on [-1,1] to the corresponding derivative of the printed alpha. The response-only quantitative rates and sharpness claim are not formalized because the PDF does not request them.

11.1 Audited informal answer

11.1.1 Faithful restatement

Let

$$E = C^\infty([-1, 1]; \mathbb{R}), \qquad \|h\|_\infty = \max_{x \in [-1,1]} |h(x)|.$$

For a sequence $(f_n) \subset E$ and $g \in E$, the notation

$$(f_n) \Rightarrow g$$

means

$$\forall k \in \mathbb{N}, \qquad \lim_{n \to \infty} |f_n^{(k)} - g^{(k)}|_\infty = 0.$$

Define

$$\alpha(x) = \begin{cases} \exp\Bigl(-\dfrac{x^2}{1-x^2}\Bigr), & |x| < 1, \\ 0, & x = \pm 1. \end{cases}$$

Determine whether $f_n \Rightarrow \alpha$ for each of the two sequences

$$f_n^{(a)}(x) = \exp\bigl(-x^2 - x^4 - \cdots - x^{2n}\bigr)$$

and

$$f_n^{(b)}(x) = \left(1 - \frac{x^2 + x^4 + \cdots + x^{2n}}{n}\right)^n.$$

The visual PDF confirms that in part b the whole sum $x^2 + x^4 + \cdots + x^{2n}$ is divided by n. There is no apparent mathematical typo. The phrase "equip E with the uniform norm" is to be read together with the displayed definition: $\Rightarrow$ is convergence for all seminorms $h \mapsto |h^{(k)}|_\infty$, i.e. the usual C^∞-topology.

11.1.2 Final result

Theorem Both sequences converge to α in the sense of the problem:

$$\boxed{f_n^{(a)} \Rightarrow \alpha} \qquad \text{and} \qquad \boxed{f_n^{(b)} \Rightarrow \alpha}.$$

More precisely, for every $k, M \in \mathbb{N}$,

$$\max_{0 \le j \le k} \left\| \bigl(f_n^{(a)} - \alpha\bigr)^{(j)} \right\|_\infty = O_{k,M}(n^{-M}),$$

whereas, for every fixed k,

$$\max_{0 \le j \le k} \left| \bigl(f_n^{(b)} - \alpha\bigr)^{(j)} \right|_\infty = O_k(n^{-1}).$$

The rate $O(n^{-1})$ in the second assertion is sharp already for uniform convergence of the functions.

11.2 1. Reduction to functions on ([0,1])

Set

$$P_n(t) = \sum_{j=1}^{n} t^j, \qquad 0 \le t \le 1,$$

and, for $0 \le t < 1$,

$$P(t) = \sum_{j=1}^{\infty} t^j = \frac{t}{1-t}.$$

Define

$$A_n(t) = e^{-P_n(t)}, \qquad B_n(t) = \left(1 - \frac{P_n(t)}{n}\right)^n,$$

and

$$\beta(t) = \begin{cases} e^{-P(t)} = e^{-t/(1-t)}, & 0 \le t < 1, \\ 0, & t = 1. \end{cases}$$

Then

$$f_n^{(a)}(x) = A_n(x^2), \qquad f_n^{(b)}(x) = B_n(x^2), \qquad \alpha(x) = \beta(x^2).$$

Thus it suffices to prove

$$A_n \longrightarrow \beta \quad \text{and} \quad B_n \longrightarrow \beta \qquad \text{in } C^\infty([0,1]).$$

Indeed, for every smooth U and every $k \ge 0$, repeated differentiation gives

$$\frac{d^k}{dx^k} U(x^2) = k! \sum_{\ell=0}^{\lfloor k/2 \rfloor} \frac{(2x)^{k-2\ell}}{(k-2\ell)!, \ell} U^{(k-\ell)}(x^2).$$

Consequently,

$$\left| \frac{d^k}{dx^k} \big(U_n(x^2) - U(x^2)\big) \right\|_\infty \le C_k \max_{0 \le j \le k} |U_n^{(j)} - U^{(j)}|_{\infty,[0,1]}.$$

11.2.1 1.1. Smoothness and flatness of β

For $t < 1$, put

$$u = \frac{1}{1-t}.$$

Then

$$\beta(t) = e^{1-u} = e, e^{-u}.$$

Since $u' = u^2$, induction gives polynomials Q_r such that

$$\beta^{(r)}(t) = \beta(t), Q_r! \left(\frac{1}{1-t}\right), \qquad 0 \le t < 1.$$

Indeed, $Q_0 = 1$, and if the formula holds at order r, then

$$Q_{r+1}(u) = u^2\big(Q_r'(u) - Q_r(u)\big).$$

For every polynomial Q,

$$e^{-u}Q(u) \longrightarrow 0 \qquad (u \to +\infty).$$

Hence

$$\lim_{t\to 1^-} \beta^{(r)}(t) = 0 \qquad \text{for every } r \geq 0.$$

Extending every derivative by the value 0 at $t = 1$ gives a C^∞ function. For example, differentiability at the endpoint follows successively from the mean value theorem. Thus

$$\beta \in C^\infty([0,1]), \qquad \beta^{(r)}(1) = 0 \quad (r \geq 0).$$

In particular, $\alpha = \beta \circ (x \mapsto x^2)$ belongs to E and is flat at both endpoints.

11.3 2. The first sequence

We prove

$$A_n = e^{-P_n} \longrightarrow \beta = e^{-P} \quad \text{in } C^\infty([0,1]).$$

11.3.1 2.1. Elementary derivative bounds

For $r \geq 0$ and $0 \leq t \leq 1$,

$$|P_n^{(r)}(t)| \leq n^{r+1}.$$

For $r = 0$ this is $P_n(t) \leq n$. For $r \geq 1$,

$$P_n^{(r)}(t) = \sum_{j=r}^{n} (j)_r t^{j-r},$$

where

$$(j)_r = j(j-1)\cdots(j-r+1),$$

and hence

$$|P_n^{(r)}(t)| \leq \sum_{j=1}^{n} j^r \leq n^{r+1}.$$

Repeated use of the product and chain rules shows that every term in $A_n^{(r)}$ is of the form

$$c, e^{-P_n} \prod_{i=1}^{m} P_n^{(s_i)},$$

where

$$s_i \geq 1, \qquad s_1 + \cdots + s_m = r, \qquad m \leq r.$$

Therefore

$$\boxed{|A_n^{(r)}(t)| \leq C_r n^{2r} e^{-P_n(t)} \leq C_r n^{2r}.} \tag{2.1}$$

11.3.2 2.2. Boundary layer

Let

$$\rho_n = n^{-1/2}, \qquad J_n = [1 - \rho_n, 1].$$

We first show that

$$P_n(t) \geq c\sqrt{n} \qquad (t \in J_n) \tag{2.2}$$

for an absolute $c > 0$ and all sufficiently large n.

Choose

$$m = \left\lfloor \frac{1}{4\rho_n} \right\rfloor .$$

Then $m \leq n$ for large n. If $t \geq 1 - \rho_n$ and $1 \leq j \leq m$, then, using

$$\log(1 - s) \geq -2s \qquad (0 \leq s \leq \tfrac{1}{2}),$$

we get

$$t^j \geq (1 - \rho_n)^m \geq e^{-2m\rho_n} \geq e^{-1/2}.$$

Moreover $m \geq (8\rho_n)^{-1}$ for large n. Thus

$$P_n(t) \geq \sum_{j=1}^{m} t^j \geq \frac{e^{-1/2}}{8\rho_n} = c\sqrt{n}.$$

Combining (2.1) and (2.2),

$$\sup_{t \in J_n} |A_n^{(r)}(t)| \leq C_r n^{2r} e^{-c\sqrt{n}}. \tag{2.3}$$

For β, the representation

$$\beta^{(r)}(t) = \beta(t) Q_r! \left(\frac{1}{1-t}\right)$$

and $1/(1-t) \geq \sqrt{n}$ on J_n imply

$$\sup_{t \in J_n} |\beta^{(r)}(t)| \leq C_r e^{-c_r \sqrt{n}}. \tag{2.4}$$

Thus all derivatives of both A_n and β are uniformly negligible in the boundary layer.

11.3.3 2.3. Interior region

Let

$$I_n = [0, 1 - \rho_n].$$

For $t < 1$,

$$P(t) - P_n(t) = \sum_{j=n+1}^{\infty} t^j = \frac{t^{n+1}}{1-t} =: R_n(t).$$

Hence

$$\beta(t) = A_n(t) e^{-R_n(t)}$$

and

$$A_n(t) - \beta(t) = A_n(t)\bigl(1 - e^{-R_n(t)}\bigr). \tag{2.5}$$

For $r \geq 0$, Leibniz's rule applied to

$$R_n(t) = t^{n+1}(1-t)^{-1}$$

gives

$$R_n^{(r)}(t) = \sum_{a=0}^{r} \binom{r}{a} (n+1)_a t^{n+1-a} (r-a)! (1-t)^{-(r-a+1)}.$$

On I_n, $1 - t \geq \rho_n$ and $t \leq 1 - \rho_n$. Therefore

$$|R_n^{(r)}(t)| \leq C_r n^r \rho_n^{-(r+1)} (1-\rho_n)^{n+1-r}.$$

Since

$$(1-\rho_n)^{n+1-r} \leq e^{-\rho_n(n+1-r)} \leq e^{-\sqrt{n}/2}$$

for large n, we obtain

$$\max_{0 \leq r \leq k} |R_n^{(r)}|_{\infty, I_n} \leq C_k n^{C_k} e^{-\sqrt{n}/2}. \tag{2.6}$$

Every derivative of $1 - e^{-R_n}$ is a finite sum of products of derivatives of R_n, with at least one such factor. Thus (2.6) implies

$$\max_{0\le r\le k} \left|\left(1 - e^{-R_n}\right)^{(r)}\right|_{\infty, I_n} \le C_k n^{C_k} e^{-c\sqrt{n}}. \tag{2.7}$$

Applying Leibniz's rule to (2.5), using (2.1) and (2.7), gives

$$\max_{0\le r\le k} |A_n^{(r)} - \beta^{(r)}|_{\infty, I_n} \le C_k n^{C_k} e^{-c_k\sqrt{n}}. \tag{2.8}$$

Together with the boundary estimates (2.3)–(2.4), this proves

$$\boxed{|A_n - \beta|_{C^k([0,1])} \le C_k n^{C_k} e^{-c_k\sqrt{n}}.} \tag{2.9}$$

In particular, for every M,

$$|A_n - \beta|_{C^k([0,1])} = O_{k,M}(n^{-M}).$$

After composition with x^2, this proves part a:

$$\boxed{f_n^{(a)} \Rightarrow \alpha.}$$

11.4 3. A weighted binomial approximation

To handle the second sequence uniformly up to the endpoint, we need a version of

$$\left(1 - \frac{y}{n}\right)^n \longrightarrow e^{-y}$$

that remains valid after differentiation and permits y to range over ([0,n]).

11.4.1 Lemma 3.1

For every pair of nonnegative integers (m,L), there is a constant $C_{m,L}$ such that, for all sufficiently large n,

$$\boxed{\sup_{0\le y\le n} (1+y)^L \left|\frac{d^m}{dy^m}\left(1 - \frac{y}{n}\right)^n - \frac{d^m}{dy^m} e^{-y}\right| \le \frac{C_{m,L}}{n}.} \tag{3.1}$$

Proof For $n \geq m$,

$$\frac{d^m}{dy^m}\left(1-\frac{y}{n}\right)^n = (-1)^m c_{n,m}\left(1-\frac{y}{n}\right)^{n-m},$$

where

$$c_{n,m} = \frac{(n)_m}{n^m} = \prod_{j=0}^{m-1}\left(1-\frac{j}{n}\right).$$

Also

$$\frac{d^m}{dy^m}e^{-y} = (-1)^m e^{-y}.$$

Since all factors in $c_{n,m}$ lie in ([0,1]),

$$0 \leq 1 - c_{n,m} \leq \sum_{j=0}^{m-1}\frac{j}{n} \leq \frac{C_m}{n}. \tag{3.2}$$

Put

$$h_{n,m}(y) = \left(1-\frac{y}{n}\right)^{n-m}.$$

Region $0 \leq y \leq n/2$ Set $z = y/n$ and

$$a = -(n-m)\log(1-z),$$

so that $h_{n,m}(y) = e^{-a}$. For $n \geq 2m$,

$$a \geq (n-m)z \geq \frac{y}{2}.$$

Moreover,

$$a - y = (n-m)\big(-\log(1-z) - z\big) - mz.$$

For $0 \leq z \leq 1/2$,

$$0 \leq -\log(1-z) - z = \sum_{\ell=2}^{\infty}\frac{z^\ell}{\ell} \leq \frac{z^2}{1-z} \leq 2z^2.$$

Consequently,

$$|a-y| \leq C_m\left(\frac{y}{n}+\frac{y^2}{n}\right).$$

By the mean value theorem applied to $s \mapsto e^{-s}$,

$$|e^{-a} - e^{-y}| \leq e^{-\min(a,y)}|a-y| \leq \frac{C_m}{n}(y+y^2)e^{-y/2}.$$

Multiplication by $(1+y)^L$ preserves the $O(1/n)$ bound, because every polynomial times $e^{-y/2}$ is bounded on $[0,\infty)$.

Also

$$h_{n,m}(y) \le e^{-y/2},$$

so (3.2) gives the same weighted $O(1/n)$ estimate for

$$(c_{n,m}-1)h_{n,m}(y).$$

Region $n/2 \le y \le n$ Here

$$h_{n,m}(y) \le 2^{-(n-m)}, \qquad e^{-y} \le e^{-n/2}.$$

Thus

$$(1+y)^L\big(h_{n,m}(y)+e^{-y}\big) \le (1+n)^L\big(2^{-(n-m)}+e^{-n/2}\big) = O_{m,L}(n^{-1}).$$

Combining both regions and (3.2) proves the lemma. □

11.5 4. Derivatives of the geometric partial sums

The following estimate allows us to compose Lemma 3.1 with P_n.

11.5.1 Lemma 4.1

For every $r \ge 0$, there is $C_r > 0$, independent of n, such that

$$\boxed{|P_n^{(r)}(t)| \le C_r\big(1+P_n(t)\big)^{r+1}} \tag{4.1}$$

for all $n \ge 1$ and $0 \le t \le 1$.

Proof The case $r=0$ is immediate. Suppose $r \ge 1$.

If $0 \le t \le 1/2$, then

$$P_n^{(r)}(t) \le \sum_{j=r}^{\infty}(j)_r 2^{-(j-r)} =: C_r,$$

and (4.1) follows.

Now suppose $t > 1/2$, and set

$$\delta = 1-t, \qquad L = \min(n,\delta^{-1}).$$

First,

$$P_n^{(r)}(t) \le \sum_{j=1}^{n} j^r \le C_r n^{r+1}. \tag{4.2}$$

Also, since $t^{-r} \le 2^r$ and $t^j \le e^{-\delta j}$,

$$P_n^{(r)}(t) \le 2^r \sum_{j=1}^{\infty} j^r e^{-\delta j} \le C_r \delta^{-(r+1)}. \tag{4.3}$$

Therefore

$$P_n^{(r)}(t) \le C_r L^{r+1}. \tag{4.4}$$

It remains to compare L to $P_n(t)$. If $L \le 8$, then trivially

$$L \le 8(1 + P_n(t)).$$

If $L > 8$, choose

$$m = \left\lfloor \frac{L}{4} \right\rfloor.$$

Then

$$m \ge \frac{L}{8}, \qquad m \le n, \qquad m\delta \le \frac{1}{4}.$$

For $1 \le j \le m$,

$$t^j = (1-\delta)^j \ge e^{-2\delta j} \ge e^{-1/2}.$$

Hence

$$P_n(t) \ge \sum_{j=1}^{m} t^j \ge e^{-1/2} m \ge cL.$$

Combining this with (4.4) proves (4.1). □

11.6 5. The second sequence

Let

$$G_n(y) = \left(1 - \frac{y}{n}\right)^n, \qquad 0 \le y \le n,$$

and

$$G(y) = e^{-y}.$$

Then

$$B_n = G_n \circ P_n, \qquad A_n = G \circ P_n.$$

We prove that for every fixed k,

$$|B_n - A_n|_{C^k([0,1])} = O_k(n^{-1}). \tag{5.1}$$

For $k = 0$, Lemma 3.1 with $m = L = 0$ immediately gives

$$|B_n - A_n|_\infty \le \frac{C}{n}.$$

Now let $k \ge 1$. Repeated application of the chain rule shows that the k-th derivative of

$$(G_n - G) \circ P_n$$

is a finite linear combination, with coefficients depending only on k, of terms

$$\big(G_n^{(m)} - G^{(m)}\big)(P_n(t)) \prod_{i=1}^{m} P_n^{(r_i)}(t),$$

where

$$1 \le m \le k, \qquad r_i \ge 1, \qquad r_1 + \cdots + r_m = k.$$

By Lemma 4.1,

$$\prod_{i=1}^{m} |P_n^{(r_i)}(t)| \le C_k\big(1 + P_n(t)\big)^{\sum_i (r_i+1)} = C_k\big(1 + P_n(t)\big)^{k+m}.$$

Because $m \le k$,

$$\prod_{i=1}^{m} |P_n^{(r_i)}(t)| \le C_k\big(1 + P_n(t)\big)^{2k}. \tag{5.2}$$

Applying Lemma 3.1 with weight exponent $L = 2k$ gives

$$\Big|G_n^{(m)}(P_n(t)) - G^{(m)}(P_n(t))\Big| \le \frac{C_k}{n\big(1 + P_n(t)\big)^{2k}}. \tag{5.3}$$

The product of (5.2) and (5.3) is $O_k(1/n)$, uniformly in t. There are only finitely many chain-rule terms, so (5.1) follows.

From the first part,

$$|A_n - \beta|_{C^k([0,1])} = O_{k,M}(n^{-M})$$

for every M. Taking $M = 2$, for example,

$$\boxed{|B_n - \beta|_{C^k([0,1])} \le \frac{C_k}{n}.} \tag{5.4}$$

Composing with x^2 now yields

$$\boxed{f_n^{(b)} \Rightarrow \alpha.}$$

11.7 6. Boundary checks, small cases, and sharpness

11.7.1 6.1. At $x = 0$

For every n,

$$f_n^{(a)}(0) = f_n^{(b)}(0) = \alpha(0) = 1.$$

All three functions are even, so all odd derivatives at 0 vanish.

For the first sequence, the agreement is much stronger. Near 0,

$$\sum_{j=n+1}^{\infty} x^{2j} = O(x^{2n+2}),$$

and therefore

$$f_n^{(a)}(x) - \alpha(x) = O(x^{2n+2}).$$

Thus the derivatives at 0 agree exactly through order $2n + 1$.

11.7.2 6.2. At $x = \pm 1$

Since α is flat,

$$\alpha^{(k)}(\pm 1) = 0 \qquad (k \geq 0).$$

For the first sequence,

$$f_n^{(a)}(\pm 1) = e^{-n}.$$

Every fixed-order derivative at ± 1 is e^{-n} times a polynomial in n, and hence tends to 0.

For the second sequence,

$$1 - \frac{x^2 + x^4 + \cdots + x^{2n}}{n}$$

has a simple zero at each of $x = \pm 1$. Hence $f_n^{(b)}$ has a zero of order n at each endpoint. In particular,

$$\left(f_n^{(b)}\right)^{(k)}(\pm 1) = 0 \qquad \text{whenever } n > k.$$

11.7.3 6.3. Sharpness for part b

Fix $0 < t < 1$. Since $P_n(t) \to P(t)$ exponentially fast and remains bounded,

$$n \log\left(1 - \frac{P_n(t)}{n}\right) = -P_n(t) - \frac{P_n(t)^2}{2n} + O_t(n^{-2}).$$

It follows that

$$B_n(t) = e^{-P_n(t)}\left(1 - \frac{P_n(t)^2}{2n} + O_t(n^{-2})\right).$$

Consequently,

$$\boxed{n\big(B_n(t) - \beta(t)\big) \longrightarrow -\frac{1}{2}P(t)^2 e^{-P(t)}.} \tag{6.1}$$

For example, take $t = 1/2$, corresponding to $x = 1/\sqrt{2}$. Then

$$P(1/2) = 1, \qquad \beta(1/2) = e^{-1},$$

and

$$n\left(f_n^{(b)}!\left(\frac{1}{\sqrt{2}}\right) - \alpha\left(\frac{1}{\sqrt{2}}\right)\right) \longrightarrow -\frac{1}{2e}.$$

Therefore

$$|f_n^{(b)} - \alpha|_\infty \neq o(n^{-1}),$$

so the $O(n^{-1})$ rate is optimal.

11.7.4 6.4. Role of the domain

The restriction to ([-1,1]) is essential. If $|x| > 1$, then

$$x^2 + x^4 + \cdots + x^{2n} \asymp_x x^{2n},$$

so

$$\left|1 - \frac{x^2 + x^4 + \cdots + x^{2n}}{n}\right|^n \longrightarrow +\infty.$$

Thus the second sequence has no analogous convergence on any interval containing a point with $|x| > 1$.

The endpoint values $\alpha(\pm 1) = 0$ are forced by the interior limit and, more strongly, make α flat and C^∞ at the endpoints.

11.8 Conclusion

For both proposed approximation schemes, not only the functions but every fixed derivative converges uniformly on the full closed interval:

$$\boxed{\forall k \in \mathbb{N}, \quad \lim_{n\to\infty} \left\| \left(f_n^{(a)}\right)^{(k)} - \alpha^{(k)} \right\|_\infty = 0,}$$

and

$$\boxed{\forall k \in \mathbb{N}, \quad \lim_{n\to\infty} \left| \left(f_n^{(b)}\right)^{(k)} - \alpha^{(k)} \right|_\infty = 0.}$$

Hence the answer to both parts is affirmative.

Formal verification. Lean source; online type-check.

12 Question Q764

Informal verdict. **A — Complete solution**, with high confidence.

Lean coverage. Full canonical Q764 coverage is now ingested. Q764.Line.lineKCenter_correct and lineKCenter_work_le connect the actual counted line run to the original Finset Real objective and prove <=100*(n*n+k*n) work; exhaustiveMetric_correct and exhaustiveMetric_work_le give the actual exact finite-metric enumeration and its 24*(n choose k)*n*k+24*$n+1$ bound; vertexCover_iff_oneTwo_radius_one, vertexCover_polyReduces_metricKCenter, and vertexCover_inP_of_metricKCenter_inP give the concrete 1–2 metric equivalence, polynomial materialization, and unconditional decision-solver transfer. No P != NP premise or unconditional impossibility theorem occurs.

12.1 Audited informal answer

12.1.1 1. Faithful restatement

Let $1 \le k \le n$.

a Given distinct real numbers

$$X = x_1 < x_2 < \cdots < x_n,$$

and $F \subseteq X$, define

$$\delta(X, F) = \max_{x\in X} \min_{y\in F} |x - y|.$$

Find an efficient algorithm producing a k-element subset $F \subseteq X$ minimizing $\delta(X, F)$.

b Solve the analogous problem when X is an arbitrary n-point subset of a metric space, with (|x-y|) replaced by the metric distance.

There are three minor ambiguities in the wording.

1. The statement only says $k \leq n$, but k must be a positive integer, since the minimum over an empty F is otherwise undefined.
2. The minimizing subset need not be unique, so "the subset F" should be read as "a minimizing subset F."
3. "Efficient" is not formally defined. Below, arithmetic and comparisons of input distances are counted at unit cost. For rational inputs, the algorithms also have polynomial bit complexity.

12.1.2 2. Final result

Theorem

1. **Ordered real line.** An optimal k-element set can be found exactly in

$$O(n^2 + kn) = O(n^2)$$

time and $O(n^2 + kn) = O(n^2)$ memory. The algorithm is a bottleneck dynamic program over consecutive blocks.

1. **Arbitrary finite metric.**

- The problem is exactly the finite **discrete metric k-center problem**.
- Its decision version is NP-complete, even when every nonzero distance is either 1 or 2. Consequently, unless $P = NP$, there is no polynomial-time exact algorithm when k is part of the input.
- An exact exhaustive algorithm takes

$$O\left(\binom{n}{k} nk\right)$$

time.

- There is an $O(nk)$-time polynomial algorithm returning a set F with

$$\delta(X, F) \leq 2, \delta_{\text{opt}}.$$

Unless $P = NP$, no polynomial algorithm can guarantee a factor strictly smaller than 2.

12.2 Part a: points on the real line

12.2.1 3. The optimal cost of one consecutive block

For $1 \le i \le j \le n$, define

$$c(i,j) = \min_{i \le \ell \le j} \max_{i \le t \le j} |x_t - x_\ell|.$$

Thus $c(i,j)$ is the smallest radius with which one center, constrained to be one of

$$x_i, x_{i+1}, \dots, x_j,$$

can cover that whole block.

For a fixed center x_ℓ, the farthest point of the block is one of its two endpoints, so

$$\max_{i \le t \le j} |x_t - x_\ell| = \max x_\ell - x_i, ; x_j - x_\ell.$$

Writing

$$m_{ij} = \frac{x_i + x_j}{2},$$

we have

$$\max x_\ell - x_i, x_j - x_\ell = \frac{x_j - x_i}{2} + |x_\ell - m_{ij}|\,.$$

Consequently:

Lemma 1 An optimal one-center location for the block $x_i, \dots, x_j$ is any x_ℓ nearest to the midpoint $(x_i + x_j)/2$, and

$$c(i,j) = \frac{x_j - x_i}{2} + \min_{i \le \ell \le j} \left| x_\ell - \frac{x_i + x_j}{2} \right|. \tag{1}$$

In particular, the center is constrained to lie in X; this is not the continuous k-center problem.

12.2.2 4. Reduction to consecutive blocks

The order on the real line provides the crucial structure.

Lemma 2: consecutive-cell lemma For every set of centers

$$F = f_1 < \cdots < f_s \subseteq X,$$

the points of X can be assigned to nearest centers so that the points assigned to each f_t form a consecutive block of X.

Proof Break ties consistently, for example in favor of the smaller center. The points at least as close to f_t as to f_{t+1} lie to the left of the midpoint

$$\frac{f_t + f_{t+1}}{2}.$$

Thus the nearest-center index is nondecreasing as one moves from left to right through X. Its level sets are therefore consecutive intervals of indices.

Every center belongs to its own cell, since its distance to itself is zero, so all cells are nonempty. □

It follows that an optimal solution is equivalent to partitioning X into at most k consecutive nonempty blocks and placing an optimal one-center solution in every block.

More precisely, if

$$0 = i_0 < i_1 < \cdots < i_s = n,$$

the associated blocks are

$$x_{i_{t-1}+1}, \ldots, x_{i_t}, \qquad 1 \le t \le s.$$

Proposition 3: bottleneck-partition characterization The optimal value is

$$R_k = \min_{\substack{1 \le s \le k \\ 0 = i_0 < i_1 < \cdots < i_s = n}} ; \max_{1 \le t \le s} c(i_{t-1}+1, i_t). \tag{2}$$

Proof Given any set of at most k centers, Lemma 2 produces at most k consecutive cells. On a cell ([i,j]), its chosen center has radius at least $c(i,j)$. Therefore its total radius is at least the right-hand side of 2.

Conversely, given a partition occurring on the right-hand side, choose in every block a center realizing $c(i,j)$. Every point is then within the displayed maximum radius of its block center. □

Using fewer than k centers causes no difficulty: arbitrary unused points of X can be added as extra centers, and doing so cannot increase the radius. Thus the minimum over "at most k" and the minimum over "exactly k" coincide.

12.2.3 5. Dynamic programming recurrence

Let

$$\rho_p(j)$$

be the optimal radius for the prefix

$$x_1, \ldots, x_j$$

using at most p centers. Set

$$\rho_p(0) = 0, \qquad \rho_0(j) = +\infty \quad (j \geq 1).$$

The last block of a partition of $x_1, \ldots, x_j$ must have the form

$$x_i, x_{i+1}, \ldots, x_j$$

for some $1 \leq i \leq j$. Hence:

Lemma 4: dynamic-programming recurrence For $p, j \geq 1$,

$$\boxed{\rho_p(j) = \min_{1 \leq i \leq j} \max\bigl\{\rho_{p-1}(i-1), c(i,j)\bigr\}.} \tag{3}$$

Proof If the last block starts at i, the preceding prefix uses at most $p-1$ centers and has optimal cost $\rho_{p-1}(i-1)$; the final block costs $c(i,j)$. The total bottleneck cost is their maximum.

Conversely, every partition counted by $\rho_p(j)$ has a uniquely determined last-block starting index i, so every possible solution appears in the minimum. □

A direct evaluation of 3 would take $O(kn^2)$ time. We now exploit monotonicity to reduce the dynamic-programming part to $O(kn)$.

12.2.4 6. Monotonicity of the transition

For fixed p and j, put

$$A_i = \rho_{p-1}(i-1), \qquad B_i = c(i,j).$$

Lemma 5 As i increases:

1. A_i is nondecreasing;

2. B_i is nonincreasing.

Moreover, for fixed i, $c(i,j)$ is nondecreasing in j.

Proof The value A_i concerns a prefix ending at $i-1$. Enlarging that prefix cannot decrease the required radius, so A_i is nondecreasing.

For B_i, removing the leftmost point of a block cannot increase the optimum. If an optimal center for ([i,j]) is not x_i, retain it. If it is x_i, replace it by x_{i+1}; every remaining point lies to the right, and is no farther from x_{i+1} than from x_i. Thus

$$c(i+1,j) \le c(i,j).$$

Similarly, adding a point on the right cannot reduce the one-center cost. If an optimal center for ([i,j+1]) is one of $x_i,\dots,x_j$, restricting to the smaller block proves the assertion. If it is x_{j+1}, then x_j is at least as good for all points $x_i,\dots,x_j$. Hence

$$c(i,j) \le c(i,j+1).$$

□

Let

$$t = t_p(j) = \min i \in 1,\dots,j : A_i \ge B_i,$$

with the convention $t = j+1$ if this set is empty.

For $i < t$,

$$\max(A_i, B_i) = B_i,$$

which is nonincreasing in i. For $i \ge t$,

$$\max(A_i, B_i) = A_i,$$

which is nondecreasing in i. Therefore:

Lemma 6: crossing lemma A minimizing index in 3 is always one of

$$t_p(j) - 1, \qquad t_p(j),$$

after discarding indices outside $1,\dots,j$.

Furthermore, for fixed p,

$$t_p(1) \le t_p(2) \le \cdots \le t_p(n). \tag{4}$$

Proof The first assertion follows from the decreasing-then-increasing behavior just described.

For the second, if $i < t_p(j)$, then

$$A_i < c(i,j) \le c(i,j+1),$$

so $i < t_p(j+1)$. Hence the first crossing cannot move to the left. □

Thus, while j runs from 1 to n, a single pointer tracking $t_p(j)$ moves right at most n times.

12.2.5 7. Precomputing all block costs in $O(n^2)$

By 1, the optimal center for ([i,j]) is a point nearest to

$$m_{ij} = \frac{x_i + x_j}{2}.$$

For a fixed i, these midpoints are nondecreasing as j increases. If ties are resolved in favor of the rightmost nearest point, the corresponding optimal-center index is also nondecreasing.

Indeed, between consecutive points $x_\ell, x_{\ell+1}$, the point $x_{\ell+1}$ is at least as close as x_ℓ to a number m exactly when

$$m \ge \frac{x_\ell + x_{\ell+1}}{2}.$$

Once this becomes true, it remains true as m moves to the right.

Therefore a moving pointer computes all $c(i,j)$, for fixed i, in $O(n)$ time. Repeating for all i gives $O(n^2)$.

12.2.6 8. Exact quadratic-time algorithm

The following pseudocode uses one-based indices. `C[i,j]` stores $c(i,j)$, and `M[i,j]` stores a corresponding optimal center index.

```
QUADRATIC-LINE-K-CENTER(x[1..n], k)

# Phase 1: all one-block costs and centers
for i = 1,...,n:
    ell = i
    for j = i,...,n:
        # Move ell to the rightmost point nearest to (x[i]+x[j])/2.
```

```
        # The doubled comparison avoids division by 2.
        while ell < j and
              abs(2*x[ell+1] - x[i] - x[j])
                <= abs(2*x[ell] - x[i] - x[j]):
            ell = ell + 1

        M[i,j] = ell
        C[i,j] = max(x[ell] - x[i], x[j] - x[ell])


# Phase 2: bottleneck dynamic programming
prev[0] = 0
for j = 1,...,n:
    prev[j] = +infinity

for p = 1,...,k:
    cur[0] = 0
    t = 1

    for j = 1,...,n:
        # After this loop, t is the first crossing index;
        # if no crossing exists, t = j+1.
        while t <= j and prev[t-1] < C[t,j]:
            t = t + 1

        candidates = empty set
        if t <= j:
            insert t into candidates
        if t >= 2:
            insert t-1 into candidates

        choose i in candidates minimizing
            max(prev[i-1], C[i,j])

        cur[j] = max(prev[i-1], C[i,j])
        parent[p,j] = i

    prev = cur
```

```
# Phase 3: reconstruct the consecutive blocks
F = empty set
p = k
j = n

while j > 0:
    i = parent[p,j]
    insert x[M[i,j]] into F
    j = i - 1
    p = p - 1

# The dynamic program may have used fewer than k nonempty blocks.
while |F| < k:
    insert any point of X \ F into F

return F and prev[n]
```

Theorem 7: correctness and complexity The algorithm returns a k-element set $F \subseteq X$ satisfying

$$\delta(X, F) = R_k.$$

Its running time is

$$O(n^2 + kn),$$

and its memory requirement is

$$O(n^2 + kn).$$

Proof The first phase correctly computes every $c(i, j)$ and a realizing center by Lemma 1 and the monotone-midpoint argument.

For each fixed p, the pointer `t` is exactly the crossing index from Lemma 6. Since it is nondecreasing in j, it makes at most n increments. Lemma 6 shows that checking `t-1` and `t` evaluates recurrence 3 exactly. Thus the dynamic program computes every $\rho_p(j)$, in particular

$$\texttt{prev[n]} = \rho_k(n) = R_k.$$

Following the parent pointers reconstructs a partition realizing this value. The center `M[i,j]` realizes the one-block cost for each reconstructed block. Hence the resulting set of centers covers X with radius R_k.

If fewer than k centers occur, adding arbitrary unused points cannot increase the radius. It cannot decrease it below R_k, since R_k was already the minimum over all sets of at most k centers. Thus the final k-element set remains optimal.

The cost table takes $O(n^2)$ time and space. For every p, the dynamic-programming scan takes $O(n)$, giving $O(kn)$. The parent table uses $O(kn)$ space. □

12.2.7 9. A simpler independent decision algorithm

For a proposed radius r, one can decide in linear time whether k centers suffice.

Take the leftmost uncovered point x_i. Choose as center the rightmost x_j satisfying

$$x_j \leq x_i + r.$$

This center covers all points through $x_j + r$. Repeat with the next uncovered point.

This greedy rule uses the minimum possible number of radius-r centers. Indeed, every feasible cover must have some center $x_\ell \leq x_i + r$ covering x_i. The greedy center satisfies $x_j \geq x_\ell$, so

$$x_\ell + r \leq x_j + r;$$

the competing center cannot cover any point farther to the right than the greedy center. Induction on the uncovered suffix proves optimality.

Since an optimal radius is necessarily one of the pairwise distances

$$x_j - x_i,$$

sorting all $O(n^2)$ candidate distances and binary-searching with this decision procedure gives a simpler $O(n^2 \log n)$-time exact algorithm. The preceding dynamic program removes the logarithmic factor.

12.3 Part b: an arbitrary finite metric space

Let ((X,d)) be a finite metric space with $|X| = n$, and write

$$d(x, F) = \min_{f \in F} d(x, f).$$

12.3.1 10. Threshold-graph characterization

For $r \geq 0$, define a graph G_r on vertex set X by declaring distinct (u,v) adjacent when

$$d(u, v) \leq r.$$

A subset $F \subseteq X$ satisfies

$$\delta(X, F) \leq r$$

if and only if every vertex is either in F or adjacent in G_r to a vertex of F. In other words, F is a dominating set of G_r.

Let $\gamma(G)$ denote the minimum size of a dominating set of G, and let

$$D = d(x, y) : x, y \in X.$$

Because X is finite, $\delta(X, F)$ is always an element of D. Therefore:

Proposition 8 The exact optimal radius is

$$\boxed{R_k^{\text{met}} = \min r \in D : \gamma(G_r) \leq k.} \tag{5}$$

Moreover, the optimal k-element sets are precisely the k-element dominating sets of $G_{R_k^{\text{met}}}$.

This is a complete mathematical characterization, but finding a minimum dominating set is computationally difficult.

12.3.2 11. An exact finite algorithm

Assuming the distance matrix is available, one may simply enumerate all k-element subsets:

```
EXACT-METRIC-K-CENTER(X, d, k)

best_radius = +infinity
best_set = none

for every k-element subset F of X:
    r = 0
    for every x in X:
        r = max(r, min_{f in F} d(x,f))

    if r < best_radius:
```

```
        best_radius = r
        best_set = F

return best_set and best_radius
```

There are $\binom{n}{k}$ subsets, and evaluating one requires $O(nk)$ distance comparisons. Thus the running time is

$$O\left(\binom{n}{k}nk\right). \tag{6}$$

For every fixed k, this is polynomial in n, but it is exponential when k is part of the input.

12.3.3 12. NP-completeness of the general metric problem

Consider the decision problem:

Given a finite metric ((X,d)), an integer k, and a radius r, does there exist $F \subseteq X$, $|F| \leq k$, such that $\delta(X, F) \leq r$?

It belongs to NP: a proposed F can be checked in $O(nk)$ time.

We prove NP-hardness using the standard NP-completeness of **VERTEX COVER**, one of Karp's classical NP-complete problems. ([Informatique Cornell Bowers][1])

Reduction Let $G = (V, E)$ be a nontrivial instance of VERTEX COVER, with integer k. Construct a graph H whose vertex set is the disjoint union

$$W = V \sqcup E,$$

where each edge $e \in E$ is regarded as a new "edge-vertex."

The adjacencies of H are:

1. all vertices of V are pairwise adjacent, so V induces a clique;
2. an edge-vertex $e = u, v$ is adjacent exactly to u and v;
3. distinct edge-vertices are not adjacent.

Now define a distance on W by

$$d_H(a, b) = \begin{cases} 0, & a = b, \\ 1, & a \neq b \text{ and } a, b \text{ are adjacent in } H, \\ 2, & \text{otherwise.} \end{cases} \tag{7}$$

Lemma 9 The function d_H is a metric.

Proof Symmetry and positive definiteness are immediate. For the triangle inequality, all nonzero distances are at least 1, while every distance is at most 2. Thus, for three pairwise distinct points,

$$d_H(a, c) \leq 2 \leq d_H(a, b) + d_H(b, c).$$

Cases with repeated points are immediate. $\square$

A set has covering radius at most 1 in this metric exactly when it is a dominating set of H.

Lemma 10 The original graph G has a vertex cover of size at most k if and only if H has a dominating set of size at most k.

Proof Suppose $C \subseteq V$ is a vertex cover. Every edge-vertex $e = u, v$ is adjacent to at least one point of C. Since V is a clique, every vertex of V is in C or adjacent to every member of C. Thus C dominates H.

Conversely, let D dominate H. Whenever an edge-vertex

$$e = u, v$$

belongs to D, replace it by either endpoint, say u. Repeating this operation produces a set

$$C \subseteq V, \qquad |C| \leq |D|.$$

Every edge-vertex e is incident to a vertex of C:

- if $e \in D$, an endpoint was inserted when e was replaced;
- if $e \notin D$, then e had to be dominated by one of its endpoints in D, since edge-vertices are not adjacent to one another.

Therefore C contains an endpoint of every edge of G, so C is a vertex cover. $\square$

The construction is polynomial. Thus the metric k-center decision problem is NP-hard even for the fixed threshold $r = 1$ and metrics whose nonzero distances belong to ({1,2}). Together with membership in NP:

Theorem 11 The decision version of the arbitrary-metric problem is NP-complete, even for 1-2 metrics.

Consequently, the exact optimization problem in Q764b is NP-hard. Unless $P = NP$, there is no polynomial-time exact algorithm for arbitrary finite metrics when k is part of the input.

The distinction between "at most k" and "exactly k" is immaterial: a dominating set of size smaller than k can be augmented by arbitrary vertices.

12.3.4 13. A polynomial-time factor-2 algorithm

Although exact optimization is NP-hard, there is a very simple optimal-threshold approximation.

Farthest-first algorithm Choose an arbitrary first center c_1. Having selected

$$C_t = c_1, \ldots, c_t,$$

choose

$$c_{t+1} \in \arg\max_{x \in X} d(x, C_t).$$

```
FARTHEST-FIRST(X, d, k)

choose any c[1] in X
F = {c[1]}

for every x in X:
    nearest[x] = d(x, c[1])

for t = 2,...,k:
    choose c[t] maximizing nearest[x]
    insert c[t] into F

    for every x in X:
        nearest[x] = min(nearest[x], d(x,c[t]))

return F
```

It performs $O(n)$ work for each new center, hence $O(nk)$ distance operations.

Theorem 12 If F is returned by farthest-first and R^* is the exact optimal radius, then

$$\delta(X, F) \leq 2R^*.$$

Proof Let

$$r = \delta(X, F),$$

and choose $c_{k+1} \in X$ satisfying

$$d(c_{k+1}, F) = r.$$

We claim that the $k+1$ points

$$c_1, c_2, \ldots, c_k, c_{k+1}$$

are pairwise at distance at least r.

Indeed, when c_t was selected, its distance from the previously selected centers was the then-current maximum covering distance. These maximum distances form a nonincreasing sequence, whose final value is r. Hence for $s < t \leq k$,

$$d(c_s, c_t) \geq r.$$

The same assertion holds for c_{k+1} by its definition.

Now let F^* be an optimal set of k centers with radius R^*. Assign each of the $k+1$ displayed points to an optimal center at distance at most R^*. By the pigeonhole principle, two of them, say c_s, c_t, are assigned to the same optimal center f. The triangle inequality gives

$$d(c_s, c_t) \leq d(c_s, f) + d(f, c_t) \leq 2R^*.$$

Since $d(c_s, c_t) \geq r$, it follows that

$$r \leq 2R^*.$$

□

12.3.5 14. The factor 2 cannot be improved in polynomial time

The same reduction used for NP-hardness gives a gap instance.

For the 1-2 metric constructed from a VERTEX COVER instance:

- if a vertex cover of size at most k exists, then the optimal metric k-center radius is 1;

- otherwise, no radius-1 solution exists, while every nonempty set of centers covers all points with radius at most 2, so the optimum is 2.

Suppose a polynomial algorithm had approximation factor

$$2-\varepsilon$$

for some fixed $\varepsilon > 0$. On an instance of optimum 1, it would return a solution of radius strictly smaller than 2. Since all possible radii are (0,1,2), its radius would be at most 1. On an instance of optimum 2, every feasible solution has radius 2. The algorithm would therefore decide VERTEX COVER in polynomial time.

Thus:

Theorem 13 Unless $P = NP$, no polynomial-time algorithm for arbitrary finite metrics can guarantee an approximation factor strictly smaller than 2.

Hence the farthest-first approximation guarantee is best possible in the standard worst-case complexity sense.

12.4 15. Boundary and sanity checks

$k = n$ Taking $F = X$ gives

$$\delta(X, X) = 0.$$

The dynamic program returns the partition into singleton blocks.

$k = 1$ **on the line** Only the two extreme points matter. An optimal center is a point of X nearest to

$$\frac{x_1 + x_n}{2},$$

and the optimum is

$$\min_{1 \le \ell \le n} \max x_\ell - x_1, x_n - x_\ell.$$

Nonuniqueness For

$$X = 0, 2, \qquad k = 1,$$

both ($\{0\}$) and ($\{2\}$) are optimal, with radius 2.

A two-cluster example For

$$X = 0, 1, 10, 11, \qquad k = 2,$$

the optimal consecutive partition is

$$0, 1 \mid 10, 11.$$

Each block has discrete one-center radius 1, so an optimal answer is

$$F = 1, 11, \qquad \delta(X, F) = 1.$$

Why the order is decisive On the line, nearest-center cells are consecutive intervals, producing the dynamic program. In an arbitrary metric, the radius-r neighborhoods can encode essentially arbitrary graph domination instances, which is exactly what causes NP-completeness.

12.5 Final answer

For **Q764**a, the problem admits an exact quadratic-time solution. Compute the optimal one-center radius $c(i, j)$ for every consecutive block, then use the bottleneck recurrence

$$\rho_p(j) = \min_{1 \le i \le j} \max \rho_{p-1}(i-1), c(i, j).$$

Monotonicity reduces each dynamic-programming layer to linear time, giving total complexity

$$O(n^2 + kn) = O(n^2).$$

For **Q764**b, the exact problem is NP-complete even for metrics with distances only 1 and 2. An exact solution can be obtained by enumerating all k-subsets in

$$O\left(\binom{n}{k} nk\right)$$

time. In polynomial time, farthest-first selection gives the sharp guarantee

$$\delta(X, F) \le 2, \delta_{\text{opt}},$$

and no factor below 2 is possible unless $P = NP$.

[1]: https://www.cs.cornell.edu/courses/cs722/2000sp/karp.pdf "cs722"

Formal verification. Lean source; online type-check.

13 Question Q766

Informal verdict. **A — Complete solution**, with high confidence.

Lean coverage. The retry archive closes canonical Q766. exists_integrable_monotone_rearrangement constructs a generalized inverse-CDF q from Integrable f alone and proves measurability, monotonicity, interval integrability, and equal pushforward. phiF_min_and_argmin and phiF_canonical then give phiF f = Phi q for every tuple, an optimal partition and attained exact value J q n s, sufficiency for every optimal block-median tuple, and an unrestricted iff classification for every tuple after a sorting permutation. The general block-cost equality retains constants, atoms, repeated centres, unused coordinates, and zero-length blocks.

13.1 Audited informal answer

Q766 — Complete solution

1. Faithful restatement

Let $f : (0,1) \to \mathbb{R}$ be continuous and integrable, and let $n \geq 1$. For $X = (x_1, \ldots, x_n) \in \mathbb{R}^n$, define

$$\delta(X, y) = \min_{1 \leq k \leq n} |y - x_k|, \qquad \phi(X) = \int_0^1 \delta(X, f(t))\, dt.$$

The problem asks for the minimum of ϕ and for the complete set of n-tuples at which it is attained. The statement explicitly permits adding assumptions on f when useful and notes that the case $n = 1$ was treated in an earlier examination problem. There are no subparts, footnotes, or apparent typographical errors. Luc Pommeret I interpret "integrable" in the standard French sense

$$\int_0^1 |f(t)|\, dt < \infty.$$

This is the natural necessary condition: if only a conditionally convergent improper integral were meant and $\int_0^1 |f| = \infty$, then $\phi(X) = +\infty$ for every finite tuple X.

1. Distribution and increasing rearrangement

Let λ denote Lebesgue measure on $(0,1)$, and let

$$\mu = f_{\#}\lambda$$

be the image probability measure of f; explicitly,

$$\mu(B) = \lambda\big(\{t \in (0,1) : f(t) \in B\}\big).$$

Then

$$\phi(x_1, \dots, x_n) = \int_{\mathbb{R}} \min_{1 \le k \le n} |y - x_k| \, d\mu(y).$$

Thus the problem depends only on the distribution of f, not on the order in which its values occur. Let

$$F(y) = \mu((-\infty, y])$$

and define the generalized quantile, or nondecreasing rearrangement, by

$$q(u) = \inf\{y \in \mathbb{R} : F(y) \ge u\}, \qquad 0 < u < 1.$$

Then q is nondecreasing, $q \in L^1(0,1)$, and $q_{\#}\lambda = \mu$. Hence

$$\phi(X) = \int_0^1 \min_{1 \le k \le n} |q(u) - x_k| \, du.$$

Set

$$H(s) = \int_0^s q(u) \, du, \qquad 0 \le s \le 1.$$

The function H is continuous on $[0,1]$ and convex. For $0 \le a \le b \le 1$, define

$$K(a,b) = H(a) + H(b) - 2H\left(\frac{a+b}{2}\right).$$

1. Main theorem: exact minimum and complete classification

Let

$$\Delta_n = \{(s_0, \dots, s_n) : 0 = s_0 \le s_1 \le \cdots \le s_n = 1\}$$

and, for $s \in \Delta_n$, put

$$J_n(s) = \sum_{i=1}^{n} K(s_{i-1}, s_i).$$

Theorem 1 For every integrable measurable $f : (0,1) \to \mathbb{R}$,

$$\boxed{\min_{X \in \mathbb{R}^n} \phi(X) = \min_{s \in \Delta_n} \sum_{i=1}^{n} \left[H(s_{i-1}) + H(s_i) - 2H\left(\frac{s_{i-1} + s_i}{2}\right) \right].} \tag{3.1}$$

Both minima are attained. More precisely, let

$$\mathcal{S}_n = \operatorname*{argmin}_{s \in \Delta_n} J_n(s).$$

For $s \in \mathcal{S}_n$ and every positive-length block $s_{i-1} < s_i$, set

$$\theta_i = \frac{s_{i-1} + s_i}{2}$$

and define

$$M_i(s) = [q(\theta_i -), q(\theta_i +)],$$

where $q(\theta_i -)$ and $q(\theta_i +)$ are the one-sided limits of the monotone function q. Choose any

$$c_i \in M_i(s)$$

on every positive block; on a zero-length block one may choose c_i arbitrarily. Then

$$(c_1, \dots, c_n)$$

is a minimizer of ϕ. Conversely, every minimizing tuple, after reordering its coordinates increasingly, arises in this way from an element of $\mathcal{S}_n$. Thus (3.1), together with the median intervals $M_i(s)$, is a complete description of both the minimum and its entire attainment set.

1. Proof of Theorem 1

4.1 The one-centre problem We first recall and prove the elementary median principle. Let ν be a finite positive measure of total mass p, with finite first moment, and let

$$G(c) = \int_{\mathbb{R}} |y - c| \, d\nu(y).$$

The function G is convex. Its one-sided derivatives are

$$G'_-(c) = \nu((-\infty, c)) - \nu([c, \infty)),$$

and

$$G'_+(c) = \nu((-\infty, c]) - \nu((c, \infty)).$$

Indeed, these formulas follow by applying dominated convergence to the one-sided difference quotients of $|y - c|$. A point c minimizes G if and only if

$$G'_-(c) \le 0 \le G'_+(c),$$

equivalently,

$$\nu((-\infty, c)) \le \frac{p}{2}, \qquad \nu((c, \infty)) \le \frac{p}{2}. \tag{4.1}$$

These are exactly the median inequalities. Now restrict q to a quantile interval (a, b), where $a < b$, and let

$$m = \frac{a + b}{2}.$$

Since q is nondecreasing, the complete median interval of q on (a, b) is

$$[q(m-), q(m+)].$$

For any c in this interval,

$$q(u) \le c \quad \text{for a.e. } u \in (a, m),$$

and

$$q(u) \ge c \quad \text{for a.e. } u \in (m, b).$$

Because both halves have the same length,

$$\begin{aligned}
\min_{c \in \mathbb{R}} \int_a^b |q(u) - c| \, du &= \int_a^m (c - q(u)) \, du + \int_m^b (q(u) - c) \, du \\
&= \int_m^b q(u) \, du - \int_a^m q(u) \, du \qquad (4.2) \\
&= H(a) + H(b) - 2H(m) \\
&= K(a, b).
\end{aligned}$$

For $a = b$, both sides are 0.

4.2 From centres to an ordered quantile partition Fix $X = (x_1, \dots, x_n)$. Since ϕ is invariant under permutations, assume

$$x_1 \le x_2 \le \cdots \le x_n.$$

For each $y \in \mathbb{R}$, choose the smallest index realizing the nearest-centre distance:

$$r(y) = \min \left\{ i : |y - x_i| = \min_j |y - x_j| \right\}.$$

The map r is nondecreasing. Indeed, suppose $y < z$, but $r(y) = j > i = r(z)$. Then $x_i \le x_j$. The inequality saying that x_j is at least as close to y as x_i gives

$$y \ge \frac{x_i + x_j}{2},$$

while the inequality saying that x_i is at least as close to z as x_j gives

$$z \le \frac{x_i + x_j}{2},$$

contradicting $y < z$. Since both q and r are nondecreasing, $r \circ q$ is a nondecreasing integer-valued function on $(0, 1)$. Consequently its level sets form, up to endpoints, consecutive intervals

$$(s_{i-1}, s_i), \qquad 0 = s_0 \le s_1 \le \cdots \le s_n = 1.$$

Using (4.2),

$$\begin{aligned} J_n(s) &= \sum_{i=1}^n \min_{c \in \mathbb{R}} \int_{s_{i-1}}^{s_i} |q(u) - c| \, du \\ &\le \sum_{i=1}^n \int_{s_{i-1}}^{s_i} |q(u) - x_i| \, du \\ &= \int_0^1 \min_j |q(u) - x_j| \, du \\ &= \phi(X). \end{aligned}$$

Therefore

$$\min_{s \in \Delta_n} J_n(s) \le \inf_{X \in \mathbb{R}^n} \phi(X). \tag{4.3}$$

4.3 From a quantile partition to centres Conversely, take any $s \in \Delta_n$. For every positive block choose a median $c_i \in M_i(s)$; on empty blocks choose anything. Then, pointwise,

$$\min_j |q(u) - c_j| \le |q(u) - c_i|, \qquad s_{i-1} < u < s_i.$$

Hence

$$\begin{aligned}
\phi(c_1, \ldots, c_n) &= \int_0^1 \min_j |q(u) - c_j|\, du \\
&\le \sum_{i=1}^n \int_{s_{i-1}}^{s_i} |q(u) - c_i|\, du \\
&= J_n(s).
\end{aligned}$$

Taking the infimum over X and then the minimum over s,

$$\inf_X \phi(X) \le \min_{s \in \Delta_n} J_n(s). \tag{4.4}$$

Combining (4.3) and (4.4) proves equality. Since H is continuous, J_n is continuous on the compact simplex Δ_n. Thus the right-hand minimum is attained. Choosing medians on an optimal partition gives a tuple attaining the same value, so the left-hand minimum is also attained.

4.4 Equality cases Suppose first that $s \in \mathcal{S}_n$ and the c_i are block medians. We obtained

$$\phi(c_1, \ldots, c_n) \le J_n(s) = \min_X \phi(X).$$

The reverse inequality follows from the definition of the minimum, so equality holds: every such tuple is minimizing. Conversely, let X be minimizing and form its nearest-centre quantile partition s as in §4.2. Then

$$\min_s J_n(s) \le J_n(s) \le \phi(X) = \min_X \phi(X) = \min_s J_n(s).$$

All inequalities are equalities. In particular $s \in \mathcal{S}_n$, and on each nonempty cell

$$\int_{s_{i-1}}^{s_i} |q(u) - x_i|\, du = \min_c \int_{s_{i-1}}^{s_i} |q(u) - c|\, du.$$

Therefore x_i is a median on that block, i.e.

$$x_i \in M_i(s).$$

This proves the full classification.

1. Specialization to the continuous function in Q766

Continuity has a useful additional consequence. Lemma 2: the distribution of a nonconstant continuous f has no gaps Let

$$I = f((0,1)).$$

Since $(0,1)$ is connected and f is continuous, I is an interval. Moreover,

$$\operatorname{supp}\mu = \overline{I}.$$

Indeed, if $y = f(t_0)$ and U is a neighbourhood of y, continuity gives an interval $V \ni t_0$ such that $f(V) \subset U$. Thus

$$\mu(U) \geq \lambda(V) > 0.$$

The same argument applies to points in $\overline{I}$ by first choosing a value of f inside the prescribed neighbourhood. Consequently, if f is nonconstant, $\operatorname{supp}\mu$ is a nondegenerate interval, possibly unbounded. There are no open gaps of zero μ-measure inside the support. A jump of the quantile q would correspond to such a gap. Hence:

$$\boxed{f \text{ nonconstant and continuous } \implies q \text{ is continuous on } (0,1).} \tag{5.1}$$

The distribution may have atoms—f can be constant on an interval—but atoms produce flat portions of q, not jumps. Therefore every positive quantile block has a unique median:

$$M_i(s) = \{q(\theta_i)\}.$$

Theorem 3: complete answer for the original continuous f Constant case If $f(t) = c$ for all t, then

$$\phi(X) = \min_k |c - x_k|.$$

Thus

$$\boxed{\min \phi = 0,}$$

and

$$\boxed{\operatorname{Argmin}\phi = \{X \in \mathbb{R}^n : \text{some coordinate of } X \text{ equals } c\}.}$$

The other coordinates are arbitrary. Nonconstant case Assume f is nonconstant. Let q, H, J_n, and $\mathcal{S}_n$ be as above. Then

$$\boxed{\min_{X\in\mathbb{R}^n} \phi(X) = \min_{0=s_0\le\cdots\le s_n=1} \sum_{i=1}^{n} \left[H(s_{i-1}) + H(s_i) - 2H\left(\frac{s_{i-1}+s_i}{2}\right)\right].} \tag{5.2}$$

Every optimal partition is in fact strict:

$$0 = s_0 < s_1 < \cdots < s_{n-1} < s_n = 1. \tag{5.3}$$

The complete attainment set is

$$\boxed{\operatorname{Argmin}\phi = \bigcup_{s\in\mathcal{S}_n} \left\{ \left(q\left(\tfrac{s_{\sigma(1)-1}+s_{\sigma(1)}}{2}\right), \dots, q\left(\tfrac{s_{\sigma(n)-1}+s_{\sigma(n)}}{2}\right)\right) : \sigma \in \mathfrak{S}_n \right\}.} \tag{5.4}$$

Equivalently, for an increasing ordering of the coordinates,

$$\boxed{x_i = q\left(\frac{s_{i-1}+s_i}{2}\right), \qquad 1 \le i \le n,} \tag{5.5}$$

for some $s \in \mathcal{S}_n$. All optimal coordinates are distinct, and the minimizer set is compact.

Proof of strictness and distinctness Write

$$e_n = \min_{X\in\mathbb{R}^n} \phi(X).$$

For nonconstant continuous f,

$$e_n < e_{n-1} \qquad (n \ge 2). \tag{5.6}$$

To prove this, take an optimal $(n-1)$-tuple A. Its set of centres is finite, whereas $\operatorname{supp}\mu$ is a nondegenerate interval. Choose

$$y_0 \in \operatorname{supp}\mu \setminus A$$

and let

$$d = \operatorname{dist}(y_0, A) > 0.$$

Choose $0 < r < d/3$. Since y_0 lies in the support,

$$\mu((y_0 - r, y_0 + r)) > 0.$$

After adding y_0 as a new centre, the distance decreases on this interval by at least

$$(d - r) - r = d - 2r > 0$$

and does not increase elsewhere. This proves (5.6). If an optimal quantile partition for e_n had a zero-length block, deleting it would give an $(n-1)$-block partition of the same cost, contradicting (5.6). This proves (5.3). If two optimal centres coincided, the tuple would effectively use at most $n - 1$ distinct centres, again contradicting (5.6). Thus all centres are distinct. Finally, $\mathcal{S}_n$ is a compact subset of Δ_n. Since it does not meet the boundary where a block has zero length, all its block lengths are uniformly bounded below. Hence all the midpoint parameters stay in a compact subinterval of $(0, 1)$, on which q is bounded and continuous. Formula (5.4) then shows that the minimizer set is compact.

1. Necessary equations for an optimum

Let f be nonconstant and continuous, and let

$$s \in \mathcal{S}_n, \qquad \theta_i = \frac{s_{i-1} + s_i}{2}, \qquad x_i = q(\theta_i).$$

Since q is continuous on $(0, 1)$,

$$H'(u) = q(u), \qquad 0 < u < 1.$$

Every internal breakpoint s_i is an unconstrained interior variable. Differentiating J_n with respect to s_i gives

$$\begin{aligned} 0 &= \frac{\partial J_n}{\partial s_i} \\ &= \left[q(s_i) - q\left(\frac{s_{i-1} + s_i}{2}\right) \right] + \left[q(s_i) - q\left(\frac{s_i + s_{i+1}}{2}\right) \right]. \end{aligned}$$

Thus every optimum satisfies

$$\boxed{2q(s_i) = x_i + x_{i+1}, \qquad 1 \le i < n.} \tag{6.1}$$

In value space, $q(s_i)$ is therefore exactly the midpoint between the adjacent optimal centres. This is the Voronoi boundary condition. Together, (5.5) and (6.1) say:

each centre is the median of the distribution in its Voronoi cell;

each boundary is the arithmetic midpoint of the adjacent centres.

These are the one-dimensional L^1 centroidal Voronoi equations. They are necessary, but in general not sufficient for global optimality: the finite-dimensional function J_n need not be convex.

Atomless form Suppose additionally that μ is atomless. Then F is continuous and strictly increasing on its support, and $q = F^{-1}$. Let

$$b_0 = -\infty, \qquad b_i = \frac{x_i + x_{i+1}}{2} \quad (1 \le i < n), \qquad b_n = +\infty.$$

Equation (6.1) gives

$$s_i = F(b_i).$$

Since $x_i = q((s_{i-1} + s_i)/2)$,

$$F(x_i) = \frac{s_{i-1} + s_i}{2}.$$

Therefore every ordered optimal tuple satisfies

$$\boxed{2F(x_i) = F(b_{i-1}) + F(b_i), \qquad 1 \le i \le n,} \tag{6.2}$$

with $F(-\infty) = 0$ and $F(+\infty) = 1$. Equivalently,

$$\mu((b_{i-1}, x_i)) = \mu((x_i, b_i)).$$

Thus x_i bisects the probability mass of its Voronoi cell.

1. Exact dynamic-programming algorithm

Formula (5.2) admits an exact Bellman recursion. For $t \in [0, 1]$, define

$$V_0(0) = 0, \qquad V_0(t) = +\infty \quad (t > 0),$$

and for $k \ge 1$,

$$\boxed{V_k(t) = \min_{0 \le r \le t} \big(V_{k-1}(r) + K(r, t)\big).} \tag{7.1}$$

Then

$$\boxed{e_n = V_n(1).} \tag{7.2}$$

All optimal partitions can be recovered by backtracking all minimizers r in (7.1). Once the breakpoints are known, the centres are given by the block medians, and for the nonconstant continuous case by

$$x_i = q\left(\frac{s_{i-1} + s_i}{2}\right).$$

A grid discretization of (7.1) with M quantile grid points has the standard $O(nM^2)$ complexity, before any convexity or Monge-type acceleration.

1. Boundary and test cases

8.1 The case $n = 1$ There is no partition variable:

$$s_0 = 0, \qquad s_1 = 1.$$

Thus

$$e_1 = H(1) - 2H\left(\frac{1}{2}\right) = \int_{1/2}^{1} q(u)\,du - \int_{0}^{1/2} q(u)\,du.$$

The complete median interval is

$$[q(1/2-), q(1/2+)].$$

For the nonconstant continuous f, q is continuous, so the unique minimizer is

$$\boxed{x_1 = q(1/2),}$$

the unique median of the distribution of f.

8.2 Affine functions Let

$$f(t) = \alpha t + \beta, \qquad \alpha \neq 0.$$

The distribution is uniform on an interval of length $|\alpha|$. Writing its lower endpoint as

$$A = \min(\beta, \beta + \alpha),$$

the quantile is

$$q(u) = A + |\alpha| u.$$

A direct calculation gives

$$K(a,b) = \frac{|\alpha|}{4}(b-a)^2.$$

Consequently,

$$e_n = \frac{|\alpha|}{4} \min_{\substack{h_i \geq 0 \\ \sum h_i = 1}} \sum_{i=1}^{n} h_i^2 = \frac{|\alpha|}{4n}.$$

Equality in Cauchy–Schwarz forces

$$h_i = \frac{1}{n}.$$

Thus the unique increasing optimal tuple is

$$\boxed{x_i = A + \frac{2i-1}{2n}|\alpha|, \qquad 1 \leq i \leq n,}$$

and the full attainment set consists of its permutations.
8.3 Decay as $n \to \infty$ For every integrable f,

$$e_n \longrightarrow 0.$$

Indeed, given $\varepsilon > 0$, choose $R > 0$ such that

$$\int_{|y|>R} |y| \, d\mu(y) < \varepsilon.$$

Place n equally spaced centres in $[-R, R]$, including both endpoints. For $|y| \leq R$, the nearest-centre distance is at most $R/(n-1)$, while outside $[-R, R]$ it is at most $|y|$. Therefore

$$e_n \leq \frac{R}{n-1} + \int_{|y|>R} |y| \, d\mu(y) < \frac{R}{n-1} + \varepsilon.$$

For nonconstant continuous f, the sequence is therefore strictly decreasing and tends to zero:

$$0 < e_n < e_{n-1}, \qquad e_n \downarrow 0.$$

1. Smooth high-resolution asymptotic

The permission in the statement to add assumptions allows a more explicit asymptotic answer. Theorem 4 Assume that the quantile q extends to a C^1 function on $[0,1]$ and that

$$q'(u) > 0 \qquad (0 \le u \le 1).$$

Then

$$\boxed{e_n = \frac{1}{4n}\left(\int_0^1 \sqrt{q'(u)}\,du\right)^2 (1+o(1)).} \tag{9.1}$$

Proof Put

$$g = q', \qquad m_0 = \min_{[0,1]} g > 0, \qquad M_0 = \max_{[0,1]} g,$$

and let ω be the modulus of continuity of g. For a block $[a,b]$, write

$$h = b - a, \qquad c = \frac{a+b}{2}.$$

Since $q(c)$ is the block median,

$$\begin{aligned} K(a,b) &= \int_0^{h/2} \big(q(c+v) - q(c-v)\big)\,dv \\ &= \int_0^{h/2} \int_{c-v}^{c+v} g(r)\,dr\,dv. \end{aligned}$$

Hence

$$\left| K(a,b) - \frac{g(c)h^2}{4} \right| \le \frac{\omega(h)h^2}{4}. \tag{9.2}$$

Let

$$A = \int_0^1 \sqrt{g(u)}\,du.$$

For the upper bound, choose the breakpoints so that

$$\int_{s_{i-1}}^{s_i} \sqrt{g(u)}\,du = \frac{A}{n}.$$

The largest block length is $O(1/n)$. By (9.2), uniformly in i,

$$K(s_{i-1}, s_i) = \frac{A^2}{4n^2}(1 + o(1)).$$

Summing gives

$$e_n \le \frac{A^2}{4n}(1 + o(1)). \tag{9.3}$$

For the lower bound, an equal-length partition gives

$$e_n \le \frac{M_0}{4n}.$$

On the other hand, every block of length h costs at least

$$\frac{m_0 h^2}{4}.$$

Thus, for an optimal partition, its largest block length tends to zero. Let $h_i = s_i - s_{i-1}$ and $c_i = (s_{i-1} + s_i)/2$. From (9.2),

$$e_n \ge \frac{1 - o(1)}{4} \sum_{i=1}^{n} g(c_i) h_i^2.$$

Cauchy–Schwarz gives

$$\sum_{i=1}^{n} g(c_i) h_i^2 \ge \frac{1}{n} \left(\sum_{i=1}^{n} \sqrt{g(c_i)}\, h_i \right)^2.$$

The sum in parentheses is a Riemann sum for A, because the mesh tends to zero. Therefore

$$e_n \ge \frac{A^2}{4n}(1 - o(1)). \tag{9.4}$$

Combining (9.3) and (9.4) proves (9.1).

Density form If μ has a positive continuous density p on a compact interval $[A, B]$, then

$$q'(u) = \frac{1}{p(q(u))}.$$

Changing variables $u = F(y)$ gives

$$\int_0^1 \sqrt{q'(u)}\, du = \int_A^B \sqrt{p(y)}\, dy.$$

Hence

$$\boxed{e_n = \frac{1}{4n}\left(\int_A^B \sqrt{p(y)}\,dy\right)^2 (1+o(1)).} \tag{9.5}$$

If $f \in C^1([0,1])$ is strictly monotone with f' nowhere zero, then q is f or its reversal, and

$$\boxed{e_n = \frac{1}{4n}\left(\int_0^1 \sqrt{|f'(t)|}\,dt\right)^2 (1+o(1)).} \tag{9.6}$$

For an affine f, (9.6) gives the exact value $|\alpha|/(4n)$.

1. Uniqueness is not guaranteed

Even smooth strict monotonicity of f does not force a unique optimal n-point set. Consider first the nondecreasing step function

$$q_0(u) = \begin{cases} -1, & 0 < u < 1/5, \\ 0, & 1/5 \le u \le 4/5, \\ 1, & 4/5 < u < 1. \end{cases}$$

For $n = 2$, denote by $J_{q_0}(s)$ the cost of the partition $(0, s, 1)$. Directly,

$$J_{q_0}(1/5) = \frac{1}{5}, \qquad J_{q_0}(1/2) = \frac{2}{5}.$$

For two quantiles $q, \widetilde{q}$, their primitives satisfy

$$\sup_t |H_q(t) - H_{\widetilde{q}}(t)| \le \|q - \widetilde{q}\|_{L^1},$$

and consequently

$$\sup_{0 \le s \le 1} |J_q(s) - J_{\widetilde{q}}(s)| \le 8\|q - \widetilde{q}\|_{L^1}. \tag{10.1}$$

Choose a C^∞, strictly increasing function q satisfying

$$q(1-u) = -q(u)$$

and

$$\|q - q_0\|_{L^1} < \frac{1}{80}.$$

Such a q is obtained, for example, by symmetric mollification of q_0, followed by adding a sufficiently small positive multiple of $2u - 1$. By (10.1),

$$J_q(1/5) < \frac{3}{10} < J_q(1/2).$$

Thus 1/2 is not an optimal breakpoint. But symmetry gives

$$J_q(s) = J_q(1 - s).$$

Therefore every minimizing breakpoint $s_* \neq 1/2$ has a distinct minimizing partner $1 - s_*$. Since q is strictly increasing, the corresponding unordered pairs of centres are distinct. Taking $f = q$ gives a smooth strictly increasing example with at least two different optimal two-point sets. Hence the occurrence of the possibly non-singleton set $\mathcal{S}_n$ in (5.4) is genuinely necessary.

1. Role and necessity of the hypotheses

Absolute integrability. This is necessary for a finite problem. If

$$\int_0^1 |f(t)| \, dt = \infty,$$

then for $M = \max_i |x_i|$,

$$\delta(X, f(t)) \geq (|f(t)| - M)_+,$$

so $\phi(X) = +\infty$ for every X. Continuity. It is not needed for the general quantile formula or existence theorem. It is used to show that the support is an interval with no gaps, hence that q is continuous and every positive block has a unique median. Without continuity, median intervals can be nontrivial. For example, for the distribution

$$\frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1,$$

every $x \in [-1, 1]$ is a one-centre minimizer. Nonconstancy. It is necessary for strict decrease of e_n, strict positivity of block lengths, and distinctness of optimal centres. Atomlessness. It is needed only for the simple equal-mass equations (6.2). The exact quantile formula remains valid in the presence of atoms. Compactness of the image. No such assumption is required. The support may be unbounded; finite first moment suffices.

1. Topological, countability, and choice assumptions

The proof takes place entirely in the ordinary Euclidean spaces $\mathbb{R}$, $\mathbb{R}^n$, and the finite-dimensional simplex Δ_n.

The separation assumptions are only the usual Hausdorff structure of Euclidean space.

Existence uses the compactness of the closed simplex Δ_n and continuity of J_n. No compactness of $\operatorname{supp}\mu$ is used.

All spaces involved are metrizable and second countable. Compactness may equivalently be handled by sequential compactness; nets or filters are unnecessary.

No countability assumption on level sets or fibres of f is used.

The quantile is canonically defined, so no choice principle is required there. Choosing one median on each of finitely many blocks uses only finite choice.

No interchange of a non-absolutely convergent integral or unjustified limiting operation occurs.

Final answer For a nonconstant continuous integrable f, let q be its nondecreasing rearrangement and $H(s) = \int_0^s q$. Then

$$\boxed{\min_X \phi(X) = \min_{0=s_0<\cdots<s_n=1} \sum_{i=1}^{n} \left[H(s_{i-1}) + H(s_i) - 2H\left(\frac{s_{i-1}+s_i}{2}\right)\right].}$$

The minimum is attained. If $\mathcal{S}_n$ denotes the set of optimal partitions, the complete set of minimizing tuples is

$$\boxed{\bigcup_{s\in\mathcal{S}_n} \mathfrak{S}_n \cdot \left(q\left(\frac{s_0+s_1}{2}\right), \ldots, q\left(\frac{s_{n-1}+s_n}{2}\right)\right).}$$

Every optimal partition satisfies

$$\boxed{2q(s_i) = q\left(\frac{s_{i-1}+s_i}{2}\right) + q\left(\frac{s_i+s_{i+1}}{2}\right), \qquad 1 \le i < n.}$$

These formulas give the exact minimum, a complete classification of every point of attainment, an exact Bellman algorithm, and—in the smooth positive-density case—the asymptotic value

$$\boxed{e_n \sim \frac{1}{4n}\left(\int \sqrt{p}\right)^2.}$$

Formal verification. Lean source; online type-check.

14 Question Q776

Informal verdict. **A — Complete solution**, with high confidence.

Lean coverage (partial). Lean proves the exact Hadamard integral representation, the dominant-saddle extremal lemma, the sharp exponential envelope for every fixed m, and the complete Bessel identity for m=2. The general oscillatory Poincare expansion, algebraic prefactor, correction coefficient, and uniform remainder remain under Aristotle continuation.

14.1 Audited informal answer

Q776 — Restatement For each fixed integer $m \geq 2$, define the entire function

$$f_m(x) = \sum_{n=0}^{\infty} \frac{x^n}{(n!)^m}.$$

The problem asks for a simple asymptotic equivalent of $f_m(x)$ as $x \to -\infty$. The editorial remark also records the positive-axis asymptotic

$$f_m(x) \sim (2\pi)^{(1-m)/2} m^{-1/2} x^{(1-m)/(2m)} e^{mx^{1/m}} \qquad (x \to +\infty).$$

This is the complete statement on page 14 of the canonical PDF. Luc Pommeret There is no apparent typo. There is, however, a genuine ambiguity in the word "equivalent": on the negative axis the dominant term is oscillatory and has infinitely many zeros, so it cannot be a global equivalent in the strict ratio sense. The correct global result is an additive Poincaré asymptotic expansion.

Final theorem Fix $m \geq 2$. Put

$$t = -x, \qquad R = t^{1/m}, \qquad \alpha = \frac{\pi}{m},$$

and define

$$\Psi_m(R) = mR\sin\alpha - \frac{m-1}{2}\alpha = mR\sin\frac{\pi}{m} - \frac{(m-1)\pi}{2m}.$$

Then, as $x \to -\infty$, equivalently $R \to +\infty$,

$$\boxed{\begin{aligned} f_m(x) &= \frac{2(2\pi)^{(1-m)/2}}{\sqrt{m}}\, t^{(1-m)/(2m)} e^{mt^{1/m}\cos(\pi/m)} \\ &\quad \times \left[\cos\Psi_m(t^{1/m}) + \frac{m^2-1}{24m\, t^{1/m}} \cos\left(\Psi_m(t^{1/m}) - \frac{\pi}{m}\right) + O_m\!\left(t^{-2/m}\right)\right]. \end{aligned}}$$

In particular, the leading global estimate is

$$f_m(x) = \frac{2(2\pi)^{(1-m)/2}}{\sqrt{m}} (-x)^{(1-m)/(2m)} e^{m(-x)^{1/m}\cos(\pi/m)} \left[\cos\left(m(-x)^{1/m} \sin\frac{\pi}{m} - \frac{(m-1)\pi}{2m}\right) + O_m\Big((-$$

The constants implicit in the O-terms depend on the fixed integer m, but not on x. More generally, there are real constants $c_{m,k}$, with

$$c_{m,0} = 1, \qquad c_{m,1} = \frac{m^2-1}{24m},$$

such that, for every fixed $N \geq 1$,

$$\boxed{\begin{aligned} f_m(-R^m) &= \frac{2(2\pi)^{(1-m)/2}}{\sqrt{m}}\, R^{-(m-1)/2} e^{mR\cos(\pi/m)} \\ &\quad \times \left[\sum_{k=0}^{N-1} \frac{c_{m,k}}{R^k} \cos\left(\Psi_m(R) - \frac{k\pi}{m}\right) + O_{m,N}(R^{-N})\right]. \end{aligned}}$$

This is a full asymptotic expansion in descending integral powers of $R = (-x)^{1/m}$.

Proof

1. An exact torus integral representation

Let $d = m-1$, and for $\theta = (\theta_1, \ldots, \theta_d) \in [-\pi, \pi]^d$, set

$$S(\theta) = \sum_{j=1}^{d} e^{i\theta_j} + e^{-i(\theta_1 + \cdots + \theta_d)}.$$

For every $r \in \mathbb{C}$,

$$\boxed{f_m(r^m) = \frac{1}{(2\pi)^d} \int_{[-\pi,\pi]^d} e^{rS(\theta)}\, d\theta.}$$

Indeed, expanding all exponentials gives

$$e^{rS(\theta)} = \prod_{j=1}^{d} \left(\sum_{n_j \geq 0} \frac{r^{n_j} e^{i n_j \theta_j}}{n_j!} \right) \left(\sum_{n_m \geq 0} \frac{r^{n_m} e^{-i n_m (\theta_1 + \cdots + \theta_d)}}{n_m!} \right).$$

The expansion is absolutely and uniformly convergent on the compact torus. Integration with respect to each θ_j forces

$$n_j = n_m \qquad (1 \leq j \leq d).$$

Consequently,

$$\frac{1}{(2\pi)^d} \int e^{rS(\theta)}\, d\theta = \sum_{n=0}^{\infty} \frac{r^{mn}}{(n!)^m} = f_m(r^m).$$

For the negative axis choose

$$r = Re^{i\alpha}, \qquad \alpha = \frac{\pi}{m},$$

so that $r^m = -R^m$.

1. Identification of the dominant saddle points for $m \geq 3$

We first prove an elementary extremal lemma. Lemma Let $z_1, \ldots, z_m$ be complex numbers of modulus 1 satisfying

$$z_1 z_2 \cdots z_m = -1.$$

For $m \geq 3$,

$$\mathrm{Re}(z_1 + \cdots + z_m) \leq m \cos \frac{\pi}{m}.$$

Equality holds exactly in the following two cases:

$$z_1 = \cdots = z_m = e^{i\pi/m},$$

or

$$z_1 = \cdots = z_m = e^{-i\pi/m}.$$

Proof The constraint set is compact, so a maximum exists. Write $z_j = e^{iu_j}$. Tangent variations preserving the product have the form

$$u_j \longmapsto u_j + st_j, \qquad \sum_{j=1}^{m} t_j = 0.$$

At a critical point,

$$\sum_{j=1}^{m} (-\sin u_j) t_j = 0$$

for every (t_j) of sum zero. Hence all the numbers $\sin u_j$ are equal. Thus, for some β, each u_j is congruent modulo 2π either to β or to $\pi - \beta$. Suppose p of the angles are of the first type and $m - p$ are of the second type. Then

$$\operatorname{Re} \sum_{j=1}^{m} z_j = (2p - m) \cos \beta.$$

If $0 < p < m$, then

$$\operatorname{Re} \sum z_j \leq |2p - m| \leq m - 2.$$

On the other hand,

$$m - m \cos \frac{\pi}{m} = 2m \sin^2 \frac{\pi}{2m} < \frac{\pi^2}{2m} < 2 \qquad (m \geq 3),$$

so

$$m - 2 < m \cos \frac{\pi}{m}.$$

Therefore no mixed configuration can maximize the real part. At a maximizer all the z_j are equal, say $z_j = z$. The product condition gives $z^m = -1$. Among the m-th roots of -1, the largest real part is $\cos(\pi/m)$, attained exactly by $e^{\pm i\pi/m}$. $\square$ Apply the lemma with

$$z_j = e^{i(\alpha + \theta_j)} \quad (1 \leq j \leq d), \qquad z_m = e^{i(\alpha - \theta_1 - \cdots - \theta_d)}.$$

Their product is

$$z_1 \cdots z_m = e^{im\alpha} = -1,$$

and

$$\operatorname{Re}(rS(\theta)) = R \operatorname{Re}(z_1 + \cdots + z_m).$$

Thus the maximum of the real part of the exponent is

$$mR\cos\alpha,$$

and there are exactly two maximizers on the torus:

$$\theta^{(+)} = (0, \dots, 0)$$

and

$$\theta^{(-)} = (-2\alpha, \dots, -2\alpha).$$

Outside fixed disjoint neighborhoods of these points, compactness gives some $\eta_m > 0$ such that

$$\mathrm{Re}(rS(\theta)) \le mR\cos\alpha - \eta_m R.$$

The corresponding part of the integral is therefore

$$O_m\left(e^{mR\cos\alpha - \eta_m R}\right),$$

which is exponentially smaller than every term retained below.

1. Local expansion at a saddle

We analyze the saddle at the origin. Introduce

$$v_j = u_j \quad (1 \le j \le d), \qquad v_m = -\sum_{j=1}^{d} u_j.$$

Then $\sum_{j=1}^{m} v_j = 0$, and locally

$$S(u) = \sum_{j=1}^{m} e^{iv_j}.$$

Taylor expansion gives

$$S(u) = m - \frac{1}{2}Q(u) + H_3(u) + H_4(u) + H_5(u) + O_m(|u|^6),$$

where

$$Q(u) = \sum_{j=1}^{m} v_j^2,$$

$$H_3(u) = -\frac{i}{6}\sum_{j=1}^{m} v_j^3, \qquad H_4(u) = \frac{1}{24}\sum_{j=1}^{m} v_j^4,$$

and H_5 is homogeneous and odd. The quadratic form is

$$Q(u) = \sum_{j=1}^{d} u_j^2 + \left(\sum_{j=1}^{d} u_j\right)^2 = u^\mathsf{T} A u,$$

with

$$A = I_d + \mathbf{1}\mathbf{1}^\mathsf{T}.$$

Its eigenvalues are 1, with multiplicity $d-1$, and m, with multiplicity 1. Hence

$$\det A = m.$$

Let $\rho \to \infty$ in a fixed closed subsector of $\operatorname{Re}\rho > 0$. A standard scaling $u = |\rho|^{-1/2} y$, made rigorous by splitting into $|y| \le |\rho|^{\varepsilon}$ and its complement, yields

$$\int_{\text{near } 0} e^{\rho(S(u)-m)}\, du = \frac{(2\pi)^{d/2}}{\sqrt{m}}\, \rho^{-d/2} \times \left(1 + \frac{c_{m,1}}{\rho} + O_m(|\rho|^{-2})\right).$$

For completeness, we calculate $c_{m,1}$. After the Gaussian scaling, the potential contribution of order $\rho^{-1/2}$ is odd and integrates to zero. The contribution of order ρ^{-1} is

$$H_4 + \frac{1}{2} H_3^2.$$

Let $V = (V_1, \dots, V_m)$ be the centered Gaussian vector on the hyperplane $\sum V_j = 0$ corresponding to the density $e^{-Q/2}$. Its covariance matrix is

$$\mathbb{E}(V_j V_k) = \delta_{jk} - \frac{1}{m}.$$

In particular,

$$v := \mathbb{E}(V_j^2) = \frac{m-1}{m}, \qquad c := \mathbb{E}(V_j V_k) = -\frac{1}{m} \quad (j \neq k).$$

By Gaussian pairings,

$$\mathbb{E}(V_j^4) = 3v^2,$$

so

$$\mathbb{E}\sum_{j=1}^{m} V_j^4 = \frac{3(m-1)^2}{m}.$$

Also,

$$\mathbb{E}(V_j^6) = 15v^3,$$

and, for $j \neq k$,

$$\mathbb{E}(V_j^3 V_k^3) = 9v^2c + 6c^3.$$

Therefore

$$\begin{aligned}\mathbb{E}\left(\sum_{j=1}^{m} V_j^3\right)^2 &= 15mv^3 + m(m-1)(9v^2c + 6c^3)\\ &= \frac{6(m-1)(m-2)}{m}.\end{aligned}$$

It follows that

$$\begin{aligned}c_{m,1} &= \frac{1}{24}\mathbb{E}\sum_j V_j^4 - \frac{1}{72}\mathbb{E}\left(\sum_j V_j^3\right)^2\\ &= \frac{(m-1)^2}{8m} - \frac{(m-1)(m-2)}{12m}\\ &= \boxed{\frac{m^2-1}{24m}}.\end{aligned}$$

The same scaling argument, carried to arbitrary order, gives

$$\int_{\text{near } 0} e^{\rho(S(u)-m)}\,du \sim \frac{(2\pi)^{d/2}}{\sqrt{m}}\,\rho^{-d/2}\sum_{k=0}^{\infty}\frac{c_{m,k}}{\rho^k}.$$

Only integral powers occur: at every half-integral order the polynomial in the scaled variables is odd, so its Gaussian integral vanishes.

1. The two conjugate saddle contributions for $m \geq 3$

At the first saddle take

$$\rho = r = Re^{i\alpha}.$$

Its contribution is

$$\frac{(2\pi)^{-d/2}}{\sqrt{m}}\, e^{mr} r^{-d/2} \left(1 + \frac{c_{m,1}}{r} + O_m(R^{-2})\right).$$

For the second saddle, write

$$\theta = \theta^{(-)} + u.$$

An exact identity holds:

$$S(\theta^{(-)} + u) = e^{-2i\alpha} S(u).$$

Indeed, every one of the m exponential terms acquires the same factor $e^{-2i\alpha}$. Consequently,

$$rS(\theta^{(-)} + u) = \overline{r}\, S(u),$$

because

$$re^{-2i\alpha} = Re^{-i\alpha} = \overline{r}.$$

Thus the second saddle contributes the complex conjugate

$$\frac{(2\pi)^{-d/2}}{\sqrt{m}}\, e^{m\overline{r}} \overline{r}^{-d/2} \left(1 + \frac{c_{m,1}}{\overline{r}} + O_m(R^{-2})\right).$$

Adding the two contributions and absorbing the exponentially smaller remainder gives

$$f_m(-R^m) = \frac{(2\pi)^{-d/2}}{\sqrt{m}} \Bigg[e^{mr} r^{-d/2} \left(1 + \frac{c_{m,1}}{r} + O_m(R^{-2})\right) \\ + e^{m\bar{r}} \bar{r}^{-d/2} \left(1 + \frac{c_{m,1}}{\bar{r}} + O_m(R^{-2})\right) \Bigg].$$

Now

$$e^{mr} r^{-d/2} = R^{-d/2} e^{mR\cos\alpha} e^{\,i(mR\sin\alpha - d\alpha/2)}.$$

Therefore

$$f_m(-R^m) = \frac{2(2\pi)^{-d/2}}{\sqrt{m}} R^{-d/2} e^{mR\cos\alpha}$$
$$\times \left[\cos\left(mR\sin\alpha - \frac{d}{2}\alpha \right) + \frac{c_{m,1}}{R} \cos\left(mR\sin\alpha - \frac{d}{2}\alpha - \alpha \right) + O_m(R^{-2}) \right].$$

Since $d = m - 1$, this is the claimed formula for $m \geq 3$.

1. The exceptional boundary case $m = 2$

Here $\alpha = \pi/2$, so $\operatorname{Re} r = 0$; the real-part maximum used above is no longer isolated. A stationary-phase argument is required. The integral representation becomes

$$f_2(-R^2) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{2iR\cos\theta}\, d\theta.$$

Equivalently,

$$f_2(-R^2) = J_0(2R).$$

The stationary points are $\theta = 0$ and $\theta = \pi$. Near $\theta = 0$, make the exact substitution

$$u = 2\sin\frac{\theta}{2}.$$

Then

$$2\cos\theta = 2 - u^2, \qquad d\theta = \frac{du}{\sqrt{1 - u^2/4}}.$$

With a smooth cutoff equal to 1 near the saddle, the local amplitude $a(u)$ satisfies

$$a(0) = 1, \qquad a''(0) = \frac{1}{4}.$$

For a smooth compactly supported amplitude,

$$\int_{\mathbb{R}} e^{-iRu^2} a(u)\, du = e^{-i\pi/4} \sqrt{\frac{\pi}{R}} \left[a(0) - \frac{i}{4R} a''(0) + O(R^{-2}) \right].$$

One direct proof uses the Fourier transform and the exact Fresnel identity

$$\int_{\mathbb{R}} e^{-iRu^2+i\xi u}\,du = e^{-i\pi/4}\sqrt{\frac{\pi}{R}}e^{i\xi^2/(4R)}.$$

Thus the contribution of $\theta = 0$ is

$$\frac{e^{i(2R-\pi/4)}}{2\sqrt{\pi R}}\left(1 - \frac{i}{16R} + O(R^{-2})\right).$$

The contribution of $\theta = \pi$ is its complex conjugate. Away from the two stationary points, repeated integration by parts gives an error smaller than any prescribed power of R^{-1}. Hence

$$\boxed{f_2(-R^2) = \frac{1}{\sqrt{\pi R}}\left[\cos\left(2R - \frac{\pi}{4}\right) + \frac{1}{16R}\sin\left(2R - \frac{\pi}{4}\right) + O(R^{-2})\right].}$$

This is exactly the general formula, because

$$\left.\frac{m^2-1}{24m}\right|_{m=2} = \frac{1}{16},$$

and

$$\cos\left(\Psi_2(R) - \frac{\pi}{2}\right) = \sin\Psi_2(R).$$

Thus all $m \geq 2$ are covered.

Why the leading term is not a global strict equivalent Let

$$\mathcal{A}_m(R) = \frac{2(2\pi)^{(1-m)/2}}{\sqrt{m}}R^{-(m-1)/2}e^{mR\cos(\pi/m)}.$$

The leading approximation would be

$$\mathcal{A}_m(R)\cos\Psi_m(R).$$

For integers k large enough, define

$$R_k^* = \frac{\frac{\pi}{2} + k\pi + \frac{(m-1)\pi}{2m}}{m\sin(\pi/m)}.$$

Then

$$\Psi_m(R_k^*) = \frac{\pi}{2} + k\pi,$$

so the proposed leading expression is exactly zero. But the two-term expansion gives

$$f_m(-(R_k^*)^m) = \mathcal{A}_m(R_k^*)\left[\frac{m^2-1}{24mR_k^*}\cos\left(\frac{\pi}{2}+k\pi-\frac{\pi}{m}\right)+O_m((R_k^*)^{-2})\right]$$
$$= (-1)^k\mathcal{A}_m(R_k^*)\left[\frac{m^2-1}{24mR_k^*}\sin\frac{\pi}{m}+O_m((R_k^*)^{-2})\right],$$

which is nonzero for all sufficiently large k. Therefore

$$f_m(-R^m) \sim \mathcal{A}_m(R)\cos\Psi_m(R)$$

is not valid globally in the strict quotient sense. It is valid along every approach for which

$$R\,|\cos\Psi_m(R)| \longrightarrow \infty.$$

In particular, it holds uniformly on sets satisfying

$$|\cos\Psi_m(R)| \geq \delta > 0.$$

The additive expansion stated in the theorem is valid without excluding any values of R.

Checks and boundary behavior $m = 2$

$$f_2(-t) = J_0(2\sqrt{t}),$$

and the theorem gives

$$f_2(-t) = \frac{t^{-1/4}}{\sqrt{\pi}}\left[\cos\left(2\sqrt{t}-\frac{\pi}{4}\right)+\frac{1}{16\sqrt{t}}\sin\left(2\sqrt{t}-\frac{\pi}{4}\right)+O(t^{-1})\right],$$

the classical Bessel expansion. $m = 3$ Since $\cos(\pi/3) = 1/2$ and $\sin(\pi/3) = \sqrt{3}/2$,

$$f_3(-t) = \frac{e^{\frac{3}{2}t^{1/3}}}{\pi\sqrt{3}\,t^{1/3}}\left[\cos\left(\frac{3\sqrt{3}}{2}t^{1/3}-\frac{\pi}{3}\right)\right.$$
$$\left.+\frac{1}{9t^{1/3}}\cos\left(\frac{3\sqrt{3}}{2}t^{1/3}-\frac{2\pi}{3}\right)+O(t^{-2/3})\right].$$

Thus $f_3(-t)$ oscillates with an exponentially increasing envelope. Sign changes At points where $\Psi_m(R) = k\pi$,

$$f_m(-R^m) = (-1)^k \mathcal{A}_m(R)(1 + O_m(R^{-1})).$$

Hence f_m changes sign infinitely often on the negative real axis, for every $m \geq 2$. Necessity of the hypotheses The integer condition on m is essential to the elementary m-torus representation used in the proof. The case $m = 1$, excluded in the statement, is qualitatively different:

$$f_1(x) = e^x, \qquad f_1(x) = e^x \quad (x \to -\infty).$$

The estimates above are for fixed m. They are not uniform as $m \to \infty$: the two dominant saddles are separated by $2\pi/m$ and coalesce in that additional limit.

Formal verification. Lean source; online type-check.

15 Question Q781

Informal verdict. A — Complete solution, with **high confidence**.

Lean coverage. Lean proves the full arbitrary-index compact-topological-group translation-separation theorem without metrizability or countability assumptions.

15.1 Audited informal answer

Q781 — Complete solution

1. Faithful restatement and interpretation

Let G be a compact topological group, and let $(A_i)_{i \in I}$ be a family of subsets of G. The family is called separable by translations if there are elements $t_i \in G$ such that the left translates $t_i A_i$, $i \in I$, are pairwise disjoint. Given a family $(U_i)_{i \in I}$ of open subsets of G, assume that every finite subfamily is separable by translations. Prove that the whole family is separable by translations. The PDF first says merely "let G be a group" and then speaks of compactness and open subsets. The intended meaning must therefore be "compact topological group." The word "separable" is the specially defined translation-packing property, not separability in the sense of possessing a countable dense subset.

The parenthetical remark permits assuming metrizability and countability if needed, but neither is necessary. There are no further subparts or footnotes. Luc Pommeret+1

1. The theorem

Theorem Let G be a compact topological group, let I be an arbitrary index set, and let $U_i \subseteq G$ be open for every $i \in I$. Suppose that for every finite $F \subseteq I$, there exist elements $t_i^{(F)} \in G$, $i \in F$, such that

$$t_i^{(F)} U_i \cap t_j^{(F)} U_j = \varnothing \qquad (i, j \in F,\ i \neq j).$$

Then there exist elements $t_i \in G$, $i \in I$, such that

$$t_i U_i \cap t_j U_j = \varnothing \qquad (i, j \in I,\ i \neq j).$$

Thus Q781 has an affirmative answer in its full stated generality: neither metrizability of G nor countability of I is needed.

1. The disjointness condition is closed

The crucial point is that, because the U_i are open, collision of two translated sets is an open condition on the two translation parameters. Fix distinct $i, j \in I$. Define

$$R_{ij} = \bigl\{(s,t) \in G \times G :\ sU_i \cap tU_j = \varnothing\bigr\}.$$

We claim that R_{ij} is closed in $G \times G$. Indeed, for $s, t \in G$,

$$\begin{aligned} sU_i \cap tU_j \neq \varnothing &\iff \exists\, u \in U_i,\ v \in U_j \text{ such that } su = tv \\ &\iff \exists\, u \in U_i,\ v \in U_j \text{ such that } s^{-1}t = uv^{-1} \\ &\iff s^{-1}t \in U_i U_j^{-1}. \end{aligned}$$

The subset $U_i U_j^{-1}$ is open. To see this, inversion is a homeomorphism, so U_j^{-1} is open, and

$$U_i U_j^{-1} = \bigcup_{u \in U_i} u U_j^{-1}$$

is a union of open left translates. The map

$$q : G \times G \longrightarrow G, \qquad q(s,t) = s^{-1}t,$$

is continuous. Consequently,

$$(G \times G) \setminus R_{ij} = q^{-1}(U_i U_j^{-1})$$

is open. Hence R_{ij} is closed. Notice the useful direction of semicontinuity here: a limit of pairs of translations whose corresponding open sets are disjoint still gives disjoint open sets. In contrast, a limit of intersecting translated open sets need not remain intersecting.

1. Compactness of the full parameter space

Consider the product space

$$X = G^I = \prod_{i \in I} G$$

with the product topology. A point $x \in X$ will be written

$$x = (x_i)_{i \in I}.$$

By Tychonoff's theorem, X is compact. For every unordered pair $\{i, j\} \subseteq I$, $i \neq j$, define

$$K_{ij} = \big\{x = (x_k)_{k \in I} \in X : x_i U_i \cap x_j U_j = \varnothing\big\}.$$

Let

$$p_{ij} : X \longrightarrow G \times G, \qquad p_{ij}(x) = (x_i, x_j)$$

be the coordinate projection. Then

$$K_{ij} = p_{ij}^{-1}(R_{ij}).$$

Since R_{ij} is closed and p_{ij} is continuous, K_{ij} is closed in X. We now prove that the family

$$\{K_{ij} : i, j \in I,\ i \neq j\}$$

has the finite-intersection property. Take finitely many pair constraints,

$$K_{i_1 j_1}, \ldots, K_{i_m j_m}.$$

Let

$$F = \{i_1, j_1, \ldots, i_m, j_m\} \subseteq I.$$

This is a finite set of indices. By hypothesis, the finite family $(U_i)_{i\in F}$ can be separated: there exist $s_i \in G$, $i \in F$, such that

$$s_iU_i \cap s_jU_j = \varnothing \qquad (i, j \in F,\ i \neq j).$$

Extend this finite assignment to a point $x \in G^I$ by setting

$$x_i = \begin{cases} s_i, & i \in F, \\ e, & i \notin F, \end{cases}$$

where e is the identity of G. Then

$$x \in K_{i_1j_1} \cap \cdots \cap K_{i_mj_m}.$$

Thus every finite intersection of the K_{ij} is nonempty. Because X is compact and the K_{ij} are closed with the finite-intersection property,

$$\bigcap_{\substack{i,j\in I \\ \neq j}} K_{ij} \neq \varnothing.$$

Choose

$$(t_i)_{i\in I} \in \bigcap_{i\neq j} K_{ij}.$$

Then, by the definition of K_{ij},

$$t_iU_i \cap t_jU_j = \varnothing \qquad (i \neq j).$$

Therefore the entire family $(U_i)_{i\in I}$ is separable by translations. □

1. The countable metrizable proof

The optional assumptions in the PDF allow a purely sequential formulation. Suppose $I = \mathbb{N}$ and G is compact metrizable. For each $n \geq 1$, choose translations

$$t_1^{(n)}, \ldots, t_n^{(n)}$$

separating $U_1, \ldots, U_n$, and extend this to a point $x^{(n)} \in G^{\mathbb{N}}$ by setting

$$x_k^{(n)} = \begin{cases} t_k^{(n)}, & k \leq n, \\ e, & k > n. \end{cases}$$

If d is a compatible metric on G, the product topology on $G^{\mathbb{N}}$ is induced by, for example,

$$D(x,y) = \sum_{k=1}^{\infty} 2^{-k} \min\{1, d(x_k, y_k)\}.$$

The countable product $G^{\mathbb{N}}$ is compact metrizable. Equivalently, one may obtain a convergent subsequence by successively extracting subsequences on which the first coordinate converges, then the second, and so on, and taking the diagonal subsequence. Thus, after passing to a subsequence,

$$x^{(n_r)} \longrightarrow x = (t_k)_{k\geq 1} \quad \text{in } G^{\mathbb{N}}.$$

Fix $i \neq j$. Since $n_r \to \infty$, for every sufficiently large r,

$$x_i^{(n_r)} U_i \cap x_j^{(n_r)} U_j = \varnothing.$$

The relation R_{ij} proved closed above, so passage to the limit gives

$$t_i U_i \cap t_j U_j = \varnothing.$$

Since i, j were arbitrary, the limit tuple separates the entire family. Why this is not the general proof For a general compact group, compactness is not a sequential notion. A compact space need not be sequentially compact, and an uncountable product G^I generally need not be metrizable or first countable. For example, the compact Boolean group

$$H = \{0,1\}^{\mathcal{P}(\mathbb{N})}$$

is not sequentially compact. Define $x_n \in H$ by

$$x_n(A) = \begin{cases} 1, & n \in A, \\ 0, & n \notin A. \end{cases}$$

Given any subsequence (x_{n_k}), take

$$A = \{n_{2k} : k \geq 1\}.$$

Then the A-coordinate of x_{n_k} alternates between 0 and 1, so the subsequence does not converge. Thus a proof that simply says “extract a convergent subsequence” would be invalid without the metrizability assumptions. The finite-intersection argument above is genuinely non-sequential and works in the full generality of Q781.

1. Audit of the topological and set-theoretic assumptions

Separation assumptions No Hausdorff, regularity, normality, or first-countability axiom is used in the argument itself. The proof only uses:

continuity of multiplication and inversion;

openness of the U_i;

compactness of G^I.

Under the usual convention, a "compact group" is a compact Hausdorff topological group, but Hausdorffness is not needed for the closed-constraint argument. Compactness The general proof uses the arbitrary-product form of Tychonoff's theorem:

The product of an arbitrary family of compact spaces is compact in the product topology.

Compactness is then used through its equivalent finite-intersection formulation: a family of closed subsets of a compact space whose finite intersections are all nonempty has nonempty total intersection. Countability No countability assumption on I is used. No second-countability or separability assumption on G is used. Countability becomes relevant only if one wishes to replace the product-compactness proof by the sequential diagonal argument. Choice In ordinary ZFC, Tychonoff's theorem is available. Foundationally, for compact Hausdorff spaces, the ultrafilter lemma—equivalently the Boolean prime ideal theorem—suffices for the required product compactness; full choice is more than is needed for that step. No choice is needed to extend a finite tuple to all of I: the identity e supplies a canonical value on every unused coordinate. In the countable sequential proof, literally choosing one finite separator for every n uses a countable selection principle, which is again automatic in ZFC. The precise role of openness Openness is used exactly to prove that

$$\{(s,t) : sU_i \cap tU_j = \varnothing\}$$

is closed. In fact, the same proof works under the weaker pairwise assumption

$$U_i U_j^{-1} \text{ is open for every } i \neq j.$$

Thus openness of each individual U_i is sufficient but is not the logically weakest possible condition.

1. Boundary cases

If $I = \varnothing$ or $|I| = 1$, there are no pairwise constraints, so the conclusion is immediate. If some $U_i = \varnothing$, that index causes no constraint because every translate of U_i is empty. Such indices can be ignored and then assigned any translation, for example e. If G is a finite discrete group of order N, the hypothesis already implies that there are at most N nonempty U_i. Indeed, $N+1$ pairwise disjoint nonempty subsets cannot exist inside G. The set of nonempty indices is therefore finite, and applying the hypothesis to that finite subfamily immediately gives the conclusion. This agrees with the general compactness proof.

1. Why compactness cannot be omitted

The assertion fails for a noncompact group even when:

the group is discrete, metrizable, locally compact, and countable;

I is countable;

every U_i is open.

Take

$$G = (\mathbb{Z}, +)$$

with the discrete topology. Let the index set be

$$I = \{L, R\} \cup \mathbb{N},$$

and define

$$U_L = \{m \in \mathbb{Z} : m \leq 0\}, \qquad U_R = \{m \in \mathbb{Z} : m \geq 0\},$$

and

$$U_n = \{0\} \qquad (n \in \mathbb{N}).$$

Every finite subfamily can be separated. Indeed, suppose it contains k of the singleton sets, indexed in some order by $n_1, \dots, n_k$. Use the translations

$$t_L = 0, \qquad t_R = k+1, \qquad t_{n_r} = r \quad (1 \leq r \leq k).$$

Then

$$t_L + U_L = (-\infty, 0] \cap \mathbb{Z},$$

$$t_R + U_R = [k+1, \infty) \cap \mathbb{Z},$$

and the singleton translates are

$$\{1\}, \ldots, \{k\}.$$

These sets are pairwise disjoint. The same assignment works if one or both of L, R are absent from the finite subfamily. However, the entire family cannot be separated. Suppose translations $a, b, c_n \in \mathbb{Z}$ existed. Disjointness of the two rays would require

$$a < b.$$

For the singleton $\{c_n\}$ to avoid both rays, one must have

$$a < c_n < b.$$

Moreover, the c_n must be pairwise distinct. But the finite interval

$$\{a+1, \ldots, b-1\}$$

contains only finitely many integers. It cannot contain all the distinct c_n. This contradiction proves that compactness is essential for a theorem of this generality.

1. Why arbitrary non-open subsets cannot be allowed

The result also fails for general subsets of a compact group, even if all the subsets are closed. Let

$$G = \mathbb{T} = \mathbb{R}/\mathbb{Z}$$

and choose an index set I with cardinality strictly greater than $|\mathbb{T}|$, for instance

$$I = \mathcal{P}(\mathbb{T}).$$

For every $i \in I$, put

$$A_i = \{0\}.$$

Every finite subfamily is separable: simply choose finitely many distinct elements $t_i \in \mathbb{T}$. A separation of the entire family would require the singleton translates

$$t_i A_i = \{t_i\}, \qquad i \in I,$$

to be pairwise disjoint. Thus $i \mapsto t_i$ would be an injection from I into $\mathbb{T}$, which is impossible because

$$|I| = |\mathcal{P}(\mathbb{T})| > |\mathbb{T}|.$$

Here G is compact Hausdorff and every A_i is closed, but the A_i are not open. Hence the openness/closed-collision condition cannot simply be discarded.

Final conclusion The answer to Q781 is yes. For each pair $i \neq j$, disjointness of $t_i U_i$ and $t_j U_j$ is a closed condition on the parameter tuple $(t_k)_{k \in I} \in G^I$. The finite-separability hypothesis says that these closed conditions have the finite-intersection property. Tychonoff compactness of G^I therefore supplies one tuple satisfying all pairwise conditions simultaneously.

Formal verification. Lean source; online type-check.

16 Question Q788

Informal verdict. A — Complete solution, with **high confidence** in the two requested regimes.

Lean coverage (partial). Lean proves the deterministic range for every configuration, the complete n=2 distribution and upper-edge asymptotic, and two quadratic-form ingredients. The arbitrary-n probability asymptotics and Brownian-bridge limit remain under Aristotle continuation.

16.1 Audited informal answer

Q788 — rigorous solution

1. Faithful restatement

Identify the unit circle $\mathbb{U} \subset \mathbb{R}^2$ with

$$\mathbb{T} := \{z \in \mathbb{C} : |z| = 1\}.$$

Let $Z_1, \ldots, Z_n$ be independent Haar-uniform random points of $\mathbb{T}$, and put

$$P_n(z) := \prod_{j=1}^{n} (z - Z_j), \qquad D_n := \max_{|z|=1} |P_n(z)|.$$

Thus D_n is the maximum, over $M \in \mathbb{U}$, of the product of the Euclidean chord distances MM_j. The problem states that $D_n \in [2, 2^n]$ almost surely and asks for estimates of

$$\mathbb{P}(D_n \geq \alpha)$$

in the two regimes:
α fixed and $n \to \infty$;
n fixed and $\alpha \uparrow 2^n$.

There is one evident typographical omission in the PDF: in the sentence $\mathbb{P}(D \in [2, 2^n]) = 1$, the variable D should be D_n. There is no other substantive ambiguity. Luc Pommeret

1. Probability space and main result

Work on

$$\Omega = [0, 2\pi)^{\mathbb{N}}, \qquad \mathcal{F} = \mathcal{B}([0, 2\pi))^{\otimes \mathbb{N}}, \qquad \mathbb{P} = \left(\frac{d\theta}{2\pi}\right)^{\otimes \mathbb{N}}.$$

Let Θ_j be the coordinate maps and $Z_j = e^{i\Theta_j}$. Then the Z_j are independent and Haar-uniform on $\mathbb{T}$. The map

$$(\theta_1, \dots, \theta_n) \longmapsto \max_{t \in [0, 2\pi]} \prod_{j=1}^{n} |e^{it} - e^{i\theta_j}|$$

is continuous, since it is the maximum over a compact set of a jointly continuous function. Hence D_n is a well-defined random variable. Define

$$\kappa_n := \frac{\sqrt{n}}{\Gamma\left(\frac{n+1}{2}\right)} \left(\frac{2}{\pi}\right)^{\frac{n-1}{2}}.$$

Theorem Let $n \geq 2$. *a* Fixed α, $n \to \infty$ If $\alpha \leq 2$, then

$$\mathbb{P}(D_n \geq \alpha) = 1.$$

If $\alpha > 2$, then for every $A > 0$ there is a constant $C_{\alpha,A} < \infty$ such that

$$\boxed{\mathbb{P}(D_n < \alpha) \leq C_{\alpha,A} n^{-A}.}$$

Consequently,

$$\boxed{\mathbb{P}(D_n \geq \alpha) = 1 - O_{\alpha,A}(n^{-A}) \qquad \text{for every } A > 0.}$$

There is also a complementary explicit lower estimate: for every fixed $\alpha > 2$, there is $c_\alpha > 0$ such that, for all sufficiently large n,

$$\boxed{\mathbb{P}(D_n < \alpha) \geq n! \left(\frac{c_\alpha}{\pi n \log n}\right)^n \geq \exp\bigl(-n \log\log n - O_\alpha(n)\bigr).}$$

Thus the exceptional probability tends to zero faster than every negative power of n. The natural, nondegenerate scale is exponential in $\sqrt{n}$: there is a strictly positive random variable I^*, expressible as the maximum of a Brownian-bridge functional, such that

$$\boxed{\frac{2\log D_n}{\sqrt{n}} \xrightarrow{d} I^*.}$$

Hence, at every continuity point $x \geq 0$ of the law of I^*,

$$\mathbb{P}\Bigl(D_n \geq e^{x\sqrt{n}/2}\Bigr) \longrightarrow \mathbb{P}(I^* \geq x).$$

b Fixed n, $\alpha \uparrow 2^n$ Put

$$\delta := 1 - \frac{\alpha}{2^n} = \frac{2^n - \alpha}{2^n}.$$

Then, as $\delta \downarrow 0$,

$$\boxed{\mathbb{P}(D_n \geq 2^n(1-\delta)) = \kappa_n \, \delta^{\frac{n-1}{2}} \bigl(1 + O_n(\delta)\bigr).}$$

Equivalently,

$$\boxed{\mathbb{P}(D_n \geq \alpha) \sim \frac{\sqrt{n}}{\Gamma\left(\frac{n+1}{2}\right)} \left(\frac{2}{\pi}\right)^{\frac{n-1}{2}} \left(\frac{2^n - \alpha}{2^n}\right)^{\frac{n-1}{2}}.}$$

Moreover,

$$\mathbb{P}(D_n = 2^n) = 0.$$

1. Fixed-level estimates

3.1 A deterministic Fourier consequence of $D_n \leq \alpha$ For a deterministic configuration $Z_j = e^{i\theta_j}$, define

$$u(t) := \log |P_n(e^{it})| = \sum_{j=1}^{n} \log |e^{it} - e^{i\theta_j}|.$$

The logarithmic singularities are integrable. We use the classical L^1-Fourier expansion

$$\log |1 - e^{ix}| = \log\Big(2 \left|\sin \frac{x}{2}\right|\Big) = -\sum_{r=1}^{\infty} \frac{\cos(rx)}{r}.$$

For completeness, this follows by first writing, for $0 < \rho < 1$,

$$\log |1 - \rho e^{ix}| = -\sum_{r=1}^{\infty} \frac{\rho^r \cos(rx)}{r},$$

and then letting $\rho \uparrow 1$ in $L^1([0, 2\pi])$. The singularity is bounded by an integrable multiple of $1 + |\log |x||$. In particular,

$$\frac{1}{2\pi} \int_0^{2\pi} \log |e^{it} - e^{i\theta}| \, dt = 0.$$

Therefore

$$\frac{1}{2\pi} \int_0^{2\pi} u(t) \, dt = 0.$$

Suppose now that $D_n \leq \alpha$. Then $u(t) \leq \log \alpha$. Writing $u = u_+ - u_-$, the zero-mean identity gives

$$\int u_+ = \int u_-,$$

and hence

$$\int_0^{2\pi} |u(t)| \, dt = 2 \int_0^{2\pi} u_+(t) \, dt \leq 4\pi \log \alpha.$$

Let

$$S_r := \sum_{j=1}^{n} Z_j^r.$$

The r-th Fourier coefficient of u, for $r \geq 1$, is

$$\widehat{u}(r) = -\frac{1}{2r}\sum_{j=1}^{n} e^{-ir\theta_j} = -\frac{\overline{S_r}}{2r}.$$

Consequently,

$$\frac{|S_r|}{2r} = |\widehat{u}(r)| \le \frac{1}{2\pi}\int_0^{2\pi} |u(t)|\, dt \le 2\log\alpha.$$

We have proved: Lemma 1 For every $r \ge 1$,

$$\boxed{D_n \le \alpha \quad \Longrightarrow \quad |S_r| \le 4r\log\alpha.}$$

Thus a uniformly bounded maximum forces every fixed collection of power sums to lie in a fixed bounded region.

3.2 A fixed-dimensional small-ball estimate Lemma 2 Fix an integer $K \ge 1$ and positive constants $R_1, \ldots, R_K$. Then there is $C = C(K, R_1, \ldots, R_K)$ such that

$$\mathbb{P}\big(|S_r| \le R_r \text{ for } 1 \le r \le K\big) \le Cn^{-K}.$$

Proof Introduce the $2K$-dimensional random vector

$$Y(\theta) := (\cos\theta, \sin\theta, \ldots, \cos K\theta, \sin K\theta) \in \mathbb{R}^{2K}.$$

Then

$$(\Re S_1, \Im S_1, \ldots, \Re S_K, \Im S_K) = \sum_{j=1}^{n} Y(\Theta_j).$$

Let

$$\varphi(\xi) := \mathbb{E}\big[e^{i\xi\cdot Y(\Theta_1)}\big] = \frac{1}{2\pi}\int_0^{2\pi} e^{i\xi\cdot Y(\theta)}\, d\theta$$

be the characteristic function. Behavior near the origin Orthogonality of the trigonometric system gives

$$\mathbb{E}Y = 0, \qquad \mathbb{E}(YY^{\mathsf{T}}) = \frac{1}{2}I_{2K}.$$

Taylor expansion therefore yields

$$\varphi(\xi) = 1 - \frac{1}{4}|\xi|^2 + O_K(|\xi|^3).$$

Hence there are $c, \rho > 0$, depending only on K, such that

$$|\varphi(\xi)| \le e^{-c|\xi|^2}, \qquad |\xi| \le \rho.$$

Behavior away from the origin If $|\varphi(\xi)| = 1$, equality in the triangle inequality for the defining integral implies that

$$e^{i\xi \cdot Y(\theta)}$$

is constant for almost every θ, hence everywhere. Thus the trigonometric polynomial $\xi \cdot Y(\theta)$ is constant modulo 2π, and by continuity is constant as a real function. Since it has no constant Fourier coefficient, it must vanish identically. Linear independence of the sine and cosine functions then gives $\xi = 0$. Therefore

$$|\varphi(\xi)| < 1 \qquad (\xi \neq 0).$$

We also need integrability at infinity. Write $\xi = Rv$, where $R = |\xi|$ and $|v| = 1$, and let

$$p_v(\theta) := v \cdot Y(\theta).$$

This is a nonzero real trigonometric polynomial of degree at most K. There is a constant $c_K > 0$ such that

$$\max_{1 \le q \le 2K+1} |p_v^{(q)}(\theta)| \ge c_K$$

for every unit vector v and every θ. Indeed, otherwise compactness would give a nonzero p_v whose first $2K+1$ derivatives vanish at some point. Then $p_v - p_v(\theta_0)$, after multiplication by $e^{iK\theta}$, would give a polynomial of degree at most $2K$ with a zero of order greater than $2K$, forcing p_v to be constant, hence zero, a contradiction. Partitioning the circle into a number of intervals depending only on K, on each of which one of these derivatives stays bounded away from zero, the one-dimensional van der Corput estimate gives

$$|\varphi(\xi)| \le C_K (1 + |\xi|)^{-1/(2K+1)}.$$

Indeed, on an interval where $|p_v^{(q)}| \ge c_K/2$, van der Corput gives

$$\left| \int e^{iRp_v(\theta)}\, d\theta \right| \le C_K R^{-1/q} \le C_K R^{-1/(2K+1)}.$$

Choose an integer

$$N_0 > 2K(2K+1).$$

Then $|\varphi|^{N_0} \in L^1(\mathbb{R}^{2K})$. Since $|\varphi| < 1$ away from zero and tends to zero at infinity, there is $q \in (0,1)$ such that

$$\sup_{|\xi| \geq \rho} |\varphi(\xi)| \leq q.$$

For $n \geq N_0$,

$$\begin{aligned}\int_{\mathbb{R}^{2K}} |\varphi(\xi)|^n \, d\xi &\leq \int_{|\xi|<\rho} e^{-cn|\xi|^2} \, d\xi + q^{n-N_0} \int_{|\xi| \geq \rho} |\varphi(\xi)|^{N_0} \, d\xi \\ &\leq C_K n^{-K} + C_K q^n \leq C'_K n^{-K}.\end{aligned}$$

Thus the n-fold convolution of the law of Y has a bounded continuous density

$$f_n(x) = \frac{1}{(2\pi)^{2K}} \int_{\mathbb{R}^{2K}} e^{-i\xi \cdot x} \varphi(\xi)^n \, d\xi,$$

and

$$\|f_n\|_\infty \leq C_K n^{-K}.$$

Finally, the set

$$B = \{(w_1, \ldots, w_K) \in \mathbb{C}^K : |w_r| \leq R_r\}$$

has finite $2K$-dimensional volume. Hence

$$\mathbb{P}\big((S_1, \ldots, S_K) \in B\big) \leq \mathrm{vol}(B) \|f_n\|_\infty \leq C n^{-K}.$$

This proves the lemma. □

3.3 Superpolynomial convergence in part a Fix $\alpha > 2$. By Lemma 1,

$$\{D_n < \alpha\} \subseteq \bigcap_{r=1}^{K} \{|S_r| \leq 4r \log \alpha\}.$$

Lemma 2 therefore gives

$$\mathbb{P}(D_n < \alpha) \leq C_{\alpha,K} n^{-K}.$$

Given $A > 0$, choose an integer $K > A$. Then

$$\mathbb{P}(D_n < \alpha) = O_{\alpha,A}(n^{-A}),$$

which proves the asserted upper estimate for the exceptional probability. No limit and expectation have been interchanged here: the only limiting operation used in the probabilistic estimate was Fourier inversion after proving that $\varphi^n \in L^1$.

3.4 A constructive lower bound for the exceptional probability We now show that $D_n < \alpha$ is not impossible and give an explicit lower estimate. Let

$$\omega_j = e^{2\pi ij/n}, \qquad 0 \le j < n,$$

be the regular n-gon. Lemma 3: stability of the regular polygon There is an absolute constant $C > 0$ with the following property. Suppose

$$Z_j = e^{i(2\pi j/n+\varepsilon_j)}, \qquad |\varepsilon_j| \le \eta.$$

Then

$$D_n \le (2 + n\eta)\exp\bigl(Cn\eta\log(en)\bigr).$$

Proof Fix $z = e^{it}$ and put

$$a_j := |z - \omega_j|.$$

Choose j_0 so that a_{j_0} is minimal. Since

$$|Z_j - \omega_j| \le |\varepsilon_j| \le \eta,$$

we have

$$|z - Z_j| \le a_j + \eta.$$

For $j \ne j_0$,

$$a_j + \eta \le a_j e^{\eta/a_j}.$$

Consequently,

$$\prod_{j=0}^{n-1} |z - Z_j| \le (a_{j_0} + \eta) \prod_{j \ne j_0} a_j \exp\left(\eta \sum_{j \ne j_0} \frac{1}{a_j}\right).$$

Now

$$\prod_{j=0}^{n-1} a_j = |z^n - 1| \le 2,$$

while

$$\prod_{j \ne j_0} a_j = \left| \frac{z^n - 1}{z - \omega_{j_0}} \right| = \left| \sum_{\ell=0}^{n-1} z^{n-1-\ell} \omega_{j_0}^{\ell} \right| \le n,$$

with the quotient interpreted by continuity when $z = \omega_{j_0}$. Hence

$$(a_{j_0} + \eta) \prod_{j \ne j_0} a_j \le 2 + n\eta.$$

It remains to estimate the inverse distances. Since j_0 is a nearest grid point, the grid points at cyclic distance k from j_0 have angular distance at least $(2k-1)\pi/n$ from t. Using

$$2\sin(x/2) \ge \frac{2x}{\pi}, \qquad 0 \le x \le \pi,$$

we obtain

$$\sum_{j \ne j_0} \frac{1}{a_j} \le Cn \sum_{k=1}^{n} \frac{1}{k} \le Cn \log(en).$$

Combining these estimates and maximizing over $z \in \mathbb{T}$ proves the lemma. $\square$ Fix $\alpha > 2$. Choose $c_\alpha > 0$ sufficiently small that

$$2e^{2Cc_\alpha} < \alpha.$$

Set

$$\eta_n := \frac{c_\alpha}{n \log n}.$$

Lemma 3 gives, for all sufficiently large n,

$$D_n \le \left(2 + \frac{c_\alpha}{\log n}\right) \exp\left(Cc_\alpha \frac{\log(en)}{\log n}\right) < \alpha.$$

Consider the n disjoint arcs of angular half-width η_n centered at the regular n-gon points. The probability that the n labelled random points occupy these arcs with exactly one point in each is

$$n!\left(\frac{\eta_n}{\pi}\right)^n = n!\left(\frac{c_\alpha}{\pi n \log n}\right)^n.$$

On this event the roots may be relabelled so that Lemma 3 applies. Thus

$$\mathbb{P}(D_n < \alpha) \geq n!\left(\frac{c_\alpha}{\pi n \log n}\right)^n.$$

Using $n! \geq (n/e)^n$,

$$\mathbb{P}(D_n < \alpha) \geq \left(\frac{c_\alpha}{e\pi \log n}\right)^n = \exp\bigl(-n \log\log n - O_\alpha(n)\bigr).$$

3.5 The actual probabilistic scale A known theorem for this precise random-polynomial model gives a sharper description on the scale on which D_n fluctuates. For

$$P_N(z) = \prod_{k=1}^{N}(z - z_k)^{m_k}, \qquad s_N^2 = \sum_{k=1}^{N} m_k^2,$$

the theorem assumes

$$\sum_{k=1}^{N} \exp\left(-\varepsilon \frac{s_N}{m_k}\right) \longrightarrow 0 \qquad (\varepsilon > 0)$$

and concludes that

$$\frac{1}{s_N} \log \max_{|z|=1} |P_N(z)|^2 \xrightarrow{d} I^*,$$

where $I^* > 0$ almost surely and is obtained from a Brownian bridge. Here $m_k = 1$, so $s_N = \sqrt{N}$, and the condition becomes

$$N e^{-\varepsilon\sqrt{N}} \longrightarrow 0.$$

It is therefore satisfied, giving

$$\frac{2 \log D_n}{\sqrt{n}} \xrightarrow{d} I^*.$$

This is the theorem of Tucci and Whiting, and its hypotheses match the present model exactly. arXiv+1 This external theorem is not needed for the preceding superpolynomial fixed-α estimate.

1. The upper-edge asymptotic

We now fix $n \geq 2$ and let

$$\alpha = 2^n(1-\delta), \qquad \delta \downarrow 0.$$

4.1 Reduction by rotation Because D_n is unchanged when all roots are multiplied by the same unit complex number, define

$$x_1 := 0, \qquad x_j := \operatorname{Arg}(Z_j\overline{Z_1}) \in (-\pi, \pi], \quad 2 \leq j \leq n.$$

The torus transformation

$$(\Theta_1, \ldots, \Theta_n) \longmapsto (\Theta_1, \Theta_2 - \Theta_1, \ldots, \Theta_n - \Theta_1)$$

preserves Haar measure. Hence

$$x_2, \ldots, x_n$$

are independent and uniform on $(-\pi, \pi]$, independently of the discarded global rotation. Set $x = (x_2, \ldots, x_n) \in \mathbb{R}^{n-1}$.

4.2 Localization near a coincident configuration Suppose

$$D_n \geq 2^n(1-\delta).$$

Then for some $z \in \mathbb{T}$,

$$\prod_{j=1}^{n} \frac{|z - Z_j|}{2} \geq 1 - \delta.$$

Every factor belongs to $[0,1]$. Therefore each factor is itself at least $1-\delta$. Let $-e^{ic}$ denote z, so that e^{ic} is its antipode. If $d_j \in [0, \pi]$ is the circular angular distance between Z_j and e^{ic}, then

$$\frac{|z - Z_j|}{2} = \cos\frac{d_j}{2}.$$

Thus

$$\cos\frac{d_j}{2} \geq 1 - \delta.$$

Using, for $0 \leq y \leq \pi/2$,

$$1 - \cos y \geq \frac{2y^2}{\pi^2},$$

we obtain

$$d_j = O(\sqrt{\delta}).$$

All roots therefore lie in an arc of length $O(\sqrt{\delta})$. In particular,

$$|x| = O_n(\sqrt{\delta}).$$

This also shows that, for sufficiently small δ, there is no ambiguity from the choice of angular lifts.

4.3 Quadratic approximation For $|u| < \pi$, define

$$h(u) := -\log\cos\frac{u}{2}.$$

Near zero,

$$h(u) = \frac{u^2}{8} + O(u^4).$$

For $x_1 = 0$ and $x = (x_2, \ldots, x_n)$ sufficiently small,

$$\frac{D_n}{2^n} = \max_{c\in\mathbb{R}} \prod_{j=1}^{n} \cos\frac{x_j - c}{2}.$$

Hence

$$\Phi(x) := -\log\frac{D_n}{2^n} = \min_{c\in\mathbb{R}} \sum_{j=1}^{n} h(x_j - c).$$

Let

$$\bar{x} := \frac{1}{n}\sum_{j=1}^{n} x_j = \frac{1}{n}\sum_{j=2}^{n} x_j$$

and define

$$Q(x) := \min_{c\in\mathbb{R}} \sum_{j=1}^{n} (x_j - c)^2 = \sum_{j=1}^{n} (x_j - \bar{x})^2.$$

The minimizing c for the function involving h lies between the smallest and largest x_j, because h' is odd and strictly increasing near zero. It is therefore $O(|x|)$. The Taylor expansion of h, evaluated at this minimizer and at $c = \bar{x}$, gives the two-sided estimate

$$\boxed{\Phi(x) = \frac{1}{8}Q(x) + O_n(|x|^4).}$$

Since Q is positive definite on the variables $x_2, \ldots, x_n$, and its smallest eigenvalue is $1/n$, we have

$$|x|^2 \leq nQ(x).$$

Consequently the remainder may equivalently be written

$$\Phi(x) = \frac{1}{8}Q(x) + O_n(Q(x)^2).$$

4.4 The relevant ellipsoid Writing $m = n - 1$, one has

$$Q(x) = \sum_{j=2}^{n} x_j^2 - \frac{1}{n}\left(\sum_{j=2}^{n} x_j\right)^2 = x^{\mathsf{T}}Ax,$$

where

$$A = I_m - \frac{1}{n}J_m$$

and J_m is the all-ones matrix. The eigenvalues of A are
1, with multiplicity $m - 1$;
$1/n$, in the direction $(1, \ldots, 1)$.
Therefore

$$\det A = \frac{1}{n}.$$

The event in question is

$$\Phi(x) \leq -\log(1 - \delta) = \delta + O(\delta^2).$$

The localization already proved gives $Q(x) = O_n(\delta)$. Substituting this into the quadratic approximation shows that there is $C_n < \infty$ such that, for all sufficiently small δ,

$$\{Q(x) \leq 8\delta - C_n\delta^2\} \subseteq \left\{\frac{D_n}{2^n} \geq 1 - \delta\right\} \subseteq \{Q(x) \leq 8\delta + C_n\delta^2\}.$$

Thus the desired probability is the normalized volume of an ellipsoid, up to relative error $O_n(\delta)$. Let

$$v_m = \frac{\pi^{m/2}}{\Gamma(m/2+1)}$$

be the volume of the Euclidean unit ball in $\mathbb{R}^m$. Then

$$\mathrm{vol}\{x : x^{\mathsf{T}} Ax \le 8\delta\} = \frac{v_m (8\delta)^{m/2}}{\sqrt{\det A}} = \sqrt{n}\, v_m (8\delta)^{m/2}.$$

Because the relative-angle vector is uniform with density $(2\pi)^{-m}$,

$$\begin{aligned}\mathbb{P}(D_n \ge 2^n(1-\delta)) &= \frac{\sqrt{n}\, v_m 8^{m/2}}{(2\pi)^m} \delta^{m/2}\big(1 + O_n(\delta)\big) \\ &= \frac{\sqrt{n}}{\Gamma(m/2+1)} \left(\frac{2}{\pi}\right)^{m/2} \delta^{m/2}\big(1 + O_n(\delta)\big).\end{aligned}$$

Since $m = n - 1$, this is exactly

$$\mathbb{P}(D_n \ge 2^n(1-\delta)) = \frac{\sqrt{n}}{\Gamma\big(\frac{n+1}{2}\big)} \left(\frac{2}{\pi}\right)^{\frac{n-1}{2}} \delta^{\frac{n-1}{2}}\big(1 + O_n(\delta)\big).$$

This proves part b.

4.5 The exact endpoint Every factor $|z - Z_j|$ is at most 2. Therefore $D_n = 2^n$ can hold only if there is some $z \in \mathbb{T}$ such that

$$|z - Z_j| = 2$$

for every j. This is equivalent to

$$Z_1 = \cdots = Z_n = -z.$$

Conversely, coincident roots do give $D_n = 2^n$. Since the Z_j are independent and have nonatomic distributions,

$$\mathbb{P}(Z_1 = \cdots = Z_n) = 0,$$

so

$$\mathbb{P}(D_n = 2^n) = 0.$$

1. Boundary and small-n checks

5.1 Exact calculation for $n = 2$ Let $\Delta \in [0, \pi]$ be the smaller angular separation of the two roots. Since the angular difference is uniform, Δ is uniform on $[0, \pi]$. After rotating the roots to $e^{\pm i\Delta/2}$,

$$\begin{aligned} D_2 &= \max_t |e^{it} - e^{i\Delta/2}||e^{it} - e^{-i\Delta/2}| \\ &= 2 \max_t | \cos(\Delta/2) - \cos t| \\ &= 2\big(1 + \cos(\Delta/2)\big) \\ &= 4 \cos^2(\Delta/4). \end{aligned}$$

Hence, for $2 \le \alpha \le 4$,

$$\boxed{\mathbb{P}(D_2 \ge \alpha) = \frac{4}{\pi} \arccos\left(\frac{\sqrt{\alpha}}{2}\right).}$$

As $\alpha \uparrow 4$,

$$\mathbb{P}(D_2 \ge \alpha) \sim \frac{2}{\pi}\sqrt{4 - \alpha}.$$

Our general formula gives

$$\kappa_2 = \frac{\sqrt{2}}{\Gamma(3/2)} \left(\frac{2}{\pi}\right)^{1/2} = \frac{4}{\pi},$$

and, since $\delta = (4 - \alpha)/4$,

$$\kappa_2 \delta^{1/2} = \frac{4}{\pi} \frac{\sqrt{4 - \alpha}}{2} = \frac{2}{\pi}\sqrt{4 - \alpha},$$

in exact agreement. 5.2 The first higher-dimensional case For $n = 3$,

$$\kappa_3 = \frac{\sqrt{3}}{\Gamma(2)} \frac{2}{\pi} = \frac{2\sqrt{3}}{\pi}.$$

Thus

$$\mathbb{P}(D_3 \ge 8(1 - \delta)) = \frac{2\sqrt{3}}{\pi}\delta + O(\delta^2).$$

The linear power is consistent with the fact that, after quotienting out global rotation, two independent relative angles must fall into a two-dimensional ellipsoid of radius $O(\sqrt{\delta})$.

1. Why the hypotheses matter

The independence assumption is essential. Let U be uniform and set

$$Z_j = e^{i(U+2\pi j/n)}, \qquad 0 \le j < n.$$

Each Z_j is individually uniform, but the points are dependent. Their polynomial is

$$P_n(z) = z^n - e^{inU},$$

so

$$D_n = 2$$

identically. Thus, for every fixed $\alpha > 2$,

$$\mathbb{P}(D_n \ge \alpha) = 0,$$

the opposite of the independent result. Uniformity is also essential. If the Z_j are independent but all have the degenerate distribution $Z_j = 1$, then

$$D_n = 2^n$$

identically. In particular, both the $\sqrt{n}$-scale law and the upper-edge asymptotic fail. Finally, $n \ge 2$ is relevant at the endpoint. For $n = 1$,

$$D_1 = 2$$

deterministically, whereas for every $n \ge 2$ the event $D_n = 2^n$ has probability zero.

Final answer For every fixed $\alpha > 2$,

$$\boxed{\mathbb{P}(D_n \ge \alpha) = 1 - O_{\alpha,A}(n^{-A}) \quad \text{for every } A > 0,}$$

with the explicit complementary construction

$$\mathbb{P}(D_n < \alpha) \ge \exp\bigl(-n \log\log n - O_\alpha(n)\bigr).$$

The typical size is $D_n = \exp(\Theta_{\text{law}}(\sqrt{n}))$. For fixed n and $\alpha \uparrow 2^n$,

$$\boxed{\mathbb{P}(D_n \ge \alpha) \sim \frac{\sqrt{n}}{\Gamma\left(\frac{n+1}{2}\right)} \left(\frac{2}{\pi}\right)^{\frac{n-1}{2}} \left(\frac{2^n - \alpha}{2^n}\right)^{\frac{n-1}{2}}.}$$

Formal verification. Lean source; online type-check.

17 Question Q803

Informal verdict. **A — Complete solution**, with **high confidence**.

Lean coverage. For every strictly positive summable real sequence, Lean defines the full achievement set of all Boolean-selected subseries, including finite selections, and proves the exact finite-stage characterization: Sigma$a$ is the intersection of F_n + [0,r_n]; r_n + Delta_n is the attained longest interval length in the n-th approximant and decreases to the attained longest interval length in Sigma$a$; hence interior Sigma$a$ is empty iff Delta_n tends to zero, equivalently iff for every epsilon > 0 some r_n + Delta_n < epsilon. This is unconditional and stronger than the printed hypothesis because no monotonicity assumption is needed, so it specializes directly to every positive decreasing summable sequence in Q803.

17.1 Audited informal answer

Q803 — Exact finite-stage characterization

1. Faithful restatement

Let

$$a_1 \geq a_2 \geq \cdots > 0, \qquad \sum_{n=1}^{\infty} a_n < \infty.$$

Define the set of all subsums

$$\Sigma(a) := \left\{ \sum_{n=1}^{\infty} \varepsilon_n a_n : \varepsilon_n \in \{0,1\} \right\}.$$

Finite extracted series correspond to eventually-zero sequences (ε_n). Including or excluding the empty subseries, whose sum is 0, does not affect the interior. The question asks for a characterization of those sequences for which

$$\operatorname{int} \Sigma(a) = \varnothing.$$

This is exactly the complete Q803 statement in the PDF; there are no subparts, footnotes, or further qualifications. In standard French usage, "décroissante" means nonincreasing. There is no apparent typo or substantive ambiguity. Luc Pommeret

1. The exact characterization

For $n \geq 0$, put

$$r_n := \sum_{k>n} a_k, \qquad r_0 = \sum_{k\geq 1} a_k,$$

and let

$$F_n := \left\{ \sum_{k=1}^{n} \varepsilon_k a_k : \varepsilon_k \in \{0,1\} \right\}$$

be the finite set of subsums of the first n terms. Write its distinct elements in increasing order:

$$F_n = \{f_{n,0} < f_{n,1} < \cdots < f_{n,m_n}\}.$$

Partition this ordered set into maximal consecutive blocks

$$f_{n,p}, f_{n,p+1}, \ldots, f_{n,q}$$

such that

$$f_{n,j+1} - f_{n,j} \leq r_n \qquad (p \leq j < q).$$

Thus a new block begins whenever a consecutive gap is strictly larger than r_n. Define

$$\Delta_n := \max_{\text{such maximal blocks } [p,q]} \big(f_{n,q} - f_{n,p}\big).$$

Theorem The sequence

$$\ell_n := r_n + \Delta_n$$

is nonincreasing and converges. Moreover,

$$\boxed{\lim_{n\to\infty} \ell_n = \max\big\{\, b-a : [a,b] \subseteq \Sigma(a) \,\big\}.}$$

In particular, because $r_n \to 0$,

$$\boxed{\operatorname{int}\Sigma(a) = \varnothing \quad \Longleftrightarrow \quad \lim_{n\to\infty} \Delta_n = 0.}$$

Equivalently,

$$\boxed{\operatorname{int}\Sigma(a)=\varnothing \quad\Longleftrightarrow\quad \forall\varepsilon>0\ \exists n \quad r_n+\Delta_n<\varepsilon.}$$

Thus the answer is: the subsum set has empty interior precisely when every r_n-connected cluster of finite subsums of the first n terms has diameter tending to zero. This finite-chain criterion is essentially the analytic characterization recorded as Proposition 2.8 in the achievement-set literature. arXiv+1

1. Proof of the characterization

3.1. Compactness of the subsum set Let

$$\Omega := \{0,1\}^{\mathbb{N}}$$

and define

$$\pi : \Omega \longrightarrow \mathbb{R}, \qquad \pi(\varepsilon) := \sum_{n=1}^{\infty} \varepsilon_n a_n.$$

Equip Ω with the usual product metric, for example

$$d(\varepsilon,\eta) = \sum_{n=1}^{\infty} 2^{-n}|\varepsilon_n - \eta_n|.$$

If $\varepsilon_k = \eta_k$ for $1 \le k \le N$, then

$$|\pi(\varepsilon) - \pi(\eta)| \le \sum_{k>N} a_k = r_N.$$

Since $r_N \to 0$, the map π is uniformly continuous. The space Ω is compact: from any sequence in Ω, one first extracts a subsequence on which the first coordinate is constant, then a further subsequence on which the second coordinate is constant, and so on; the diagonal subsequence converges. Hence

$$\Sigma(a) = \pi(\Omega)$$

is compact.

3.2. Canonical finite approximations Define

$$I_n := F_n + [0,r_n] = \bigcup_{f\in F_n} [f, f+r_n].$$

Each I_n is a finite union of closed bounded intervals. We claim that

$$I_{n+1} \subseteq I_n.$$

Indeed,

$$F_{n+1} = F_n \cup (F_n + a_{n+1}), \qquad r_n = a_{n+1} + r_{n+1}.$$

For each $f \in F_n$, the two child intervals at level $n+1$ are

$$[f, f + r_{n+1}]$$

and

$$[f + a_{n+1}, f + a_{n+1} + r_{n+1}] = [f + a_{n+1}, f + r_n].$$

Both are contained in $[f, f + r_n]$. Therefore $I_{n+1} \subseteq I_n$. We now prove the fundamental identity

$$\boxed{\Sigma(a) = \bigcap_{n=0}^{\infty} I_n.}$$

The inclusion $\Sigma(a) \subseteq I_n$ is immediate: if

$$x = \sum_{k=1}^{\infty} \varepsilon_k a_k,$$

then

$$x = \underbrace{\sum_{k=1}^{n} \varepsilon_k a_k}_{\in F_n} + \underbrace{\sum_{k>n} \varepsilon_k a_k}_{\in [0, r_n]}.$$

Conversely, suppose $x \in I_n$ for every n. For each n, choose a binary word

$$w^{(n)} = (\varepsilon_1^{(n)}, \ldots, \varepsilon_n^{(n)})$$

such that

$$0 \le x - \sum_{k=1}^{n} \varepsilon_k^{(n)} a_k \le r_n.$$

Extend $w^{(n)}$ by zeros to an element $\widetilde{w}^{(n)} \in \Omega$. By the diagonal compactness argument, a subsequence

$$\widetilde{w}^{(n_j)}$$

converges to some $\varepsilon \in \Omega$. Continuity of π gives

$$\pi\big(\widetilde{w}^{(n_j)}\big) \longrightarrow \pi(\varepsilon).$$

But

$$\Big|x - \pi\big(\widetilde{w}^{(n_j)}\big)\Big| \leq r_{n_j} \longrightarrow 0,$$

so $\pi(\varepsilon) = x$. Hence $x \in \Sigma(a)$.

3.3. A lemma on decreasing finite unions of intervals We use the following general fact. Lemma Let

$$W_0 \supseteq W_1 \supseteq W_2 \supseteq \cdots$$

be compact subsets of $\mathbb{R}$, each of which is a finite union of closed intervals. Let L_n be the maximum length of a connected component of W_n. Then L_n is nonincreasing, and if

$$W := \bigcap_{n=0}^{\infty} W_n, \qquad L := \lim_{n\to\infty} L_n,$$

then

$$L = \max\{b - a : [a, b] \subseteq W\}.$$

Proof Every component of W_{n+1}, being connected and contained in W_n, lies in a single component of W_n. Therefore

$$L_{n+1} \leq L_n.$$

Suppose first that $[a, b] \subseteq W$. For every n, the connected interval $[a, b]$ lies in one component of W_n. Hence

$$b - a \leq L_n$$

for every n, and therefore

$$b - a \leq L.$$

It remains to construct an interval of length L in W. For each n, choose the leftmost component

$$C_n = [u_n, v_n]$$

of W_n whose length is L_n. All pairs (u_n, v_n) lie in a fixed compact rectangle. Passing to a subsequence,

$$u_{n_j} \longrightarrow u, \qquad v_{n_j} \longrightarrow v.$$

Since

$$v_{n_j} - u_{n_j} = L_{n_j} \longrightarrow L,$$

we have

$$v - u = L.$$

Fix m. For sufficiently large j, $n_j \geq m$, so

$$C_{n_j} \subseteq W_{n_j} \subseteq W_m.$$

For $t \in [0, 1]$, the point

$$x_j(t) := (1-t)u_{n_j} + tv_{n_j}$$

belongs to W_m, and

$$x_j(t) \longrightarrow (1-t)u + tv.$$

Since W_m is closed,

$$(1-t)u + tv \in W_m.$$

Thus $[u, v] \subseteq W_m$ for every m, and hence

$$[u, v] \subseteq W.$$

Its length is L. $\square$

3.4. Computing the components of I_n Recall

$$I_n = \bigcup_{j=0}^{m_n} [f_{n,j}, f_{n,j} + r_n].$$

Two consecutive intervals overlap or touch exactly when

$$f_{n,j+1} \le f_{n,j} + r_n,$$

that is,

$$f_{n,j+1} - f_{n,j} \le r_n.$$

Consequently, if

$$f_{n,p}, \dots, f_{n,q}$$

is one maximal r_n-connected block, the corresponding component of I_n is

$$[f_{n,p}, f_{n,q} + r_n],$$

whose length is

$$f_{n,q} - f_{n,p} + r_n.$$

Therefore the maximum component length of I_n is exactly

$$L_n = r_n + \Delta_n = \ell_n.$$

Applying the preceding lemma to the decreasing sequence (I_n), and using

$$\Sigma(a) = \bigcap_n I_n,$$

we obtain

$$\max\{b - a : [a, b] \subseteq \Sigma(a)\} = \lim_{n\to\infty} (r_n + \Delta_n).$$

Finally, $r_n \to 0$, so

$$\lim_{n\to\infty} \Delta_n = \lim_{n\to\infty} (r_n + \Delta_n).$$

The set $\Sigma(a)$ has nonempty interior exactly when it contains a nondegenerate interval. This proves

$$\operatorname{int} \Sigma(a) = \varnothing \iff \Delta_n \longrightarrow 0.$$

□

1. Consequence: the empty-interior case is exactly the Cantor-set case

Under the original hypotheses, $\Sigma(a)$ has no isolated points. Indeed, take

$$x = \sum_{n=1}^{\infty} \varepsilon_n a_n \in \Sigma(a).$$

Either infinitely many ε_n are 0, in which case one can change a sufficiently far 0 into a 1, or infinitely many are 1, in which case one can change a sufficiently far 1 into a 0. This produces a distinct element of $\Sigma(a)$ at distance $a_n \to 0$ from x. Thus $\Sigma(a)$ is compact and perfect. If its interior is empty, it is nowhere dense. Moreover, any connected subset of $\mathbb{R}$ containing two distinct points contains the whole interval between them; hence an empty-interior $\Sigma(a)$ is totally disconnected. Therefore, under the hypotheses of Q803,

$$\boxed{\operatorname{int}\Sigma(a) = \varnothing \quad \Longleftrightarrow \quad \Sigma(a) \text{ is a Cantor set.}}$$

1. The classical term-versus-tail regimes

The exact criterion above is needed because comparisons between a_n and r_n do not decide the mixed case. They do, however, settle the two extreme regimes. 5.1. If the tail eventually dominates the term Suppose

$$a_n \leq r_n$$

for all $n > N$. At such an index n, for each $f \in F_{n-1}$,

$$[f, f + r_n] \quad \text{and} \quad [f + a_n, f + a_n + r_n]$$

overlap, and their union is

$$[f, f + a_n + r_n] = [f, f + r_{n-1}].$$

Hence

$$I_n = I_{n-1}.$$

It follows that

$$I_n = I_N \qquad (n \geq N),$$

and therefore

$$\Sigma(a) = I_N.$$

Thus $\Sigma(a)$ is a finite union of nondegenerate closed intervals and has nonempty interior. In particular,

$$\boxed{\operatorname{int}\Sigma(a)=\varnothing \implies a_n>r_n \text{ for infinitely many } n.}$$

If $a_n \le r_n$ for every n, then $I_n = I_0$ for every n, so

$$\boxed{\Sigma(a)=\left[0,\sum_{n=1}^{\infty}a_n\right].}$$

5.2. If the term eventually dominates the tail Suppose

$$a_n > r_n$$

for all $n > N$. Consider the tail achievement set

$$\Sigma_N := \left\{\sum_{n>N}\varepsilon_n a_n : \varepsilon_n \in \{0,1\}\right\}.$$

Its natural binary coding is injective. Indeed, if two binary sequences first differ at index $j > N$, say $\varepsilon_j = 1$ and $\eta_j = 0$, then

$$\sum_{n>N}(\varepsilon_n-\eta_n)a_n \ge a_j-\sum_{n>j}a_n = a_j-r_j>0.$$

Thus Σ_N is homeomorphic to the binary Cantor space and in particular contains no interval. Now

$$\Sigma(a) = F_N + \Sigma_N$$

is a finite union of translates of the compact nowhere-dense set Σ_N. Hence it too has empty interior. Therefore

$$\boxed{a_n>r_n \text{ eventually} \implies \operatorname{int}\Sigma(a)=\varnothing.}$$

The converse is false: empty interior can occur even though $a_n \le r_n$ infinitely often.

1. Why the mixed case cannot be decided by the signs of $a_n - r_n$

Consider, for $0 < q \le 2/3$, the strictly decreasing multigeometric sequence

$$a_{2k+1} = 3q^k, \qquad a_{2k+2} = 2q^k, \qquad k \ge 0.$$

The tail after the first term in the k-th block is

$$r_{2k+1} = 2q^k + \sum_{j>k} 5q^j = q^k \left(2 + \frac{5q}{1-q}\right),$$

whereas the tail after the second term is

$$r_{2k+2} = \sum_{j>k} 5q^j = q^k \frac{5q}{1-q}.$$

For both $q = \frac{1}{5}$ and $q = \frac{1}{4}$, one has

$$a_{2k+1} < r_{2k+1}, \qquad a_{2k+2} > r_{2k+2}.$$

Thus the complete pattern of inequalities is identical: every odd index is tail-dominated, and every even index is term-dominated. Nevertheless, the corresponding achievement sets have opposite interior behavior.

6.1. The case $q = \frac{1}{5}$: empty interior After m complete blocks there are at most 4^m distinct prefix subsums, because each block contributes one of

$$0,\ 2,\ 3,\ 5$$

times the appropriate power of $1/5$. The remaining tail is

$$r_{2m} = \sum_{k=m}^{\infty} 5 \cdot 5^{-k} = \frac{25}{4} 5^{-m}.$$

Thus I_{2m} is covered by at most 4^m intervals, each of length r_{2m}. Its largest component therefore has length at most the sum of these lengths:

$$\ell_{2m} \le 4^m r_{2m} = \frac{25}{4} \left(\frac{4}{5}\right)^m \longrightarrow 0.$$

Since (ℓ_n) is nonincreasing,

$$\ell_n \longrightarrow 0.$$

By the theorem,

$$\boxed{\operatorname{int} \Sigma\left(3, 2; \frac{1}{5}\right) = \varnothing.}$$

6.2. The case $q = \frac{1}{4}$: a nondegenerate interval Now let

$$a_{2k+1} = \frac{3}{4^k}, \qquad a_{2k+2} = \frac{2}{4^k}.$$

A complete block contributes a digit in

$$D := \{0, 2, 3, 5\}.$$

After m blocks, the finite block subsums are

$$B_m = \left\{ \sum_{k=0}^{m-1} \frac{d_k}{4^k} : d_k \in D \right\}.$$

Scale them by 4^{m-1}:

$$A_m := 4^{m-1} B_m = \left\{ \sum_{k=0}^{m-1} d_k 4^{m-1-k} : d_k \in D \right\} \subseteq \mathbb{Z}.$$

We claim that A_m contains every integer in the interval

$$[\alpha_m, \beta_m], \qquad \alpha_m := \frac{2(4^m - 1)}{3}, \qquad \beta_m := 4^m - 1.$$

For $m = 1$,

$$A_1 = D$$

contains $2, 3$, and indeed

$$\alpha_1 = 2, \qquad \beta_1 = 3.$$

Suppose the claim holds for m. Since

$$A_{m+1} = 4A_m + D,$$

the set A_{m+1} contains every integer from

$$4\alpha_m + 2$$

to

$$4\beta_m + 3.$$

To see this, consider residues modulo 4:
residues 2 and 3 are obtained using the digits 2 and 3;
a multiple $4t$ in the stated range is obtained using digit 0;

a number $4t+1$ is written as $4(t-1)+5$.

The endpoint recurrences are

$$4\alpha_m + 2 = \alpha_{m+1}, \qquad 4\beta_m + 3 = \beta_{m+1}.$$

This proves the claim. The tail after m blocks is

$$r_{2m} = \sum_{k=m}^{\infty} \frac{5}{4^k} = \frac{20}{3}\, 4^{-m}.$$

After multiplying by 4^{m-1}, its length becomes

$$4^{m-1} r_{2m} = \frac{5}{3}.$$

Since A_m contains every consecutive integer from α_m through β_m, the intervals of length $5/3$ starting at these integers overlap and cover

$$\left[\alpha_m, \beta_m + \frac{5}{3}\right].$$

Scaling back, I_{2m} contains

$$J_m = \left[\frac{\alpha_m}{4^{m-1}}, \frac{\beta_m + 5/3}{4^{m-1}}\right].$$

The endpoints simplify to

$$J_m = \left[\frac{8}{3}\left(1 - 4^{-m}\right),\, 4 + \frac{8}{3}\, 4^{-m}\right].$$

Hence

$$\left[\frac{8}{3}, 4\right] \subseteq J_m \subseteq I_{2m}$$

for every m. Because the even-indexed approximants are cofinal in the decreasing family (I_n),

$$\bigcap_{m=1}^{\infty} I_{2m} = \bigcap_{n=1}^{\infty} I_n = \Sigma(a).$$

Therefore

$$\boxed{\left[\frac{8}{3}, 4\right] \subseteq \Sigma\left(3, 2; \frac{1}{4}\right).}$$

In particular,

$$\boxed{\operatorname{int}\Sigma\left(3,2;\frac{1}{4}\right)\neq\varnothing.}$$

Thus even the complete pattern of inequalities

$$a_n \lesseqgtr r_n$$

does not characterize empty interior.

1. Boundary tests

7.1. Geometric sequences Let

$$a_n = q^{n-1}, \qquad 0 < q < 1.$$

Then

$$r_n = \frac{q^n}{1-q}.$$

If $q < \frac{1}{2}$, then

$$a_n > r_n$$

for every n, so the achievement set has empty interior. If $q \geq \frac{1}{2}$, then

$$a_n \leq r_n$$

for every n, and therefore

$$\Sigma(a) = \left[0, \frac{1}{1-q}\right].$$

At the boundary $q = \frac{1}{2}$, equality $a_n = r_n$ holds at every index, and

$$\Sigma(a) = [0, 2].$$

This shows that the strictness of the tail comparison matters.

7.2. Finite support If only finitely many a_n are nonzero, then for sufficiently large n,

$$r_n = 0, \qquad I_n = F_n.$$

The achievement set is finite and therefore has empty interior. The original assumption $a_n > 0$ for every n excludes this degenerate case.

1. A useful easily checked sufficient condition

Since I_n is a union of at most 2^n intervals of length r_n, its largest component satisfies

$$\ell_n \leq 2^n r_n.$$

Consequently,

$$\boxed{\liminf_{n\to\infty} 2^n r_n = 0 \quad \Longrightarrow \quad \operatorname{int}\Sigma(a) = \varnothing.}$$

More sharply, one may replace 2^n by the actual number $|F_n|$:

$$\liminf_{n\to\infty} |F_n|\, r_n = 0 \quad \Longrightarrow \quad \operatorname{int}\Sigma(a) = \varnothing.$$

These are sufficient, not necessary, conditions.

1. Exact finite-stage computation

For rational input data and exactly known tails, the numbers

$$\ell_n = r_n + \Delta_n$$

can be computed without numerical tolerances. PythonExécuterfrom fractions import Fraction from collections.abc import Sequence, Iterator

def interval_moduli(terms: Sequence[Fraction], tails: Sequence[Fraction],) -> Iterator[tuple[int, Fraction]]: """ Compute ell_n = the largest component length of

I_n = F_n + [0, r_n],

where: terms[n-1] = a_n, tails[n] = r_n = sum_{k>n} a_k.

The array tails must have length len$terms$ + 1. Use Fraction values to make all comparisons exact. """ terms = [Fraction$x$ for x in terms] tails = [Fraction$x$ for x in tails]

if len$tails$!= len$terms$ + 1: raise ValueError("tails must have length len$terms$ + 1")

finite_sums = {Fraction0}

for n, a_n in enumerate$terms, start = 1$: old_sums = tuple$finite_sums$ finite_sums.update$x + a_n for x in old_sums$

values = sorted$finite_sums$ r_n = tails[n]

left = values[0] previous = values[0] best = Fraction0

for x in values[1:]: if x - previous <= r_n: previous = x else: best = max$best, previous - left + r_n$ left = previous = x best = max$best, previous - left + r_n$ yield n, best At stage n, the worst-case number of distinct finite subsums is 2^n, so the straightforward method is exponential. Finite computations can reveal candidate behavior, but a proof of empty interior still requires a rigorous estimate forcing $\ell_n \to 0$, while a proof of nonempty interior requires a persistent positive lower bound or an explicitly exhibited common interval.

1. Hypothesis and topology audit

Topological assumptions Everything takes place in $\mathbb{R}$ with its usual Euclidean topology. It is Hausdorff, second-countable, complete, and metrizable. Connected subsets of $\mathbb{R}$ are intervals; this is the only order-topological fact used about connectedness. Compactness Compactness of $\{0,1\}^{\mathbb{N}}$ was used only in its countable metric form and was proved by a diagonal subsequence argument. Compactness of bounded endpoint sequences was used through Bolzano–Weierstrass. No nets, filters, or general uncountable Tychonoff theorem are needed. Countability The proof is entirely sequential and depends on the sequence being indexed by $\mathbb{N}$. There are no uncountable families of subsums or covers. Choice Within ordinary ZFC, no substantial form of the axiom of choice is used. Choices from finite sets may be made canonically—for example, by taking the lexicographically least coding word or the leftmost longest component. No uncountable choice is involved. Baire category and measure The main characterization does not use the Baire category theorem or Lebesgue measure. The finite-union argument in the eventually fast regime uses only the elementary fact that a finite union of closed nowhere-dense sets is nowhere dense. Necessity of the stated sequence hypotheses

Summability is essential to this formulation. It gives $r_n \to 0$, convergence of every subseries, uniform convergence of the coding map, and compactness of $\Sigma(a)$. For a divergent positive series, the set of finite subsums is unbounded and infinite subseries may diverge.

Monotonicity is not needed for the exact criterion. The proof works for every summable nonnegative sequence in its given order. Since the subsum set is invariant under permutations, monotonicity merely gives a canonical ordering.

Strict positivity can be weakened to nonnegativity. Zero terms may simply be deleted. Strict positivity is used when asserting that an infinite achievement set has no isolated points.

Even signed absolutely summable sequences reduce to the positive case. If $\sum |a_n| < \infty$, then

$$\Sigma(a) = \sum_{a_n<0} a_n + \Sigma(|a_n|),$$

because at a negative coordinate selecting a_n is equivalent, after translation, to not selecting $|a_n|$. Thus signed achievement sets are translates of positive ones and have the same interior behavior.

1. Final classification

Define r_n, F_n, and Δ_n as above. Then

$$\boxed{\operatorname{int}\Sigma(a) = \varnothing \iff \lim_{n\to\infty} \Delta_n = 0,}$$

where Δ_n is the largest span of a maximal chain of consecutive n-term subsums whose successive gaps are at most the remaining tail r_n. More strongly,

$$\boxed{\max\{\,|J| : J \subseteq \Sigma(a) \text{ is an interval}\,\} = \lim_{n\to\infty}\bigl(r_n + \Delta_n\bigr).}$$

This is a complete literal answer to Q803. It is, however, an exponentially complicated finite-stage criterion rather than a simple condition involving only the individual comparisons a_n versus r_n. Current surveys distinguish this exact analytic characterization from the still-unresolved search for an easy structural test separating Cantor sets from Cantorvals in the mixed regime. arXiv+1

Formal verification. Lean source; online type-check.

18 Question Q804

Informal verdict. A — Complete solution, with **high confidence** in both requested subparts.

Lean coverage (partial). Lean proves d4=sqrt(8/3), d6=sqrt2, the exact cube edge and a strictly better eight-point configuration $cube_not_optimal$, hence the negative answer required by part a, plus a second dodecahedral counterexample and the icosahedral lower bound. Its d is defined as sSup (Separations n), but attainment and equivalence with the printed maximum are not yet proved. The remaining canonical obligations are exactly that maximum/attainment bridge for every n >= 2 and the unconditional sharp limit sqrt$n$dn=sqrt(8*pi/sqrt3). The matching d12 upper bound and a full Platonic classification are optional and are not continuation requirements.

18.1 Audited informal answer

Q804 — Complete solution

1. Restatement and convention

Let

$$S = S^2 = \{x \in \mathbb{R}^3 : \|x\| = 1\}.$$

For every finite set $X \subset S$ with at least two elements, define

$$\delta(X) = \min_{\substack{x,y \in X \\ x \neq y}} \|x - y\|,$$

where $\|\cdot\|$ is the ordinary Euclidean distance in $\mathbb{R}^3$. For $n \geq 2$, let

$$d(n) = \max_{\substack{X \subset S \\ |X| = n}} \delta(X).$$

The problem asks:

If there is a regular polyhedron with n vertices inscribed in S, is $d(n)$ equal to its edge length?

Determine the limit of $\sqrt{n}\, d(n)$ as $n \to \infty$. Luc Pommeret

There is no apparent typo. The only genuine convention issue is the phrase "regular polyhedron." I use its standard convex meaning, namely the five Platonic solids. If star polyhedra were admitted, the word "edge" would no longer necessarily refer to the shortest distance between vertices. The maximum defining $d(n)$ exists: on the compact space $(S^2)^n$, the function

$$(x_1, \ldots, x_n) \longmapsto \min_{i<j} \|x_i - x_j\|$$

is continuous.

1. Final result

Under the convex Platonic convention:

n	regular polyhedron	edge length on S^2	equal to $d(n)$?
4	tetrahedron	$\sqrt{\frac{8}{3}}$	yes
6	octahedron	$\sqrt{2}$	yes
8	cube	$\sqrt{\frac{4}{3}}$	no
12	icosahedron	$\sqrt{2-\frac{2}{\sqrt{5}}}$	yes
20	dodecahedron	$\sqrt{2-\frac{2\sqrt{5}}{3}}$	no.

Thus the answer to part a, as a general assertion, is no. For part b,

$$\boxed{\lim_{n\to\infty}\sqrt{n}\,d(n)=\sqrt{\frac{8\pi}{\sqrt{3}}}}$$

and numerically

$$\sqrt{\frac{8\pi}{\sqrt{3}}}=3.809251227\ldots.$$

Part a: the regular polyhedra

1. Inner-product formulation

For a finite $X\subset S^2$, put

$$s(X)=\max_{\substack{x,y\in X\\ x\neq y}}\langle x,y\rangle.$$

Since

$$\|x-y\|^2=2-2\langle x,y\rangle$$

for unit vectors,

$$\delta(X)^2=2-2s(X).$$

Thus maximizing the minimum distance is equivalent to minimizing the largest inner product. We first record an elementary fact. Lemma 3.1: an obtuse set has at most $d+1$ elements Let $v_1,\ldots,v_m$ be nonzero vectors in $\mathbb{R}^d$ such that

$$\langle v_i, v_j \rangle < 0 \qquad (i \neq j).$$

Then $m \leq d+1$. Proof If $m \geq d+2$, then the m vectors

$$(v_i, 1) \in \mathbb{R}^{d+1}$$

are linearly dependent. Hence there are real numbers a_i, not all zero, such that

$$\sum_i a_i v_i = 0, \qquad \sum_i a_i = 0.$$

The second equality implies that the coefficients have both signs. Set

$$P = \{i : a_i > 0\}, \qquad N = \{j : a_j < 0\}, \qquad b_j = -a_j > 0.$$

Then

$$w := \sum_{i \in P} a_i v_i = \sum_{j \in N} b_j v_j.$$

Taking the scalar product of the two expressions for w,

$$\|w\|^2 = \sum_{\substack{i \in P \\ j \in N}} a_i b_j \langle v_i, v_j \rangle < 0,$$

a contradiction. □

1. The tetrahedron: $d(4) = \sqrt{8/3}$

Let $x_1, \ldots, x_4 \in S^2$. Then

$$0 \leq \left\| \sum_{i=1}^{4} x_i \right\|^2 = 4 + 2 \sum_{1 \leq i < j \leq 4} \langle x_i, x_j \rangle.$$

Consequently,

$$\sum_{i<j} \langle x_i, x_j \rangle \geq -2.$$

There are six pairs, so at least one pair satisfies

$$\langle x_i, x_j \rangle \geq -\frac{1}{3}.$$

Thus

$$s(X) \geq -\frac{1}{3}$$

and

$$\delta(X)^2 \leq 2 + \frac{2}{3} = \frac{8}{3}.$$

Equality is attained by

$$\frac{1}{\sqrt{3}}(1,1,1), \quad \frac{1}{\sqrt{3}}(1,-1,-1), \quad \frac{1}{\sqrt{3}}(-1,1,-1), \quad \frac{1}{\sqrt{3}}(-1,-1,1),$$

whose distinct pairwise inner products are all $-1/3$. These are the vertices of a regular tetrahedron. Hence

$$\boxed{d(4) = \sqrt{\frac{8}{3}}}.$$

1. The octahedron: $d(6) = \sqrt{2}$

For six points $x_1, \ldots, x_6 \in S^2$, Lemma 3.1 shows that they cannot all have negative pairwise inner products, since an obtuse set in $\mathbb{R}^3$ has at most four elements. Therefore

$$s(X) \geq 0,$$

and hence

$$\delta(X)^2 \leq 2.$$

The six points

$$\{\pm e_1, \pm e_2, \pm e_3\}$$

have pairwise inner products 0 or -1. Thus their largest distinct inner product is 0. They form a regular octahedron, and

$$\boxed{d(6) = \sqrt{2}}.$$

1. The cube is not optimal for $n = 8$

The vertices of the inscribed cube are

$$\frac{1}{\sqrt{3}}(\pm 1, \pm 1, \pm 1).$$

The largest inner product between distinct vertices is $1/3$, attained by adjacent vertices. Its edge length is therefore

$$\sqrt{2 - \frac{2}{3}} = \sqrt{\frac{4}{3}}.$$

We now give a strictly better eight-point configuration. Set

$$u = \frac{1}{1 + 2\sqrt{2}} = \frac{2\sqrt{2} - 1}{7}, \qquad h = \sqrt{u}, \qquad r = \sqrt{1 - u}.$$

For $k = 0, 1, 2, 3$, define

$$A_k = \left(r \cos \frac{k\pi}{2}, r \sin \frac{k\pi}{2}, h \right)$$

and

$$B_k = \left(r \cos \left(\frac{\pi}{4} + \frac{k\pi}{2} \right), r \sin \left(\frac{\pi}{4} + \frac{k\pi}{2} \right), -h \right).$$

These are the vertices of a square antiprism on S^2. Within either square, the largest distinct inner product is

$$h^2 + r^2 \cos \frac{\pi}{2} = u.$$

Between the two squares, the largest inner product occurs when the azimuthal difference is $\pi/4$:

$$r^2 \cos \frac{\pi}{4} - h^2 = \frac{1 - u}{\sqrt{2}} - u.$$

The definition $1 - u = 2\sqrt{2}\, u$ gives

$$\frac{1 - u}{\sqrt{2}} - u = u.$$

All other inner products are smaller. Hence this configuration has

$$s(X) = u.$$

Since

$$u = \frac{1}{1 + 2\sqrt{2}} < \frac{1}{3},$$

we obtain

$$d(8) \geq \sqrt{2 - 2u} > \sqrt{2 - \frac{2}{3}} = \sqrt{\frac{4}{3}}.$$

Therefore

$$\boxed{\text{the inscribed cube is not optimal.}}$$

Numerically, the square-antiprism construction gives minimum distance

$$\sqrt{2 - \frac{2}{1 + 2\sqrt{2}}} = 1.2155625\ldots,$$

whereas the cube edge has length $1.1547005\ldots$.

1. The icosahedron: exact optimality for $n = 12$

We use a self-contained linear-programming bound on the sphere. Lemma 7.1: spherical polynomial bound Let P_k be the degree-k Legendre polynomial, normalized by $P_k(1) = 1$. Suppose

$$f(t) = \sum_{k=0}^{m} f_k P_k(t),$$

where

$$f_0 > 0, \qquad f_k \geq 0 \quad (k \geq 1).$$

Suppose also that

$$f(t) \leq 0 \qquad (-1 \leq t \leq s).$$

Then every N-point set $X \subset S^2$ satisfying

$$\langle x, y \rangle \leq s \qquad (x \neq y)$$

obeys

$$N \leq \frac{f(1)}{f_0}.$$

Proof The spherical-harmonic addition formula gives

$$P_k(\langle x, y\rangle) = \frac{4\pi}{2k+1} \sum_{\ell=-k}^{k} Y_{k,\ell}(x) Y_{k,\ell}(y)$$

for any real orthonormal basis of degree-k spherical harmonics. Consequently,

$$\sum_{x,y \in X} P_k(\langle x, y\rangle) = \frac{4\pi}{2k+1} \sum_{\ell=-k}^{k} \left(\sum_{x \in X} Y_{k,\ell}(x) \right)^2 \geq 0.$$

Therefore

$$\sum_{x,y \in X} f(\langle x, y\rangle) \geq f_0 N^2.$$

On the other hand, every off-diagonal term is nonpositive, so

$$\sum_{x,y \in X} f(\langle x, y\rangle) \leq N f(1).$$

Thus $f_0 N^2 \leq N f(1)$, proving the result. □ Application to twelve points Put

$$a = \frac{1}{\sqrt{5}}, \qquad f(t) = (t+1)(t+a)^2(t-a).$$

For $-1 \leq t \leq a$,

$$t+1 \geq 0, \qquad (t+a)^2 \geq 0, \qquad t-a \leq 0,$$

so $f(t) \leq 0$. Using

$$\begin{aligned} P_0(t) &= 1, \\ P_1(t) &= t, \\ P_2(t) &= \frac{3t^2-1}{2}, \\ P_3(t) &= \frac{5t^3-3t}{2}, \\ P_4(t) &= \frac{35t^4-30t^2+3}{8}, \end{aligned}$$

one obtains

$$f = \sum_{k=0}^{4} f_k P_k$$

with

$$\begin{aligned}
f_0 &= \frac{2(5+\sqrt{5})}{75}, \\
f_1 &= \frac{2(5+\sqrt{5})}{25}, \\
f_2 &= \frac{46+14\sqrt{5}}{105}, \\
f_3 &= \frac{2(5+\sqrt{5})}{25}, \\
f_4 &= \frac{8}{35}.
\end{aligned}$$

All coefficients are positive, and direct calculation gives

$$f(1) = 12 f_0.$$

The lemma therefore shows that a code with all distinct inner products at most $1/\sqrt{5}$ has at most twelve points. We must still rule out twelve points having all inner products strictly less than $1/\sqrt{5}$. Suppose such a configuration existed. For $N = 12$, both inequalities in the proof of Lemma 7.1 would be equalities:

$$f_0 N^2 = \sum_{i,j} f(\langle x_i, x_j \rangle) = N f(1).$$

Since every off-diagonal summand is nonpositive, each one must vanish. The roots of f below a are

$$-1, \qquad -a.$$

Thus every distinct pair would have negative inner product, contradicting Lemma 3.1. Hence every twelve-point set satisfies

$$s(X) \geq \frac{1}{\sqrt{5}}.$$

Now let

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad R = \sqrt{1+\varphi^2}.$$

The twelve normalized vectors

$$\frac{1}{R}(0, \pm 1, \pm\varphi), \qquad \frac{1}{R}(\pm 1, \pm\varphi, 0), \qquad \frac{1}{R}(\pm\varphi, 0, \pm 1)$$

are the vertices of a regular icosahedron. Their distinct pairwise inner products belong to

$$\left\{ -1, -\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right\}.$$

Thus their largest inner product is $1/\sqrt{5}$, and

$$\boxed{d(12) = \sqrt{2 - \frac{2}{\sqrt{5}}}}.$$

So the regular icosahedron is optimal.

1. The dodecahedron is not optimal for $n = 20$

Let

$$\varphi = \frac{1+\sqrt{5}}{2}.$$

A standard realization of the dodecahedron is

$$\frac{1}{\sqrt{3}} \begin{pmatrix} (\pm 1, \pm 1, \pm 1), \\ (0, \pm\varphi^{-1}, \pm\varphi), \\ (\pm\varphi^{-1}, \pm\varphi, 0), \\ (\pm\varphi, 0, \pm\varphi^{-1}) \end{pmatrix},$$

with the signs chosen independently in each row. Using

$$\varphi^2 = \varphi + 1, \qquad \varphi + \varphi^{-1} = \sqrt{5},$$

a direct calculation shows that the distinct pairwise inner products are

$$-1, \quad -\frac{\sqrt{5}}{3}, \quad -\frac{1}{3}, \quad \frac{1}{3}, \quad \frac{\sqrt{5}}{3}.$$

Thus the largest inner product is $\sqrt{5}/3$, and the dodecahedral edge length is

$$\sqrt{2-\frac{2\sqrt{5}}{3}}.$$

We construct twenty points with a smaller maximum inner product. Let

$$N=(0,0,1), \qquad S=(0,0,-1).$$

For $k=0,\ldots,5$, put

$$E_k=\left(\cos\frac{k\pi}{3},\sin\frac{k\pi}{3},0\right),$$

and

$$U_k=\left(\sqrt{\frac{2}{3}}\cos\left(\frac{\pi}{6}+\frac{k\pi}{3}\right),\sqrt{\frac{2}{3}}\sin\left(\frac{\pi}{6}+\frac{k\pi}{3}\right),\frac{1}{\sqrt{3}}\right),$$

$$L_k=\left(\sqrt{\frac{2}{3}}\cos\left(\frac{\pi}{6}+\frac{k\pi}{3}\right),\sqrt{\frac{2}{3}}\sin\left(\frac{\pi}{6}+\frac{k\pi}{3}\right),-\frac{1}{\sqrt{3}}\right).$$

The set

$$X=\{N,S\}\cup\{E_k,U_k,L_k:0\le k\le 5\}$$

has twenty elements. The relevant maximum inner products are:

pair type	maximum inner product
E_j,E_k	$\frac{1}{2}$
U_j,U_k or L_j,L_k	$\frac{2}{3}$
E_j,U_k or E_j,L_k	$\sqrt{\frac{2}{3}}\cos\frac{\pi}{6}=\frac{1}{\sqrt{2}}$
U_j,L_k	$\frac{1}{3}$
N,U_k or S,L_k	$\frac{1}{\sqrt{3}}$.

All remaining types give no larger value. Therefore

$$s(X)=\frac{1}{\sqrt{2}}, \qquad \delta(X)=\sqrt{2-\sqrt{2}}.$$

But

$$\frac{1}{\sqrt{2}}<\frac{\sqrt{5}}{3},$$

because $1/2 < 5/9$. Hence

$$\sqrt{2-\sqrt{2}}\sqrt{2-\frac{2\sqrt{5}}{3}}.$$

Thus

$$\boxed{\text{the inscribed regular dodecahedron is not optimal.}}$$

Part b: the asymptotic limit

1. Angular separation

Let

$$\rho(x,y) = \arccos\langle x, y\rangle$$

be geodesic distance on the unit sphere, and define

$$\theta_n = \max_{\substack{X\subset S^2\\ |X|=n}} \min_{x\neq y} \rho(x,y).$$

Because

$$\|x-y\| = 2\sin\frac{\rho(x,y)}{2},$$

we have

$$d(n) = 2\sin\frac{\theta_n}{2}.$$

The caps of angular radius $\theta_n/2$ centered at an optimal set of n points have disjoint interiors. Since such a cap has area

$$2\pi\left(1-\cos\frac{\theta_n}{2}\right),$$

we obtain

$$n\,2\pi\left(1-\cos\frac{\theta_n}{2}\right) \le 4\pi.$$

In particular,

$$\theta_n \longrightarrow 0.$$

Consequently,

$$\frac{d(n)}{\theta_n} = \frac{2\sin(\theta_n/2)}{\theta_n} \longrightarrow 1.$$

It remains to determine the asymptotics of θ_n.

1. The planar packing constant

We first prove the needed planar result rather than invoking it as a black box. Lemma 10.1: density of a periodic separated set Let $Y \subset \mathbb{R}^2$ be periodic under a lattice Λ, with q points modulo Λ, and suppose distinct points of Y are at distance at least 1. If F is a fundamental parallelogram of Λ, then

$$\frac{q}{|F|} \leq \frac{2}{\sqrt{3}}.$$

Proof Consider the Voronoi tessellation of Y modulo Λ, hence on the flat torus $\mathbb{R}^2/\Lambda$. Let the q Voronoi cells have $k_1, \ldots, k_q$ sides. Every Voronoi side is the perpendicular bisector of two sites at distance at least 1. Hence each cell contains the disk of radius $1/2$ centered at its site. A convex k-gon containing a disk of radius r centered at an interior point has area at least

$$a_r(k) = kr^2 \tan\frac{\pi}{k}.$$

Indeed, move its supporting lines inward until tangent to the disk. If the consecutive outward normals have angular gaps $\alpha_1, \ldots, \alpha_m$, with $m \leq k$ and $\sum \alpha_i = 2\pi$, the resulting tangential polygon has area

$$r^2 \sum_{i=1}^{m} \tan\frac{\alpha_i}{2} \geq mr^2 \tan\frac{\pi}{m} \geq kr^2 \tan\frac{\pi}{k}.$$

For $r = 1/2$, put

$$a(k) = \frac{k}{4}\tan\frac{\pi}{k}.$$

As a function of real $x \geq 3$,

$$a'(x) = \frac{1}{4}\left(\tan\frac{\pi}{x} - \frac{\pi}{x}\sec^2\frac{\pi}{x}\right) < 0,$$

and

$$a''(x) = \frac{\pi^2}{2x^3}\sec^2\frac{\pi}{x}\tan\frac{\pi}{x} > 0.$$

Thus a is decreasing and convex. Let $V, E, F = q$ be the numbers of vertices, edges and cells in the torus tessellation. Every vertex has degree at least three, so

$$3V \leq 2E.$$

Euler's relation on the torus gives

$$V - E + q = 0.$$

Hence $E \leq 3q$, and therefore

$$\sum_{i=1}^{q} k_i = 2E \leq 6q.$$

By convexity and monotonicity of a,

$$\begin{aligned} |F| &\geq \sum_{i=1}^{q} a(k_i) \\ &\geq q\, a\left(\frac{1}{q}\sum_i k_i\right) \\ &\geq q\, a(6) = q\frac{\sqrt{3}}{2}. \end{aligned}$$

Thus $q/|F| \leq 2/\sqrt{3}$. ☐ Lemma 10.2: separated points in a planar domain Let $\Omega \subset \mathbb{R}^2$ be bounded, measurable, and have boundary of area zero. Let $M_\Omega(r)$ be the largest cardinality of a subset of Ω whose mutual distances are at least r. Then

$$\boxed{\lim_{r\downarrow 0} r^2 M_\Omega(r) = \frac{2}{\sqrt{3}}|\Omega|.}$$

Proof: lower bound Take the triangular lattice of minimum spacing r. Its fundamental parallelogram has area

$$\frac{\sqrt{3}}{2}r^2.$$

Averaging over all translations of this lattice, the average number of lattice points lying in Ω equals

$$\frac{|\Omega|}{(\sqrt{3}/2)r^2} = \frac{2|\Omega|}{\sqrt{3}\,r^2}.$$

Some translation therefore contains at least this many points in Ω. Hence

$$\liminf_{r\downarrow 0} r^2 M_\Omega(r) \geq \frac{2}{\sqrt{3}}|\Omega|.$$

Proof: upper bound First consider a square of side L with separation 1. Repeat the configuration periodically with period $L+1$ in both coordinate directions. The resulting periodic set remains 1-separated, and therefore Lemma 10.1 gives

$$M_{[0,L]^2}(1) \leq \frac{2}{\sqrt{3}}(L+1)^2.$$

Now cover Ω, for arbitrary $\varepsilon > 0$, by finitely many grid squares with disjoint interiors and total area at most $|\Omega| + \varepsilon$. Assign boundary points to one square. Applying the preceding square estimate after scaling by $1/r$ gives

$$\limsup_{r\downarrow 0} r^2 M_\Omega(r) \leq \frac{2}{\sqrt{3}}(|\Omega| + \varepsilon).$$

Letting $\varepsilon \downarrow 0$ proves the upper bound. $\square$ The coefficient $2/\sqrt{3}$ is the point density of the triangular lattice with unit minimum spacing.

1. Transfer to a smooth surface

We prove a slightly more general statement. Proposition 11.1 Let M be a compact smooth Riemannian surface of area A. Let $P_M(r)$ be the maximum number of points of M with pairwise geodesic distance at least r. Then

$$\boxed{\lim_{r\downarrow 0} r^2 P_M(r) = \frac{2A}{\sqrt{3}}.}$$

Proof Fix $\eta > 0$. By using sufficiently small normal-coordinate neighborhoods and a finite geodesic subdivision of M, we may partition M, up to finitely many piecewise smooth boundaries, into cells

$$Q_j = \phi_j(\Omega_j), \qquad 1 \leq j \leq m,$$

such that for $u, v \in \Omega_j$,

$$(1-\eta)|u-v| \leq d_M(\phi_j(u), \phi_j(v)) \leq (1+\eta)|u-v|,$$

and for every measurable $E \subset \Omega_j$,

$$(1-\eta)|E| \le \operatorname{area}(\phi_j(E)) \le (1+\eta)|E|.$$

These estimates follow from the continuity of the metric tensor in normal coordinates; the cells may be chosen inside strongly convex coordinate balls. Upper bound Let $X \subset M$ be r-separated, and assign each boundary point to one cell. The preimages of the points of $X \cap Q_j$ are Euclidean $r/(1+\eta)$-separated. Lemma 10.2 gives

$$\limsup_{r\downarrow 0} r^2 |X \cap Q_j| \le \frac{2}{\sqrt{3}}(1+\eta)^2 |\Omega_j|.$$

Summing over j, and using

$$\sum_j |\Omega_j| \le \frac{A}{1-\eta},$$

we obtain

$$\limsup_{r\downarrow 0} r^2 P_M(r) \le \frac{2A}{\sqrt{3}} \frac{(1+\eta)^2}{1-\eta}.$$

Lower bound Remove from every Q_j a geodesic collar of width $r/2$ along its boundary, and denote the remaining set by $Q_{j,r}$. Its preimage $\Omega_{j,r}$ satisfies

$$|\Omega_{j,r}| \longrightarrow |\Omega_j|.$$

Inside $\Omega_{j,r}$, use a triangular lattice of spacing

$$\frac{r}{1-\eta}.$$

By the averaging argument from Lemma 10.2, we can choose at least

$$\frac{2}{\sqrt{3}} \frac{(1-\eta)^2}{r^2} |\Omega_{j,r}|$$

points. Their images are r-separated within Q_j. Points chosen in distinct cells are also r-separated: every path joining them must travel at least $r/2$ from the first point to the boundary of its cell and at least $r/2$ from the boundary of the second cell to the second point. Thus

$$\liminf_{r\downarrow 0} r^2 P_M(r) \ge \frac{2}{\sqrt{3}}(1-\eta)^2 \sum_j |\Omega_j| \ge \frac{2A}{\sqrt{3}} \frac{(1-\eta)^2}{1+\eta}.$$

Finally let $\eta \downarrow 0$. □ For the unit sphere, $A = 4\pi$, so

$$P_{S^2}(r) \sim \frac{8\pi}{\sqrt{3}\, r^2}.$$

1. From the packing count to θ_n

Set

$$C = \frac{8\pi}{\sqrt{3}}.$$

We have

$$r^2 P_{S^2}(r) \longrightarrow C.$$

By definition,

$$P_{S^2}(r) \geq n \quad \Longleftrightarrow \quad \theta_n \geq r.$$

This asymptotic inverse relation yields

$$n\theta_n^2 \longrightarrow C.$$

For completeness, fix $\varepsilon > 0$. For all sufficiently small r,

$$\frac{C-\varepsilon}{r^2} \leq P_{S^2}(r) \leq \frac{C+\varepsilon}{r^2}.$$

Since $P_{S^2}(\theta_n) \geq n$,

$$n\theta_n^2 \leq C + \varepsilon$$

for large n. Conversely, set

$$r_n = \sqrt{\frac{C-2\varepsilon}{n}}.$$

For large n,

$$P_{S^2}(r_n) \geq \frac{C-\varepsilon}{r_n^2} = n\,\frac{C-\varepsilon}{C-2\varepsilon} n.$$

Hence $\theta_n \geq r_n$, so

$$n\theta_n^2 \geq C - 2\varepsilon.$$

Letting $\varepsilon \downarrow 0$,

$$\sqrt{n}\,\theta_n \longrightarrow \sqrt{\frac{8\pi}{\sqrt{3}}}.$$

Finally, since $d(n)/\theta_n \to 1$,

$$\boxed{\lim_{n\to\infty} \sqrt{n}\,d(n) = \sqrt{\frac{8\pi}{\sqrt{3}}}.}$$

This agrees with the general best-packing formula for rectifiable sets, whose specialization to S^2 gives the same constant. arXiv

1. Boundary checks and normalization

For comparison:

$$d(2) = 2$$

using antipodal points, and

$$d(3) = \sqrt{3}$$

using an equilateral triangle on a great circle. Indeed,

$$0 \le \|x_1 + x_2 + x_3\|^2$$

implies that some pair has inner product at least $-1/2$. The radius-one hypothesis fixes the scale. On a sphere of radius R, all chord distances are multiplied by R, and the limiting constant becomes

$$R\sqrt{\frac{8\pi}{\sqrt{3}}}.$$

The distinction between chord and angular distance is essential for the exact finite values, but not for the leading asymptotic term, because

$$2\sin(\theta/2) \sim \theta \qquad (\theta \to 0).$$

Formal verification. Lean source; online type-check.

19 Question Q805

Informal verdict. A — Complete solution, with **high confidence** in the classification.

Lean coverage. Lean proves the complete classification of planar norms linearly equivalent to their duals, including the reflection and even-order rotational normal forms, automatic rotational symmetry, double duality, orthogonal classification, and power-bounded cosquare normal form.

19.1 Audited informal answer

Q805 — complete classification

1. Faithful restatement

Equip $\mathbb{R}^2$ with its usual Euclidean inner product. For a norm N on $\mathbb{R}^2$, define

$$N^*(x) = \sup_{y \in \mathbb{R}^2 \setminus \{0\}} \frac{\langle x, y \rangle}{N(y)}.$$

Characterize those norms N for which there is an invertible linear map $u \in GL_2(\mathbb{R})$ such that

$$N^* = N \circ u.$$

The PDF contains no subparts, footnotes, or additional qualifications. The absence of an absolute value in the displayed definition is harmless: replacing y by $-y$ shows that the displayed supremum equals the usual dual norm with $|\langle x, y \rangle|$. Luc Pommeret

1. Statement of the classification

For $\theta \in \mathbb{R}$, let R_θ denote counterclockwise rotation through angle θ, and let

$$F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

be reflection in the first coordinate axis. For a compact convex set $K \subset \mathbb{R}^2$ containing 0 in its interior, write

$$K^\circ = \{x \in \mathbb{R}^2 : \langle x, y \rangle \leq 1 \text{ for every } y \in K\}$$

for its Euclidean polar. Main theorem Let N be a norm on $\mathbb{R}^2$, and let

$$B = \{x : N(x) \leq 1\}$$

be its unit ball. The following are equivalent.

There exists $u \in GL_2(\mathbb{R})$ such that $N^* = N \circ u$.

There exists $L \in GL_2(\mathbb{R})$ such that, for $C = LB$, one of the following holds: Reflection type

$$C = FC^\circ. \tag{R}$$

Rotational type

$$C = R_{\pi/m} C^\circ \tag{C$_m$}$$

for some even integer $m \geq 2$.

In case (C_m), one automatically has

$$R_{2\pi/m} C = C. \tag{2.1}$$

Conversely, every centrally symmetric convex body satisfying either (R) or (C_m) is the unit ball of a norm with the required property. A norm may belong to both families; the theorem is an exhaustive classification, not necessarily a disjoint partition. More explicitly, if $C = LB$ and $C = QC^\circ$, where $Q = F$ or $Q = R_{\pi/m}$, then one may take

$$\boxed{u = L^{-1} Q L^{-T}.} \tag{2.2}$$

Thus every admissible norm can, after a linear change of coordinates, be made self-dual either by a reflection or by a half-step rotation relative to an even cyclic symmetry.

1. Translation into convex geometry

We begin with elementary polar calculus. Lemma 3.1 — Polar identities Let $K \subset \mathbb{R}^2$ be compact, convex, and contain 0 in its interior. For $A \in GL_2(\mathbb{R})$ and $r > 0$,

$$(AK)^\circ = A^{-T} K^\circ, \qquad (rK)^\circ = r^{-1} K^\circ, \qquad K^{\circ\circ} = K. \tag{3.1}$$

Proof The first two identities follow directly from the definition. For example,

$$x \in (AK)^\circ \iff \langle x, Ay \rangle \leq 1 \quad (y \in K)$$

which is equivalent to

$$A^T x \in K^\circ,$$

hence $x \in A^{-T} K^\circ$. For the bipolar identity, $K \subset K^{\circ\circ}$ is immediate. Let $x \notin K$. Since K is compact, there is $z \in K$ minimizing $\|x - z\|_2$. Put $p = x - z \neq 0$. Convexity implies, for every $y \in K$,

$$\langle p, y - z \rangle \leq 0.$$

Indeed, $z + t(y - z) \in K$ for $0 \leq t \leq 1$, and differentiating the squared distance from x at $t = 0$ gives this inequality. Consequently,

$$\sup_{y \in K} \langle p, y \rangle \leq \langle p, z \rangle < \langle p, x \rangle.$$

Because 0 lies in the interior of K,

$$s := \sup_{y \in K} \langle p, y \rangle > 0.$$

Then $p/s \in K^\circ$, whereas

$$\left\langle \frac{p}{s}, x \right\rangle > 1.$$

Thus $x \notin K^{\circ\circ}$. Hence $K^{\circ\circ} = K$. □

Lemma 3.2 — Unit balls The unit ball of N^* is B°. The unit ball of $N \circ u$ is $u^{-1} B$. Consequently,

$$N^* = N \circ u \quad \Longleftrightarrow \quad B^\circ = u^{-1} B \quad \Longleftrightarrow \quad B = u B^\circ. \tag{3.2}$$

Proof By homogeneity,

$$N^*(x) = \sup_{y \in B} \langle x, y \rangle.$$

Thus $N^*(x) \leq 1$ precisely when $\langle x, y \rangle \leq 1$ for all $y \in B$, namely when $x \in B^\circ$. The second assertion is immediate:

$$N(ux) \leq 1 \iff ux \in B \iff x \in u^{-1} B.$$

This proves (3.2). □ So the problem is precisely:

Characterize the centrally symmetric convex bodies $B \subset \mathbb{R}^2$ that are linearly equivalent to their polar.

The nontrivial part is to put the linear equivalence into canonical form.

1. A compactness constraint on the duality map

Let

$$\mathrm{Aut}(B) = \{A \in GL_2(\mathbb{R}) : AB = B\}.$$

Lemma 4.1 The group $\mathrm{Aut}(B)$ is compact. Proof Choose $0 < r < R$ such that

$$rD \subset B \subset RD,$$

where D is the Euclidean unit disk. If $A \in \mathrm{Aut}(B)$ and $\|x\|_2 \leq 1$, then $rx \in B$, so

$$rAx = A(rx) \in B \subset RD.$$

Therefore

$$\|A\|_{\mathrm{op}} \leq \frac{R}{r}.$$

The same estimate applies to A^{-1}, since $A^{-1}B = B$. Hence the group is uniformly bounded and uniformly bounded away from singular matrices. It is closed in the space of 2×2 matrices and therefore compact by Heine–Borel. □ Now suppose that

$$B = uB^{\circ}. \tag{4.1}$$

Taking polars gives

$$B^{\circ} = u^{-T}B.$$

Substitution into (4.1) yields

$$B = uu^{-T}B.$$

Thus the matrix

$$S := uu^{-T} \tag{4.2}$$

belongs to $\mathrm{Aut}(B)$. In particular, all powers of S are bounded. This seemingly modest observation almost completely determines the possible congruence type of u.

1. Canonical form of an admissible duality

Write

$$H = \frac{u + u^T}{2}, \qquad A = \frac{u - u^T}{2}.$$

Thus H is symmetric and A is alternating. In dimension two,

$$A = tJ, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

after a suitable congruence and scaling. Recall that under a change $u \mapsto TuT^T$,

$$(TuT^T)(TuT^T)^{-T} = T(uu^{-T})T^{-1}. \tag{5.1}$$

Hence power-boundedness of uu^{-T} is preserved. Also, replacing u by $-u$ does not change uB°, since $B^\circ = -B^\circ$. Lemma 5.1 — Bilinear normal form Suppose $u \in GL_2(\mathbb{R})$ and uu^{-T} is power-bounded. Then there exist $T \in GL_2(\mathbb{R})$, $c > 0$, and $Q \in O(2)$ such that

$$Q = cTuT^T. \tag{5.2}$$

Moreover:

if $\det u > 0$, Q may be chosen to be a rotation;

if $\det u < 0$, Q may be chosen to be a reflection.

Proof We distinguish the possible ranks and signatures of H. Case 1: H is definite After replacing u by $-u$ if necessary and applying a congruence, we may suppose

$$H = I.$$

Then

$$u = I + tJ = \begin{pmatrix} 1 & -t \\ t & 1 \end{pmatrix}.$$

Since

$$(I + tJ)(I + tJ)^T = (1 + t^2)I,$$

the matrix

$$Q = \frac{I + tJ}{\sqrt{1 + t^2}}$$

is an element of $SO(2)$.

Case 2: $H = 0$ Then $u = tJ$ with $t \neq 0$. After multiplying by a positive scalar and possibly replacing u by $-u$, this becomes J, which is rotation through $\pi/2$.

Case 3: H has rank one After congruence and changing the sign of u, we may write

$$u = \begin{pmatrix} 1 & -t \\ t & 0 \end{pmatrix}, \qquad t \neq 0.$$

A direct computation gives

$$uu^{-T} = \begin{pmatrix} -1 & -2/t \\ 0 & -1 \end{pmatrix}.$$

This is a nontrivial Jordan block with eigenvalue -1, so its powers have off-diagonal entries growing linearly in magnitude. It is not power-bounded. Hence this case is impossible.

Case 4: H is indefinite After congruence,

$$u = \begin{pmatrix} 1 & -t \\ t & -1 \end{pmatrix}, \qquad t \neq \pm 1.$$

Here

$$uu^{-T} = \frac{1}{1-t^2} \begin{pmatrix} 1+t^2 & 2t \\ 2t & 1+t^2 \end{pmatrix}. \tag{5.3}$$

If $t \neq 0$, its trace is

$$\frac{2(1+t^2)}{1-t^2}.$$

Its absolute value is strictly larger than 2, so the two eigenvalues are real reciprocal numbers, one of absolute value greater than 1. Thus the powers are unbounded. Consequently $t = 0$, and

$$u = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = F,$$

an orthogonal reflection. These four cases exhaust the possible symmetric parts H, proving the lemma. □

Corollary 5.2 — Orthogonal reduction of the body If $B = uB^\circ$, then there are $L \in GL_2(\mathbb{R})$ and $Q \in O(2)$ such that

$$C := LB \qquad \text{satisfies} \qquad C = QC^\circ. \tag{5.4}$$

Proof Choose T, c, Q as in Lemma 5.1 and put

$$L = \sqrt{c}T.$$

Then, using $B = uB^\circ$,

$$QC^\circ = cTuT^T\left(c^{-1/2}T^{-T}B^\circ\right) = \sqrt{c}TuB^\circ = \sqrt{c}TB = C.$$

□ Thus it remains only to understand orthogonal self-polarities.

1. Classification of the orthogonal map

Suppose that

$$C = QC^\circ, \qquad Q \in O(2). \tag{6.1}$$

Since Q is orthogonal,

$$(QK)^\circ = QK^\circ.$$

Taking polars in (6.1) gives

$$C^\circ = QC. \tag{6.2}$$

Combining (6.1) and (6.2),

$$Q^2C = C. \tag{6.3}$$

6.1. Orientation-reversing case If $\det Q = -1$, then Q is a Euclidean reflection. After an orthogonal change of coordinates it is F. This gives precisely the reflection class

$$C = FC^\circ.$$

No further restriction follows, because $F^2 = I$.

6.2. Orientation-preserving case Suppose that

$$Q = R_\alpha.$$

Equation (6.3) gives

$$R_{2\alpha}C = C. \tag{6.4}$$

Also $C = -C$, so $R_\pi C = C$. Let G be the subgroup of rotations generated by $R_{2\alpha}$ and R_π. Infinite symmetry group If G is infinite, it is dense in the circle group of rotations. Since C is closed and invariant under every element of G, it is invariant under the closure of G, hence under every Euclidean rotation. Therefore $C = rD$ for some $r > 0$. But $C = QC^\circ$ then gives

$$rD = \frac{1}{r}D,$$

so $r = 1$. Thus $C = D$. The disk already belongs to (C_2), since

$$D = R_{\pi/2}D^\circ.$$

Finite symmetry group Suppose G is finite. Then for some even integer m,

$$G = \left\langle R_{2\pi/m} \right\rangle. \tag{6.5}$$

The integer m is even because $R_\pi \in G$. Put

$$\delta = \frac{2\pi}{m}.$$

Since $Q^2 = R_{2\alpha} \in G$, the class of α modulo δ satisfies

$$2\alpha \equiv 0 \pmod{\delta}.$$

Thus

$$\alpha \equiv 0 \quad \text{or} \quad \alpha \equiv \frac{\delta}{2} \pmod{\delta}. \tag{6.6}$$

If $\alpha \equiv 0 \pmod{\delta}$, then $Q \in G$, hence $QC = C$. From $C = QC^\circ$ we get $C = C^\circ$. But the only Euclidean self-polar convex body is the unit disk. Indeed, if $C = C^\circ$, then every $x \in C$ satisfies

$$\|x\|_2^2 = \langle x, x\rangle \le 1,$$

so $C \subset D$. Taking polars gives

$$D = D^\circ \subset C^\circ = C,$$

hence $C = D$. In the other case, choose $H \in G$ such that

$$HQ = R_{\delta/2} = R_{\pi/m}.$$

Since $HC = C$, equation $C = QC^\circ$ becomes

$$C = R_{\pi/m}C^\circ.$$

This is exactly the rotational normal form (C_m). Moreover,

$$R_{2\pi/m}C = C$$

follows from (6.3). This proves the main theorem.

1. A completely explicit radial-function characterization

The preceding theorem reduces the problem to two canonical functional equations. For a centrally symmetric convex body C, let

$$e_\theta = (\cos\theta, \sin\theta),$$

and define its radial and support functions by

$$\rho(\theta) = \max\{r \geq 0 : re_\theta \in C\},$$

$$h(\theta) = \max_{x\in C}\langle e_\theta, x\rangle.$$

The radial function of the polar is

$$\rho_{C^\circ}(\theta) = \frac{1}{h(\theta)}. \tag{7.1}$$

Indeed, $te_\theta \in C^\circ$ exactly when

$$t\,h(\theta) \leq 1.$$

If $C = QC^\circ$, then $C^\circ = Q^T C$, and therefore

$$\frac{1}{h(\theta)} = \rho_{C^\circ}(\theta) = \rho_C(Qe_\theta). \tag{7.2}$$

Hence: Reflection type

$$\boxed{h(\theta)\rho(-\theta) = 1 \quad \text{for every } \theta.} \tag{7.3}$$

Rotational type For some even $m \geq 2$,

$$\boxed{h(\theta)\rho\Big(\theta + \frac{\pi}{m}\Big) = 1 \quad \text{for every } \theta.} \tag{7.4}$$

Equation (7.4) automatically implies

$$\rho\left(\theta+\frac{2\pi}{m}\right)=\rho(\theta).$$

Conversely, either (7.3) or (7.4) implies the corresponding equality of bodies, because two star-shaped bodies containing the origin are equal when their radial functions agree. This gives a literal one-variable parametrization. Namely, begin with a positive continuous π-periodic function ρ such that

$$C_\rho=\{re_\theta : 0\le r\le \rho(\theta)\}$$

is convex. Compute

$$h_\rho(\theta)=\max_{\varphi\in\mathbb{R}}\rho(\varphi)\cos(\theta-\varphi).$$

Then C_ρ gives a solution exactly when, after a linear change of coordinates, either

$$h_\rho(\theta)\rho(-\theta)=1,$$

or, for an even m,

$$h_\rho(\theta)\rho\left(\theta+\frac{\pi}{m}\right)=1.$$

1. Examples and boundary cases

8.1. Ellipsoidal norms Let A be symmetric positive definite and

$$N_A(x)=\sqrt{x^TAx}.$$

Then

$$N_A^*(x)=\sqrt{x^TA^{-1}x}.$$

Taking

$$u=A^{-1}$$

gives

$$N_A(ux)=\sqrt{x^TA^{-1}x}=N_A^*(x).$$

After a linear change of coordinates, the unit ball becomes the Euclidean disk.

8.2. Regular even polygons Let $m \geq 4$ be even and let $P_m(r)$ be the regular m-gon with vertices

$$re_{2\pi k/m}, \qquad 0 \leq k < m.$$

Its polar has vertices in the directions

$$\frac{(2k+1)\pi}{m}$$

and circumradius

$$\frac{1}{r\cos(\pi/m)}.$$

Choose

$$r = \frac{1}{\sqrt{\cos(\pi/m)}}.$$

Then the polar has the same circumradius, and

$$P_m(r)^\circ = R_{\pi/m} P_m(r).$$

Since $P_m(r)$ is invariant under $R_{2\pi/m}$, it also satisfies

$$P_m(r) = R_{\pi/m} P_m(r)^\circ.$$

Thus every even $m \geq 4$ occurs. The case $m = 2$ is the quarter-turn relation

$$C = R_{\pi/2} C^\circ.$$

These are traditionally the planar bodies self-polar with respect to an alternating area form; the Euclidean disk and a suitably scaled regular hexagon are examples.

8.3. The square and the ℓ_∞ norm For

$$N(x, y) = \max(|x|, |y|),$$

the dual norm is

$$N^*(x, y) = |x| + |y|.$$

With

$$u = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \sqrt{2}\, R_{\pi/4},$$

we have

$$N(u(x,y)) = \max(|x-y|, |x+y|) = |x| + |y|.$$

Thus $N^* = N \circ u$. This is the $m = 4$ rotational case before normalization of the scalar factor. For $N = \|\cdot\|_1$, one may take

$$u = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

8.4. A genuinely reflection-type example Let

$$v_0 = \left(1, \frac{1}{5}\right), \qquad v_1 = (1, 0), \qquad v_2 = \left(\frac{3}{4}, -\frac{5}{4}\right),$$

and put

$$v_{i+3} = -v_i \qquad (0 \le i \le 2).$$

Let

$$K = \operatorname{conv}\{v_0, \dots, v_5\}.$$

The polar vertices corresponding to the first three consecutive edges are

$$w_0 = (1, 0), \qquad w_1 = \left(1, -\frac{1}{5}\right), \qquad w_2 = \left(-\frac{3}{4}, -\frac{5}{4}\right),$$

and the remaining three are their negatives. These are exactly the images under F of the six vertices of K. Hence

$$K^\circ = FK$$

and therefore

$$K = FK^\circ.$$

This example does not possess an orientation-preserving duality. Indeed, the absolute determinants of consecutive vertex pairs are

$$\frac{1}{5}, \quad \frac{5}{4}, \quad \frac{7}{5}, \quad \frac{1}{5}, \quad \frac{5}{4}, \quad \frac{7}{5}. \tag{8.1}$$

A linear automorphism of K must permute the vertices cyclically or anticyclically and has determinant of absolute value 1, since it preserves the area of K. The cyclic word in (8.1), whose three entries are distinct, has only the two rotational symmetries corresponding to the identity and central inversion and has no reflection symmetry. Consequently,

$$\operatorname{Aut}(K) = \{\pm I\}.$$

If $vK^\circ = K$, then, since $FK^\circ = K$,

$$vF^{-1}K = K.$$

Thus $vF \in \operatorname{Aut}(K)$, so $v = \pm F$. Every duality has negative determinant. This shows that the reflection branch cannot be omitted.

8.5. The classical ℓ_p^2 norms Let $1 < p < \infty$, and let $q = p/(p-1)$. The dual of ℓ_p^2 is ℓ_q^2. The norm ℓ_p^2 has the required property exactly when $p = 2$. Indeed, if $p \neq 2$, one of p, q is greater than 2 and the other lies between 1 and 2. Suppose $p > 2 > q$. Near $(1, 0)$, the boundary of the p-ball is

$$x = (1 - |y|^p)^{1/p}.$$

For $p > 2$, this is twice continuously differentiable at $y = 0$. The corresponding boundary of the q-ball is

$$x = (1 - |y|^q)^{1/q} = 1 - \frac{1}{q}|y|^q + o(|y|^q),$$

and

$$\frac{x(y) - x(0) - x'(0)y}{y^2} \sim -\frac{1}{q}|y|^{q-2} \longrightarrow -\infty.$$

Thus the q-sphere is not C^2 at the coordinate axes, whereas the p-sphere is C^2 everywhere. An invertible linear map preserves the property of being a C^2 embedded curve, so the two balls cannot be linearly equivalent. Hence, among $1 < p < \infty$, only $p = 2$ works. The endpoint norms $p = 1, \infty$ do work, as shown above. This also demonstrates that smoothness and strict convexity alone are far from sufficient.

1. Polygonal decision algorithm

For a polygonal norm the property is exactly decidable using finite linear algebra. Assume that the nonredundant vertices of B are

$$v_0, \ldots, v_{2n-1}$$

in cyclic order.

For each i, compute the polar vertex w_i by solving

$$\langle w_i, v_i \rangle = 1, \qquad \langle w_i, v_{i+1} \rangle = 1.$$

Any linear map carrying B° onto B must carry the cyclically ordered polar vertices either cyclically or anticyclically onto the vertices of B.

For each sign $\varepsilon \in \{1, -1\}$ and each shift k, define the unique linear map u satisfying

$$uw_0 = v_k, \qquad uw_1 = v_{k+\varepsilon}.$$

Verify whether

$$uw_i = v_{k+\varepsilon i}$$

for every i, with indices modulo $2n$.

There are $4n$ candidates, and each is checked using $O(n)$ exact arithmetic operations. Thus the naive complexity is $O(n^2)$. If the vertices are rational, the entire test can be carried out over $\mathbb{Q}$. No computation is used in the proof of the classification theorem.

1. Topological and logical assumptions

The proof uses only finite-dimensional Euclidean topology. Separation The identity $K^{\circ\circ} = K$ uses finite-dimensional strict separation of a point from a compact convex set. It was proved directly using a nearest point, so no Hahn–Banach theorem is required. Compactness Compactness is used in three places:

the unit ball of a norm is compact by Heine–Borel;

support functions attain their maxima;

$\mathrm{Aut}(B)$ is compact, which forces uu^{-T} to be power-bounded.

No weak topology or Banach–Alaoglu theorem occurs. Countability and sequential arguments No countability assumption is imposed on any family. The only sequence considered is the countable sequence of powers of a rotation. When this subgroup is dense, invariance passes to its closure by taking convergent sequences. This is legitimate because $SO(2)$ and $\mathbb{R}^2$ are metrizable; no nets or general nonsequential compactness arguments are involved. Choice No form of the axiom of choice beyond finite choices is needed. There is no Zorn lemma, no selection of representatives from an infinite family, and no nonconstructive compactness principle. Smoothness and strict convexity Neither is assumed. The regular polygon examples show that nonsmooth and non-strictly-convex norms occur naturally. Invertibility of u The explicit hypothesis $u \in GL_2(\mathbb{R})$ is actually redundant. If a linear map u satisfied $N^* = N \circ u$, then $ux = 0$ would imply $N^*(x) = 0$, hence $x = 0$. Thus u is injective and therefore invertible in dimension two. Dependence on dimension The reduction of the symmetric part of u into the four cases above is specifically two-dimensional. In higher dimensions the congruence classification of bilinear forms and the possible compact "cosquares" uu^{-T} are substantially richer.

1. Final form of the answer

A norm N on $\mathbb{R}^2$ is linearly isometric to its dual precisely when, after a nonsingular linear change of coordinates, its unit ball C is one of the following:

$$\boxed{C = FC^\circ}$$

for a reflection F, or

$$\boxed{C = R_{\pi/m}C^\circ}$$

for some even integer $m \geq 2$; in the latter case C necessarily has m-fold rotational symmetry. Equivalently, in radial/support coordinates, the complete alternatives are

$$\boxed{h(\theta)\rho(-\theta) = 1}$$

or

$$\boxed{h(\theta)\rho\Big(\theta + \frac{\pi}{m}\Big) = 1 \quad (m \geq 2 \text{ even}).}$$

These conditions are both necessary and sufficient.

Formal verification. Lean source; online type-check.

20 Question Q830

Informal verdict. **A — Complete solution**, with **high confidence** under the response's explicit convention that a solution is a C^2 function on an open interval and that $yy'' = f$ means $y(x)y''(x) = f(x)$.

Lean coverage. Full canonical coverage under the explicit classical C2 pointwise convention. For part a, Lean proves that the explicit inverse-integral family gives entire solutions of y y_second = 1 and that every interval solution is its restriction, hence exactly the maximal solutions. For the open-ended part b, Lean proves forcing continuity and zero compatibility, simple and tangent zero jet constraints, the fundamental integral identity, the full f = 0 classification, and a smooth nonuniqueness example. The answer's additional local sufficiency and endpoint-asymptotic theorems are optional strengthenings, not requirements of the printed question.

20.1 Audited informal answer

Q830

1. Faithful restatement and interpretation

The PDF asks:

a Determine the maximal solutions of $y\,y'' = 1$. b Given a map $f : \mathbb{R} \to \mathbb{R}$, what can be said about the solutions of $y\,y'' = f$?

There is no further hypothesis, footnote, or contextual qualification attached to Q830. The printed equation in part b is $y\,y'' = f$, not $y\,y'' = f(y)$; I therefore read it pointwise as

$$y(x)y''(x) = f(x).$$

Part b genuinely leaves unspecified the domain and regularity of a "solution," whether global or local solutions are sought, and what kind of classification is expected. Luc Pommeret+1 I use the following natural convention. A classical solution on an open interval I is a function $y \in C^2(I, \mathbb{R})$ satisfying the equation at every point of I. A solution is maximal if it is not the restriction of a classical solution to a strictly larger open interval.

1. Part a: complete classification of the maximal solutions

Define

$$\Psi(q) := \sqrt{2} \int_0^q e^{s^2}\, ds = \sqrt{\frac{\pi}{2}}\,\mathrm{erfi}(q), \qquad q \in \mathbb{R}.$$

The function Ψ is odd, C^∞, strictly increasing, and maps $\mathbb{R}$ bijectively onto $\mathbb{R}$. Theorem 1 The maximal classical solutions of

$$y\,y'' = 1$$

are precisely the functions parametrized by

$$\boxed{x = \tau + a\Psi(q), \qquad y = \varepsilon a e^{q^2}, \qquad q \in \mathbb{R},}$$

where

$$a > 0, \qquad \tau \in \mathbb{R}, \qquad \varepsilon \in \{-1, 1\}.$$

Equivalently,

$$\boxed{y(x) = \varepsilon a \exp\left(\Psi^{-1}\left(\frac{x-\tau}{a}\right)^2\right), \qquad x \in \mathbb{R}.}$$

Every maximal solution is therefore defined on all of $\mathbb{R}$. Its sign is constant, and $|y|$ has a unique minimum a, attained at $x = \tau$. Proof Let $y \in C^2(I)$ solve $yy'' = 1$ on an interval I. Since the left-hand side would vanish wherever $y = 0$, one has

$$y(x) \neq 0 \qquad (x \in I).$$

Thus y has a constant sign. Put

$$\varepsilon = \operatorname{sgn} y, \qquad u = \varepsilon y = |y| > 0.$$

Because $y = \varepsilon u$ and $y'' = \varepsilon u''$,

$$u\,u'' = 1.$$

Multiplication by u'/u, or direct differentiation, gives the first integral

$$\frac{d}{dx}\left(\frac{1}{2}u'^2 - \log u\right) = u'\left(u'' - \frac{1}{u}\right) = 0.$$

Hence

$$\frac{1}{2}u'^2 - \log u = C$$

for some constant C. Set

$$a = e^{-C} > 0.$$

Then

$$u'^2 = 2\log\frac{u}{a}.$$

In particular,

$$u \geq a.$$

Define

$$q = \frac{u'}{\sqrt{2}}.$$

The first integral becomes

$$u = ae^{q^2}.$$

Moreover,

$$q' = \frac{u''}{\sqrt{2}} = \frac{1}{\sqrt{2}\,u}.$$

Consequently,

$$\frac{d}{dx}\big(a\Psi(q)\big) = a\sqrt{2}\,e^{q^2}q' = a\sqrt{2}\,\frac{u}{a}\frac{1}{\sqrt{2}u} = 1.$$

Therefore there exists $\tau \in \mathbb{R}$ such that

$$x = \tau + a\Psi(q).$$

Since $y = \varepsilon u$, this gives

$$y = \varepsilon ae^{q^2}.$$

Conversely, let

$$x = \tau + a\Psi(q), \qquad y = \varepsilon ae^{q^2}.$$

Because

$$\frac{dx}{dq} = a\sqrt{2}\,e^{q^2} > 0,$$

q is a smooth function of x. Differentiating,

$$y' = \frac{2\varepsilon aqe^{q^2}}{a\sqrt{2}e^{q^2}} = \varepsilon\sqrt{2}\,q,$$

and hence

$$y'' = \varepsilon\sqrt{2}\,\frac{dq}{dx} = \frac{\varepsilon}{ae^{q^2}} = \frac{1}{y}.$$

Thus $yy'' = 1$. Finally, $\Psi(q) \to \pm\infty$ as $q \to \pm\infty$, so the parametrization gives a solution on all of $\mathbb{R}$. Every local solution obtained above is a restriction of one of these entire solutions. Therefore these and only these are maximal. $\square$ Initial-value parametrization For arbitrary data

$$y(x_0) = y_0 \neq 0, \qquad y'(x_0) = v_0,$$

the parameters are

$$\varepsilon = \operatorname{sgn} y_0, \qquad q_0 = \frac{\varepsilon v_0}{\sqrt{2}},$$

$$a = |y_0|e^{-v_0^2/2}, \qquad \tau = x_0 - a\Psi(q_0).$$

Thus every admissible initial condition determines exactly one maximal solution.

1. Part b: general classical-solution theory

Because f is allowed to be completely arbitrary, no universal explicit formula can exist. Indeed, for every prescribed $Y \in C^2(I)$, choosing

$$f = YY''$$

makes Y a solution. Thus arbitrary growth, oscillation, zero sets, and flat behavior can be encoded in f. There is nevertheless a precise structural theory. It consists of:

ordinary existence and uniqueness wherever $y \neq 0$;

exact compatibility conditions at zeros;

an integral identity controlling consecutive zeros;

a sharp classification of finite singular endpoints when f has a nonzero limit.

3.1 Immediate necessary conditions Let $y \in C^2(I)$ satisfy

$$yy'' = f.$$

Then necessarily

$$\boxed{f \in C(I)}$$

because $f = yy''$ is a product of continuous functions. Moreover,

$$\boxed{y(c) = 0 \quad \Longrightarrow \quad f(c) = 0.}$$

Thus

$$Z(y) \subseteq Z(f).$$

In particular, if f is discontinuous on every nonempty open interval—for example $f = \mathbf{1}_{\mathbb{Q}}$—then there is no classical solution on any nonempty open interval.

3.2 Regular branches: existence, uniqueness, and continuation Assume now that $f \in C(I)$, where I is an open interval. Theorem 2 Given

$$x_0 \in I, \qquad y_0 \neq 0, \qquad v_0 \in \mathbb{R},$$

there exists a unique maximal zero-free branch

$$y : J \longrightarrow \mathbb{R}, \qquad x_0 \in J \subset I,$$

such that

$$y(x_0) = y_0, \qquad y'(x_0) = v_0, \qquad yy'' = f.$$

On J, it satisfies the Volterra equation

$$\boxed{y(x) = y_0 + v_0(x - x_0) + \int_{x_0}^{x} (x - t)\frac{f(t)}{y(t)}\,dt.}$$

If a finite endpoint $c \in I$ of J exists, then

$$\boxed{\liminf_{x \to c,\ x \in J} |y(x)| = 0.}$$

Proof Away from $y = 0$, the equation is the first-order system

$$\begin{cases} y' = v, \\ v' = \dfrac{f(x)}{y}. \end{cases}$$

Near (x_0, y_0, v_0), choose a rectangle on which

$$|y| \geq \frac{|y_0|}{2}.$$

On this rectangle the vector field

$$F(x,y,v) = \left(v, \frac{f(x)}{y}\right)$$

is continuous and locally Lipschitz in (y,v). The usual contraction argument applied to

$$Y(x) = Y_0 + \int_{x_0}^{x} F(t, Y(t))\,dt$$

gives a unique local solution. Uniqueness allows all local continuations that remain in $y \neq 0$ to be united into one maximal zero-free branch. Integrating $y'' = f/y$ twice gives the displayed Volterra equation. Now suppose that $c \in I$ is a finite endpoint and that

$$\liminf_{x \to c} |y(x)| > 0.$$

Then $|y| \geq \delta > 0$ near c. Since f is bounded near c,

$$|y''| \leq \frac{\sup |f|}{\delta}.$$

Hence y' has a finite limit at c, and then y also has a finite nonzero limit. The local existence theorem at c extends the branch past c, contradicting maximality. □ A zero-free branch need not have a unique maximal classical continuation through a zero; this distinction is essential and will be illustrated below.

3.3 A fundamental integral identity For every classical solution,

$$(yy')' = y'^2 + yy'' = y'^2 + f.$$

Hence, for $a < b$ in the solution interval,

$$\boxed{\int_a^b f(x)\,dx = y(b)y'(b) - y(a)y'(a) - \int_a^b y'(x)^2\,dx.}$$

Consequences If $y(a) = y(b) = 0$, then

$$\boxed{\int_a^b f(x)\,dx = -\int_a^b y'(x)^2\,dx \leq 0.}$$

Equality occurs exactly when $y \equiv 0$ on $[a,b]$, in which case $f \equiv 0$ there. Thus a nonzero solution cannot have two zeros $a < b$ unless

$$\int_a^b f < 0.$$

In particular, if $f \geq 0$, a solution cannot have two distinct zeros unless it vanishes identically, together with f, between those zeros. If y is T-periodic, then

$$\int_x^{x+T} f(t)\,dt = -\int_x^{x+T} y'(t)^2\,dt \leq 0.$$

1. Exact local analysis at a zero

Let c be a zero of a classical solution, and write

$$p = y'(c), \qquad q = y''(c).$$

Taylor's formula with Peano remainder gives

$$y(c+h) = ph + \frac{1}{2}qh^2 + o(h^2).$$

4.1 Necessary jet conditions Simple zero: $p \neq 0$ Since $y''(c+h) \to q$,

$$\frac{f(c+h)}{h} = \frac{y(c+h)}{h}\,y''(c+h) \longrightarrow pq.$$

Therefore f is differentiable at c, and

$$\boxed{f'(c) = y'(c)y''(c) = pq.}$$

Tangential zero: $p = 0$ Then

$$\frac{f(c+h)}{h^2} = \frac{y(c+h)}{h^2}\,y''(c+h) \longrightarrow \frac{1}{2}q^2.$$

Hence

$$\boxed{f(c+h) = \frac{1}{2}q^2h^2 + o(h^2).}$$

If $q \neq 0$, then f is positive on both sides of c, sufficiently close to c. If $q = 0$, necessarily

$$\boxed{f(c+h) = o(h^2).}$$

If zeros of y accumulate at c, then necessarily

$$y'(c) = y''(c) = 0.$$

Indeed, a simple or nondegenerate quadratic zero is isolated.

4.2 Complete characterization of simple zeros Theorem 3 Let f be defined near c. There exists a local classical solution with

$$y(c) = 0, \qquad y'(c) = p \neq 0$$

if and only if

$$\boxed{g(h) := \frac{f(c+h)}{h}}$$

extends continuously to $h = 0$. When this condition holds, for every prescribed $p \neq 0$ there is a unique such local solution, and

$$\boxed{y''(c) = \frac{g(0)}{p}.}$$

Proof Necessity was proved above. For sufficiency, translate c to 0, so that

$$f(h) = h\, g(h)$$

with g continuous. Seek

$$y(h) = h\, z(h), \qquad z(0) = p.$$

The equation is equivalent to

$$y''(h) = \frac{g(h)}{z(h)}.$$

Twice integrating from 0, using $y(0) = 0$ and $y'(0) = p$, gives

$$hz(h) = ph + \int_0^h (h-s)\frac{g(s)}{z(s)}\, ds,$$

or equivalently

$$\boxed{z(h) = p + h\int_0^1 (1-r)\frac{g(hr)}{z(hr)}\, dr.}$$

Choose $\delta > 0$ and consider the complete metric space

$$\mathcal{B} = \left\{ z \in C([-\delta, \delta]) : \|z - p\|_\infty \le \frac{|p|}{2} \right\}.$$

Every $z \in \mathcal{B}$ satisfies $|z| \ge |p|/2$. Define

$$(Tz)(h) = p + h \int_0^1 (1-r) \frac{g(hr)}{z(hr)} \, dr.$$

If $M = \sup_{|h| \le \delta} |g(h)|$, then

$$\|Tz - p\|_\infty \le \frac{\delta M}{|p|}.$$

For sufficiently small δ, T maps $\mathcal{B}$ into itself. Moreover,

$$\|Tz_1 - Tz_2\|_\infty \le \frac{2\delta M}{p^2} \|z_1 - z_2\|_\infty.$$

Again decreasing δ, this is a strict contraction. Hence T has a unique fixed point z. Set $y(h) = hz(h)$. The integral equation becomes

$$y(h) = ph + \int_0^h (h-s) \frac{g(s)}{z(s)} \, ds.$$

The integrand is continuous, so $y \in C^2$ and

$$y''(h) = \frac{g(h)}{z(h)}.$$

Therefore

$$y(h)y''(h) = hz(h) \frac{g(h)}{z(h)} = hg(h) = f(h).$$

Finally,

$$y''(0) = \frac{g(0)}{p}.$$

Uniqueness follows from uniqueness of the fixed point. □ Thus simple zeros are completely understood.

4.3 Complete characterization of nondegenerate tangent zeros Theorem 4 Let f be defined near c. There is a local classical solution satisfying

$$y(c) = y'(c) = 0, \qquad y''(c) \ne 0$$

if and only if

$$\boxed{\frac{f(c+h)}{h^2}}$$

extends continuously to $h = 0$ with a strictly positive value

$$A > 0.$$

When this holds, there are exactly two local solutions with a nondegenerate tangent zero at c. They satisfy respectively

$$\boxed{y''(c) = +\sqrt{2A} \qquad \text{and} \qquad y''(c) = -\sqrt{2A}.}$$

Proof Necessity follows from

$$\frac{f(c+h)}{h^2} \longrightarrow \frac{1}{2}y''(c)^2.$$

For sufficiency and uniqueness, translate c to 0 and write

$$f(h) = h^2 g(h), \qquad g(0) = A > 0.$$

Choose

$$B = \pm\sqrt{\frac{A}{2}}.$$

We seek a solution of the form

$$y(h) = h^2 z(h), \qquad z(h) \longrightarrow B.$$

It suffices first to work on $h > 0$. Put

$$h = e^{-t}, \qquad Z(t) = z(e^{-t}), \qquad G(t) = g(e^{-t}).$$

Then $t \to +\infty$ corresponds to $h \to 0^+$. A direct calculation gives

$$y'' = Z'' - 3Z' + 2Z,$$

where primes now denote t-derivatives. The equation becomes

$$Z\big(Z'' - 3Z' + 2Z\big) = G(t).$$

Write

$$Z = B + w.$$

Because $A = 2B^2$, the equation is

$$w'' - 3w' + 4w = H(t, w),$$

where

$$H(t, \xi) = \frac{G(t)}{B + \xi} - 2B + 2\xi.$$

As $t \to \infty$,

$$H(t, 0) \to 0, \qquad \partial_\xi H(t, 0) = -\frac{G(t)}{B^2} + 2 \to 0.$$

Let

$$\omega = \frac{\sqrt{7}}{2}, \qquad K(r) = \frac{1}{\omega} e^{-3r/2} \sin(\omega r), \qquad r \geq 0.$$

For a continuous function R tending to 0, the unique solution tending to 0 at $+\infty$ of

$$w'' - 3w' + 4w = R$$

is

$$w(t) = \int_0^\infty K(r) R(t + r)\, dr.$$

Consider the Banach space $C_0([T, \infty))$ of continuous functions tending to 0, with the supremum norm, and define

$$(\mathcal{T} w)(t) = \int_0^\infty K(r) H(t + r, w(t + r))\, dr.$$

Since $H(t, 0) \to 0$ and its derivative with respect to its second variable tends uniformly to 0 near $\xi = 0$, one may choose T large and a small ball

$$\|w\|_\infty \leq \rho < \frac{|B|}{2}$$

such that $\mathcal{T}$ maps this ball into itself and is a strict contraction. Therefore there is a unique $w \in C_0([T, \infty))$ solving the equation. Returning to $h = e^{-t}$ gives a unique right-hand solution with

$$\frac{y(h)}{h^2} \to B.$$

The same argument applied to $f(-h)$ gives the unique left-hand solution. They glue in C^2 at 0, because

$$y(0) = y'(0) = 0, \qquad y''(0) = 2B = \pm\sqrt{2A}.$$

Conversely, any solution with this second derivative gives $Z(t) \to B$ and $Z'(t) \to 0$; variation of constants forces it to satisfy the same fixed-point equation, so uniqueness follows from the contraction. □ For example, when $f(x) = x^2$, the two tangent solutions are

$$y(x) = \pm\frac{x^2}{\sqrt{2}}.$$

4.4 Degenerate zeros The remaining case is

$$y(c) = y'(c) = y''(c) = 0.$$

Necessarily

$$f(c+h) = o(h^2),$$

but no finite-order jet condition can classify all such solutions. For instance, with

$$\phi(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

the function ϕ is C^∞ and flat at 0. Taking

$$f = \phi\phi'',$$

both ϕ and various sign-gluings of ϕ provide solutions with a completely flat zero. Thus the degenerate case genuinely contains arbitrary smooth flat behavior.

1. Finite endpoints of zero-free branches

Let y be a maximal zero-free branch on an interval ending at a finite point c, and put

$$u = |y| > 0.$$

Then

$$u\,u'' = f.$$

5.1 Positive limiting forcing repels the solution from zero Theorem 5 If

$$f(c) > 0,$$

then c cannot be a finite endpoint of a maximal zero-free branch. Consequently, if

$$f(x) > 0 \qquad (x \in I),$$

then every solution with nonzero initial value exists uniquely on all of I. In particular, if $I = \mathbb{R}$, every such solution is entire. Proof Suppose, for definiteness, that the branch is defined to the left of c. By continuity, for some $m > 0$,

$$f(x) \geq m$$

near c. Hence

$$u'' \geq \frac{m}{u} > 0.$$

Thus u' is increasing. The continuation criterion gives

$$\liminf_{x \uparrow c} u(x) = 0.$$

If $u' \geq 0$ somewhere sufficiently near c, monotonicity of u' would imply that u is thereafter nondecreasing and bounded away from zero. Thus $u' < 0$ throughout a final interval. Multiplying

$$u'' \geq \frac{m}{u}$$

by the negative quantity u' reverses the inequality:

$$u'u'' \leq m\frac{u'}{u}.$$

Integration from x_0 to x gives

$$\frac{1}{2}\big(u'(x)^2 - u'(x_0)^2\big) \leq m \log \frac{u(x)}{u(x_0)}.$$

The right-hand side tends to $-\infty$ along any sequence for which $u(x) \to 0$, whereas the left-hand side is bounded below by $-u'(x_0)^2/2$, a contradiction.

□ Part a is the special case $f \equiv 1$, for which the additional autonomous first integral makes the solutions explicitly computable.

5.2 Negative limiting forcing: universal collapse law Theorem 6 Suppose c is a finite right endpoint of a maximal zero-free branch and

$$f(c) = -a < 0.$$

Then

$$|y(x)| \longrightarrow 0 \qquad (x \uparrow c),$$

and

$$\boxed{|y(x)| \sim (c-x)\sqrt{2a \log \frac{1}{c-x}}.}$$

Moreover,

$$\boxed{|y'(x)| \sim \sqrt{2a \log \frac{1}{c-x}} \longrightarrow +\infty.}$$

The analogous statement holds at a left endpoint. Proof Put $u = |y| > 0$. Near c,

$$u'' = \frac{f(x)}{u} < 0,$$

so u is strictly concave and u' is decreasing. The continuation criterion gives $\liminf u = 0$. Concavity then implies that u is eventually strictly decreasing and

$$u(x) \to 0.$$

Let $x = x(s)$ be the inverse of $s = u(x)$ near $s = 0$, and define

$$p(s) = -u'(x(s)) > 0.$$

Since $ds/dx = u' = -p$,

$$\frac{dp}{ds} = \frac{u''}{p}.$$

Therefore

$$\frac{d}{ds}p(s)^2 = 2u'' = \frac{2f(x(s))}{s}.$$

Fix $s_0 > 0$. Integration gives

$$p(s)^2 = p(s_0)^2 - 2\int_s^{s_0} \frac{f(x(r))}{r}\,dr.$$

Since $f(x(r)) \to -a$,

$$\boxed{p(s)^2 \sim 2a \log \frac{1}{s}.}$$

Next,

$$c - x(s) = \int_0^s \frac{dr}{p(r)}.$$

Using the preceding asymptotic,

$$c - x(s) \sim \frac{1}{\sqrt{2a}} \int_0^s \frac{dr}{\sqrt{\log(1/r)}}.$$

By l'Hôpital's rule,

$$\int_0^s \frac{dr}{\sqrt{\log(1/r)}} \sim \frac{s}{\sqrt{\log(1/s)}}.$$

Thus

$$c - x(s) \sim \frac{s}{\sqrt{2a\log(1/s)}}.$$

Taking logarithms shows

$$\log \frac{1}{s} \sim \log \frac{1}{c - x(s)}.$$

Inverting the previous relation yields

$$s \sim (c - x(s))\sqrt{2a \log \frac{1}{c - x(s)}}.$$

Since $s = u(x) = |y(x)|$, this is the asserted formula. The derivative asymptotic follows from the formula for $p(s)$. □

5.3 The case $f(c) = 0$ is genuinely nonuniform Neither of the previous alternatives extends to $f(c) = 0$. For example, for $x < 0$, let

$$y(x) = -x + (-x)^{3/2}.$$

Then

$$y''(x) = \frac{3}{4\sqrt{-x}}, \qquad f(x) = y(x)y''(x) = \frac{3}{4}\big(\sqrt{-x} - x\big).$$

Defining $f(0) = 0$ gives a continuous forcing, while

$$y(x) \to 0, \qquad y'(x) \to -1, \qquad y''(x) \to +\infty.$$

Thus the branch cannot extend as a C^2 solution through 0, despite $f(0) = 0$ and $f \geq 0$ near 0. This shows that strict positivity in Theorem 5 is necessary.

1. Exhaustive global structure and failure of uniqueness at flat zeros

Let $y \in C^2(I)$ be a solution and let

$$Z = y^{-1}(0).$$

Then Z is closed. On every connected component of $I \setminus Z$,

y has constant sign;

it is the unique regular branch determined by any one of its nonzero initial data.

Conversely, all classical solutions are obtained by choosing such regular branches on the components of $I \setminus Z$, setting $y = 0$ on $Z \subseteq f^{-1}(0)$, and imposing C^2-matching at the boundary points. The simple and nondegenerate quadratic matching conditions are exactly those in Theorems 3 and 4. Degenerate flat zeros allow additional freedom. In particular, maximal classical continuation through a zero need not be unique. Take again

$$\phi(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0, \end{cases} \qquad f = \phi\phi''.$$

Define

$$y_1(x) = \phi(x)$$

and

$$y_2(x) = \begin{cases} \phi(x), & x \geq 0, \\ -\phi(x), & x < 0. \end{cases}$$

All derivatives of ϕ vanish at 0, so both $y_1, y_2 \in C^\infty(\mathbb{R})$. For $x \neq 0$,

$$y_2''(x) = \operatorname{sgn}(x)\phi''(x),$$

and consequently

$$y_2 y_2'' = \phi\phi'' = f.$$

The two solutions agree on $[0, \infty)$ but differ on $(-\infty, 0)$. Thus the same regular branch and the same initial data at any $x_0 > 0$ admit two distinct entire continuations through the flat zero at 0.

1. Boundary cases

$f \equiv 0$ Every classical solution of

$$yy'' = 0$$

is affine:

$$\boxed{y(x) = Ax + B.}$$

Indeed, $y'' = 0$ wherever $y \neq 0$. It also vanishes on the interior of the zero set of y, and continuity of y'' then implies $y'' = 0$ at boundary points of that zero set. $f \equiv -1$ This gives the opposite behavior from part a. Define

$$\Theta(q) = \sqrt{2}\int_0^q e^{-s^2}\,ds = \sqrt{\frac{\pi}{2}}\,\operatorname{erf}(q).$$

Its range is

$$\left(-\sqrt{\frac{\pi}{2}}, \sqrt{\frac{\pi}{2}}\right).$$

All maximal solutions are

$$x = \tau + a\Theta(q), \qquad y = \varepsilon a e^{-q^2}, \qquad q \in \mathbb{R},$$

with $a > 0$ and $\varepsilon = \pm 1$. Their maximal domain is the finite interval

$$\boxed{\left(\tau - a\sqrt{\frac{\pi}{2}},\ \tau + a\sqrt{\frac{\pi}{2}}\right).}$$

At both endpoints $y \to 0$ and $|y'| \to \infty$, exactly as predicted by Theorem 6. Initial data at a zero For

$$f(x) = x^2,$$

both

$$y_+(x) = \frac{x^2}{\sqrt{2}}, \qquad y_-(x) = -\frac{x^2}{\sqrt{2}}$$

satisfy the equation and have

$$y_\pm(0) = y'_\pm(0) = 0.$$

Thus the hypothesis $y_0 \neq 0$ in the ordinary uniqueness theorem cannot be omitted.

1. Final classification

For part a, every maximal solution is entire and is given uniquely by

$$\boxed{x = \tau + a\sqrt{2}\int_0^q e^{s^2}\,ds, \qquad y = \varepsilon a e^{q^2}, \quad a > 0,\ \tau \in \mathbb{R},\ \varepsilon = \pm 1.}$$

For part b, under the natural classical interpretation $y \in C^2$:

f must be continuous on the solution interval.

Away from zeros of y, the equation has the usual unique maximal zero-free branch for every nonzero initial value.

Zeros can occur only where $f = 0$.

Simple zeros are possible exactly when $f(x)/(x-c)$ extends continuously.

Nondegenerate tangent zeros are possible exactly when $f(x)/(x-c)^2$ extends continuously with a positive value; there are then exactly two local tangent solutions.

Degenerate flat zeros may exhibit nonunique continuation.

Between two zeros,

$$\int_a^b f = -\int_a^b y'^2 \leq 0.$$

A finite zero-free endpoint with $f(c) > 0$ is impossible.

If $f(c) < 0$, the branch collapses according to

$$|y(x)| \sim |x-c|\sqrt{2|f(c)|\log\frac{1}{|x-c|}}.$$

These statements exhaust the universal structure available for an arbitrary prescribed forcing f.

Formal verification. Lean source; online type-check.

21 Question Q831

Informal verdict. **A — Complete solution**, with **high confidence**.

Lean coverage. Lean unconditionally proves existence and uniqueness of the universal integer polynomial for every n, m, and i. It also kernel-checks, for every positive m and primitive m-th root, the finite root-of-unity coefficient and expanded-sum formulas, which apply over the complex numbers for every canonical m >= 2, plus the complete m = 2 family, boundary cases, Frobenius specialization, and worked examples. The unformalized companion/resultant and Newton/partition presentations are alternative descriptions, not additional requirements of Q831.

21.1 Audited informal answer

Q831 — Complete solution

1. Faithful restatement

Let $\Sigma_1, \dots, \Sigma_n$ be the elementary symmetric polynomials in $X_1, \dots, X_n$. For each integer $m \geq 2$, express explicitly

$$\Sigma_i(X_1^m, X_2^m, \dots, X_n^m)$$

in terms of the original elementary symmetric polynomials $\Sigma_1(X), \dots, \Sigma_n(X)$. The PDF contains no subparts or additional hypotheses. There is no apparent typo. The only minor ambiguity is the word "explicit"; below it is answered in three strong senses: a closed coefficient formula, an integral determinant/resultant formula, and a finite partition formula. Luc Pommeret

1. Notation and main theorem

Write

$$e_r = \Sigma_r(X_1, \dots, X_n), \qquad e_0 = 1,$$

and set $e_r = 0$ when $r < 0$ or $r > n$. Define

$$e_i^{[m]} := \Sigma_i(X_1^m, \dots, X_n^m).$$

Thus

$$e_i^{[m]} = \sum_{1\le j_1<\cdots<j_i\le n} (X_{j_1}\cdots X_{j_i})^m.$$

Theorem For every $n \ge 1$, $m \ge 1$, and $0 \le i \le n$, there is a unique polynomial

$$\Phi_{m,i}^{(n)} \in \mathbb{Z}[Y_1, \ldots, Y_n]$$

such that

$$e_i^{[m]} = \Phi_{m,i}^{(n)}(e_1, \ldots, e_n).$$

It has the following explicit descriptions. 2.1 Root-of-unity formula Let ζ be any primitive m-th root of unity, and put $Y_0 = 1$. Then

$$\boxed{\Phi_{m,i}^{(n)}(Y_1, \ldots, Y_n) = (-1)^{(m+1)i} \sum_{\substack{0\le k_0,\ldots,k_{m-1}\le n\\ k_0+\cdots+k_{m-1}=mi}} \zeta^{\sum_{r=0}^{m-1} rk_r} \prod_{r=0}^{m-1} Y_{k_r}.} \tag{2.1}$$

Although roots of unity occur in the displayed expression, all its coefficients are rational integers; it is independent of the chosen primitive root ζ. Equivalently, if

$$E(z) = 1 + Y_1 z + \cdots + Y_n z^n,$$

then

$$\boxed{\Phi_{m,i}^{(n)} = (-1)^{(m+1)i}[z^{mi}] \prod_{r=0}^{m-1} E(\zeta^r z).} \tag{2.2}$$

2.2 Companion-matrix and resultant formulas Let

$$P_Y(Z) = Z^n - Y_1 Z^{n-1} + Y_2 Z^{n-2} - \cdots + (-1)^n Y_n,$$

and let $C = C(Y)$ be its companion matrix:

$$C = \begin{pmatrix} 0 & 0 & \cdots & 0 & (-1)^{n+1}Y_n \\ 1 & 0 & \cdots & 0 & (-1)^n Y_{n-1} \\ 0 & 1 & \cdots & 0 & (-1)^{n-1}Y_{n-2} \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & Y_1 \end{pmatrix}.$$

Then

$$\boxed{\det(TI_n - C^m) = \sum_{i=0}^{n} (-1)^i \Phi_{m,i}^{(n)} T^{n-i}.} \tag{2.3}$$

Consequently,

$$\boxed{\Phi_{m,i}^{(n)} = (-1)^i [T^{n-i}] \det(TI_n - C^m).} \tag{2.4}$$

Equivalently, with the convention that the resultant of a monic polynomial P with G is the determinant of multiplication by G on the quotient by P,

$$\boxed{\mathrm{Res}_Z\big(Z^n - Y_1 Z^{n-1} + \cdots + (-1)^n Y_n,\, T - Z^m\big) = \sum_{i=0}^{n} (-1)^i \Phi_{m,i}^{(n)} T^{n-i}.} \tag{2.5}$$

This is a single integral determinant and involves neither division nor roots of unity.

1. Proof of the theorem

3.1 The companion-matrix identity Work first in the universal polynomial ring

$$A = \mathbb{Z}[Y_1, \dots, Y_n].$$

The matrix C is the matrix, in the basis

$$1, Z, \dots, Z^{n-1},$$

of multiplication by Z in the free A-module

$$A[Z]/(P_Y).$$

Indeed, multiplication by Z shifts the first $n-1$ basis vectors, while the relation

$$Z^n = Y_1 Z^{n-1} - Y_2 Z^{n-2} + \cdots + (-1)^{n+1} Y_n$$

gives precisely the last column of C. Now specialize

$$Y_r = e_r(X_1, \dots, X_n).$$

Then

$$P_Y(Z) = \prod_{j=1}^{n} (Z - X_j).$$

Over the field $K = \mathbb{Q}(X_1, \ldots, X_n)$, the elements $X_1, \ldots, X_n$ are pairwise distinct, and the Chinese remainder theorem gives

$$K[Z]/(P_Y) \cong \prod_{j=1}^{n} K, \qquad f(Z) \longmapsto \big(f(X_1), \ldots, f(X_n)\big).$$

Under this isomorphism, multiplication by Z becomes multiplication by the diagonal matrix

$$\operatorname{diag}(X_1, \ldots, X_n).$$

Therefore multiplication by Z^m has eigenvalues

$$X_1^m, \ldots, X_n^m,$$

and hence

$$\det(TI_n - C^m) = \prod_{j=1}^{n} (T - X_j^m).$$

Expanding the right-hand side gives

$$\prod_{j=1}^{n} (T - X_j^m) = \sum_{i=0}^{n} (-1)^i \Sigma_i(X_1^m, \ldots, X_n^m) T^{n-i}.$$

Comparing coefficients proves (2.3)–(2.5), and in particular proves the existence of integral polynomials $\Phi_{m,i}^{(n)}$. 3.2 Uniqueness The elementary symmetric polynomials $e_1, \ldots, e_n$ are algebraically independent over $\mathbb{Z}$. Here is a direct verification. Use the lexicographic order $X_1 > \cdots > X_n$. Then

$$\operatorname{LM}(e_r) = X_1 X_2 \cdots X_r.$$

Consequently, for $\alpha = (\alpha_1, \ldots, \alpha_n)$,

$$\operatorname{LM}\left(e_1^{\alpha_1} \cdots e_n^{\alpha_n}\right) = X_1^{b_1} \cdots X_n^{b_n}, \qquad b_j = \sum_{r=j}^{n} \alpha_r.$$

The vector $(b_1, \ldots, b_n)$ determines α, since

$$\alpha_j = b_j - b_{j+1}, \qquad b_{n+1} = 0.$$

Thus distinct monomials in the e_j's have distinct leading monomials after substitution, so no nonzero polynomial in $Y_1, \ldots, Y_n$ can vanish after substituting $Y_j = e_j$. This proves uniqueness. 3.3 Derivation of the root-of-unity formula We have

$$E(z) := \sum_{k=0}^{n} e_k z^k = \prod_{j=1}^{n} (1 + X_j z).$$

For a primitive m-th root ζ,

$$\prod_{r=0}^{m-1} (1 - \zeta^r u) = 1 - u^m.$$

Taking $u = -zX_j$, we obtain

$$\prod_{r=0}^{m-1} (1 + \zeta^r z X_j) = 1 - (-zX_j)^m = 1 + (-1)^{m+1} z^m X_j^m.$$

Therefore

$$\begin{aligned}\prod_{r=0}^{m-1} E(\zeta^r z) &= \prod_{r=0}^{m-1} \prod_{j=1}^{n} (1 + \zeta^r z X_j) \\ &= \prod_{j=1}^{n} \left(1 + (-1)^{m+1} z^m X_j^m\right) \\ &= \sum_{i=0}^{n} (-1)^{(m+1)i} e_i^{[m]} z^{mi}.\end{aligned}$$

Hence

$$e_i^{[m]} = (-1)^{(m+1)i} [z^{mi}] \prod_{r=0}^{m-1} E(\zeta^r z),$$

which is (2.2). Expanding each factor,

$$E(\zeta^r z) = \sum_{k=0}^{n} e_k \zeta^{rk} z^k,$$

gives (2.1). The left side is already known from the companion formula to be an integral polynomial in the e_j's. By algebraic independence, the root-of-unity expression is therefore precisely the same integral polynomial $\Phi^{(n)}_{m,i}$. This also proves that it is independent of the chosen primitive root.

1. A formula involving only integer sums and partitions

The determinant formula is division-free. There is also a fully expanded formula through the power sums. Let

$$p_q = X_1^q + \cdots + X_n^q.$$

For $q \geq 1$, define the Newton polynomial

$$\boxed{N_q(Y) = \sum_{\substack{a_1,\ldots,a_q \geq 0 \\ a_1+2a_2+\cdots+qa_q=q}} (-1)^{q-A} \frac{q(A-1)!}{a_1! \cdots a_q!} Y_1^{a_1} \cdots Y_q^{a_q},} \tag{4.1}$$

where

$$A = a_1 + \cdots + a_q, \qquad Y_r = 0 \quad (r > n).$$

Then

$$p_q = N_q(e_1, \ldots, e_n).$$

The apparently fractional coefficient in (4.1) is an integer, because

$$\frac{q(A-1)!}{a_1! \cdots a_q!} = \sum_{\substack{1 \leq s \leq q \\ a_s > 0}} s \, \frac{(A-1)!}{a_1! \cdots (a_s-1)! \cdots a_q!}.$$

The first Newton polynomials are

$$\begin{aligned}
N_1 &= Y_1, \\
N_2 &= Y_1^2 - 2Y_2, \\
N_3 &= Y_1^3 - 3Y_1Y_2 + 3Y_3, \\
N_4 &= Y_1^4 - 4Y_1^2Y_2 + 2Y_2^2 + 4Y_1Y_3 - 4Y_4.
\end{aligned}$$

Now let $\lambda \vdash i$ be a partition of i. If $m_r(\lambda)$ denotes the multiplicity of r in λ, put

$$z_\lambda = \prod_{r \geq 1} r^{m_r(\lambda)} m_r(\lambda)!.$$

Then

$$\boxed{\Phi^{(n)}_{m,i} = \sum_{\lambda\vdash i} \frac{(-1)^{i-\ell(\lambda)}}{z_\lambda} \prod_{a=1}^{\ell(\lambda)} N_{m\lambda_a}.} \tag{4.2}$$

Although rational coefficients occur termwise in (4.2), their sum is the integral polynomial given by the companion determinant. Proof All power-series calculations below are formal, in $\mathbb{Q}[X_1, \ldots, X_n][[u]]$; no convergence is involved. First,

$$\begin{aligned} \log\left(\sum_{r=0}^{n} e_r u^r\right) &= \sum_{j=1}^{n} \log(1 + X_j u) \\ &= \sum_{q\geq 1} \frac{(-1)^{q-1}}{q} p_q u^q. \end{aligned} \tag{4.3}$$

On the other hand,

$$\log\left(1 + \sum_{r=1}^{n} e_r u^r\right) = \sum_{A\geq 1} \frac{(-1)^{A-1}}{A} \left(\sum_{r=1}^{n} e_r u^r\right)^A.$$

Extracting the coefficient of u^q gives exactly (4.1). Next,

$$\begin{aligned} \sum_{i=0}^{n} e_i^{[m]} t^i &= \prod_{j=1}^{n} (1 + X_j^m t) \\ &= \exp\left(\sum_{r\geq 1} \frac{(-1)^{r-1}}{r} p_{mr} t^r\right). \end{aligned} \tag{4.4}$$

Expanding the exponential and collecting terms according to partitions of i gives

$$e_i^{[m]} = \sum_{\lambda\vdash i} \frac{(-1)^{i-\ell(\lambda)}}{z_\lambda} \prod_{a=1}^{\ell(\lambda)} p_{m\lambda_a}.$$

Substituting $p_q = N_q(e)$ proves (4.2). An equivalent recursive form is

$$\boxed{\Phi_{m,0} = 1, \qquad i\Phi_{m,i} = \sum_{r=1}^{i} (-1)^{r-1} N_{mr} \Phi_{m,i-r}.} \tag{4.5}$$

For example,

$$\begin{aligned}\Phi_{m,1} &= N_m,\\ \Phi_{m,2} &= \frac{N_m^2 - N_{2m}}{2},\\ \Phi_{m,3} &= \frac{N_m^3 - 3N_m N_{2m} + 2N_{3m}}{6}.\end{aligned} \tag{4.6}$$

1. Particularly simple formulas for $m = 2$ and $m = 3$

5.1 Squares Taking $\zeta = -1$ in (2.2) gives

$$e_i^{[2]} = (-1)^i [z^{2i}] E(z) E(-z).$$

Thus

$$e_i^{[2]} = (-1)^i \sum_{\substack{a+b=2i\\ 0\le a,b\le n}} (-1)^b e_a e_b.$$

Pairing the terms $a = i - j,\ b = i + j$ yields the especially simple identity

$$\boxed{\Sigma_i(X_1^2, \ldots, X_n^2) = \Sigma_i^2 + 2 \sum_{j=1}^{\min(i,n-i)} (-1)^j \Sigma_{i-j} \Sigma_{i+j},} \tag{5.1}$$

where $\Sigma_0 = 1$. For example,

$$\begin{aligned}\Sigma_1(X_1^2, \ldots, X_n^2) &= \Sigma_1^2 - 2\Sigma_2,\\ \Sigma_2(X_1^2, \ldots, X_n^2) &= \Sigma_2^2 - 2\Sigma_1\Sigma_3 + 2\Sigma_4,\\ \Sigma_3(X_1^2, \ldots, X_n^2) &= \Sigma_3^2 - 2\Sigma_2\Sigma_4 + 2\Sigma_1\Sigma_5 - 2\Sigma_6.\end{aligned}$$

Terms whose indices exceed n are understood to be zero. 5.2 Cubes Let $\omega^3 = 1$, $\omega \neq 1$. Formula (2.1) becomes

$$\boxed{\Sigma_i(X_1^3, \ldots, X_n^3) = \sum_{\substack{0\le a,b,c\le n\\ a+b+c=3i}} \omega^{b+2c} \Sigma_a \Sigma_b \Sigma_c.} \tag{5.2}$$

Grouping equal monomials gives a formula with visibly integral coefficients:

$$\boxed{\begin{aligned}\Sigma_i(X_1^3,\ldots,X_n^3) = \Sigma_i^3 & \\ &+ 3 \sum_{\substack{0\le a,c\le n\\ a\ne c,\ 2a+c=3i}} \Sigma_a^2\Sigma_c \\ &+ 6 \sum_{\substack{0\le a<b<c\le n\\ a+b+c=3i\\ a\equiv b\equiv c \pmod 3}} \Sigma_a\Sigma_b\Sigma_c \\ &- 3 \sum_{\substack{0\le a<b<c\le n\\ a+b+c=3i\\ \{a,b,c\} \bmod 3=\{0,1,2\}}} \Sigma_a\Sigma_b\Sigma_c.\end{aligned}} \tag{5.3}$$

Indeed, for a multiset $\{a, a, c\}$ the three root-of-unity weights sum to 3. For three distinct indices whose sum is divisible by 3, their residues are either all equal, giving weight sum 6, or are $0, 1, 2$, giving

$$3\omega + 3\omega^2 = -3.$$

For instance,

$$\Sigma_1(X_1^3,\ldots,X_n^3) = \Sigma_1^3 - 3\Sigma_1\Sigma_2 + 3\Sigma_3.$$

For $n = 4$,

$$\begin{aligned}\Sigma_2(X_1^3,\ldots,X_4^3) = \Sigma_2^3 - 3\Sigma_1\Sigma_2\Sigma_3 + 3\Sigma_1^2\Sigma_4 \\ - 3\Sigma_2\Sigma_4 + 3\Sigma_3^2.\end{aligned}$$

1. Boundary cases and characteristic

The formulas are universal polynomial identities over $\mathbb{Z}$. Consequently, they remain valid after specialization in every commutative ring, with no restriction on characteristic. Several checks follow immediately. Top elementary symmetric polynomial

$$\Sigma_n(X_1^m,\ldots,X_n^m) = (X_1\cdots X_n)^m = \Sigma_n^m.$$

Thus

$$\Phi_{m,n}^{(n)} = Y_n^m.$$

First elementary symmetric polynomial

$$\Sigma_1(X_1^m, \ldots, X_n^m) = X_1^m + \cdots + X_n^m = N_m(\Sigma_1, \ldots, \Sigma_n).$$

One variable For $n = 1$,

$$\Phi_{m,1}^{(1)}(Y_1) = Y_1^m.$$

The omitted case $m = 1$ The same formulas give

$$\Phi_{1,i}^{(n)} = Y_i.$$

Positive characteristic Let the coefficient ring have characteristic $p > 0$. For $a \geq 0$ and $q \geq 1$,

$$\Sigma_i(X_1^{p^a q}, \ldots, X_n^{p^a q}) = (\Sigma_i(X_1^q, \ldots, X_n^q))^{p^a}.$$

Therefore, as polynomials over $\mathbb{F}_p$,

$$\boxed{\Phi_{p^a q,i}^{(n)} \equiv \left(\Phi_{q,i}^{(n)}\right)^{p^a} \pmod{p}.}$$

In particular,

$$\Phi_{p^a,i}^{(n)} \equiv Y_i^{p^a} \pmod{p}.$$

For example, in characteristic 2, formula (5.1) reduces to

$$\Sigma_i(X_1^2, \ldots, X_n^2) = \Sigma_i^2,$$

as required by Frobenius. The recurrence (4.5) involves division by i, so it must not be used literally in characteristic dividing i. The companion determinant (2.4), resultant (2.5), and root-of-unity polynomial after integral simplification have no such restriction.

1. Exact reproducible computation

The following SymPy code uses the division-free companion-matrix formula. PythonExécuterfrom ___future___ import annotations

from itertools import combinations import sympy as sp

def q831_polynomials(n: int, m: int): """ Return (e, Phi), where

Phi[i] = Sigma_i$X_1^m, ..., X_n^m$

expressed as a polynomial in e = (e1, ..., en). """ if not isinstance(n, int) or not isinstance(m, int) or n < 1 or m < 1: raise ValueError("n and m must be positive integers")

e = tuple(sp.Symbol(f"e{j}") for j in range(1, n + 1)) T = sp.Symbol("T")

21.2 Companion matrix of

21.3 Z^n - e1 Z^$n-1$ + e2 Z^$n-2$ - ... + (-1)^n en.

C = sp.zeros$n$

for r in range(n - 1): C[r + 1, r] = 1

for r in range$n$: j = n - r C[r, n - 1] = (-1) ** (j - 1) * e[j - 1]

characteristic = sp.Poly(sp.expand((T * sp.eye$n$ - C**m).det()), T,)

Phi = tuple(sp.expand((-1) ** i

- characteristic.coeff_monomial(T ** (n - i))

) for i in range(n + 1))

return e, Phi

def elementary(values, k: int): """Elementary symmetric polynomial of degree k in values.""" if k == 0: return sp.Integer1

return sum((sp.prod(values[j] for j in subset) for subset in combinations(range$len(values)$, k)), sp.Integer0,)

def verify_q831(n: int, m: int) -> bool: """ Independently substitute the elementary symmetric polynomials in x_1, ..., x_n and verify every identity symbolically. """ e, Phi = q831_polynomials(n, m) x = tuple(sp.Symbol(f"x{j}") for j in range(1, n + 1))

substitution = { e[j - 1]: elementary(x, j) for j in range(1, n + 1) }

xm = tuple(value**m for value in x)

return all(sp.expand(Phi[i].subs$substitution$ - elementary(xm, i)) == 0 for i in range(n + 1))

21.4 Example:

assert verify_q831(4, 3)

e, Phi = q831_polynomials(4, 3) for i, expression in enumeratePhi: print(i, expression) For $n = 4, m = 3$, this returns

$$
\begin{aligned}
\Phi_{3,0} &= 1, \\
\Phi_{3,1} &= e_1^3 - 3e_1e_2 + 3e_3, \\
\Phi_{3,2} &= e_2^3 - 3e_1e_2e_3 + 3e_1^2e_4 - 3e_2e_4 + 3e_3^2, \\
\Phi_{3,3} &= e_3^3 - 3e_2e_3e_4 + 3e_1e_4^2, \\
\Phi_{3,4} &= e_4^3.
\end{aligned}
$$

Final conclusion Q831 has a complete affirmative answer. For every $m \geq 2$ and $1 \leq i \leq n$, the polynomial

$$\Sigma_i(X_1^m, \ldots, X_n^m)$$

is given explicitly by the root-of-unity sum (2.1), equivalently by the integral companion determinant/resultant (2.3)–(2.5), or by the finite partition expansion (4.2). All formulas are universal over $\mathbb{Z}$, hence valid without exception over every commutative ring and in every characteristic.

Formal verification. Lean source; online type-check.

22 Question Q838

Informal verdict. A — Complete and apparently correct, with **high confidence**.

Lean coverage. The currently imported and CI-verified baseline proves the oscillating integrable law for every positive finite-dimensional real normed space using nonempty codebooks with card <= k. The audited continuation module Q838Exact.lean closes the sole canonical gap: it defines the literal card = k functional, proves equality with Q838.mqe for every k >= 1 by finite-set enlargement in the infinite ambient space, links it to ordinary expectations for the constructed integrable law, transfers liminf 0 and limsup top in every positive dimension and arbitrary norm, proves finiteness for every k >= 1, and derives an unconditional no-ENNReal-limit corollary. Positive Zador theory remains optional and outside the completion gate. The full module is not yet ingested or independently CI-checked in the campaign repository.

22.1 Audited informal answer

Q838 — Complete answer

1. Faithful restatement and source audit

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $X : \Omega \to \mathbb{R}^n$ be a random vector, and fix a norm $\|\cdot\|$ on $\mathbb{R}^n$. For a finite set $\Sigma \subset \mathbb{R}^n$, write

$$d_\Sigma(x) = \inf_{a \in \Sigma} \|x - a\|,$$

and define the k-point mean quantization error

$$m_k(X) = \inf_{\#\Sigma=k} \mathbb{E}\, d_\Sigma(X).$$

The question asks for general asymptotic results on m_k, and in particular whether, for every integrable X, the sequence $k^{1/n} m_k(X)$ must have a limit. The PDF also says that for the uniform distribution on $[-1,1]^n$ with the Euclidean norm, an earlier exercise proved

$$k^{1/n} m_k \longrightarrow 2^{-n}.$$

That assertion is indeed printed in the source. Luc Pommeret As printed, it is false for every $n \geq 2$. The correct limit has the form $2q_{n,\|\cdot\|}$, where $q_{n,\|\cdot\|}$ is the norm-dependent quantization constant defined below. For $n = 1$, $q_{1,|\cdot|} = 1/4$, so $2q_{1,|\cdot|} = 1/2 = 2^{-1}$; this probably explains why the erroneous expression is not detected in dimension one. The exact intended correction cannot be inferred from the PDF alone. There are no dependence or independence assumptions: only one random vector is involved, and m_k depends solely on its law.

1. Main theorem

Let $\mu = \mathcal{L}(X)$, and write its Lebesgue decomposition as

$$\mu = f(x)\, dx + \mu_s,$$

where $f\, dx$ is the absolutely continuous part and μ_s is singular. Put

$$p = \frac{n}{n+1}.$$

Let U be the uniform probability measure on the unit cube $C = [0,1]^n$, and define

$$q_{n,\|\cdot\|} := \lim_{k\to\infty} k^{1/n} \inf_{\#\Sigma \leq k} \int_C d_\Sigma(x)\, dx.$$

The existence and positivity of this limit will be proved below. Theorem For every norm on $\mathbb{R}^n$, the following statements hold.

Universal lower bound. If X is integrable, then

$$\boxed{\liminf_{k\to\infty} k^{1/n} m_k(X) \geq q_{n,\|\cdot\|} \left(\int_{\mathbb{R}^n} f(x)^{\,n/(n+1)}\, dx \right)^{(n+1)/n}.}$$

The right-hand side is interpreted as $+\infty$ when the integral diverges.

Zador asymptotic under a higher moment. If, for some $\eta > 0$,

$$\mathbb{E}\|X\|^{1+\eta} < \infty,$$

then the limit exists and

$$\boxed{\lim_{k\to\infty} k^{1/n} m_k(X) = q_{n,\|\cdot\|} \left(\int_{\mathbb{R}^n} f(x)^{n/(n+1)}\, dx \right)^{(n+1)/n}.} \tag{2.1}$$

In particular, the singular part of the law does not contribute to the leading $k^{-1/n}$ term.

Integrability alone is not sufficient. For every $n \geq 1$ and every norm on $\mathbb{R}^n$, there is an integrable discrete random vector X such that

$$\boxed{\liminf_{k\to\infty} k^{1/n} m_k(X) = 0, \qquad \limsup_{k\to\infty} k^{1/n} m_k(X) = +\infty.} \tag{2.2}$$

Thus the final question in Q838 has a definitive negative answer, even if limits in the extended real line are allowed. Formula (2.1) is the $r = 1$ form of the classical Zador quantization theorem. The Euclidean r-power version under an $r + \varepsilon$ moment condition, including the contribution of the absolutely continuous and singular parts, appears explicitly in Bucklew and Wise's theorem. Minds@UW+1 A self-contained proof of the form needed here follows.

1. Preliminary facts

For a finite Borel measure ν with finite first moment, define

$$E_k(\nu) := \inf_{\#\Sigma \leq k} \int_{\mathbb{R}^n} d_\Sigma(x)\, d\nu(x).$$

For a probability measure $\mu = \mathcal{L}(X)$, this is $m_k(X)$. The infimum over $\#\Sigma = k$ equals the infimum over $\#\Sigma \leq k$: a set with fewer than k points can be enlarged by adding arbitrary new points, and doing so cannot increase the distance. We shall repeatedly use the following elementary properties. Measure and spatial scaling For $c, t \geq 0$,

$$E_k(c\nu) = cE_k(\nu).$$

If $T(x) = a + tx$ and $T_\#\nu$ is the pushforward, then

$$E_k(T_\#\nu) = tE_k(\nu).$$

Union of codebooks For finite measures ν, ρ and positive integers k, ℓ,

$$E_{k+\ell}(\nu + \rho) \leq E_k(\nu) + E_\ell(\rho). \tag{3.1}$$

Indeed, take the union of a k-point codebook for ν and an ℓ-point codebook for ρ. Monotonicity under restriction If $0 \leq \nu \leq \rho$, then

$$E_k(\nu) \leq E_k(\rho). \tag{3.2}$$

Stability under transport The function $x \mapsto d_\Sigma(x)$ is 1-Lipschitz. Hence, if ν, ρ have the same total mass and π is any coupling of them,

$$\left| \int d_\Sigma \, d\nu - \int d_\Sigma \, d\rho \right| \leq \int \|x - y\| \, d\pi(x, y).$$

Taking infima over Σ gives

$$|E_k(\nu) - E_k(\rho)| \leq \inf_\pi \int \|x - y\| \, d\pi(x, y). \tag{3.3}$$

No optimal codebook is needed in the argument; arbitrarily near-optimal ones suffice.

1. The uniform-cube constant

Let

$$a_k := E_k(\mathbf{1}_C \, dx), \qquad C = [0, 1]^n.$$

Since C has volume 1, $\mathbf{1}_C \, dx$ is the uniform probability measure. 4.1 Existence of the limit Partition C into m^n congruent subcubes of side $1/m$. In each subcube place a scaled copy of an N-point codebook for C. Scaling gives

$$a_{m^n N} \leq \frac{1}{m} a_N. \tag{4.1}$$

Define

$$q := \inf_{N \geq 1} N^{1/n} a_N.$$

Clearly,

$$k^{1/n} a_k \geq q.$$

Conversely, fix N and put

$$m = \left\lfloor (k/N)^{1/n} \right\rfloor.$$

Then $m^n N \le k$, so by monotonicity and (4.1),

$$k^{1/n} a_k \le \frac{k^{1/n}}{m}\, a_N.$$

As $k \to \infty$,

$$\frac{k^{1/n}}{m} \longrightarrow N^{1/n}.$$

Consequently,

$$\limsup_{k\to\infty} k^{1/n} a_k \le N^{1/n} a_N.$$

Taking the infimum over N gives

$$\limsup_{k\to\infty} k^{1/n} a_k \le q.$$

Therefore

$$\boxed{\lim_{k\to\infty} k^{1/n} a_k = q_{n,\|\cdot\|}.} \tag{4.2}$$

4.2 Positivity and elementary bounds Let

$$\beta = \lambda_n\{x : \|x\| \le 1\}$$

be the Lebesgue volume of the norm unit ball. If $A \subset \mathbb{R}^n$ is measurable with volume v, then for every $a \in \mathbb{R}^n$,

$$\begin{aligned} \int_A \|x-a\|\,dx &= \int_0^\infty \lambda_n\big(A \setminus B(a,t)\big)\,dt \\ &\ge \int_0^{(v/\beta)^{1/n}} \big(v - \beta t^n\big)\,dt \\ &= \frac{n}{n+1}\beta^{-1/n} v^{1+1/n}. \end{aligned} \tag{4.3}$$

For a k-point codebook, partition C into its Voronoi cells, whose volumes are $v_1, \dots, v_r$, with $r \le k$. Applying (4.3) and convexity gives

$$\begin{aligned}\int_C d_\Sigma(x)\,dx &\geq \frac{n}{n+1}\beta^{-1/n}\sum_{i=1}^{r} v_i^{1+1/n}\\ &\geq \frac{n}{n+1}\beta^{-1/n}k^{-1/n}.\end{aligned}$$

Thus

$$\boxed{q_{n,\|\cdot\|} \geq \frac{n}{n+1}\beta^{-1/n} > 0.} \tag{4.4}$$

A regular cubical grid gives

$$\boxed{q_{n,\|\cdot\|} \leq \int_{[-1/2,1/2]^n} \|x\|\,dx < \infty.} \tag{4.5}$$

1. Compactly supported laws

We first prove the asymptotic formula for compactly supported measures. 5.1 Piecewise constant densities on many small cubes Put

$$p = \frac{n}{n+1}; \qquad \frac{1}{p} = 1 + \frac{1}{n}.$$

Consider a density

$$g_k = \sum_{i=1}^{M_k} c_{i,k}\mathbf{1}_{Q_{i,k}},$$

where the $Q_{i,k}$ are cubes of a common side length h_k, contained in a fixed bounded region. Assume

$$L_k := k^{1/n}h_k \longrightarrow \infty. \tag{5.1}$$

Then

$$M_k = O(h_k^{-n}) = O(k/L_k^n) = o(k). \tag{5.2}$$

Set

$$s_{i,k} = c_{i,k}^p|Q_{i,k}|, \qquad S_k = \sum_i s_{i,k} = \int g_k^p,$$

and

$$A_{i,k} = c_{i,k}|Q_{i,k}|^{1+1/n}.$$

Since $p(1+1/n) = 1$,

$$A_{i,k}^p = s_{i,k}. \tag{5.3}$$

We claim that, provided the total masses remain bounded,

$$\boxed{k^{1/n}E_k(g_k\,dx) - q_{n,\|\cdot\|}S_k^{1/p} \longrightarrow 0.} \tag{5.4}$$

Upper bound Fix $\varepsilon > 0$. By (4.2), there is an integer R such that

$$a_r \leq (q+\varepsilon)r^{-1/n} \qquad (r \geq R). \tag{5.5}$$

If $S_k > 0$, put

$$w_{i,k} = \frac{s_{i,k}}{S_k}.$$

Give R points to every positive-density cube, and distribute the remaining points proportionally to $w_{i,k}$. More precisely, with

$$K_k = k - RM_k,$$

set

$$r_{i,k} = R + \lfloor K_k w_{i,k} \rfloor.$$

By (5.2), $K_k/k \to 1$, and $\sum_i r_{i,k} \leq k$. Moreover,

$$r_{i,k} \geq K_k w_{i,k}.$$

Using scaled copies of unit-cube codebooks,

$$E_k(g_k dx) \leq \sum_i A_{i,k} a_{r_{i,k}}.$$

Consequently,

$$\begin{aligned} E_k(g_k dx) &\leq (q+\varepsilon)\sum_i A_{i,k} r_{i,k}^{-1/n} \\ &\leq (q+\varepsilon)K_k^{-1/n}\sum_i A_{i,k} w_{i,k}^{-1/n}. \end{aligned}$$

From (5.3),

$$A_{i,k} = (S_k w_{i,k})^{1/p},$$

and hence

$$A_{i,k} w_{i,k}^{-1/n} = S_k^{1/p} w_{i,k}.$$

Therefore

$$\limsup_{k\to\infty} k^{1/n} E_k(g_k dx) \leq (q+\varepsilon) S_k^{1/p} + o(1). \tag{5.6}$$

Lower bound: the boundary firewall Let Σ be a global codebook with at most k points. Assign every point of Σ to at most one cube. For every cube $Q_{i,k}$, place a finite δ_k-net on its boundary. For $n \geq 2$, such a net can be chosen with at most

$$C\left(\frac{h_k}{\delta_k}\right)^{n-1}$$

points. For $n = 1$, the two endpoints suffice. Choose, for $n \geq 2$,

$$\ell_k = L_k^{1/(2(n-1))}, \qquad \delta_k = \frac{k^{-1/n}}{\ell_k}.$$

Then

$$k^{1/n}\delta_k = \ell_k^{-1} \longrightarrow 0, \tag{5.7}$$

while the total number of firewall points is

$$O\left(M_k\left(\frac{h_k}{\delta_k}\right)^{n-1}\right) = O\left(\frac{k}{L_k^n}(L_k\ell_k)^{n-1}\right) = O(kL_k^{-1/2}) = o(k). \tag{5.8}$$

If a nearest point of the global codebook to $x \in Q_{i,k}$ is not assigned to $Q_{i,k}$, the line segment joining x to that point meets $\partial Q_{i,k}$. A firewall point lies within $C\delta_k$ of the intersection point. Therefore, if $\Gamma_{i,k}$ consists of the codepoints assigned to $Q_{i,k}$ together with its firewall,

$$d_\Sigma(x) \geq d_{\Gamma_{i,k}}(x) - C\delta_k \qquad (x \in Q_{i,k}). \tag{5.9}$$

Let $r_{i,k}$ be an upper bound for $\#\Gamma_{i,k}$. By (5.8),

$$\sum_i r_{i,k} \leq k + o(k). \tag{5.10}$$

Scaling the unit-cube problem and using the defining inequality

$$a_r \geq q r^{-1/n},$$

we get

$$E_k(g_k dx) \geq q \sum_i A_{i,k} r_{i,k}^{-1/n} - o(k^{-1/n}). \tag{5.11}$$

For positive A_i, r_i, Hölder's inequality gives

$$\sum_i A_i r_i^{-1/n} \geq \left(\sum_i A_i^p\right)^{1/p} \left(\sum_i r_i\right)^{-1/n}. \tag{5.12}$$

Indeed,

$$\sum_i A_i^p = \sum_i (A_i r_i^{-1/n})^p r_i^{1-p} \leq \left(\sum_i A_i r_i^{-1/n}\right)^p \left(\sum_i r_i\right)^{1-p}.$$

Using (5.3), (5.10), and (5.12) in (5.11),

$$\liminf_{k\to\infty} k^{1/n} E_k(g_k dx) \geq q S_k^{1/p} + o(1). \tag{5.13}$$

Together, (5.6) and (5.13) prove (5.4).

5.2 Arbitrary compactly supported densities Let $f \geq 0$ be integrable and supported in a fixed bounded cube. For a grid of mesh h, let f_h be the function which, on each grid cube Q, is equal to the average of f on Q:

$$f_h|_Q = \frac{1}{|Q|} \int_Q f.$$

Then

$$\|f_h - f\|_{L^1} \longrightarrow 0 \qquad (h \downarrow 0). \tag{5.14}$$

This follows first for continuous functions by uniform continuity and then for L^1 functions by approximation, since the averaging operators are L^1-contractions. Write

$$\varepsilon(h) = \|f_h - f\|_1.$$

Choose $h_k \downarrow 0$ so slowly that

$$L_k := k^{1/n} h_k \longrightarrow \infty \quad \text{and} \quad L_k \varepsilon(h_k) \longrightarrow 0. \tag{5.15}$$

Such a choice is always possible. For example, choose numbers $\delta_j \downarrow 0$ such that $\varepsilon(h) \le j^{-2}$ for $h \le \delta_j$, and let L_k be the largest j such that

$$jk^{-1/n} \le \min(\delta_j, j^{-1}).$$

Then set $h_k = L_k k^{-1/n}$. On each grid cube, the measures $f\,dx$ and $f_{h_k}\,dx$ have the same total mass. Couple their common part identically, and couple the remaining equal masses within the same cube. Since the diameter of a grid cube is at most Ch_k,

$$W_1(f\,dx, f_{h_k}\,dx) \le Ch_k \|f - f_{h_k}\|_1.$$

By (3.3) and (5.15),

$$k^{1/n} \left| E_k(f\,dx) - E_k(f_{h_k}\,dx) \right| \longrightarrow 0. \tag{5.16}$$

Furthermore, for $0 < p < 1$,

$$|u^p - v^p| \le |u - v|^p,$$

and on a set of finite volume,

$$\int |f - f_{h_k}|^p \le C \|f - f_{h_k}\|_1^p.$$

Thus

$$\int f_{h_k}^p \longrightarrow \int f^p. \tag{5.17}$$

Applying (5.4) to f_{h_k} and using (5.16)–(5.17) yields

$$\boxed{\lim_{k\to\infty} k^{1/n} E_k(f\,dx) = q \left(\int f^p \right)^{1/p}.} \tag{5.18}$$

5.3 Compactly supported singular measures First suppose that a finite measure ν is supported by a compact Lebesgue-null set K. Let $N_K(h)$ denote the number of grid cubes of side h meeting K. Their union lies in a Ch-neighborhood K^{Ch} of K, so

$$N_K(h) h^n \le \lambda_n(K^{Ch}).$$

Since K is compact and null,

$$\lambda_n(K^{Ch}) \longrightarrow 0.$$

Given $\varepsilon > 0$, put

$$h_k = \left(\frac{2\varepsilon}{k}\right)^{1/n}.$$

For all sufficiently large k,

$$N_K(h_k) \le \frac{k}{2}.$$

Placing one point in every cube meeting K gives covering radius at most Ch_k. Therefore

$$E_k(\nu) \le C\nu(\mathbb{R}^n)h_k,$$

and consequently

$$\limsup_{k\to\infty} k^{1/n}E_k(\nu) \le C\nu(\mathbb{R}^n)\varepsilon^{1/n}.$$

Letting $\varepsilon \downarrow 0$,

$$\boxed{k^{1/n}E_k(\nu) \longrightarrow 0.} \tag{5.19}$$

Now let ν be any singular measure supported in a fixed compact set K_0. There is a null Borel set N carrying all its mass. By inner regularity, choose increasing compact sets $K_j \subset N$ such that

$$\nu(K_0 \setminus K_j) \longrightarrow 0.$$

The restriction $\nu|_{K_j}$ satisfies (5.19). For every finite measure ρ supported in K_0, a regular grid gives a universal estimate

$$E_\ell(\rho) \le C_{K_0}\rho(\mathbb{R}^n)\ell^{-1/n}. \tag{5.20}$$

Allocate $(1-\varepsilon)k$ points to $\nu|_{K_j}$ and εk points to the remaining measure. From (3.1), (5.19), and (5.20),

$$\limsup_{k\to\infty} k^{1/n}E_k(\nu) \le C_{K_0}\nu(K_0 \setminus K_j)\varepsilon^{-1/n}.$$

For instance, choose

$$\varepsilon = \nu(K_0 \setminus K_j)^{n/(n+1)}.$$

The right-hand side then tends to zero as $j \to \infty$. Hence every compactly supported singular measure satisfies (5.19).

5.4 The compact-support formula Let

$$\mu = f\,dx + \mu_s$$

be compactly supported. By monotonicity,

$$E_k(\mu) \geq E_k(f\,dx),$$

so (5.18) gives the lower bound. For the upper bound, allocate $(1-\varepsilon)k$ points to $f\,dx$ and εk points to μ_s. Equations (3.1), (5.18), and (5.19) give

$$\limsup_{k\to\infty} k^{1/n} E_k(\mu) \leq (1-\varepsilon)^{-1/n} q \left(\int f^p\right)^{1/p}.$$

Letting $\varepsilon \downarrow 0$,

$$\boxed{\lim_{k\to\infty} k^{1/n} E_k(\mu) = q \left(\int f^p\right)^{1/p}} \tag{5.21}$$

for every compactly supported finite measure.

1. Unbounded laws and the moment condition

We need a uniform tail estimate. 6.1 A Pierce-type bound Let $s > 1$. There is a constant $C = C(n, \|\cdot\|, s)$ such that, for every finite measure ν,

$$\boxed{E_k(\nu) \leq C k^{-1/n} \nu(\mathbb{R}^n)^{1-1/s} \left(\int \|x\|^s \, d\nu(x)\right)^{1/s}.} \tag{6.1}$$

Proof It suffices first to consider a probability measure P satisfying

$$\int \|x\|^s \, dP(x) \leq 1.$$

Partition space into

$$A_0 = \{\|x\| \leq 1\}, \qquad A_j = \{2^{j-1} < \|x\| \leq 2^j\}, \quad j \geq 1.$$

Put

$$p_j = P(A_j), \qquad R_0 = 1, \quad R_j = 2^j, \qquad a_j = p_j R_j.$$

Let again $p = n/(n+1)$. We claim

$$S := \sum_{j \geq 0} a_j^p \leq C. \tag{6.2}$$

For $j \geq 1$, put $b_j = p_j R_j^s$. Then $\sum_j b_j \leq C_s$, and

$$a_j^p = b_j^p R_j^{-p(s-1)}.$$

Since $1 - p = 1/(n+1)$, write

$$R_j^{-p(s-1)} = \left(R_j^{-n(s-1)}\right)^{1-p}.$$

Hölder's inequality gives

$$\sum_{j \geq 1} a_j^p \leq \left(\sum_j b_j\right)^p \left(\sum_j R_j^{-n(s-1)}\right)^{1-p} < \infty,$$

proving (6.2). Set

$$w_j = \frac{a_j^p}{S}.$$

Use one codepoint at the origin. For every j with $(k-1)w_j \geq 1$, allocate

$$\ell_j = \lfloor (k-1) w_j \rfloor$$

additional codepoints to the ball of radius R_j. Such a ball can be covered with ℓ_j points and covering radius at most

$$C R_j \ell_j^{-1/n}.$$

For the remaining shells, use the origin. On the allocated shells,

$$\begin{aligned}
\sum_{\text{allocated}} p_j C R_j \ell_j^{-1/n} &\leq C k^{-1/n} \sum_j a_j w_j^{-1/n} \\
&= C k^{-1/n} S^{1/p} \sum_j w_j \\
&\leq C k^{-1/n}.
\end{aligned}$$

On the unallocated shells, $w_j < (k-1)^{-1}$, and therefore

$$w_j^{1/p} = w_j^{1+1/n} \le (k-1)^{-1/n} w_j.$$

Hence

$$\sum_{\text{unallocated}} a_j = S^{1/p} \sum_{\text{unallocated}} w_j^{1/p} \le C k^{-1/n}.$$

This proves $E_k(P) \le Ck^{-1/n}$ in the normalized case. Scaling space by

$$\sigma = \left(\int \|x\|^s \, dP \right)^{1/s}$$

gives the probability version of (6.1). Normalizing a finite measure by its total mass gives the displayed finite-measure estimate.

6.2 Proof of the full asymptotic formula Assume now

$$\int \|x\|^{1+\eta} \, d\mu(x) < \infty.$$

Set $s = 1 + \eta$. First, the density term is finite. Indeed, with

$$w(x) = 1 + \|x\|^s,$$

Hölder's inequality gives

$$\begin{aligned} \int f^p &= \int (fw)^p w^{-p} \\ &\le \left(\int fw \right)^p \left(\int w^{-p/(1-p)} \right)^{1-p}. \end{aligned}$$

Here

$$\frac{p}{1-p} = n,$$

and

$$\int_{\mathbb{R}^n} (1 + \|x\|^s)^{-n} \, dx < \infty$$

because $s > 1$. Let

$$\mu_R = \mu|_{\{\|x\| \le R\}}, \qquad \tau_R = \mu|_{\{\|x\| > R\}}.$$

By monotonicity and the compact-support formula,

$$\liminf_{k\to\infty} k^{1/n} E_k(\mu) \geq q \left(\int_{\{\|x\|\leq R\}} f^p \right)^{1/p}.$$

Letting $R \to \infty$ and using monotone convergence,

$$\liminf_{k\to\infty} k^{1/n} E_k(\mu) \geq q \left(\int f^p \right)^{1/p}. \tag{6.3}$$

For the upper bound, fix $0 < \varepsilon < 1$. Allocate

$$k_1 = \lfloor (1-\varepsilon)k \rfloor$$

points to μ_R, and the remaining k_2 points to τ_R. From (3.1),

$$E_k(\mu) \leq E_{k_1}(\mu_R) + E_{k_2}(\tau_R).$$

The compact-support formula gives

$$\lim_{k\to\infty} k^{1/n} E_{k_1}(\mu_R) = (1-\varepsilon)^{-1/n} q \left(\int_{\{\|x\|\leq R\}} f^p \right)^{1/p}.$$

By (6.1),

$$k^{1/n} E_{k_2}(\tau_R) \leq C\varepsilon^{-1/n} \left(\int_{\{\|x\|>R\}} \|x\|^s \, d\mu(x) \right)^{1/s}.$$

The last expression tends to zero as $R \to \infty$. Thus

$$\limsup_{k\to\infty} k^{1/n} E_k(\mu) \leq (1-\varepsilon)^{-1/n} q \left(\int f^p \right)^{1/p}.$$

Finally let $\varepsilon \downarrow 0$. Together with (6.3), this proves (2.1). The same argument also proves the universal lower bound without any moment beyond what is needed to make m_k finite: apply the compact-support result to μ_R and let $R \to \infty$.

1. The uniform cube and the error in the PDF

Let X be uniform on $[-1,1]^n$. Its density is

$$f(x) = 2^{-n} \mathbf{1}_{[-1,1]^n}(x).$$

Therefore

$$\int f(x)^p\,dx = 2^n(2^{-n})^p$$
$$= 2^{n(1-p)} = 2^{n/(n+1)}.$$

Raising to the power $1/p = (n+1)/n$ gives

$$\left(\int f^p\right)^{1/p} = 2.$$

Hence the correct answer is

$$\boxed{\lim_{k\to\infty} k^{1/n} m_k = 2q_{n,\|\cdot\|}.} \tag{7.1}$$

For the Euclidean norm, $\beta = \omega_n$, the volume of the Euclidean unit ball. From (4.4),

$$2q_{n,2} \geq \frac{2n}{n+1}\omega_n^{-1/n}. \tag{7.2}$$

For example, when $n = 2$,

$$2q_{2,2} \geq \frac{4}{3\sqrt{\pi}}\frac{1}{4} = 2^{-2}.$$

Thus the printed value 2^{-n} is impossible already in dimension two.

1. Counterexample under mere integrability

We now prove the strong nonconvergence assertion (2.2). Fix $n \geq 1$, fix any norm on $\mathbb{R}^n$, and choose $u \in \mathbb{R}^n$ such that

$$\|u\| = 1.$$

8.1 A decreasing summable block sequence Define recursively

$$N_1 = 1,$$

and, for $s \geq 1$,

$$L_s = 2^{2ns}N_s^2, \qquad N_{s+1} = N_s + L_s.$$

Set

$$M_s = 2^{-s}N_s^{-1/n}, \qquad h_s = \frac{M_s}{L_s}.$$

For every integer j in the s-th block,

$$N_s \le j < N_{s+1},$$

define

$$a_j = h_s.$$

Then

$$\sum_{j\ge 1} a_j = \sum_{s\ge 1} L_s h_s = \sum_{s\ge 1} M_s \le \sum_{s\ge 1} 2^{-s} = 1. \tag{8.1}$$

Moreover,

$$h_s = 2^{-(2n+1)s} N_s^{-(2+1/n)},$$

so (a_j) is nonincreasing. Put

$$A_k = \sum_{j\ge k} a_j.$$

8.2 Definition of the random vector Let

$$y_j = 4^j u, \qquad p_j = \frac{a_j}{4^j}.$$

Since $\sum_j p_j \le \frac{1}{4}\sum_j a_j < 1$, define X by

$$\mathbb{P}(X = y_j) = p_j, \qquad \mathbb{P}(X = 0) = 1 - \sum_{j\ge 1} p_j.$$

By Tonelli's theorem,

$$\mathbb{E}\|X\| = \sum_{j\ge 1} p_j \|y_j\| = \sum_{j\ge 1} a_j \le 1. \tag{8.2}$$

Thus X is integrable. 8.3 Two-sided estimate for m_k For the upper bound, use

$$\Sigma_k = \{0, y_1, \ldots, y_{k-1}\}.$$

Then

$$m_k \le \sum_{j\ge k} p_j \|y_j\| = A_k. \tag{8.3}$$

For the lower bound, consider the closed norm-balls

$$B_j = \overline{B}\left(y_j, \frac{4^j}{4}\right).$$

They are pairwise disjoint. Indeed, for $i < j$,

$$\|y_j - y_i\| = 4^j - 4^i \frac{4^j + 4^i}{4}.$$

A k-point codebook can meet at most k of these balls. If B_j contains no codepoint, then

$$d_\Sigma(y_j) \geq \frac{4^j}{4}.$$

Hence

$$\int d_\Sigma \, d\mu \geq \frac{1}{4} \sum_{j:\, B_j \cap \Sigma = \varnothing} a_j.$$

Since (a_j) is nonincreasing, deleting at most k terms leaves sum at least $\sum_{j \geq k+1} a_j$. Therefore

$$m_k \geq \frac{1}{4} A_{k+1}. \tag{8.4}$$

Combining (8.3) and (8.4),

$$\boxed{\frac{1}{4} A_{k+1} \leq m_k \leq A_k.} \tag{8.5}$$

8.4 A subsequence tending to zero At the beginning of the s-th block,

$$\begin{aligned} A_{N_s} &= \sum_{t \geq s} M_t \\ &= \sum_{t \geq s} 2^{-t} N_t^{-1/n} \\ &\leq N_s^{-1/n} \sum_{t \geq s} 2^{-t} \\ &= 2^{1-s} N_s^{-1/n}. \end{aligned}$$

Thus, by (8.3),

$$N_s^{1/n} m_{N_s} \leq 2^{1-s} \longrightarrow 0. \tag{8.6}$$

8.5 A subsequence tending to infinity Put

$$K_s = N_s + \left\lfloor \frac{L_s}{2} \right\rfloor .$$

At least $L_s/4$ terms of the s-th block occur after K_s, so

$$A_{K_s+1} \geq \frac{L_s}{4} h_s = \frac{M_s}{4}.$$

By (8.4),

$$m_{K_s} \geq \frac{M_s}{16}.$$

Also $K_s \geq L_s/2$, hence

$$\begin{aligned} K_s^{1/n} m_{K_s} &\geq \frac{1}{16} \left(\frac{L_s}{2} \right)^{1/n} 2^{-s} N_s^{-1/n} \\ &= \frac{2^{-1/n}}{16} \cdot 2^{-s} \cdot 2^{2s} N_s^{2/n} N_s^{-1/n} \\ &= \frac{2^{-1/n}}{16} 2^s N_s^{1/n} \longrightarrow +\infty. \end{aligned} \tag{8.7}$$

Equations (8.6) and (8.7) prove

$$\liminf_{k\to\infty} k^{1/n} m_k = 0, \qquad \limsup_{k\to\infty} k^{1/n} m_k = +\infty.$$

This completes the counterexample.

1. Small and boundary cases

9.1 Dimension one For the usual absolute value on $\mathbb{R}$,

$$q_{1,|\cdot|} = \frac{1}{4}.$$

Indeed, if an interval of length ℓ is represented by one point, the minimum possible integral distance is $\ell^2/4$. If the Voronoi intervals have lengths $\ell_1, \ldots, \ell_k$ with total length 1,

$$\int_0^1 d_\Sigma(x)\, dx \geq \frac{1}{4} \sum_{i=1}^{k} \ell_i^2 \geq \frac{1}{4k}.$$

Equality is attained by the midpoints of k equal subintervals. Thus, under a $1+\eta$ moment,

$$\boxed{\lim_{k\to\infty} k\, m_k(X) = \frac{1}{4}\left(\int_{\mathbb{R}} \sqrt{f(x)}\,dx\right)^2.} \tag{9.1}$$

For X uniform on $[-1,1]$, one has exactly

$$m_k = \frac{1}{2k}.$$

9.2 Finite support and atoms If X has support of cardinality r, then

$$m_k = 0 \qquad (k \geq r).$$

Its absolutely continuous density is zero, so formula (2.1) gives zero, as required. More generally, atoms and compactly supported singular components have zero contribution at the $k^{-1/n}$ scale. 9.3 Mixed distributions If

$$\mu = a\,\nu_{\mathrm{ac}} + (1-a)\nu_s$$

and ν_{ac} has density g, then $f = ag$, and

$$\left(\int (ag)^p\right)^{1/p} = a\left(\int g^p\right)^{1/p}.$$

Thus the leading coefficient is linear in the mass a of the absolutely continuous component. 9.4 Necessity of integrability for finite error If $\mathbb{E}\|X\| = \infty$, then for every finite Σ, with

$$R = \max_{a\in\Sigma} \|a\|,$$

the reverse triangle inequality gives

$$d_\Sigma(X) \geq \|X\| - R.$$

Hence $\mathbb{E}d_\Sigma(X) = \infty$, and $m_k = \infty$. The $1+\eta$ moment in the theorem is a sufficient tail condition, not a necessary one. The counterexample proves, however, that it cannot simply be replaced by the first-moment assumption for arbitrary laws.

1. Final conclusion

The mathematically correct general result is

$$\lim_{k\to\infty} k^{1/n} m_k = q_{n,\|\cdot\|}\left(\int_{\mathbb{R}^n} f(x)^{n/(n+1)}\,dx\right)^{(n+1)/n}$$

whenever X has a moment of some order strictly greater than 1. The density f is that of the absolutely continuous part of the law; singular mass is asymptotically negligible at this scale. For the uniform law on $[-1,1]^n$, the correct constant is

$$2q_{n,\|\cdot\|},$$

not 2^{-n} except accidentally when $n=1$. Finally, mere integrability does not guarantee convergence: for every dimension and every norm there is an integrable discrete law for which

$$\liminf k^{1/n} m_k = 0 \quad \text{and} \quad \limsup k^{1/n} m_k = +\infty.$$

Formal verification. Lean source; online type-check.

23 Question Q839

Informal verdict. A — Complete and apparently correct, with **high confidence**.

Lean coverage. Full canonical asymptotic coverage. The preserved base classifies every continuous solution for |q|<1, handles q=0 and the zero solution, proves the two exact logarithmic-phase identities, profile symmetry/positivity/endpoint zeros and the global obstruction to a phase-free ratio equivalent. Q839Completion.lean derives qPoch continuity, the denominator and reciprocal limits, continuity of phase and phaseNeg, exact positive-zero iff classifications in both sign regimes, the negative-q envelope bridge, and positive- and negative-q fixed-phase ratio limits and IsEquivalent theorems for every u0 in (0,1) and every nonzero continuous solution.

23.1 Audited informal answer

Q839 — Complete solution

1. Faithful restatement

Let $q \in (-1,1)$ be real, and let $f : \mathbb{R} \to \mathbb{R}$ be continuous and satisfy

$$f(x) = (1-qx)f(qx) \qquad (x \in \mathbb{R}).$$

The preceding oral exercise established that f has a power-series expansion valid on all of $\mathbb{R}$. Q839 asks for the asymptotic behavior of $f(x)$ as $x \to +\infty$. There are no subparts, footnotes, or apparent typographical ambiguities attached to Q839 in the PDF. Luc Pommeret+1 Because nonzero solutions have infinitely many positive zeros, no ordinary global equivalent $f(x) \sim g(x)$ with g eventually nonzero can exist. The correct result is an exact phase-dependent, logarithmically periodic asymptotic formula.

1. Statement of the result

For $|q| < 1$, introduce the finite and infinite q-Pochhammer products

$$(a;q)_N := \prod_{k=0}^{N-1}(1-aq^k), \qquad (a;q)_\infty := \prod_{k=0}^{\infty}(1-aq^k),$$

and in particular

$$(q;q)_N = \prod_{k=1}^{N}(1-q^k), \qquad (q;q)_0 = 1.$$

Classification of all continuous solutions Let $c = f(0)$.
If $q = 0$, then $f(x) = c$ for every x.
If $0 < |q| < 1$, then

$$\boxed{f(x) = c\,(qx;q)_\infty = c\prod_{j=1}^{\infty}(1-q^j x).}$$

In particular, f extends to an entire function and

$$\boxed{f(x) = c\sum_{m=0}^{\infty}\frac{(-1)^m q^{m(m+1)/2}}{(q;q)_m}\,x^m.}$$

Now suppose $q \neq 0$, and put

$$a := \log\frac{1}{|q|} > 0, \qquad \mathcal{E}_q(x) := x^{-1/2}\exp\left(\frac{(\log x)^2}{2a}\right).$$

This is the universal growth envelope.

Case $0 < q < 1$ For $x > 1$, write

$$\frac{\log x}{\log(1/q)} = n + u, \qquad n \in \mathbb{N}, \quad 0 \le u < 1.$$

Equivalently,

$$n = \left\lfloor \frac{\log x}{\log(1/q)} \right\rfloor, \qquad u = \left\{ \frac{\log x}{\log(1/q)} \right\}.$$

Define

$$\Phi_q(u) := q^{(u^2-u)/2} (q^u; q)_\infty (q^{1-u}; q)_\infty, \qquad 0 \le u \le 1.$$

Then the following identity is exact for every $x > 1$:

$$\boxed{f(x) = c(-1)^n \mathcal{E}_q(x) \frac{\Phi_q(u)}{(x^{-1}; q)_\infty}.} \tag{2.1}$$

Moreover,

$$\Phi_q(0) = \Phi_q(1) = 0, \qquad \Phi_q(u) > 0 \quad (0 < u < 1),$$

and $\Phi_q(1-u) = \Phi_q(u)$.
Case $-1 < q < 0$ Set

$$r := |q| = -q, \qquad s := r^2 = q^2.$$

For $x > 1$, write

$$\frac{\log x}{\log(1/s)} = n + u, \qquad n \in \mathbb{N}, \quad 0 \le u < 1.$$

Thus

$$n = \left\lfloor \frac{\log x}{2\log(1/r)} \right\rfloor, \qquad u = \left\{ \frac{\log x}{2\log(1/r)} \right\}.$$

Define

$$\begin{aligned} \Psi_q(u) := s^{u^2-u/2} &(s^u; s)_\infty (s^{1-u}; s)_\infty \\ &\times (-s^{u+1/2}; s)_\infty (-s^{1/2-u}; s)_\infty, \qquad 0 \le u \le 1. \end{aligned}$$

Then, again, the following identity is exact for every $x > 1$:

$$\boxed{f(x) = c(-1)^n \mathcal{E}_q(x) \frac{\Psi_q(u)}{(x^{-1};q)_\infty}.} \tag{2.2}$$

Here

$$\Psi_q(0) = \Psi_q(1) = 0, \qquad \Psi_q(u) > 0 \quad (0 < u < 1),$$

and $\Psi_q(1-u) = \Psi_q(u)$.

Complete asymptotic expansion For every fixed $q \in (-1,1) \setminus \{0\}$ and $x > 1$,

$$\boxed{\frac{1}{(x^{-1};q)_\infty} = \sum_{m=0}^{\infty} \frac{x^{-m}}{(q;q)_m}.} \tag{2.3}$$

Consequently, in either sign case, if $A_q = \Phi_q$ or Ψ_q and n, u are defined as above, then

$$\boxed{f(x) = c(-1)^n \mathcal{E}_q(x) A_q(u) \sum_{m=0}^{\infty} \frac{x^{-m}}{(q;q)_m}.} \tag{2.4}$$

Thus, for every integer $N \geq 1$,

$$\boxed{f(x) = c(-1)^n \mathcal{E}_q(x) A_q(u) \left(\sum_{m=0}^{N-1} \frac{x^{-m}}{(q;q)_m} + O_{q,N}(x^{-N}) \right).} \tag{2.5}$$

The error estimate is uniform over all logarithmic phases $u \in [0,1)$. In particular,

$$f(x) = c(-1)^n \mathcal{E}_q(x) A_q(u) \left(1 + \frac{1}{(1-q)x} + \frac{1}{(1-q)(1-q^2)x^2} + O_q(x^{-3}) \right).$$

All constants here are for fixed q; uniformity as $q \to 0$ or $|q| \to 1$ is not asserted.

1. Classification of the solutions

Iterating the functional equation gives, for every $N \geq 1$,

$$f(x) = \prod_{j=1}^{N} (1 - q^j x)\, f(q^N x). \tag{3.1}$$

Since $|q| < 1$, we have $q^N x \to 0$. By continuity at 0,

$$f(q^N x) \longrightarrow f(0) = c.$$

For fixed x,

$$\sum_{j=1}^{\infty} |q^j x| < \infty,$$

so the product

$$\prod_{j=1}^{\infty} (1 - q^j x)$$

converges. Passing to the limit in (3.1) gives

$$f(x) = c \prod_{j=1}^{\infty} (1 - q^j x) = c(qx; q)_{\infty}.$$

Conversely, this product satisfies

$$\begin{aligned}(1 - qx)(q^2 x; q)_{\infty} &= (1 - qx) \prod_{j=2}^{\infty} (1 - q^j x) \\ &= \prod_{j=1}^{\infty} (1 - q^j x),\end{aligned}$$

so it is indeed a solution. The convergence is uniform on every compact subset of $\mathbb{C}$, since for $|x| \leq R$,

$$\sum_{j=1}^{\infty} |q^j x| \leq R \sum_{j=1}^{\infty} |q|^j < \infty.$$

Thus the product defines an entire function. To obtain the coefficients, write

$$f(x) = \sum_{m=0}^{\infty} a_m x^m.$$

Substitution into

$$f(x) = (1 - qx) f(qx)$$

gives, for $m \geq 1$,

$$a_m = q^m a_m - q^m a_{m-1},$$

hence

$$(1-q^m)a_m = -q^m a_{m-1}.$$

Starting from $a_0 = c$,

$$a_m = c\,\frac{(-1)^m q^{1+2+\cdots+m}}{\prod_{j=1}^{m}(1-q^j)} = c\,\frac{(-1)^m q^{m(m+1)/2}}{(q;q)_m}.$$

Furthermore,

$$\left|\frac{a_{m+1}}{a_m}\right| = \frac{|q|^{m+1}}{|1-q^{m+1}|} \longrightarrow 0,$$

so the radius of convergence is infinite.

1. The reciprocal-product expansion

Let

$$G(z) := \frac{1}{(z;q)_\infty}, \qquad |z| < 1.$$

Because

$$(z;q)_\infty = (1-z)(qz;q)_\infty,$$

we have

$$(1-z)G(z) = G(qz).$$

Write

$$G(z) = \sum_{m=0}^{\infty} b_m z^m.$$

Comparison of coefficients gives $b_0 = 1$, and for $m \geq 1$,

$$b_m - b_{m-1} = q^m b_m.$$

Therefore

$$(1 - q^m) b_m = b_{m-1},$$

so

$$b_m = \frac{1}{(q;q)_m}.$$

This proves

$$\frac{1}{(z;q)_\infty} = \sum_{m=0}^{\infty} \frac{z^m}{(q;q)_m}, \qquad |z| < 1.$$

Taking $z = x^{-1}$ proves (2.3). For completeness, $(q;q)_m$ converges as $m \to \infty$ to the nonzero number $(q;q)_\infty$. Hence there is a constant C_q such that

$$\frac{1}{|(q;q)_m|} \le C_q \qquad (m \ge 0).$$

For $x \ge 2$,

$$\left| \sum_{m=N}^{\infty} \frac{x^{-m}}{(q;q)_m} \right| \le C_q \frac{x^{-N}}{1 - x^{-1}} \le 2 C_q x^{-N},$$

which proves the uniform remainder estimate in (2.5).

1. Derivation for $0 < q < 1$

Put

$$a = \log \frac{1}{q}, \qquad \frac{\log x}{a} = n + u, \quad 0 \le u < 1.$$

Thus

$$x = q^{-(n+u)}.$$

Split the product after its first n factors:

$$(qx;q)_\infty = (qx;q)_n \, (q^{n+1}x;q)_\infty.$$

Since $q^{n+1}x = q^{1-u}$,

$$(q^{n+1}x;q)_\infty = (q^{1-u};q)_\infty.$$

For the finite product,

$$
\begin{aligned}
(qx;q)_n &= \prod_{j=1}^{n}(1-q^j x) \\
&= (-1)^n x^n q^{n(n+1)/2} \prod_{j=1}^{n}\left(1-\frac{q^{-j}}{x}\right).
\end{aligned}
$$

Because $x = q^{-(n+u)}$,

$$\frac{q^{-j}}{x} = q^{n+u-j}.$$

Reversing the order of the factors,

$$\prod_{j=1}^{n}\left(1-\frac{q^{-j}}{x}\right) = \prod_{k=0}^{n-1}(1-q^{u+k}) = (q^u;q)_n.$$

Now

$$(q^u;q)_n = \frac{(q^u;q)_\infty}{(q^{n+u};q)_\infty} = \frac{(q^u;q)_\infty}{(x^{-1};q)_\infty}.$$

Therefore

$$(qx;q)_\infty = (-1)^n x^n q^{n(n+1)/2} \frac{(q^u;q)_\infty (q^{1-u};q)_\infty}{(x^{-1};q)_\infty}. \tag{5.1}$$

It remains to rewrite the elementary prefactor. Since $\log x = a(n+u)$,

$$
\begin{aligned}
\log\left(x^n q^{n(n+1)/2}\right) &= an(n+u) - \frac{a}{2}n(n+1) \\
&= \frac{(\log x)^2}{2a} - \frac{1}{2}\log x + \frac{a}{2}u(1-u).
\end{aligned}
$$

Because

$$e^{au(1-u)/2} = q^{(u^2-u)/2},$$

we get

$$x^n q^{n(n+1)/2} = \mathcal{E}_q(x)\, q^{(u^2-u)/2}.$$

Substituting into (5.1) proves (2.1).

1. Derivation for $-1 < q < 0$

Write

$$q = -r, \qquad 0 < r < 1, \qquad s = r^2.$$

Separating odd and even powers of q gives

$$\begin{aligned}(qx;q)_\infty &= (-rx;-r)_\infty \\ &= (sx;s)_\infty(-rx;s)_\infty.\end{aligned} \tag{6.1}$$

Let

$$\frac{\log x}{\log(1/s)} = n + u, \qquad 0 \le u < 1.$$

Thus $x = s^{-(n+u)}$. Applying the positive-base calculation to the first factor gives

$$(sx;s)_\infty = (-1)^n x^n s^{n(n+1)/2} \frac{(s^u;s)_\infty(s^{1-u};s)_\infty}{(x^{-1};s)_\infty}. \tag{6.2}$$

For the second factor,

$$(-rx;s)_\infty = \prod_{k=0}^{n-1}(1+rxs^k)\,(-rxs^n;s)_\infty.$$

Since $rxs^n = s^{1/2-u}$,

$$(-rxs^n;s)_\infty = (-s^{1/2-u};s)_\infty.$$

Also,

$$\begin{aligned}\prod_{k=0}^{n-1}(1+rxs^k) &= r^n x^n s^{n(n-1)/2}\prod_{k=0}^{n-1}\left(1+\frac{1}{rxs^k}\right) \\ &= r^n x^n s^{n(n-1)/2}(-s^{u+1/2};s)_n.\end{aligned}$$

Indeed,

$$\frac{1}{rxs^k} = s^{n+u-k-1/2},$$

whose exponents, in reverse order, are

$$u+\frac{1}{2},\ u+\frac{3}{2},\dots,u+n-\frac{1}{2}.$$

Moreover,

$$(-s^{u+1/2};s)_n = \frac{(-s^{u+1/2};s)_\infty}{(-s^{n+u+1/2};s)_\infty} = \frac{(-s^{u+1/2};s)_\infty}{(-r/x;s)_\infty}.$$

Hence

$$(-rx;s)_\infty = r^n x^n s^{n(n-1)/2} \frac{(-s^{u+1/2};s)_\infty (-s^{1/2-u};s)_\infty}{(-r/x;s)_\infty}. \tag{6.3}$$

Multiplying (6.2) and (6.3), the denominator is

$$(x^{-1};s)_\infty(-r/x;s)_\infty = (x^{-1};-r)_\infty = (x^{-1};q)_\infty.$$

The elementary prefactor is

$$(-1)^n x^{2n} r^n s^{n^2}.$$

Because $r = s^{1/2}$,

$$x^{2n} r^n s^{n^2} = x^{2n} s^{n^2+n/2}.$$

Let $A = \log(1/s) = 2a$. Since $\log x = A(n+u)$,

$$\begin{aligned}\log\left(x^{2n} s^{n^2+n/2}\right) &= A\left(n^2 + 2nu - \frac{n}{2}\right) \\ &= \frac{(\log x)^2}{A} - \frac{1}{2}\log x + A\left(\frac{u}{2} - u^2\right).\end{aligned}$$

Now $A = 2a$, and

$$e^{A(u/2-u^2)} = s^{u^2-u/2}.$$

Therefore

$$x^{2n} r^n s^{n^2} = \mathcal{E}_q(x) s^{u^2-u/2}.$$

This proves (2.2).

1. Properties of the phase profiles

All factors defining Φ_q and Ψ_q converge uniformly for $u \in [0,1]$, after separating finitely many initial factors when necessary. Thus both profiles are continuous. For $0 < u < 1$, every factor in $\Phi_q(u)$ is positive, so $\Phi_q(u) > 0$. At $u = 0$, the factor $(q^0;q)_\infty = (1;q)_\infty$ vanishes; at $u = 1$, the other Pochhammer factor vanishes. Symmetry is immediate. For Ψ_q, every factor is also positive in the open interval. Its symmetry needs one short computation. Put

$$B(u) := (-s^{u+1/2}; s)_\infty (-s^{1/2-u}; s)_\infty.$$

Using

$$(-z; s)_\infty = (1+z)(-sz; s)_\infty,$$

one obtains

$$\frac{B(1-u)}{B(u)} = \frac{1+s^{u-1/2}}{1+s^{1/2-u}} = s^{u-1/2}.$$

On the other hand,

$$\frac{s^{(1-u)^2-(1-u)/2}}{s^{u^2-u/2}} = s^{1/2-u}.$$

The two factors cancel, proving

$$\Psi_q(1-u) = \Psi_q(u).$$

For $q > 0$, the maximum of Φ_q is explicit. Indeed, writing $q = e^{-a}$, one has on $0 < u < 1$

$$\log \Phi_q(u) = \frac{a}{2} u(1-u) \\ + \sum_{k=0}^{\infty} \log(1 - e^{-a(k+u)}) + \sum_{k=0}^{\infty} \log(1 - e^{-a(k+1-u)}).$$

Each logarithmic summand has negative second derivative, and the first term has second derivative $-a$. Thus $\log \Phi_q$ is strictly concave. By symmetry its unique maximum is at $u = 1/2$, so

$$\boxed{\max_{0 \le u \le 1} \Phi_q(u) = q^{-1/8} (q^{1/2}; q)_\infty^2.}$$

For negative q, the exact continuous profile Ψ_q already gives the extremal normalized amplitude; no unimodality assertion is needed.

1. Zeros, signs, oscillations, and growth

Assume $c \neq 0$. From the product representation, all zeros are

$$x = q^{-j}, \qquad j = 1, 2, \dots,$$

and they are simple. Thus the positive zeros are

$$\begin{cases} q^{-j}, & 0 < q < 1, \\ |q|^{-2j}, & -1 < q < 0, \end{cases} \qquad j \geq 1.$$

For $q > 0$,

$$\operatorname{sgn} f(x) = \operatorname{sgn}(c)(-1)^n$$

on

$$q^{-n} < x < q^{-(n+1)}.$$

For $q < 0$,

$$\operatorname{sgn} f(x) = \operatorname{sgn}(c)(-1)^n$$

on

$$|q|^{-2n} < x < |q|^{-2(n+1)}.$$

Let

$$M_q := \begin{cases} \max\limits_{0 \leq u \leq 1} \Phi_q(u), & q > 0, \\ \max\limits_{0 \leq u \leq 1} \Psi_q(u), & q < 0. \end{cases}$$

Since $(x^{-1}; q)_\infty \to 1$, the complete cluster set of the normalized function is

$$\boxed{\operatorname{Clust}_{x \to \infty} \frac{f(x)}{c\,\mathcal{E}_q(x)} = [-M_q, M_q].}$$

Indeed, every sequence has a subsequence along which the logarithmic phase u converges and the parity of n is fixed. Conversely, every value in the interval is obtained because the phase profile is continuous, vanishes at the endpoints, and attains M_q. Consequently,

$$\boxed{\limsup_{x \to \infty} \frac{|f(x)|}{\mathcal{E}_q(x)} = |c| M_q, \qquad \liminf_{x \to \infty} \frac{|f(x)|}{\mathcal{E}_q(x)} = 0.}$$

The second equality follows already from the positive zeros. Moreover,

$$\boxed{\limsup_{x\to\infty} f(x) = +\infty, \qquad \liminf_{x\to\infty} f(x) = -\infty.}$$

Finally,

$$\log \mathcal{E}_q(x) = \frac{(\log x)^2}{2\log(1/|q|)} - \frac{1}{2}\log x.$$

Hence

$$\boxed{\limsup_{x\to\infty} \frac{\log(1+|f(x)|)}{(\log x)^2} = \frac{1}{2\log(1/|q|)}, \qquad \liminf_{x\to\infty} \frac{\log(1+|f(x)|)}{(\log x)^2} = 0.}$$

Thus the nonzero solution has superpolynomial peak growth but returns exactly to zero along a geometric sequence. For example, at any fixed phase $u_0 \in (0,1)$,

$$f\left(q^{-(n+u_0)}\right) \sim c(-1)^n \Phi_q(u_0) \mathcal{E}_q\left(q^{-(n+u_0)}\right)$$

when $q > 0$, while for $q < 0$,

$$f\left((q^2)^{-(n+u_0)}\right) \sim c(-1)^n \Psi_q(u_0) \mathcal{E}_q\left((q^2)^{-(n+u_0)}\right).$$

1. Boundary and hypothesis checks

The case $q = 0$ The equation becomes

$$f(x) = f(0),$$

so every solution is constant. The logarithmic-envelope formulas are not meant to include this case. The zero solution If $f(0) = 0$, the classification gives $f \equiv 0$. Two elementary examples For $q = \frac{1}{2}$,

$$f(x) = c\prod_{j=1}^{\infty}(1 - 2^{-j}x),$$

with positive zeros $2, 4, 8, \ldots$, logarithmic phase $\{\log_2 x\}$, and envelope

$$x^{-1/2}\exp\left(\frac{(\log x)^2}{2\log 2}\right).$$

For $q = -\frac{1}{2}$,

$$f(x) = c\prod_{j=1}^{\infty}(1 - (-2)^{-j}x),$$

with positive zeros $4, 16, 64, \ldots$, phase $\{\log_4 x\}$, and the same envelope. Why continuity matters Only continuity at 0 is actually needed. Without it, the solution space is much larger. For $q \neq 0$, choose $x_0 \neq 0$ whose multiplicative orbit

$$\mathcal{O} = \{q^k x_0 : k \in \mathbb{Z}\}$$

does not contain q^{-1}. Choose $v_0 = 1$, and define v_k recursively by

$$v_k = (1 - q^{k+1}x_0)v_{k+1}.$$

Set

$$g(q^k x_0) = v_k, \qquad g(x) = 0 \quad (x \notin \mathcal{O}).$$

The orbit and its complement are both invariant under multiplication by q, so g satisfies the functional equation. But as $k \to +\infty$,

$$q^k x_0 \to 0, \qquad v_k \longrightarrow \frac{1}{(qx_0; q)_\infty} \neq 0,$$

while one may set $g(0) = 0$. Thus g is discontinuous at 0. Why the strict condition $|q| < 1$ changes the problem At $q = 1$,

$$f(x) = (1 - x)f(x),$$

so $xf(x) = 0$; continuity forces $f \equiv 0$. At $q = -1$,

$$f(x) = (1 + x)f(-x), \qquad f(-x) = (1 - x)f(x),$$

and hence

$$f(x) = (1 - x^2)f(x).$$

Thus $x^2 f(x) = 0$, and continuity again forces $f \equiv 0$.

Final conclusion Every continuous solution is a scalar multiple of the Euler product

$$\prod_{j\geq 1}(1 - q^j x).$$

For $q \neq 0$, its natural magnitude is

$$x^{-1/2} \exp\left(\frac{(\log x)^2}{2\log(1/|q|)}\right),$$

but the normalized function does not converge: it has an explicit logarithmically periodic profile, changes sign after each positive zero, and admits the exact convergent inverse-power expansion (2.4). This completely determines its behavior as $x \to +\infty$.

Formal verification. Lean source; online type-check.

24 Question Q850

Informal verdict. A — Complete and apparently correct, with **high confidence**.

Lean coverage. Lean proves the explicit strictly decreasing lower bound psi$q$=10^{-q} for the Liouville number against every integer numerator and positive denominator, together with the truncation-tail estimates and a sharper antitone step bound.

24.1 Audited informal answer

Complete solution

1. Faithful restatement and two apparent typos

Let

$$x = \sum_{n\geq 1} 10^{-n!} = 0.110001000000000000000001\cdots.$$

The problem asks for explicit decreasing functions

$$\psi : \mathbb{N}^* \longrightarrow (0,1)$$

such that, for every $p \in \mathbb{Z}$ and every positive integer q,

$$\left| x - \frac{p}{q} \right| \geq \psi(q).$$

There are two apparent typographical slips in the PDF:

In the introductory inequality for an algebraic irrational α, the displayed formula contains x; it should contain α.

In both displayed inequalities, q is quantified in $\mathbb{N}$, although p/q and $\psi(q)$ require $q \geq 1$. Thus $q \in \mathbb{N}^*$ is intended.

Here "decreasing" is understood in the standard French sense of non-increasing. A strictly decreasing example will also be given. Luc Pommeret

1. Truncations of the Liouville number

For $n \geq 1$, put

$$x_n := \sum_{k=1}^{n} 10^{-k!} = \frac{A_n}{Q_n}, \qquad Q_n := 10^{n!}, \qquad A_n := \sum_{k=1}^{n} 10^{n!-k!} \in \mathbb{Z},$$

and denote the tail by

$$r_n := x - x_n = \sum_{k=n+1}^{\infty} 10^{-k!}.$$

Also set

$$T_n := 10^{-(n+1)!}.$$

The first omitted term gives

$$r_n > T_n.$$

On the other hand, since the factorial exponents form a subset of the integers beginning at $(n+1)!$,

$$r_n < \sum_{j=(n+1)!}^{\infty} 10^{-j} = \frac{10}{9}\, 10^{-(n+1)!}.$$

Thus

$$\boxed{T_n < r_n < \frac{10}{9} T_n.} \tag{2.1}$$

All series involved have positive terms, so these estimates require no rearrangement or limiting interchange.

1. The fundamental rational-separation estimate

Let $p \in \mathbb{Z}$ and $q \in \mathbb{N}^*$. If $p/q \neq x_n = A_n/Q_n$, then

$$\left|\frac{p}{q} - \frac{A_n}{Q_n}\right| = \frac{|pQ_n - qA_n|}{qQ_n} \geq \frac{1}{qQ_n},$$

because $pQ_n - qA_n$ is a nonzero integer. Notice that neither p/q nor A_n/Q_n needs to be assumed reduced. Therefore there are two cases:

if $p/q = x_n$, then

$$\left|x - \frac{p}{q}\right| = r_n;$$

if $p/q \neq x_n$, the reverse triangle inequality gives

$$\left|x - \frac{p}{q}\right| \geq \frac{1}{qQ_n} - r_n.$$

Writing $u_+ := \max(u, 0)$, we consequently have, for every $n \geq 1$,

$$\boxed{\left|x - \frac{p}{q}\right| \geq B_n(q) := \min\left\{r_n,\ \left(\frac{1}{qQ_n} - r_n\right)_+\right\}.} \tag{3.1}$$

This dichotomy covers every rational p/q.

1. An optimized explicit decreasing function

Define

$$\boxed{\Psi(q) := \sup_{n\geq 1} \min\left\{r_n,\ \left(\frac{1}{q\,10^{n!}} - r_n\right)_+\right\}.} \tag{4.1}$$

Theorem 1 The function $\Psi : \mathbb{N}^* \to (0, 1)$ is non-increasing and satisfies

$$\boxed{\left|x - \frac{p}{q}\right| \geq \Psi(q) \qquad (p \in \mathbb{Z},\ q \in \mathbb{N}^*).} \tag{4.2}$$

Proof By (3.1), the left-hand side is at least $B_n(q)$ for every n, and hence at least their supremum. For fixed n, the map

$$q \longmapsto \min\left\{r_n,\ \left(\frac{1}{qQ_n} - r_n\right)_+\right\}$$

is non-increasing. The supremum of non-increasing functions is non-increasing, so Ψ is non-increasing. It remains to show that $\Psi(q) > 0$. From (2.1),

$$qQ_n r_n < \frac{10q}{9}\,10^{n!-(n+1)!} = \frac{10q}{9}\,10^{-n\,n!} \longrightarrow 0.$$

Hence, for all sufficiently large n,

$$\frac{1}{qQ_n} \geq 2r_n.$$

For such an n, $B_n(q) = r_n > 0$. Thus $\Psi(q) > 0$. Finally,

$$\Psi(q) \leq r_1 < \frac{1}{90} < 1.$$

This proves the theorem. □ Formula (4.1) is the strongest lower bound obtained by applying the elementary separation argument simultaneously to all decimal truncations x_n.

1. A simpler closed-form answer

The preceding optimized formula can be replaced by a particularly transparent step function. For $q \geq 1$, define

$$N(q) := \min\left\{n \geq 1 : 19q \leq 9 \cdot 10^{n\,n!}\right\}, \tag{5.1}$$

and

$$\boxed{\psi_*(q) := 10^{-(N(q)+1)!}.} \tag{5.2}$$

The minimum exists because $n\,n! \to \infty$. Theorem 2 The function ψ_* is non-increasing and

$$\boxed{\left|x - \frac{p}{q}\right| \geq \psi_*(q) \qquad (p \in \mathbb{Z},\ q \in \mathbb{N}^*).} \tag{5.3}$$

Proof Let $n = N(q)$ and $T_n = 10^{-(n+1)!}$. Since

$$\frac{1}{qQ_n} = \frac{10^{n\,n!}}{q}\,T_n,$$

the definition of $N(q)$ gives

$$\frac{1}{qQ_n} \geq \frac{19}{9}T_n.$$

Together with $r_n < \frac{10}{9}T_n$, this yields

$$\frac{1}{qQ_n} - r_n > T_n.$$

We also have $r_n > T_n$. Hence (3.1) gives

$$\left| x - \frac{p}{q} \right| \geq T_n = 10^{-(N(q)+1)!} = \psi_*(q).$$

If $q_1 \leq q_2$, then $N(q_1) \leq N(q_2)$. Therefore

$$\psi_*(q_1) \geq \psi_*(q_2),$$

so ψ_* is non-increasing. $\square$ The first ranges are

$$\psi_*(q) = \begin{cases} 10^{-2}, & 1 \leq q \leq 4, \\ 10^{-6}, & 5 \leq q \leq 4736, \\ 10^{-24}, & 4737 \leq q \leq \left\lfloor \frac{9}{19} 10^{18} \right\rfloor, \end{cases}$$

and so forth. More generally, for every $a \geq 1$ and $0 < c \leq 1$,

$$\psi_{a,c}(q) := c\,10^{-a(N(q)+1)!}$$

is another explicit admissible non-increasing function.

1. Strictly decreasing examples

Because Ψ is positive and non-increasing,

$$\boxed{\widehat{\Psi}(q) := \frac{\Psi(q)}{q+1}} \tag{6.1}$$

is strictly decreasing. Indeed,

$$\widehat{\Psi}(q+1) = \frac{\Psi(q+1)}{q+2} \leq \frac{\Psi(q)}{q+2} < \frac{\Psi(q)}{q+1} = \widehat{\Psi}(q).$$

It remains an admissible lower bound because $\widehat{\Psi}(q) \leq \Psi(q)$. The same construction works with ψ_*:

$$\boxed{\widehat{\psi}_*(q) := \frac{10^{-(N(q)+1)!}}{q+1}}$$

is explicit, strictly decreasing, and valid.

1. A particularly simple example: $\psi(q) = 10^{-q}$

Although much weaker than the optimized bound, the elementary function

$$\boxed{\psi_0(q) = 10^{-q}} \tag{7.1}$$

already answers the question and is strictly decreasing. Proof The case $q = 1$ is immediate. Indeed,

$$\frac{1}{10} < x < \frac{1}{10} + \frac{1}{90} = \frac{1}{9} < \frac{1}{2},$$

so the closest integer is 0, and

$$|x - p| > \frac{1}{10} = 10^{-1} \qquad (p \in \mathbb{Z}).$$

Now suppose $q \geq 2$, and choose the unique $n \geq 1$ such that

$$(n+1)! \leq q < (n+2)!. \tag{7.2}$$

Then

$$T_n = 10^{-(n+1)!} \geq 10^{-q}. \tag{7.3}$$

If $n \geq 2$, put

$$a_n := \frac{10^{n\,n!}}{(n+2)!}.$$

We have

$$a_2 = \frac{10^4}{24} > \frac{19}{9},$$

and a_n is strictly increasing, since

$$\frac{a_{n+1}}{a_n} = \frac{10^{(n+1)(n+1)!-n\,n!}}{n+3}1.$$

Using $q < (n+2)!$, we obtain

$$\frac{1}{qQ_nT_n} = \frac{10^{n\,n!}}{q}\frac{10^{n\,n!}}{(n+2)!}\frac{19}{9}.$$

Consequently,

$$\frac{1}{qQ_n} - r_n\left(\frac{19}{9} - \frac{10}{9}\right)T_n = T_n.$$

Both alternatives in (3.1) are therefore at least T_n, and hence at least 10^{-q}. It remains to check $n = 1$, corresponding to $2 \le q \le 5$. Here $Q_1 = 10$ and $r_1 < 1/90$. For $q = 2$,

$$\frac{1}{20} - r_1\frac{1}{20} - \frac{1}{90} = \frac{7}{180}10^{-2}.$$

For $3 \le q \le 5$,

$$\frac{1}{10q} - r_1 \ge \frac{1}{50} - \frac{1}{90} = \frac{2}{225}10^{-3} \ge 10^{-q}.$$

Since also $r_1 > 10^{-2}$, the two cases in (3.1) again give the required bound. Thus

$$\left|x - \frac{p}{q}\right| \ge 10^{-q}$$

for all $p \in \mathbb{Z}$ and $q \ge 1$. □

1. Sharpness at the canonical denominators

The optimized function Ψ is exact along the natural sequence of truncation denominators. Let

$$Q_m = 10^{m!}, \qquad A_m = \sum_{k=1}^{m} 10^{m!-k!}.$$

Then

$$\frac{A_m}{Q_m} = x_m, \qquad \left|x - \frac{A_m}{Q_m}\right| = r_m. \tag{8.1}$$

For $m \ge 2$,

$$2Q_m^2 r_m < \frac{20}{9}\,10^{2m!-(m+1)!} = \frac{20}{9}\,10^{-(m-1)m!} < 1.$$

Hence

$$\frac{1}{Q_m^2} - r_m > r_m,$$

so the $n = m$ term in (4.1) equals r_m. Therefore

$$\Psi(Q_m) \geq r_m.$$

But applying (4.2) to $p = A_m$ and $q = Q_m$, and using (8.1), gives

$$\Psi(Q_m) \leq r_m.$$

Thus

$$\boxed{\Psi(10^{m!}) = r_m \qquad (m \geq 2).} \tag{8.2}$$

For the simpler function ψ_*, one has $N(Q_m) = m$. Indeed, the defining inequality holds for $n = m$, whereas for $n = m - 1$,

$$\frac{9 \cdot 10^{(m-1)(m-1)!}}{19 \cdot 10^{m!}} = \frac{9}{19} 10^{-(m-1)!} < 1.$$

Hence

$$\psi_*(Q_m) = T_m = 10^{-(m+1)!}.$$

Moreover,

$$1 < \frac{r_m}{T_m} < 1 + \frac{10}{9} 10^{-((m+2)! - (m+1)!)} = 1 + \frac{10}{9} 10^{-(m+1)(m+1)!},$$

so

$$\frac{r_m}{\psi_*(Q_m)} \longrightarrow 1.$$

Thus the coefficient 1 in ψ_* is asymptotically optimal along the canonical denominators. For example, at $q = 100 = 10^{2!}$,

$$\frac{11}{100} = x_2, \qquad \left| x - \frac{11}{100} \right| = 10^{-6} + 10^{-24} + 10^{-120} + \cdots ,$$

while

$$\psi_*(100) = 10^{-6}.$$

1. Why no fixed power-law lower bound is possible

At $q = Q_m = 10^{m!}$, the rational A_m/Q_m satisfies

$$\left| x - \frac{A_m}{Q_m} \right| = r_m < \frac{10}{9} 10^{-(m+1)!} = \frac{10}{9} Q_m^{-(m+1)}.$$

Given fixed $d \geq 1$ and $C > 0$, choosing m sufficiently large gives

$$r_m < C Q_m^{-d}.$$

Therefore no estimate

$$\left| x - \frac{p}{q} \right| \geq C q^{-d}$$

can hold for all rational p/q, exactly as expected for this Liouville number. Hence explicit answers to Q850 include, from simplest to strongest,

$$\boxed{\psi(q) = 10^{-q}},$$

$$\boxed{\psi_*(q) = 10^{-(N(q)+1)!}, \quad N(q) = \min\{n \geq 1 : 19q \leq 9 \cdot 10^{n\, n!}\},}$$

and the optimized truncation envelope

$$\boxed{\Psi(q) = \sup_{n \geq 1} \min \left\{ \sum_{k=n+1}^{\infty} 10^{-k!}, \left(\frac{1}{q 10^{n!}} - \sum_{k=n+1}^{\infty} 10^{-k!} \right)_+ \right\}}.$$

Formal verification. Lean source; online type-check.

25 Question Q855

Informal verdict. A — Complete solution. Confidence: **high (0.94)**.

Lean coverage. Full canonical Q855 coverage relative to the premise imported from R773. Lean defines the universal coefficients alphaC by the audited triangular recursion and proves, for every real lambda, every recurrence solution v satisfying vn/n^lambda -> 1, and every N, the stronger remainder vn-sum_{k<=N} alphaC(lambda,k)n^$lambda - k$=O$n^($lambda - N - 1$). The single theorem covers lambda=0. It does not re-prove R773's existence/basis theorem, which the canonical question states as prior input; it is not the permitted finite-cancellation/natural-lambda fallback.

25.1 Audited informal answer

Q855 — complete solution

1. Verified statement

On printed pages 14–15 of the canonical PDF, Q855 considers, for a fixed parameter λ, the real solutions of

$$u_{n+2} = u_{n+1} + \frac{\lambda}{n+2} u_n \qquad (n \geq 0). \tag{1}$$

The preceding response R773 is quoted as proving that the two-dimensional solution space has a basis $\{(v_n), (w_n)\}$ such that

$$v_n \sim n^\lambda, \qquad w_n \sim \frac{(-1)^n \lambda^n}{n!\, n^{\lambda+1}}. \tag{2}$$

The question asks for a complete asymptotic expansion of the dominant solution v:

$$v_n = n^\lambda + \alpha_1 n^{\lambda-1} + \cdots + \alpha_N n^{\lambda-N} + o(n^{\lambda-N})$$

for every N. Luc Pommeret+2Luc Pommeret+2 The PDF does not explicitly quantify λ. I take $\lambda \in \mathbb{R}$ fixed. There is one minor endpoint defect in the quoted premise: when $\lambda = 0$, the displayed expression for w_n is zero for $n \geq 1$, so it cannot literally be an asymptotic equivalent of a nonzero sequence. The requested conclusion at $\lambda = 0$ is nevertheless immediate and will be included.

1. Final theorem

For $x \in \mathbb{R}$ and $m \in \mathbb{N}$, write

$$\binom{x}{m} = \frac{x(x-1)\cdots(x-m+1)}{m!}.$$

Define $\alpha_0(\lambda) = 1$, and for $r \geq 1$ define recursively

$$\boxed{\alpha_r(\lambda) = \frac{1}{r} \sum_{k=0}^{r-1} \alpha_k(\lambda) \left[\left(2^{\,r+1-k} - 1\right) \binom{\lambda - k}{r+1-k} - \lambda(-2)^{r-k} \right].} \tag{3}$$

Then, for every fixed $\lambda \in \mathbb{R}$, every solution v of 1 satisfying $v_n \sim n^\lambda$, and every $N \geq 0$,

$$\boxed{v_n = \sum_{k=0}^{N} \alpha_k(\lambda) n^{\lambda-k} + O_{\lambda,N,v}\left(n^{\lambda-N-1}\right).} \tag{4}$$

This is one order stronger than requested. In particular,

$$v_n \sim n^\lambda \sum_{k\geq 0} \frac{\alpha_k(\lambda)}{n^k}$$

in the Poincaré sense. The coefficients depend only on λ, not on the choice of the normalized dominant solution: two such solutions differ by a multiple of w, which is smaller than every power of n^{-1}. The first coefficients are

$$\boxed{\alpha_1(\lambda) = \frac{\lambda(3\lambda+1)}{2},} \tag{5}$$

$$\boxed{\alpha_2(\lambda) = \frac{\lambda(\lambda-1)\left(27\lambda^2+5\lambda+2\right)}{24},} \tag{6}$$

$$\boxed{\alpha_3(\lambda) = \frac{\lambda^2(\lambda-1)(\lambda-2)(3\lambda-1)(9\lambda-1)}{48},} \tag{7}$$

and

$$\boxed{\alpha_4(\lambda) = \frac{\lambda(\lambda-1)(\lambda-2)(\lambda-3)}{5760}\left(1215\lambda^4-1890\lambda^3+485\lambda^2-18\lambda-8\right).} \tag{8}$$

1. Construction of the formal expansion

Introduce the recurrence operator

$$(L_\lambda u)_n = u_{n+2} - u_{n+1} - \frac{\lambda}{n+2} u_n.$$

We seek an approximate solution of the form

$$p(n) = n^\lambda A\left(\frac{1}{n}\right), \qquad A(x) = \sum_{k\geq 0} a_k x^k, \quad a_0 = 1.$$

Put $x = 1/n$. Direct substitution gives the exact identity

$$n^{-\lambda}(L_\lambda p)_n = \Phi_\lambda(A)(x), \tag{9}$$

where

$$\begin{aligned}\Phi_\lambda(A)(x) = (1+2x)^\lambda A\left(\frac{x}{1+2x}\right) - (1+x)^\lambda A\left(\frac{x}{1+x}\right)\\ - \frac{\lambda x}{1+2x}A(x).\end{aligned} \tag{10}$$

Since

$$(1+cx)^\lambda A\left(\frac{x}{1+cx}\right) = \sum_{k\geq 0} a_k x^k (1+cx)^{\lambda-k},$$

we have

$$\Phi_\lambda(A)(x) = \sum_{k\geq 0} a_k x^k \left((1+2x)^{\lambda-k} - (1+x)^{\lambda-k}\right) - \frac{\lambda x}{1+2x}\sum_{k\geq 0} a_k x^k. \tag{11}$$

The constant coefficient vanishes. The coefficient of x also vanishes automatically:

$$2\lambda - \lambda - \lambda = 0.$$

For $r \geq 1$, the coefficient of x^{r+1} is

$$\begin{aligned}[x^{r+1}]\Phi_\lambda(A) = \sum_{k=0}^{r} a_k\left(2^{r+1-k}-1\right)\binom{\lambda-k}{r+1-k}\\ - \lambda\sum_{k=0}^{r} a_k(-2)^{r-k}.\end{aligned} \tag{12}$$

The contribution involving a_r is

$$a_r\left((2-1)(\lambda-r)-\lambda\right) = -r\, a_r.$$

Consequently, once $a_0, \ldots, a_{r-1}$ have been fixed, there is a unique choice of a_r that makes the coefficient of x^{r+1} vanish. Solving for a_r gives exactly 3. Thus there is a unique formal series

$$A_\lambda(x) = \sum_{k\geq 0} \alpha_k(\lambda)x^k, \qquad \alpha_0 = 1, \tag{13}$$

such that

$$\Phi_\lambda(A_\lambda) = 0$$

as a formal power series. For fixed N, put

$$A_{\lambda,N}(x) = \sum_{k=0}^{N} \alpha_k(\lambda)x^k, \qquad p_N(n) = n^\lambda A_{\lambda,N}(1/n). \tag{14}$$

The coefficient of x^j in $\Phi_\lambda(A)$ depends only on $a_0, \dots, a_{j-1}$. Hence the truncation retains all cancellations through degree $N+1$:

$$\Phi_\lambda(A_{\lambda,N})(x) = O_{\lambda,N}(x^{N+2}) \qquad (x \to 0).$$

Using 9,

$$\boxed{(L_\lambda p_N)_n = O_{\lambda,N}\left(n^{\lambda-N-2}\right).} \tag{15}$$

It remains to prove that an approximate solution with such a small residual is close to the actual dominant solution.

1. Stability lemma for the recurrence

We use only the asymptotic basis 2 supplied in the statement. Lemma Let $\lambda \neq 0$, let V, W be solutions of the homogeneous recurrence 1 such that

$$V_n \sim n^\lambda, \qquad W_n \sim \frac{(-1)^n \lambda^n}{n!\, n^{\lambda+1}}. \tag{16}$$

Let $s \geq 0$. Suppose that, for all sufficiently large n,

$$y_{n+2} - y_{n+1} - \frac{\lambda}{n+2} y_n = g_n, \tag{17}$$

where

$$g_n = O(n^{\lambda-s-2}), \qquad y_n = o(n^\lambda). \tag{18}$$

Then

$$\boxed{y_n = O(n^{\lambda-s-1}).} \tag{19}$$

Proof Define the Casoratian

$$\Delta_n = V_{n+1}W_n - V_n W_{n+1}.$$

From 16,

$$\frac{V_{n+1}}{V_n} \longrightarrow 1$$

and

$$\frac{W_{n+1}}{W_n} = -\frac{\lambda}{n+1}\left(\frac{n}{n+1}\right)^{\lambda+1}(1+o(1)) = O(n^{-1}).$$

Therefore

$$\begin{aligned}
\Delta_{n+1} &= V_{n+2}W_{n+1} - V_{n+1}W_{n+2} \\
&= V_{n+1}W_{n+1}\left(\frac{V_{n+2}}{V_{n+1}} - \frac{W_{n+2}}{W_{n+1}}\right) \\
&\sim V_{n+1}W_{n+1}.
\end{aligned} \tag{20}$$

For all sufficiently large n, write

$$\begin{pmatrix} y_{n+1} \\ y_n \end{pmatrix} = \begin{pmatrix} V_{n+1} & W_{n+1} \\ V_n & W_n \end{pmatrix} \begin{pmatrix} A_n \\ B_n \end{pmatrix}. \tag{21}$$

The vector recurrence associated with 17 gives the standard variation-of-constants identities

$$A_{n+1} - A_n = g_n \frac{W_{n+1}}{\Delta_{n+1}}, \tag{22}$$

$$B_{n+1} - B_n = -g_n \frac{V_{n+1}}{\Delta_{n+1}}. \tag{23}$$

By 20,

$$\frac{W_{n+1}}{\Delta_{n+1}} = O(n^{-\lambda}), \qquad \frac{V_{n+1}}{\Delta_{n+1}} = O\left(\frac{1}{|W_{n+1}|}\right). \tag{24}$$

Consequently,

$$A_{n+1} - A_n = O(n^{-s-2}), \tag{25}$$

so A_n converges to a finite limit A_∞. We next control the B_nW_n contribution. From 23,

$$|B_nW_n| \le |B_{n_0}W_n| + C|W_n| \sum_{j=n_0}^{n-1} \frac{j^{\lambda-s-2}}{|W_{j+1}|}. \tag{26}$$

Set

$$T_j = \frac{j^{\lambda-s-2}}{|W_{j+1}|}.$$

Using $W_{j+1}/W_j = O(j^{-1})$,

$$\frac{T_{j-1}}{T_j} = \left(\frac{j-1}{j}\right)^{\lambda-s-2} \frac{|W_{j+1}|}{|W_j|} = O(j^{-1}). \tag{27}$$

Thus, for large j, $T_{j-1} \le T_j/2$, and hence

$$\sum_{j=n_0}^{n-1} T_j = O(T_{n-1}).$$

It follows from 26 that

$$B_n W_n = O(n^{\lambda-s-2}). \tag{28}$$

Here the fixed term $B_{n_0} W_n$ is harmless because W_n is smaller than every negative power of n: this follows directly from the factorial in 16. Since $V_n \sim n^{\lambda}$, 28 gives

$$\frac{B_n W_n}{V_n} = O(n^{-s-2}) \longrightarrow 0. \tag{29}$$

On the other hand, 21 gives

$$\frac{y_n}{V_n} = A_n + B_n \frac{W_n}{V_n}.$$

By hypothesis $y_n = o(n^{\lambda})$, so $y_n/V_n \to 0$. Together with 29, this forces

$$A_\infty = 0.$$

Therefore, by 25,

$$A_n = -\sum_{j=n}^{\infty} (A_{j+1} - A_j) = O(n^{-s-1}).$$

Finally,

$$y_n = A_n V_n + B_n W_n = O(n^{\lambda-s-1}) + O(n^{\lambda-s-2}),$$

which proves 19. □

1. Application of the stability lemma

Fix $N \geq 0$, and define p_N by 14. Set

$$y_n = v_n - p_N(n).$$

Since both v_n and $p_N(n)$ are equivalent to n^λ,

$$y_n = o(n^\lambda). \tag{30}$$

Moreover, $L_\lambda v = 0$, while 15 gives

$$(L_\lambda y)_n = -(L_\lambda p_N)_n = O(n^{\lambda - N - 2}). \tag{31}$$

For $\lambda \neq 0$, the stability lemma with $s = N$ yields

$$v_n - p_N(n) = O(n^{\lambda - N - 1}),$$

which is exactly 4. For $\lambda = 0$, recurrence 1 reduces to

$$u_{n+2} = u_{n+1}.$$

Thus every solution is constant from index 1 onward. The normalization $v_n \sim 1$ forces

$$v_n = 1 \qquad (n \geq 1).$$

Formula 3 gives $\alpha_r(0) = 0$ for every $r \geq 1$, so 4 also holds at $\lambda = 0$. This completes the proof for every real λ.

1. Structural properties of the coefficients

The recursion shows inductively that

$$\alpha_r(\lambda) \in \mathbb{Q}[\lambda].$$

In fact,

$$\deg \alpha_r = 2r, \qquad [\lambda^{2r}]\alpha_r(\lambda) = \frac{3^r}{2^r r!}. \tag{32}$$

Indeed, in 3 the only contribution of degree $2r$ comes from $k = r - 1$, through

$$(2^2 - 1)\binom{\lambda - r + 1}{2} = \frac{3}{2}\lambda^2 + \text{lower-degree terms}.$$

Thus, if c_r denotes the leading coefficient of α_r,

$$c_r = \frac{3}{2r} c_{r-1}, \qquad c_0 = 1,$$

which gives 32. Nonnegative integral parameters If $\lambda = m \in \mathbb{N}$, the expansion terminates. Indeed, set

$$P_m(n) = \sum_{k=0}^{m} \alpha_k(m) n^{m-k}.$$

From 15 with $N = m$,

$$L_m P_m(n) = O(n^{-2}).$$

But

$$R_m(n) = (n+2)\big(P_m(n+2) - P_m(n+1)\big) - mP_m(n)$$

is a polynomial in n, and

$$R_m(n) = (n+2) L_m P_m(n) = O(n^{-1}).$$

A polynomial tending to zero along the positive integers is identically zero. Therefore P_m is an exact solution of 1. By uniqueness of the formal recursion,

$$\boxed{\alpha_r(m) = 0 \qquad (r > m).} \tag{33}$$

Consequently,

$$\lambda(\lambda-1)\cdots(\lambda-r+1)$$

divides the polynomial $\alpha_r(\lambda)$.

1. Boundary and small-parameter checks

$\lambda = 0$ As already noted,

$$v_n = 1, \qquad \alpha_r(0) = 0 \quad (r \geq 1).$$

$\lambda = 1$ The formulas give

$$\alpha_1(1) = 2, \qquad \alpha_r(1) = 0 \quad (r \geq 2).$$

The corresponding exact dominant solution is

$$v_n = n + 2.$$

Indeed,

$$v_{n+1} + \frac{v_n}{n+2} = n + 3 + \frac{n+2}{n+2} = n + 4 = v_{n+2}.$$

$\lambda = 2$ Here

$$\alpha_1(2) = 7, \qquad \alpha_2(2) = 10, \qquad \alpha_r(2) = 0 \quad (r \geq 3).$$

Thus

$$v_n = n^2 + 7n + 10 = (n + 2)(n + 5)$$

is an exact solution. Indeed,

$$v_{n+1} = n^2 + 9n + 18$$

and

$$\frac{2v_n}{n+2} = 2(n + 5) = 2n + 10,$$

so their sum is $n^2 + 11n + 28 = v_{n+2}$. $\lambda = -1$ The recursion gives

$$\alpha_1(-1) = 1, \qquad \alpha_2(-1) = 2, \qquad \alpha_3(-1) = 5, \qquad \alpha_4(-1) = 15.$$

Hence

$$v_n = \frac{1}{n} + \frac{1}{n^2} + \frac{2}{n^3} + \frac{5}{n^4} + \frac{15}{n^5} + O(n^{-6}).$$

This also illustrates that negative integral values of λ cause no logarithmic terms in the sequence asymptotic expansion, despite the fact that logarithms can appear in the local singular form of the generating function.

1. Exact generating-function check

For completeness, let

$$U(z) = \sum_{n \geq 0} u_n z^n.$$

Multiplying 1 by $n+2$, summing, and simplifying gives

$$(1-z)U'(z) = (1+\lambda z)U(z) + (u_1 - u_0). \tag{34}$$

Since

$$H_\lambda(z) = e^{-\lambda z}(1-z)^{-\lambda-1}$$

solves the homogeneous differential equation, every solution has, locally at $z=0$,

$$\boxed{U(z) = e^{-\lambda z}(1-z)^{-\lambda-1}\left[u_0 + (u_1 - u_0)\int_0^z e^{\lambda t}(1-t)^\lambda \, dt\right].} \tag{35}$$

Formula 35 independently confirms the dominant algebraic mode and the factorially small complementary mode. The proof above avoids having to split the singularity analysis into the nonintegral and negative-integral cases.

1. Reproducible coefficient generator

The following SymPy code implements recursion 3. It is not used in the proof. PythonExécuterimport sympy as sp

lam = sp.symbols("lambda") max_order = 8

alpha = [sp.Integer1]

for r in range1, $max_order + 1$: rhs = sum(alpha[k]

- (

(2 ** (r + 1 - k) - 1)

- sp.binomial(lam - k, r + 1 - k)
- lam * (-2) ** (r - k)

) for k in range$r$)

coefficient = sp.factor(sp.expand_func(sp.simplify(rhs / r))) alpha.append$coefficient$

for r, coefficient in enumerate$alpha$: print$f"alpha_r = ", coefficient$

Final conclusion Q855 has a complete affirmative answer. For each fixed real λ, the normalized dominant solution possesses a unique full asymptotic expansion in descending integral powers of n. Its coefficients are the universal polynomials determined by 3, and the remainder after N terms is actually

$$O(n^{\lambda-N-1}),$$

rather than merely $o(n^{\lambda-N})$.

Formal verification. Lean source; online type-check.

26 Question Q865

Informal verdict. A — Complete and apparently correct, with **high confidence**.

Lean coverage. Full canonical Q865 coverage. The preserved 1,653-line Q865.lean gives the global product definition and natural-evaluation formula for every A_k, summability and the exact positive-root series for every n >= 2, and the odd-index limit for rmin. Q865Completion.lean proves polynomial nonzeroness and root existence for every m >= 2, finiteness/nonemptiness of the root-modulus set, attainment of its infimum, universal minimality, an IsLeast form, and the unconditional Q865_answer synthesis theorem. The two modules are not yet imported into lean/RMS.

26.1 Audited informal answer

Q865 — Complete solution

1. Verified statement

For $n \geq 2$, define

$$P_n(X) = X^n - \sum_{k=0}^{n-1} X^k.$$

Let ρ_n be its unique positive real root, let r_n be the smallest modulus of its complex roots, and put

$$u_n = 2^{-n-1}.$$

The earlier response R469 established an asymptotic expansion

$$\rho_n = 2\left(1 - \sum_{k=0}^{N} A_k(n)u_n^{k+1} + o(u_n^{N+1})\right),$$

where the A_k are polynomials, together with $r_n \to 1$ and

$$2n(1 - r_{2n}) \longrightarrow \log 3.$$

Q865 asks for a closed formula for A_k, whether the series for ρ_n converges exactly, and whether the analogous odd-index limit equals $\log 3$. There is no apparent ambiguity or typo in the displayed statement. Luc Pommeret

1. Final theorem

For every $k \geq 0$,

$$\boxed{A_k(X) = \frac{1}{k+1}\binom{(k+1)(X+1)-2}{k} = \frac{1}{(k+1)!}\prod_{j=0}^{k-1}((k+1)X+j)}$$

with the empty product interpreted as 1. Thus, for example,

$$A_0(X) = 1, \qquad A_1(X) = X, \qquad A_2(X) = \frac{X(3X+1)}{2}.$$

For every integer $n \geq 2$, the series in part b converges absolutely and

$$\boxed{\rho_n = 2\left(1 - \sum_{k=0}^{\infty} A_k(n) u_n^{k+1}\right).}$$

Equivalently,

$$\boxed{\rho_n = 2\left(1 - \sum_{m=1}^{\infty} \frac{1}{m}\binom{(n+1)m-2}{m-1} 2^{-m(n+1)}\right).}$$

Finally, a stronger statement than part c holds:

$$\boxed{r_m = 1 - \frac{\log 3}{m} + \frac{\frac{1}{2}(\log 3)^2 + \frac{1}{3}\log 3}{m^2} + O(m^{-3}) \qquad (m \to \infty).}$$

In particular,

$$\boxed{\lim_{n\to\infty} (2n+1)(1 - r_{2n+1}) = \log 3.}$$

1. Algebraic reduction

A direct calculation gives

$$(z-1)P_n(z) = z^{n+1} - 2z^n + 1 = 1 - z^n(2-z).$$

Consequently, for $z \neq 1$,

$$P_n(z) = 0 \quad \Longleftrightarrow \quad z^n(2-z) = 1. \tag{3.1}$$

For the positive root, write

$$\rho_n = 2(1-y_n).$$

Then (3.1) becomes

$$2^n(1-y_n)^n\, 2y_n = 1,$$

or

$$y_n(1-y_n)^n = 2^{-n-1} = u_n. \tag{3.2}$$

The relevant solution satisfies

$$0 < y_n < \frac{1}{n+1}. \tag{3.3}$$

Indeed, the function

$$F_n(y) = y(1-y)^n$$

is strictly increasing on $[0, 1/(n+1)]$, with maximum there equal to

$$\mathcal{R}_n = F_n\left(\frac{1}{n+1}\right) = \frac{n^n}{(n+1)^{n+1}}. \tag{3.4}$$

1. Inverting $y(1-y)^n = u$

We first establish the coefficient formula directly. Lemma Let $n \geq 1$ be fixed. The analytic solution near $u = 0$ of

$$Y = u(1-Y)^{-n}, \qquad Y(0) = 0, \tag{4.1}$$

has the expansion

$$Y(u) = \sum_{m=1}^{\infty} c_m(n) u^m,$$

where

$$c_m(n) = \frac{1}{m}\binom{(n+1)m-2}{m-1}. \tag{4.2}$$

Proof Put

$$\Phi(t) = (1-t)^{-n}.$$

Equation (4.1) is $Y = u\Phi(Y)$. For $m \geq 1$, Cauchy's coefficient formula gives

$$[u^m]Y(u) = \operatorname*{Res}_{u=0} \frac{Y(u)}{u^{m+1}}\, du.$$

Using the local substitution

$$u = \frac{t}{\Phi(t)},$$

we obtain

$$[u^m]Y(u) = \operatorname*{Res}_{t=0} \left(t^{-m}\Phi(t)^m - t^{-m+1}\Phi(t)^{m-1}\Phi'(t)\right) dt.$$

Now

$$\Phi^{m-1}\Phi' = \frac{1}{m}(\Phi^m)'.$$

Moreover, the residue of a derivative vanishes, so

$$0 = \operatorname*{Res}_{t=0} \left(t^{-m+1}\Phi(t)^m\right)' = -(m-1)\operatorname*{Res}_{t=0} t^{-m}\Phi(t)^m + \operatorname*{Res}_{t=0} t^{-m+1}(\Phi(t)^m)'.$$

It follows that

$$[u^m]Y(u) = \frac{1}{m}\operatorname*{Res}_{t=0} t^{-m}\Phi(t)^m = \frac{1}{m}[t^{m-1}]\Phi(t)^m.$$

Since

$$\Phi(t)^m = (1-t)^{-nm},$$

we find

$$[t^{m-1}](1-t)^{-nm} = \binom{nm+m-2}{m-1},$$

which proves (4.2). □

1. Answer to part a

Applying the lemma to (3.2), we have formally

$$y_n = \sum_{m\geq 1} \frac{1}{m}\binom{(n+1)m-2}{m-1} u_n^m.$$

Writing $m = k+1$, the coefficient of u_n^{k+1} is therefore

$$A_k(n) = \frac{1}{k+1}\binom{(n+1)(k+1)-2}{k}.$$

Since

$$(n+1)(k+1) - 2 = (k+1)n + k - 1,$$

this becomes

$$A_k(n) = \frac{1}{k+1}\binom{(k+1)n+k-1}{k}.$$

As a polynomial in an arbitrary variable X,

$$\boxed{A_k(X) = \frac{1}{k+1}\binom{(k+1)X+k-1}{k}.}$$

Expanding the binomial polynomial,

$$\binom{(k+1)X+k-1}{k} = \frac{1}{k!}\prod_{j=0}^{k-1}\bigl((k+1)X+j\bigr),$$

and hence

$$\boxed{A_k(X) = \frac{1}{(k+1)!}\prod_{j=0}^{k-1}\bigl((k+1)X+j\bigr).}$$

This is indeed a polynomial of degree k.

1. Answer to part b: convergence and exact equality

6.1 Radius of convergence For fixed n,

$$c_m(n) = \frac{((n+1)m-2)!}{m!(nm-1)!}.$$

Using

$$\log(M!) = M \log M - M + O(\log M),$$

we obtain

$$\lim_{m\to\infty} c_m(n)^{1/m} = \frac{(n+1)^{n+1}}{n^n}.$$

Thus the radius of convergence of the inverse series is exactly

$$\boxed{\mathcal{R}_n = \frac{n^n}{(n+1)^{n+1}}.} \tag{6.1}$$

This agrees with the critical value (3.4). We now check that u_n lies strictly inside that disk. Since $n/(n+1) \geq 2/3$,

$$\mathcal{R}_n = \frac{1}{n+1}\left(\frac{n}{n+1}\right)^n \geq \frac{1}{n+1}\left(\frac{2}{3}\right)^n.$$

For $n \geq 2$,

$$\frac{(2/3)^n/(n+1)}{2^{-n-1}} = \frac{2(4/3)^n}{n+1} > 1.$$

Indeed, the latter expression equals $32/27$ at $n = 2$, and its ratio at consecutive indices is

$$\frac{4(n+1)}{3(n+2)} \geq 1 \qquad (n \geq 2).$$

Therefore

$$u_n < \mathcal{R}_n \qquad (n \geq 2). \tag{6.2}$$

6.2 Identification of the sum Let $Y_n(u)$ denote the sum of the power series. The polynomial identity

$$Y_n(u)(1 - Y_n(u))^n = u \tag{6.3}$$

holds near 0, hence throughout the disk $|u| < \mathcal{R}_n$ by the identity theorem. For $0 < u < \mathcal{R}_n$, all coefficients of Y_n are positive. Moreover, $Y_n(u)$ cannot reach $1/(n+1)$, since that would give, by (6.3),

$$u = \frac{n^n}{(n+1)^{n+1}} = \mathcal{R}_n.$$

Thus

$$0 < Y_n(u) < \frac{1}{n+1} \qquad (0 < u < \mathcal{R}_n).$$

At $u = u_n$, set

$$\widetilde{\rho}_n = 2(1 - Y_n(u_n)).$$

Then $\widetilde{\rho}_n > 2n/(n+1) > 1$, and (6.3) gives

$$\widetilde{\rho}_n^n(2 - \widetilde{\rho}_n) = 1.$$

Therefore $P_n(\widetilde{\rho}_n) = 0$. It is the unique positive root: on $(1, 2)$, the function

$$x \longmapsto x^n(2 - x)$$

increases up to $2n/(n+1)$, then decreases strictly to 0; besides the extraneous solution $x = 1$, the equation $x^n(2 - x) = 1$ consequently has exactly one solution. Hence $\widetilde{\rho}_n = \rho_n$, proving

$$\boxed{\rho_n = 2\left(1 - \sum_{k=0}^{\infty} A_k(n) u_n^{k+1}\right).}$$

6.3 Recovery of the stated asymptotic expansion For completeness, the exact series implies the required remainder estimate uniformly for each fixed truncation order. For $m \geq 1$,

$$c_m(n) \leq \binom{(n+1)m}{m} \leq \big(e(n+1)\big)^m.$$

Let

$$q_n = e(n+1)u_n.$$

Then $q_n \to 0$. For every fixed N, and all sufficiently large n,

$$
\begin{aligned}
0 &\le Y_n(u_n) - \sum_{m=1}^{N+1} c_m(n) u_n^m \\
&\le \sum_{m=N+2}^{\infty} q_n^m = \frac{q_n^{N+2}}{1-q_n}.
\end{aligned}
$$

Consequently,

$$
Y_n(u_n) - \sum_{m=1}^{N+1} c_m(n) u_n^m = O_N\big((n+1)^{N+2} u_n^{N+2}\big) = o(u_n^{N+1}),
$$

because $(n+1)^{N+2} u_n \to 0$.

1. Answer to part c

To avoid conflicting indices, write m for the degree and retain r_m for the smallest root modulus. 7.1 A universal lower bound Let $s_m \in (0,1)$ be the unique solution of

$$
s_m^m (2 + s_m) = 1. \tag{7.1}
$$

Uniqueness follows because $s \mapsto s^m(2+s)$ is strictly increasing on $(0,\infty)$. If z is any root of P_m, then by (3.1),

$$
1 = |z|^m |2 - z| \le |z|^m (2 + |z|).
$$

By the definition of s_m,

$$
|z| \ge s_m.
$$

Therefore

$$
r_m \ge s_m. \tag{7.2}
$$

For even m, equality holds: the number $-s_m$ satisfies

$$
(-s_m)^m (2 + s_m) = 1,
$$

and hence is a root of P_m. Thus

$$
r_m = s_m \qquad (m \text{ even}). \tag{7.3}
$$

7.2 A near-negative root for odd m Suppose now that

$$m = 2q + 1 \geq 3.$$

Put

$$\theta_0 = \pi - \frac{\pi}{m}.$$

For $\theta \in [\theta_0, \pi]$, consider

$$H_\theta(t) = t^m |2 - te^{i\theta}|, \qquad 0 \leq t \leq 1.$$

On this interval, $\cos\theta < 0$, and

$$\frac{\partial}{\partial t} |2 - te^{i\theta}|^2 = 2t - 4\cos\theta > 0.$$

Thus H_θ is strictly increasing. Since

$$H_\theta(0) = 0, \qquad H_\theta(1) = |2 - e^{i\theta}| > 1,$$

there is a unique $t(\theta) \in (0, 1)$ such that

$$t(\theta)^m |2 - t(\theta)e^{i\theta}| = 1. \tag{7.4}$$

The function $t(\theta)$ is continuous. Let

$$\alpha(\theta) = \mathrm{Arg}\big(2 - t(\theta)e^{i\theta}\big),$$

where the argument is chosen in $(-\pi/2, 0]$. It is continuous, negative for $\theta < \pi$, and $\alpha(\pi) = 0$. Define

$$\Phi(\theta) = m\theta + \alpha(\theta).$$

At the left endpoint,

$$m\theta_0 = (m - 1)\pi = 2q\pi,$$

so

$$\Phi(\theta_0) < 2q\pi.$$

At the right endpoint,

$$\Phi(\pi) = m\pi = (2q + 1)\pi > 2q\pi.$$

The intermediate value theorem therefore gives some

$$\theta_m \in \left(\pi - \frac{\pi}{m}, \pi\right)$$

such that

$$m\theta_m + \alpha(\theta_m) = 2q\pi.$$

Let

$$t_m = t(\theta_m), \qquad z_m = t_m e^{i\theta_m}.$$

Equation (7.4) gives the modulus of $z_m^m(2 - z_m)$, while the last phase equation shows that its argument is $2q\pi$. Hence

$$z_m^m(2 - z_m) = 1.$$

Since $z_m \neq 1$, this gives $P_m(z_m) = 0$. Therefore

$$s_m \leq r_m \leq t_m < 1. \tag{7.5}$$

7.3 Asymptotics of s_m Let

$$L = \log 3, \qquad x_m = 1 - s_m.$$

From (7.1),

$$s_m^m = \frac{1}{2 + s_m} > \frac{1}{3},$$

so

$$s_m > 3^{-1/m}, \qquad x_m = O(m^{-1}).$$

Equation (7.1) becomes

$$-m \log(1 - x_m) = \log(3 - x_m).$$

Using

$$-\log(1 - x) = x + \frac{x^2}{2} + O(x^3),$$

and

$$\log(3 - x) = L - \frac{x}{3} + O(x^2),$$

we obtain

$$mx_m + \frac{m}{2}x_m^2 + O(mx_m^3) = L - \frac{x_m}{3} + O(x_m^2).$$

First this gives $x_m = L/m + O(m^{-2})$. Substituting this estimate back yields

$$x_m = \frac{L}{m} - \frac{\frac{1}{2}L^2 + \frac{1}{3}L}{m^2} + O(m^{-3}).$$

Thus

$$s_m = 1 - \frac{L}{m} + \frac{\frac{1}{2}L^2 + \frac{1}{3}L}{m^2} + O(m^{-3}). \tag{7.6}$$

7.4 Comparison of t_m and s_m For odd m, write

$$\theta_m = \pi - \delta_m, \qquad 0 < \delta_m < \frac{\pi}{m}.$$

Then

$$|2 - t_m e^{i\theta_m}| = |2 + t_m e^{-i\delta_m}|,$$

and

$$|2 + t_m e^{-i\delta_m}|^2 = (2 + t_m)^2 - 4t_m(1 - \cos\delta_m).$$

Therefore

$$\log(2 + t_m) - \log|2 - t_m e^{i\theta_m}| = O(\delta_m^2) = O(m^{-2}). \tag{7.7}$$

Define

$$G_m(x) = -m\log x - \log(2 + x).$$

We have $G_m(s_m) = 0$. Since t_m satisfies

$$-m\log t_m = \log|2 - t_m e^{i\theta_m}|,$$

equation (7.7) gives

$$G_m(t_m) = O(m^{-2}).$$

Moreover,

$$|G'_m(x)| = \frac{m}{x} + \frac{1}{2+x} \geq m \qquad (0 < x \leq 1).$$

The mean value theorem now gives

$$t_m - s_m = O(m^{-3}). \tag{7.8}$$

Combining (7.5), (7.6), and (7.8), we conclude that for odd m,

$$r_m = 1 - \frac{L}{m} + \frac{\frac{1}{2}L^2 + \frac{1}{3}L}{m^2} + O(m^{-3}).$$

For even m, the same formula follows from $r_m = s_m$. Hence it holds for all m:

$$\boxed{r_m = 1 - \frac{\log 3}{m} + \frac{\frac{1}{2}(\log 3)^2 + \frac{1}{3}\log 3}{m^2} + O(m^{-3}).}$$

Taking $m = 2n+1$ gives

$$(2n+1)(1 - r_{2n+1}) = \log 3 - \frac{\frac{1}{2}(\log 3)^2 + \frac{1}{3}\log 3}{2n+1} + O(n^{-2}),$$

and therefore

$$\boxed{\lim_{n\to\infty}(2n+1)(1 - r_{2n+1}) = \log 3.}$$

1. Boundary checks

For $n = 2$,

$$P_2(X) = X^2 - X - 1, \qquad \rho_2 = \frac{1+\sqrt{5}}{2}, \qquad r_2 = \frac{\sqrt{5}-1}{2}.$$

The latter is exactly the solution of

$$s^2(2+s) = 1,$$

as predicted by the even-degree argument. The restriction $n \geq 2$ also avoids the degenerate case $n = 1$: then $P_1(X) = X - 1$, while

$$u_1 = \frac{1}{4} = \mathcal{R}_1$$

lies exactly at the branch point of the inverse $y(1-y) = u$, rather than strictly inside its disk of convergence.

Formal verification. Lean source; online type-check.

27 Question Q877

Informal verdict. A — Complete and apparently correct, with **high confidence (0.95)**.

Lean coverage. Q877.lean unconditionally proves the finite-dimensional iid integrable matrix strong law and an entrywise weak-law form, then the singular-mean obstruction; q877a gives a deterministic x != 0 annihilated by every A_n on one probability-one event under almost-sure positive semidefiniteness; the explicit 3-by-3 Rademacher sequence is matrix-iid, integrable, entrywise bounded, centered, symmetric, rank 2 everywhere, has every partial sum singular, and rules out even an almost-sure kernel vector for A_0; q877b derives the same nonzero simultaneous conclusion from a deterministic positive-definite H with H*A_0 positive semidefinite almost surely, proving iid transfer, determinant/property-P preservation and cancellation by H internally. The source explicitly documents the printed omission of X != 0 and the p-to-N reindexing.

27.1 Audited informal answer

Q877 Restatement and a necessary correction Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $d \geq 1$, and let

$$A_0, A_1, A_2, \ldots : \Omega \longrightarrow M_d(\mathbb{R})$$

be independent and identically distributed random matrices satisfying

$$\mathbb{E}\|A_0\| < \infty$$

for one, hence every, matrix norm. Put

$$S_N := A_0 + \cdots + A_{N-1}, \qquad N \geq 1.$$

Property (P) is the existence of $c \in (0,1)$ such that

$$\mathbb{P}(\det S_N = 0) \geq c \qquad \text{for every } N \geq 1.$$

The questions are:

Prove that, when the A_n are symmetric positive semidefinite, property (P) forces a common almost-sure kernel.

Find additional hypotheses under which the same conclusion remains valid more generally.

The PDF literally omits the condition $X \neq 0$. Without it, $X = 0$ makes both conclusions vacuous. I therefore solve the clearly intended statement with a deterministic nonzero

$$X \in \mathbb{R}^d, \qquad \mathbb{P}\big(A_n X = 0 \text{ for all } n\big) = 1.$$

Here "symmetric positive" is interpreted in the standard French sense: symmetric positive semidefinite. Luc Pommeret+1

Final results The following conclusions hold. Part a: exact dichotomy For iid positive semidefinite matrices, the first-moment assumption is in fact unnecessary. Exactly one of the following occurs:

$$\boxed{\exists\, x \in \mathbb{R}^d \setminus \{0\} : \mathbb{P}(A_0 x = 0) = 1,}$$

in which case

$$\mathbb{P}(\det S_N = 0) = 1 \qquad \forall N,$$

or else

$$\boxed{\mathbb{P}(\det S_N = 0) \longrightarrow 0.}$$

Consequently, property (P) is equivalent to the existence of a deterministic nonzero common null vector. Part b: failure in general, and an exact positive theorem Symmetry, boundedness, centering, and constant rank do not suffice without positivity: there are bounded iid symmetric rank-$(d-1)$ matrices for which every S_N is singular but there is no common deterministic null vector. On the other hand, the following natural additional hypothesis gives an exact affirmative answer:

If two independent copies commute almost surely,

$$\mathbb{P}(A_0 A_1 = A_1 A_0) = 1,$$

then, without any moment assumption,

$$\limsup_{N \to \infty} \mathbb{P}(\det S_N = 0) > 0$$

holds if and only if the matrices have a deterministic nonzero common null vector.

Thus pairwise commutation of the support is a complete sufficient structural hypothesis for part b.

1. A general mean obstruction

We first isolate the elementary probabilistic mechanism behind part a. Lemma 1 Let $A_0, A_1, \dots$ be iid integrable random matrices and let

$$M := \mathbb{E}A_0.$$

If M is invertible, then

$$\mathbb{P}(\det S_N = 0) \longrightarrow 0.$$

Equivalently,

$$\limsup_{N\to\infty} \mathbb{P}(\det S_N = 0) > 0 \quad \Longrightarrow \quad \det(\mathbb{E}A_0) = 0.$$

Proof We first recall the weak law of large numbers in the finite-dimensional matrix space:

$$\frac{S_N}{N} \longrightarrow M \qquad \text{in probability.}$$

For completeness, this follows by truncation. Use the Frobenius norm and write

$$A_k^{(R)} = A_k \mathbf{1}_{\{\|A_k\|_F \le R\}}, \qquad T_k^{(R)} = A_k - A_k^{(R)}.$$

For fixed R, independence and centering give

$$\mathbb{E}\left\| \frac{1}{N}\sum_{k=0}^{N-1}\left(A_k^{(R)} - \mathbb{E}A_k^{(R)}\right)\right\|_F^2 = \frac{1}{N}\mathbb{E}\left\|A_0^{(R)} - \mathbb{E}A_0^{(R)}\right\|_F^2,$$

which tends to 0. Hence the bounded part converges in probability. For the tail,

$$\mathbb{E}\left\| \frac{1}{N}\sum_{k=0}^{N-1}\left(T_k^{(R)} - \mathbb{E}T_k^{(R)}\right)\right\|_F \le 2\mathbb{E}\|T_0^{(R)}\|_F.$$

The right-hand side tends to 0 as $R \to \infty$, by integrability. Markov's inequality completes the proof of the weak law. Now suppose M is invertible. Since $GL_d(\mathbb{R})$ is open, there exists $\delta > 0$ such that

$$\|B - M\|_F < \delta \quad \Longrightarrow \quad \det B \neq 0.$$

Because multiplication by $N \neq 0$ does not affect singularity,

$$
\begin{aligned}
\mathbb{P}(\det S_N = 0) &= \mathbb{P}\left(\det \frac{S_N}{N} = 0\right) \\
&\le \mathbb{P}\left(\left\|\frac{S_N}{N} - M\right\|_F \ge \delta\right) \longrightarrow 0.
\end{aligned}
$$

This proves the lemma. $\square$

1. Solution of part a

Assume now that

$$A_n = A_n^{\mathsf{T}} \succeq 0 \qquad \text{almost surely.}$$

By property (P) and Lemma 1, the positive semidefinite matrix

$$M = \mathbb{E}A_0$$

is singular. Choose

$$x \in \ker M \setminus \{0\}.$$

Then

$$0 = x^{\mathsf{T}} M x = \mathbb{E}\big[x^{\mathsf{T}} A_0 x\big].$$

The random variable $x^{\mathsf{T}} A_0 x$ is nonnegative. A nonnegative integrable random variable with expectation zero vanishes almost surely, so

$$x^{\mathsf{T}} A_0 x = 0 \qquad \text{almost surely.}$$

For a positive semidefinite matrix A,

$$x^{\mathsf{T}} A x = \|A^{1/2} x\|^2,$$

and therefore

$$x^{\mathsf{T}} A x = 0 \quad \Longrightarrow \quad Ax = 0.$$

Consequently,

$$\mathbb{P}(A_0 x = 0) = 1.$$

The A_n have the same law, so

$$\mathbb{P}(A_n x = 0) = 1 \qquad \forall n.$$

Since the index set is countable,

$$\mathbb{P}\big(A_n x = 0 \text{ for every } n\big) = \mathbb{P}\left(\bigcap_{n \geq 0} \{A_n x = 0\}\right) = 1.$$

This proves part a.

Stronger equivalence in the positive semidefinite case For integrable iid positive semidefinite matrices, the following four statements are equivalent:

$$\begin{array}{ll} \text{(i)} & \limsup_{N\to\infty} \mathbb{P}(\det S_N = 0) > 0, \\ \text{(ii)} & \mathbb{E}A_0 \text{ is singular}, \\ \text{(iii)} & \exists\, x \neq 0 : \ \mathbb{P}(A_0 x = 0) = 1, \\ \text{(iv)} & \mathbb{P}(\det S_N = 0) = 1 \quad \forall N. \end{array}$$

We proved $(i) \Rightarrow (ii) \Rightarrow (iii)$. If iii holds, then

$$S_N x = 0$$

almost surely, proving iv; and iv trivially implies i. Thus, in the positive semidefinite setting, a uniform lower bound cannot take a genuinely intermediate value: it forces the singularity probability to be 1 for every partial sum.

The first moment is actually unnecessary in part a For an arbitrary positive semidefinite random matrix A, define

$$B = \frac{A}{1 + \|A\|}.$$

Then B is bounded, positive semidefinite, and

$$\ker B = \ker A.$$

For positive semidefinite matrices $C_1, \ldots, C_N$,

$$\ker(C_1 + \cdots + C_N) = \ker C_1 \cap \cdots \cap \ker C_N.$$

Indeed, if $(C_1 + \cdots + C_N)x = 0$, then

$$0 = x^\mathsf{T}(C_1 + \cdots + C_N)x = \sum_{j=1}^{N} x^\mathsf{T} C_j x,$$

and every summand is nonnegative, hence every $C_j x = 0$. Consequently, if

$$B_n = \frac{A_n}{1 + \|A_n\|},$$

then for every N,

$$\ker \sum_{n=0}^{N-1} A_n = \bigcap_{n=0}^{N-1} \ker A_n = \bigcap_{n=0}^{N-1} \ker B_n = \ker \sum_{n=0}^{N-1} B_n.$$

Thus the singularity events are unchanged. The B_n are bounded, so the preceding proof applies to them. Hence the conclusion of part a holds without any moment assumption. In particular,

$$q_N := \mathbb{P}(\det S_N = 0)$$

is nonincreasing in N, and either $q_N = 1$ for every N, or $q_N \downarrow 0$.

1. Why additional assumptions are indispensable in part b

Let $(\xi_n, \eta_n)_{n \geq 0}$ be iid pairs of independent Rademacher variables:

$$\mathbb{P}(\xi_n = 1) = \mathbb{P}(\xi_n = -1) = \frac{1}{2}, \qquad \mathbb{P}(\eta_n = 1) = \mathbb{P}(\eta_n = -1) = \frac{1}{2}.$$

Define

$$A_n = \begin{pmatrix} 0 & \xi_n & \eta_n \\ \xi_n & 0 & 0 \\ \eta_n & 0 & 0 \end{pmatrix}.$$

These matrices are:
real symmetric;
bounded;
centered;
of rank 2 almost surely;
iid and possessing moments of every order.
For any $x, y \in \mathbb{R}$, put

$$M(x, y) = \begin{pmatrix} 0 & x & y \\ x & 0 & 0 \\ y & 0 & 0 \end{pmatrix}.$$

Every such matrix is singular. Moreover,

$$S_N = M\left(\sum_{n=0}^{N-1} \xi_n,\ \sum_{n=0}^{N-1} \eta_n\right),$$

so

$$\mathbb{P}(\det S_N = 0) = 1 \qquad \forall N.$$

Nevertheless, there is no nonzero deterministic common null vector. Indeed, if

$$v = (v_1, v_2, v_3)^{\mathsf{T}}$$

were annihilated almost surely, it would in particular be annihilated by the two support matrices $M(1,1)$ and $M(1,-1)$. The equations

$$M(1,1)v = 0, \qquad M(1,-1)v = 0$$

give

$$v_1 = 0, \qquad v_2 + v_3 = 0, \qquad v_2 - v_3 = 0,$$

hence $v = 0$. Thus even the conjunction

$$\text{symmetric} + \text{bounded} + \text{centered} + \text{fixed rank}$$

does not suffice. The positivity hypothesis in part a is doing genuine work.

1. A precise affirmative answer to part b: commuting matrices

We prove the following exact theorem. Theorem 2 — commuting-support classification Let $A_0, A_1, \ldots$ be iid real $d \times d$ random matrices. No moment assumption is required. Assume

$$\mathbb{P}(A_0 A_1 = A_1 A_0) = 1. \tag{C}$$

Then the following are equivalent:
$\limsup_{N\to\infty} \mathbb{P}(\det S_N = 0) > 0$;
there exists a deterministic $x \in \mathbb{R}^d \setminus \{0\}$ such that

$$\mathbb{P}(A_0 x = 0) = 1;$$

$\mathbb{P}(\det S_N = 0) = 1$ for every N.

In particular, under C, property (P) implies the desired conclusion. The proof uses two elementary lemmas.

Lemma 2 — atoms of sums of iid variables Let $Z_1, Z_2, \dots$ be iid random variables with values in a finite-dimensional real vector space, or in $\mathbb{C}$. Then either

$$Z_1 = 0 \quad \text{almost surely,}$$

or

$$\mathbb{P}(Z_1 + \cdots + Z_N = 0) \longrightarrow 0.$$

Proof Let μ be the law of Z_1, and let D be its set of atoms. The set D is countable. Put

$$a = \mu(D).$$

Case 1: $a < 1$ Write

$$\mu = a\mu_{\text{at}} + (1-a)\mu_{\text{na}},$$

where μ_{na} has no atoms. The convolution of an atomless probability measure with any probability measure is atomless, because

$$(\mu_{\text{na}} * \nu)(\{z\}) = \int \mu_{\text{na}}(\{z-y\})\,\nu(dy) = 0.$$

Therefore an atom of μ^{*N} can arise only from the term in which all N variables were sampled from the atomic component. Hence

$$\sup_z \mu^{*N}(\{z\}) \le a^N \longrightarrow 0.$$

Case 2: $a = 1$ The law is purely atomic. If it is a point mass at $z_0 \neq 0$, then

$$Z_1 + \cdots + Z_N = N z_0 \neq 0.$$

The only exceptional point mass is δ_0. Otherwise choose two distinct atoms $u \neq v$, of masses $\alpha, \beta > 0$. Set

$$r = \alpha + \beta, \qquad q = \frac{\beta}{\alpha + \beta} \in (0,1).$$

Let M_N be the number of indices $i \le N$ for which $Z_i \in \{u, v\}$. Then

$$M_N \sim \mathrm{Bin}(N,r), \qquad M_N \longrightarrow \infty \quad \text{in probability.}$$

Condition on the set of these indices and on all values outside $\{u,v\}$. Given $M_N = m$, the contribution of the selected variables is

$$mu + K(v-u), \qquad K \sim \mathrm{Bin}(m,q).$$

Because $v-u \neq 0$, for any prescribed value of the total sum there is at most one possible integer K. Consequently,

$$\mathbb{P}(Z_1 + \cdots + Z_N = 0 \mid M_N = m, \text{ other data}) \leq b_m(q),$$

where

$$b_m(q) = \max_{0 \leq k \leq m} \binom{m}{k} q^k (1-q)^{m-k}.$$

Elementary Stirling bounds give

$$b_m(q) = O_q(m^{-1/2}),$$

so $b_m(q) \to 0$. Since $M_N \to \infty$ in probability and $0 \leq b_m(q) \leq 1$,

$$\mathbb{E}[b_{M_N}(q)] \longrightarrow 0.$$

This proves the lemma. □

Lemma 3 — simultaneous triangularization Every commuting family of complex $d \times d$ matrices can be simultaneously upper triangularized. Proof First prove that a commuting family $\mathcal{A}$ has a common eigenvector. Proceed by induction on the dimension. If every member of $\mathcal{A}$ is scalar, every nonzero vector works. Otherwise choose a nonscalar $T \in \mathcal{A}$ and an eigenspace

$$E_\lambda = \ker(T - \lambda I).$$

This is a nonzero proper subspace. Since every $B \in \mathcal{A}$ commutes with T,

$$B(E_\lambda) \subset E_\lambda.$$

The restricted family on E_λ is commuting, so induction gives a common eigenvector. After finding a common invariant line, pass to the quotient and iterate. This gives a common invariant flag and hence simultaneous upper triangularization. □

Proof of Theorem 2 Let μ be the common law and let

$$K = \operatorname{supp}\mu.$$

Step 1: the support is a commuting family Assumption C, together with independence, says

$$(\mu \otimes \mu)\{(A, B) : AB = BA\} = 1.$$

If two support points $A, B \in K$ did not commute, continuity of the commutator would give neighborhoods $U \ni A$, $V \ni B$ such that no pair in $U \times V$ commutes. Since A, B are support points,

$$\mu(U)\mu(V) > 0,$$

contradicting C. Thus every two matrices in K commute. Let $\mathcal{A}$ be the complex unital algebra generated by K. It is commutative. By Lemma 3 there is $P \in GL_d(\mathbb{C})$ such that every

$$P^{-1}AP, \qquad A \in \mathcal{A},$$

is upper triangular. Write its diagonal entries as

$$\chi_1(A), \ldots, \chi_d(A).$$

Each $\chi_j : \mathcal{A} \to \mathbb{C}$ is linear and multiplicative. For every N,

$$P^{-1}S_N P = \sum_{n=0}^{N-1} P^{-1}A_n P$$

is upper triangular, with diagonal

$$\sum_{n=0}^{N-1} \chi_1(A_n), \ldots, \sum_{n=0}^{N-1} \chi_d(A_n).$$

Therefore

$$\det S_N = \prod_{j=1}^{d} \left(\sum_{n=0}^{N-1} \chi_j(A_n) \right). \tag{1}$$

Step 2: one joint character must vanish identically By the union bound and 1,

$$\mathbb{P}(\det S_N = 0) \leq \sum_{j=1}^{d} \mathbb{P}\left(\sum_{n=0}^{N-1} \chi_j(A_n) = 0\right). \tag{2}$$

For each j, the variables $\chi_j(A_n)$ are iid complex random variables. By Lemma 2, unless

$$\chi_j(A_0) = 0 \quad \text{almost surely,}$$

the j-th probability on the right-hand side of 2 tends to zero. If no $\chi_j(A_0)$ were almost surely zero, the entire right-hand side of 2 would tend to zero, contradicting

$$\limsup_N \mathbb{P}(\det S_N = 0) > 0.$$

Hence there exists a joint character χ among the χ_j such that

$$\chi(A) = 0 \qquad \text{for } \mu\text{-almost every } A.$$

Since χ is continuous, it vanishes on the support K. Step 3: a vanishing joint character yields a common kernel Let Ψ be the finite set of distinct diagonal characters occurring in the simultaneous triangularization. Choose $B \in \mathcal{A}$ such that

$$\psi(B) \neq \chi(B) \qquad \text{for every } \psi \in \Psi,\ \psi \neq \chi.$$

This is possible because the forbidden set is a finite union of proper linear hyperplanes in $\mathcal{A}$. Let W be the generalized eigenspace of B corresponding to $\chi(B)$:

$$W = \ker(B - \chi(B)I)^d.$$

It is nonzero because χ occurs among the diagonal characters. Since every element of $\mathcal{A}$ commutes with B, the space W is invariant under $\mathcal{A}$. By the choice of B, every diagonal character occurring on W is χ. Therefore, for every $A \in K$, all eigenvalues of $A|_W$ equal

$$\chi(A) = 0.$$

Thus each $A|_W$ is nilpotent. These restrictions form a commuting family of nilpotent matrices. By Lemma 3 they possess a common eigenvector $w \in W \setminus \{0\}$. Since every member is nilpotent, every corresponding eigenvalue is zero:

$$Aw = 0 \qquad \forall A \in K.$$

The vector w may initially be complex. Write $w = u + iv$, with $u, v \in \mathbb{R}^d$. Since every $A \in K$ is real,

$$Au = 0, \qquad Av = 0.$$

At least one of u, v is nonzero. Hence there exists a real nonzero vector x such that

$$Ax = 0 \qquad \forall A \in K.$$

In particular,

$$\mathbb{P}(A_0 x = 0) = 1.$$

This proves $1 \Rightarrow 2$. The implication $2 \Rightarrow 3$ follows from

$$S_N x = 0,$$

and $3 \Rightarrow 1$ is immediate. $\square$

Another broad sufficient hypothesis Part a also immediately extends to matrices which are positive with respect to a common deterministic inner product. Suppose there exists a deterministic positive definite matrix H such that

$$HA_0$$

is symmetric positive semidefinite almost surely. Then the matrices

$$B_n = HA_n$$

are iid positive semidefinite and

$$\det\left(\sum_{n=0}^{N-1} B_n\right) = \det(H)\det\left(\sum_{n=0}^{N-1} A_n\right).$$

Thus property (P) is unchanged. Part a gives $x \neq 0$ with

$$HA_n x = 0$$

almost surely for every n, and the invertibility of H gives

$$A_n x = 0.$$

So a common positive symmetrizer is another natural affirmative condition for part b.

1. A general finite-variance classification of property (P)

There is a useful structural theorem explaining both the positive result and the counterexample. Theorem 3 — affine-hull dichotomy Assume

$$\mathbb{E}\|A_0\|^2 < \infty.$$

Let

$$\mathcal{H} = \mathrm{aff}(\operatorname{supp}\mu)$$

be the affine hull of the support of the law of A_0. Then exactly one of the following occurs:

every matrix in $\mathcal{H}$ is singular, in which case

$$\mathbb{P}(\det S_N = 0) = 1 \qquad \forall N;$$

$\mathcal{H}$ contains an invertible matrix, in which case

$$\mathbb{P}(\det S_N = 0) \longrightarrow 0.$$

Consequently, under a finite second moment,

$$\boxed{(P) \iff \mathrm{aff}(\operatorname{supp}\mu) \subset \{A : \det A = 0\}.}$$

Proof Let

$$m = \mathbb{E}A_0$$

and write

$$\mathcal{H} = m + V,$$

where V is the direction space of the affine hull. If every member of $\mathcal{H}$ is singular, then

$$\frac{S_N}{N} \in \mathcal{H}$$

almost surely, since an affine subspace is closed under finite averages. Hence

$$\det \frac{S_N}{N} = 0,$$

and therefore $\det S_N = 0$ almost surely. Suppose now that the determinant does not vanish identically on $\mathcal{H}$. Define

$$Z_N = \frac{S_N - Nm}{\sqrt{N}}.$$

The multivariate central limit theorem, applied in the finite-dimensional vector space V, gives

$$Z_N \implies G,$$

where G is a centered Gaussian vector whose covariance is nondegenerate on V. Indeed, if a nonzero linear functional on V had zero variance, it would vanish almost surely on $A_0 - m$, contradicting the minimality of the affine hull. Expand the determinant along $m + V$:

$$\det(m + tz) = \sum_{r=0}^{d} t^r P_r(z),$$

where each P_r is a homogeneous polynomial of degree r on V. Since the determinant is not identically zero on $m + V$, there is a least r_0 for which

$$P_{r_0} \not\equiv 0.$$

Because

$$\frac{S_N}{N} = m + \frac{Z_N}{\sqrt{N}},$$

we obtain

$$N^{r_0/2} \det\left(\frac{S_N}{N}\right) = P_{r_0}(Z_N) + \sum_{r > r_0} N^{-(r-r_0)/2} P_r(Z_N).$$

The sequence (Z_N) is tight, so every term in the sum tends to zero in probability. Hence

$$N^{r_0/2} \det\left(\frac{S_N}{N}\right) \Longrightarrow P_{r_0}(G).$$

A nonzero polynomial on a finite-dimensional real vector space has a Lebesgue-null zero set. This follows by induction on the dimension and

Fubini's theorem: regarding the polynomial as a one-variable polynomial whose coefficients are polynomials in the other variables, outside the common zero set of the coefficients it has only finitely many roots. Since G has a density on V,

$$\mathbb{P}(P_{r_0}(G) = 0) = 0.$$

Portmanteau's theorem now gives

$$\begin{aligned}\limsup_{N\to\infty} \mathbb{P}(\det S_N = 0) &= \limsup_{N\to\infty} \mathbb{P}\left(N^{r_0/2} \det\left(\frac{S_N}{N}\right) = 0\right) \\ &\leq \mathbb{P}(P_{r_0}(G) = 0) = 0.\end{aligned}$$

This proves the dichotomy. □ The symmetric counterexample above has an affine hull equal to the two-dimensional linear space

$$\left\{ \begin{pmatrix} 0 & x & y \\ x & 0 & 0 \\ y & 0 & 0 \end{pmatrix} : x, y \in \mathbb{R} \right\},$$

which consists entirely of singular matrices but has trivial common kernel. This is precisely the obstruction in the general problem.

1. Boundary cases and the role of the hypotheses

For $d = 1$, part a reduces to a simple scalar statement. If $A_n \geq 0$, then

$$A_0 + \cdots + A_{N-1} = 0$$

if and only if every $A_n = 0$. Thus

$$\mathbb{P}(S_N = 0) = \mathbb{P}(A_0 = 0)^N.$$

A positive uniform lower bound is possible only when $A_0 = 0$ almost surely. If the A_n are positive definite almost surely, every finite sum is positive definite, so property (P) is impossible. The upper restriction $c < 1$ is immaterial: every proof only uses $c > 0$. Likewise, "for every N" can be weakened to

$$\limsup_{N\to\infty} \mathbb{P}(\det S_N = 0) > 0.$$

Independence, or at least an ergodic law of large numbers, is genuinely relevant. For example, choose one random variable $J \in \{1, 2\}$, uniformly, and set for every n

$$A_n = \begin{cases} \mathrm{diag}(0,1), & J = 1, \\ \mathrm{diag}(1,0), & J = 2. \end{cases}$$

The sequence is identically distributed, bounded, and positive semidefinite, and every partial sum is singular. But

$$\ker \mathrm{diag}(0,1) \cap \ker \mathrm{diag}(1,0) = \{0\},$$

so there is no deterministic common nonzero null vector. This sequence is perfectly dependent.

Conclusion Under the intended nonzero-vector interpretation, part a admits a complete and stronger answer:

$$\boxed{\text{For iid positive semidefinite matrices, } (P) \iff \exists\, x \neq 0 : \ A_n x = 0 \text{ a.s. for all } n.}$$

No moment assumption is actually required, and if the common kernel does not exist then

$$\mathbb{P}(\det S_N = 0) \to 0.$$

For part b, the conclusion is false even for bounded centered symmetric rank-$(d-1)$ matrices. A precise natural affirmative answer is obtained by requiring the support to be commuting:

$$\boxed{\mathbb{P}(A_0 A_1 = A_1 A_0) = 1 \quad \Longrightarrow \quad (P) \iff \exists\, x \neq 0 : \ A_n x = 0 \text{ a.s. for all } n.}$$

Under a finite second moment, property (P) has the additional exact algebraic characterization that the affine hull of the law's support consists entirely of singular matrices.

Formal verification. Lean source; online type-check.

28 Question Q885

Informal verdict. A — Complete and apparently correct. Confidence: **high (0.96)**.

Lean coverage. Generation 0 faithfully defines the half-open A_n, measurable ENNReal step function f, and actual I/J lintegrals on Real (equivalent to [0,infinity) because f vanishes below 1). It unconditionally proves rational divergence, sqrt2 irrationality/J-finiteness, and Liouville positivity/irrationality/J-divergence. It contains no exact general overlap or double-series/convergence-criterion layer.

28.1 Audited informal answer

Complete solution

1. Faithful restatement

For every positive integer n, let

$$A_n = [n, n + n^{-3}),$$

and define $f : \mathbb{R}_+ \to \mathbb{R}_+$ by

$$f(t) = \begin{cases} n, & t \in A_n, \\ 0, & t \notin \bigcup_{n\geq 1} A_n. \end{cases}$$

For $a > 0$, set

$$J(a) = \int_0^\infty t\, f(t) f(t/a)\, dt, \qquad D = \{a > 0 : J(a) < \infty\}.$$

The problem asks:
prove that, for every positive rational a,

$$\int_0^\infty f(t) f(t/a)\, dt = \infty,$$

and hence $J(a) = \infty$;
exhibit irrational a for which $J(a) < \infty$;
determine whether there exists an irrational a for which $J(a) = \infty$. Luc Pommeret+1

There is no apparent typo. The notation $[x, y[$ in the PDF means $[x, y)$. Whether $\mathbb{R}_+$ includes 0 is immaterial here, since f vanishes below 1. We shall prove substantially more: an exact convergence criterion in terms of one-sided rational approximations of order n^{-4}.

1. Preliminary facts

The intervals A_n are pairwise disjoint, because

$$n + n^{-3} \le n + 1,$$

with equality only for $n = 1$, where the right endpoint is excluded. For later use, define also

$$I(a) := \int_0^\infty f(t) f(t/a)\, dt.$$

All integrals below are understood as extended nonnegative Lebesgue integrals. Thus Tonelli's theorem is always applicable. Verification of the averaged integrability stated in the problem First,

$$\int_0^\infty f(t)\, dt = \sum_{n=1}^\infty n\, |A_n| = \sum_{n=1}^\infty \frac{1}{n^2} < \infty.$$

For $R > 0$, Tonelli's theorem and the substitution $s = t/u$ give

$$\begin{aligned}
\int_0^R u^{-2} J(u)\, du &= \int_0^\infty t f(t) \left(\int_0^R u^{-2} f(t/u)\, du \right) dt \\
&= \int_0^\infty t f(t) \left(\frac{1}{t} \int_{t/R}^\infty f(s)\, ds \right) dt \\
&= \int_0^\infty f(t) \int_{t/R}^\infty f(s)\, ds\, dt \\
&\le \left(\int_0^\infty f \right)^2 < \infty.
\end{aligned}$$

This justifies the contextual assertion that $J(a) < \infty$ for almost every $a > 0$.

1. Exact overlap formula

Fix $a > 0$. For $m, n \ge 1$, put

$$E_{m,n}(a) := A_m \cap a A_n,$$

where

$$a A_n = [an, an + a n^{-3}).$$

On $E_{m,n}(a)$,

$$f(t) = m, \qquad f(t/a) = n.$$

Since both families (A_m) and (aA_n) are pairwise disjoint, the sets $E_{m,n}(a)$ are pairwise disjoint. Therefore

$$I(a) = \sum_{m,n \geq 1} mn\,|E_{m,n}(a)| \tag{3.1}$$

and

$$J(a) = \sum_{m,n \geq 1} mn \int_{E_{m,n}(a)} t\,dt. \tag{3.2}$$

Let

$$r = \frac{m}{n}, \qquad x_+ = \max(x, 0).$$

Define

$$\Delta_{m,n}(a) = \begin{cases} \left(\min\left\{a,\, r^{-3} - n^4(a-r)\right\}\right)_+, & r \leq a, \\ \left(\min\left\{r^{-3},\, a - n^4(r-a)\right\}\right)_+, & r > a. \end{cases} \tag{3.3}$$

Then

$$|E_{m,n}(a)| = n^{-3}\Delta_{m,n}(a). \tag{3.4}$$

Indeed, suppose first that $r \leq a$, so $m \leq an$. The later left endpoint is an, and hence

$$\begin{aligned} |E_{m,n}(a)| &= \left(\min\{m + m^{-3}, an + an^{-3}\} - an\right)_+ \\ &= \left(\min\{m^{-3} - (an - m), an^{-3}\}\right)_+ \\ &= n^{-3}\left(\min\{r^{-3} - n^4(a-r), a\}\right)_+. \end{aligned}$$

The case $r > a$ is symmetric. When $E_{m,n}(a) \neq \varnothing$, it is an interval with left endpoint

$$\max(m, an) = n\max(r, a)$$

and length $n^{-3}\Delta_{m,n}(a)$. Consequently,

$$\boxed{I(a) = \sum_{m,n \geq 1} \frac{r}{n}\,\Delta_{m,n}(a)} \tag{3.5}$$

and

$$\boxed{J(a) = \sum_{m,n\geq 1} \left[r \max(r,a)\Delta_{m,n}(a) + \frac{r}{2n^4}\Delta_{m,n}(a)^2 \right].} \tag{3.6}$$

These formulas are exact. A useful convergence criterion If $E_{m,n}(a) \neq \varnothing$, then for some $t \in E_{m,n}(a)$,

$$an \leq t < m + m^{-3} \leq m + 1$$

and

$$m \leq t < an + an^{-3} \leq an + a.$$

Thus

$$an - 1 < m < an + a. \tag{3.7}$$

In particular, for every fixed n, only boundedly many m's can occur, with a bound depending only on a. Moreover, for all sufficiently large n,

$$\frac{a}{2} < \frac{m}{n} < 2a \qquad \text{whenever } \Delta_{m,n}(a) > 0. \tag{3.8}$$

Also,

$$0 \leq \Delta_{m,n}(a) \leq \max\{a, a^{-3}\}. \tag{3.9}$$

The second sum in (3.6),

$$\sum_{m,n} \frac{r}{2n^4}\Delta_{m,n}(a)^2,$$

therefore always converges: for each n there are only $O_a(1)$ relevant m's, while the summand is $O_a(n^{-4})$. On the other hand, by (3.8),

$$\frac{a^2}{2} \leq r \max(r,a) \leq 4a^2$$

for all sufficiently large relevant pairs. We have therefore proved the exact criterion

$$\boxed{J(a) < \infty \quad \Longleftrightarrow \quad \sum_{m,n\geq 1} \Delta_{m,n}(a) < \infty.} \tag{3.10}$$

The quantity $\Delta_{m,n}(a)$ measures the residual overlap after a rational m/n approximates a on the natural scale n^{-4}.

1. Part a: every positive rational gives divergence

Let

$$a = \frac{p}{q}, \qquad p, q \in \mathbb{N}^*.$$

For each $k \geq 1$, choose

$$m = pk, \qquad n = qk.$$

Then $m/n = a$, and hence

$$\Delta_{pk,qk}(a) = \min\{a, a^{-3}\} =: \delta_a > 0.$$

By the exact formula (3.5),

$$I(a) \geq \sum_{k=1}^{\infty} \frac{a}{qk}\delta_a = \frac{a\delta_a}{q}\sum_{k=1}^{\infty}\frac{1}{k} = \infty.$$

Thus

$$\boxed{\int_0^{\infty} f(t)f(t/a)\,dt = \infty \quad \text{for every } a \in \mathbb{Q}_{>0}.}$$

Furthermore, wherever $f(t)f(t/a) > 0$, one has $t \geq 1$. Therefore

$$J(a) = \int_0^{\infty} tf(t)f(t/a)\,dt \geq \int_0^{\infty} f(t)f(t/a)\,dt = \infty.$$

This proves part a. As a direct check, when $a = 1$, every pair $m = n$ contributes

$$n^2|A_n| = \frac{1}{n}$$

to $I(1)$, giving precisely the harmonic-series mechanism.

1. Part b: irrational examples with finite $J(a)$

We first give a general sufficient condition. Proposition 5.1 Suppose that there exist constants $c > 0$, $\sigma < 4$, and N_0 such that

$$\left|a - \frac{m}{n}\right| \geq \frac{c}{n^{\sigma}} \tag{5.1}$$

for every $n \geq N_0$ and every $m \geq 1$. Then $J(a) < \infty$. Proof Suppose $\Delta_{m,n}(a) > 0$, with n sufficiently large. If $r = m/n \leq a$, then by (3.3),

$$r^{-3} - n^4(a - r) > 0.$$

By (3.8), $r \geq a/2$, and therefore

$$0 < a - r < r^{-3}n^{-4} \leq 8a^{-3}n^{-4}.$$

If $r > a$, positivity of $\Delta_{m,n}(a)$ gives

$$0 < r - a < an^{-4}.$$

Thus in all cases,

$$\left|a - \frac{m}{n}\right| < C_a n^{-4}, \qquad C_a := \max\{a, 8a^{-3}\}. \tag{5.2}$$

But (5.1) and (5.2) are incompatible for sufficiently large n, since $\sigma < 4$. Hence only finitely many pairs (m, n) have nonempty overlap. Formula (3.2) then shows that $J(a) < \infty$. □ Explicit examples: quadratic surds Let $d \geq 2$ be an integer that is not a square, and set

$$a = \sqrt{d}.$$

For $r = m/n \leq 2a$,

$$\begin{aligned} |a - r| &= \frac{|a^2 - r^2|}{a + r} \\ &= \frac{|dn^2 - m^2|}{n^2(a + r)}. \end{aligned}$$

Since d is not a square,

$$dn^2 - m^2 \neq 0,$$

so its absolute value is at least 1. Since $a + r \leq 3a$,

$$\left|\sqrt{d} - \frac{m}{n}\right| \geq \frac{1}{3\sqrt{d}\, n^2}. \tag{5.3}$$

If $r > 2a$, then $|a - r| > a$, which is also bounded below by a positive multiple of n^{-2}. Consequently there exists $c_d > 0$ such that

$$\left|\sqrt{d} - \frac{m}{n}\right| \geq \frac{c_d}{n^2}$$

for all positive integers m, n. Proposition 5.1 applies with $\sigma = 2$. Therefore

$$\boxed{J(\sqrt{d}) < \infty \quad \text{for every nonsquare integer } d \geq 2.}$$

In particular,

$$\boxed{J(\sqrt{2}) < \infty.}$$

This answers part b, with infinitely many explicit examples.

1. Part c: an irrational with $J(a) = \infty$

We first isolate the general divergence mechanism. Proposition 6.1 Let $a > 0$. Suppose that there are positive integers m_k, n_k, with $n_k \to \infty$, such that

$$n_k^4 \left| a - \frac{m_k}{n_k} \right| \longrightarrow 0. \tag{6.1}$$

Then

$$J(a) = \infty.$$

Proof Put $r_k = m_k/n_k$. Condition (6.1) implies $r_k \to a$. If $r_k \leq a$, then by (3.3),

$$\Delta_{m_k,n_k}(a) = \left(\min\left\{a,\, r_k^{-3} - n_k^4(a - r_k)\right\}\right)_+.$$

If $r_k > a$, then

$$\Delta_{m_k,n_k}(a) = \left(\min\left\{r_k^{-3},\, a - n_k^4(r_k - a)\right\}\right)_+.$$

In either case,

$$\Delta_{m_k,n_k}(a) \longrightarrow \min\{a, a^{-3}\} > 0.$$

Thus the series $\sum_{m,n} \Delta_{m,n}(a)$ contains infinitely many terms bounded below by a fixed positive constant. It diverges, and (3.10) gives $J(a) = \infty$. ☐ An explicit irrational example Define

$$L := \sum_{j=1}^{\infty} 10^{-j!}. \tag{6.2}$$

For $k \geq 1$, set

$$q_k = 10^{k!}, \qquad p_k = q_k \sum_{j=1}^{k} 10^{-j!}.$$

Because $j! \le k!$, p_k is a positive integer, and

$$\frac{p_k}{q_k} = \sum_{j=1}^{k} 10^{-j!} < L.$$

The tail satisfies

$$0 < L - \frac{p_k}{q_k} = \sum_{j=k+1}^{\infty} 10^{-j!} < 2 \cdot 10^{-(k+1)!}. \tag{6.3}$$

Indeed, from the first omitted term onward, successive terms decrease by factors bounded above by 10^{-2}. Since

$$10^{-(k+1)!} = q_k^{-(k+1)},$$

we obtain

$$q_k^4 \left(L - \frac{p_k}{q_k} \right) < 2q_k^{3-k} = 2 \cdot 10^{(3-k)k!} \longrightarrow 0. \tag{6.4}$$

It remains to verify that L is irrational. Suppose instead that $L = r/s$ with positive integers r, s. Since $p_k/q_k < L$, the two rationals are distinct, and hence

$$L - \frac{p_k}{q_k} = \left| \frac{r}{s} - \frac{p_k}{q_k} \right| \ge \frac{1}{sq_k}. \tag{6.5}$$

But (6.3) gives

$$L - \frac{p_k}{q_k} < 2q_k^{-(k+1)} < \frac{1}{sq_k}$$

for all sufficiently large k, a contradiction. Thus L is irrational. Applying Proposition 6.1 with $m_k = p_k$, $n_k = q_k$, we conclude that

$$\boxed{J(L) = \infty, \qquad L = \sum_{j=1}^{\infty} 10^{-j!} \notin \mathbb{Q}.}$$

Hence the answer to part c is yes.

1. Final theorem

For the function f of Q885:

$$\boxed{\begin{array}{l}\text{(a) Every } a \in \mathbb{Q}_{>0} \text{ satisfies } \displaystyle\int_0^\infty f(t)f(t/a)\,dt = \infty, \text{ hence } J(a) = \infty. \\ \text{(b) Every } \sqrt{d},\ d \geq 2 \text{ nonsquare, satisfies } J(\sqrt{d}) < \infty. \\ \text{(c) The explicit irrational } L = \displaystyle\sum_{j=1}^{\infty} 10^{-j!} \text{ satisfies } J(L) = \infty.\end{array}}$$

More generally, the governing scale is quartic rational approximation:

$$\left|a - \frac{m}{n}\right| \asymp n^{-4}.$$

A Diophantine lower bound of order $n^{-\sigma}$ with $\sigma < 4$ forces only finitely many overlaps and hence finite $J(a)$, whereas infinitely many approximations satisfying

$$n^4 \left|a - \frac{m}{n}\right| \longrightarrow 0$$

force $J(a) = \infty$. Finally, the substitution $t = as$ yields the useful symmetry

$$J(a) = a^2 J(1/a),$$

so D is invariant under $a \mapsto 1/a$. Thus $1/\sqrt{d}$ gives further finite examples, while $1/L$ gives another irrational divergent example.

Formal verification. Lean source; online type-check.